GREEN CLIMATE FUND

Meeting of the Board
30 June – 3 July 2025
Port Moresby, Papua New Guinea
Provisional agenda item 11

GCF/B.42/02/Add.10

9 June 2025

Consideration of funding proposals – Addendum X

Funding proposal package for FP268

Summary

This addendum contains the following seven parts:

a) A funding proposal titled "Scaling-Up Resilience in Africa's Great Green Wall (SURAGGWA)";
b) No-objection letter issued by the national designated authority(ies) or focal point(s);
c) Environmental and social report(s) disclosure;
d) Secretariat's assessment;
e) Independent Technical Advisory Panel's assessment;
f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
g) Gender documentation.

Table of Contents

Funding Proposal

Programme title:	Scaling-Up Resilience in Africa's Great Green Wall (SURAGGWA)
Countries:	Burkina Faso, Chad, Djibouti, Mali, Mauritania, Niger, Nigeria, Senegal.
Accredited Entities:	**Food and Agriculture Organization of the United Nations (FAO)**
Nationally Designated Authorities (NDAs)	**Burkina Faso -** The Office of the Prime Minister of Burkina Faso **Chad** - Ministry of Environment, Fisheries and Sustainable development **Djibouti** - Ministry of Housing, Urban Development and Environment **Mali** - The Environment and Sustainable Deveopment Agency **Mauritania** - Ministry of Environment & Sustainable Development **Niger** - National Council of the Environment for Sustainable Development **Nigeria** - National Council on Climate Change **Senegal** - Ministry of Environment and Sustainable Development

GREEN CLIMATE FUND

Date of submission:	07/08/2023
Date of current submission	21/03/2025
Version number	V.6

Contents

Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the GCF Information Disclosure Policy, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

“FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]”

LIST OF ACRONYMS

AAD	Action Against Desertification
AE	Accredited Entity
AFR	African Forest Landscape Restoration Initiative
AMA	Accreditation Master Agreement
BCM	Billion Cubic Meters
CGs	Community Groups
CIU	Country Implementation Units
CRA	Climate Resilient Agriculture
CWA	Climatic Water Availability
DAE	Direct Access Entity
DEAL	Data for Environment, Agriculture and Land (FAO's Africa Open DEAL)
EE	Executing Entity
ESIA	Environment and Social Impact Assessment
ESMF	Environment and Social Management Framework
ESMP	Environmental and Social Management Plan
ESMS	Environmental and Social Management System
ESS	Environmental and Social Safeguard
EX-ACT	EX-Ante Carbon-balance Tool
FAO	Food and Agriculture Organization of the United Nations
FERM	Framework for Ecosystem Restoration Monitoring
FFS	Farmer Field Schools
FLO	Funding Liaison Officer
FP	Funding Proposal
GAP	Gender Action Plan
GCF	Green Climate Fund
GDP	Gross Domestic Produce
GEF	Global Environment Facility
GGW	Great Green Wall
GGWI	Great Green Wall Initiative
GGW-NA	Great Green Wall National Agency
GHG	Greenhouse Gas Emission
HDI	Human Development Index
IDP	Information Disclosure Policy
IDPs	Internally Displaced Persons
IFAD	International Fund for Agricultural Development
IGREENFIN	Inclusive Green Financing Initiative
IPCC	Inter-governmental Panel on Climate Change
IRMF	Integrated Results Management Framework
LTO	Lead Technical Officer (FAO)
MSME	Micro, Small and Medium Enterprise
NDA	National Designated Authority
NDC	Nationally Determined Contributions
ND-GAIN	Notre Dame Global Adaptation Index
NEXT	Nationally Determined Contribution Expert Tool
NPV	Net Present Value
NSC	National Steering Committee

NTFP	Non-Timber Forest Products
OPIM	Operational Partners Implementation Modality (FAO)
PA-GGW	Pan African Agency for the Great Green Wall
PMU	Programme Management Unit (regional level)
PSC	Programme Steering Committee
PTF	Programme Task Force
RAF	FAO Regional Office for Africa
RCP	Representative Concentration Pathway
RFP	Request for Proposals
RNE	FAO Regional Office for the Near East and North Africa
SDG	Sustainable Development Goals
SEPAL	System for Earth Observation, data access, Processing and Analysis for Land monitoring
SME	Small and Medium Enterprise
SPEI	Standardized Precipitation and Evapotranspiration Index
SURAGGWA	Scaling-Up Resilience in Africa's Great Green Wall
TOC	Theory of Change
UNCCD	United Nations Convention to Combat Desertification
UNFCCC	United Nations Framework Convention on Climate Change
USD	United States Dollar
WCGs	Women's Community Groups
WFP	World Food Programme

A. PROJECT/PROGRAMME SUMMARY

A.1. Project or programme	Programme	**A.2. Public or private sector**	Public

A.3. Request for Proposals (RFP)	*If the funding proposal is being submitted in response to a specific GCF Request for Proposals, indicate which RFP it is targeted for. Please note that there is a separate template for the Simplified Approval Process and REDD+.* Not applicable

A.4. Result area(s)			
	*Check the applicable GCF result area(s) that the overall proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of **GCF and Co-financers' contribution** devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.*		
		GCF contribution	**Co-financers' contribution**
	Mitigation total	**40%**	56 %
	☐ Energy generation and access	0 %	0 %
	☐ Low-emission transport	0 %	0 %
	☐ Buildings, cities, industries and appliances	0 %	0 %
	☒ Forestry and land use	40%	56 %
	Adaptation total	**60 %**	44 %
	☒ Most vulnerable people and communities	48 %	35 %
	☐ Health and well-being, and food and water security	0 %	0 %
	☐ Infrastructure and built environment	0 %	0 %
	☒ Ecosystems and ecosystem services	12 %	9 %

A.5. Expected mitigation outcome *(Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)*	65 million tons of CO_2 equivalent	**A.6. Expected adaptation outcome** *(Core indicator 2: direct and indirect beneficiaries reached)*	*Indicate total number of direct and indirect beneficiaries.*[1] 5,742,927	
			Indicate number of direct beneficiaries **1,881,002**	*Indicate number of indirect beneficiaries* **3,861,925**
			Indicate % of direct beneficiaries vis-à-vis total population *0.57%*	*Indicate % of indirect beneficiaries vis-à-vis total population* *1.2%*

A.7. Total financing (GCF + co-finance)	221,990,712 USD [2]	**A.9. Project size**	Medium (Upto USD 250 million)

[1] Annex 23 provides details on the E.6.
[2] Annex 17 gives the allocation by country.

A.8. Total GCF funding requested	150,000,000 USD *For multi-country proposals, please fill out annex 17.*

.10. Financial instrument(s) requested for the GCF funding	*Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.* ☒ Grant USD 150 million ☐ Loan Enter number ☐ Guarantee Enter number ☐ Equity Enter number ☐ Results-based payment Enter number		
A.11. Implementation period	*10 years*[3]	**A.12. Total lifespan**	*20 years*
A.13. Expected date of AE internal approval	*This is the date that the Accredited Entity obtained/will obtain its own approval to implement the project/ programme, if available.*	**A.14. ESS category**	*Refer to the AE's safeguard policy and GCF ESS Standards to assess your FP category.* B
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	**A.16. Has Readiness or PPF support been used to prepare this FP?**	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	**A.18. Is this FP included in the country programme?**	Yes ☒ No ☐
A.19. Complementarity and coherence	*Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.* Yes ☒ No ☐		
A.20. Executing Entity information	**Food and Agriculture Organization of the United Nations** **Government of Burkina Faso**, acting through Ministry of Environment, Water and Sanitation **Government of Chad**, acting through Ministry of Environment, Fisheries and Sustainable **Government of Djibouti**, acting through Ministry of Environment and Sustainable Development **Government of Mali**, acting through Ministry of Environment and Sustainable Development **Government of Mauritania**, acting through Ministry of Environment and Sustainable Development **Government of Niger**, acting through Ministry of Hydraulics, Sanitation and Environment **Government of Nigeria**, acting through Federal Ministry of Environment **Government of Senegal**, acting through Ministry of Environment and Sustainable Development		

A.21. Executive summary (max. 750 words, approximately 1.5 pages)

Climate change problem

i. The Sahelian countries[4] of the Great Green Wall (GGW) are among the lowest income countries and the most vulnerable to climate change. They ranked at the bottom of the Human Development Index in 2020 and a large share of their population lacks access to basic health care and education and suffers from food insecurity and degradation of the natural resources they depend on. Agriculture, livestock and forestry activities are the foundation of their economies and more than 70 per cent of rural communities depend directly on rainfed agriculture. The Sahel has experienced some of the most extreme climate events that occurred on Earth in the 20th century. Climate change is hitting the region hard, with higher temperatures and an increased frequency of extreme events, such as prolonged droughts and torrential rainfall. This climate change aggravates the anthropogenic degradation of ecosystems and the resulting depletion of vegetation and biodiversity has undermined livelihoods and increased vulnerability. This has severely impacted the Sahelian agro-silvo-pastoral landscapes, aggravating food and nutrition insecurity and compromising the sustainability of livelihoods.

Proposed interventions

ii. Scaling-Up Resilience in Africa's Great Green Wall (SURAGGWA) is a USD 221,990,212 million multi country programme to be funded by a USD 150 million grant from the Green Climate Fund (GCF) and USD 71,990,212 million co-financing to be provided by Governments counterparts through projects funded by Canada, the Global Environment Facility (GEF), the World Bank and the national governments of the eight countries. The SURAGGWA programme aims to take forward the objectives of the Great Green Wall (GGW), which is an African Union-led initiative that aims to transform Sahelian landscapes by restoring 100 million hectares of degraded lands, creating 10 million jobs and sequestering 250 million tons CO2e by 2030. The SURAGGWA programme is designed to work in synergy with major investments by GCF and other climate donors to facilitate a major paradigm shift by building ecological and climate resilience across eight Sahelian countries that are extremely vulnerable to climate change, namely, Burkina Faso, Chad, Djibouti, Mali, Mauritania, Niger, Nigeria and Senegal - while at the same time making a major contribution to climate change mitigation through carbon sequestration in restored land. The programme will also work closely with the Pan-African Agency for the Great Green Wall (PA-GGW) to improve restoration monitoring capacity across the SURAGGWA countries and the Pan-African Agency itself. SURAGGWA will build on FAO's previous support to the Sahelian countries through its highly acclaimed Action Against Desertification (AAD) initiative funded by the European Union and Turkey; and scale-up successful practices from these and other recent initiatives, including investments by GCF, GEF, IFAD and the World Bank, among others.

iii. SURAGGWA proposes to implement three key interventions that are designed to address the technical, organizational and financial barriers to alleviate climate change impacts and heighten resilience of local communities' livelihoods:

- scale-up successful land restoration practices in the eight above-mentioned Sahelian countries, using a diversity of native species to increase livelihood resilience while also sequestering carbon;
- support the development of climate-resilient, low emission non-timber forest product (NTFP) and fodder value chains benefiting vulnerable communities' livelihoods and food and nutrition security in the eight countries, creating incentives for sustainable management of restored land, and;
- strengthen the capacity of the Great Green Wall's regional and national institutions to coordinate implementation of restoration investments, monitor restoration results, and mobilize additional resources including through engaging with climate change adaptation and mitigation financing mechanisms.

iv. The SURAGGWA programme will work with community organizations to build ownership and sustainability and strengthen community awareness and capacity to engage in land restoration and NTFP value chain activities, with a special focus on women's participation. Given that there are a host of initiatives that are aimed at enhancing access to financial services in some of the selected countries, including the GCF-IFAD iGREENFIN partnership, the current programme will build linkages with these initiatives and with existing financial institutions and address the specific bottlenecks faced by the target groups in the area targeted by SURAGGWA. This includes establishing linkages with existing providers, building innovative models of value chain financing and de-risking lending to smallholders and producer organizations in the region.

[3] The SURAGGWA multi-country programme will be implemented over 10 years. However, at the country level the countries are expected to complete their activities within seven to eight years generally. Some countries are expected to start with a lag.
[4] The name Sahel refers to the semi-arid region stretching longitudinally from Senegal in West Africa to Djibouti on the East African coast and latitudinally from just north of the tropical forests to just south of the Sahara Desert (roughly between 10 ° and 20 °N). However, there is no universally defined list of countries of the Sahel.

Climate impacts/benefits

v. The SURAGGWA programme will make a direct contribution to the African Union's GGW targets, helping to realize the Green Climate Fund's pledge at the One Planet Summit (January 2021) and catalyze changes needed to achieve these ambitious targets. The proposed programme will restore approximately 1.273 million ha of degraded lands, benefiting 5.7 million vulnerable people in the GGW area[5] (Annex 23) while sequestering 65.1 million tons of CO_2 over the programme lifespan of 20 years.[6]

vi. The programme is aligned with the African Union Great Green Wall Initiative Strategy and Ten-Year Implementation Framework[7] with the following results: i) Support for developing countries through the restoration of 1.273 million hectares of degraded land ii) Support for developing countries that results in 1,489,660 direct beneficiaries adopting low-emission climate-resilient agricultural practices, and securing livelihoods while reconfiguring food systems; iii) Capacity building of DAEs participating directly in programme implementation (CSE Senegal, La Banque Agricole Senegal) and the National agencies for the Great Green Wall, improving their land restoration coordination, monitoring and reporting capacities; iv) Strengthening the capacities and provide TA for cooperatives and 1,795 SMEs; v) Supporting local and regional capacities for attracting investment in climate change mitigation and adaptation projects; vi) enhancing capacities and engagement with carbon markets for climate change mitigation and adaptation and vii) Improved local capacity for attracting investment in adaptation projects.

vii. The programme will make a significant contribution to: (i) the countries' Nationally Determined Contributions (NDC) to climate change adaptation and mitigation; (ii) the Bonn Challenge and associated African Forest Landscape Restoration Initiative (AFR100)[8] in committed[9] countries; (iii) the concerted UN system Action plan in support of the GGW, (iv) the UN Decade on Ecosystem Restoration (2021-2030) with GGW as one of its flagship projects, as well as associated Sustainable Development Goals, including SDG 1 which is aimed at eradicating poverty, SDG 2 which commits to zero hunger, gender equality (SDG 5), decent work and economic growth (SDG 8); Climate action (SDG 13); Life on land (SDG 15), and Partnerships (SDG 17).

viii. Investments in improving land and natural resource management get to the heart of what many communities in the Sahel view as both their most pressing human security concerns, and the factors that contribute to persistent conflict and competition. The programme's investments in land restoration and support to related smallholder value chains will thus contribute to climate change mitigation and adaptation, but also to community-based peace-building.

[5] Annex 23 provides details of the calculation approach for direct and indirect beneficiaries.
[6] The net result of the SURAGGWA programme is slightly lower at 65.09 million tCO2e, as it takes into account a slight increase in livestock emissions due to the fodder production contribution of the restored land, see Annex 22, carbon impact potential assessment
[7] 43834-doc-FINAL_REVISION-Integration_Document_AU_Strategy_23_Feb2024_Final.pdf
[8] https://afr100.org/
[9] AFR100 countries which are also GGW countries benefiting from SURAGGWA are the following: Burkina Faso, Chad, Mali, Niger, Nigeria, Senegal,

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

Climate change problem[10]

1. The eight countries of the Great Green Wall (GGW) included in the SURAGGWA[11] programme are among the world's poorest countries, and those most vulnerable to climate change.[12] These countries rely on agriculture (including cropping, livestock, forestry and fisheries) as a key source of their food supply, incomes and employment[13]. The GGW area hosts a large diversity of plant and animal species that help sustain human livelihoods and enhance their resilience, but a considerable share of it has been degraded and continues to lose its potential for providing the many ecosystem services that the communities and biodiversity conservation depend on. Rural livelihoods in the GGW zone are extremely vulnerable to climate change and climate variability due to the high exposure and to the countries' limited adaptive capacity.[14] Climate change has increased the frequency, intensity and duration of droughts and floods, rises in temperatures, desertification, water stress and soil erosion, all of which reduce agricultural productivity and food security (please see Annex 2 for full details of climate drivers).[15] Shocks from climate change and adverse weather conditions – resulting in the loss of assets, crops and livestock, disruptions in value chains and soaring food prices – which combined with the impact of COVID-19 in the last few years have pushed millions of smallholder farmers and pastoralists further into poverty.[16] Dryland ecosystems play a key role in adaptation when they are properly managed, restored and reduce pressure on vulnerable natural resources (water, soils, wood and non-wood products, fodder and pastoral land). Land restoration investments enhance climate change adaptation, as they increase soil fertility, preserve soil water holding capacity, reduce run-off, protect land and agricultural production from increasingly intense floods and droughts, increase infiltration of rainwater, and regenerate grasses for livestock.

2. During the last 40 years, the mean temperature rose in every country in the SURAGGWA programme, by a rate ranging from +0.14°C per decade in Djibouti to +0.45°C per decade in Chad, with the total temperature increase ranging from + 0.57°C to +1.81°C, over the entire period. During the same period, although most of the countries in the SURAGGWA multi-country programme showed constant or conflicting values in their annually accumulated precipitation trends and precipitation variability trends, annually accumulated evapotranspiration was observed to increase (ranging from +411 mm per decade in Niger to +814 mm per decade in Senegal), while climatic water balance 3, 6 and 12 month Standardized Precipitation-Evapotranspiration Index (SPEI) presented a general decrease (climatic water balance ranging from -406 mm per decade in Niger to -823 mm per decade in Senegal, and SPEI ranging between -0.0002 to -0.0010 per decade, depending on the accumulating period and the state.).[17]

Table 1 - Historical climate trends 1980-2020[18]

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[10] Apart from the climate analysis carried out specifically for the preparation of this programme (detailed in annex 2), this section also benefits from previous climate change analysis undertaken by development partners such as IFAD (GCF IGREENFIN) and the World Bank.
11 Field investments in climate change adaptation and mitigation will be made in eight GGW countries: Burkina Faso, Chad, Djibouti, Mali, Mauritania, Niger, Nigeria and Senegal. Only Nigeria and Senegal are currently classified as lower middle-income countries by the World Bank. In addition, under Output 3.1 FAO will collaborate with the Pan-African Agency for the Great Green Wall (PA-GGW) on a region-wide land degradation and restoration monitoring capacity building effort.
12 IPCC (2021), AR6 Climate Change 2021: The Physical Science Basis, Sixth Assessment Report, https://www.ipcc.ch/report/ar6/wg1/
13 UNCCD, "The Global Land Outlook, West Africa," Bonn, Germany, 2019.
14 See e.g. OECD (2018), Climate Change Profile: West African Sahel, https://www.government.nl/binaries/government/documents/publications/2019/02/05/climate-changeprofiles/West+African+Sahel.pdf. For a country-by-country characterization of rural production systems and their vulnerability, see Annex 2, pp. 162-197
15 OECD (2018), Climate Change Profile: West African Sahel, https://www.government.nl/binaries/government/documents/publications/2019/02/05/climate-changeprofiles/West+African+Sahel.pdf
16 IPCC (2019), "Food Security".
17 SPEI is a unitless standardized) index usually ranging between 2 and -2. As the absolute values are low, the variation trends are lower (in 1000th of a unit).
[18] Climate Research Unit (CRU) of the University of East Anglia (Harris et al. 2020).

Historical trends								
Index	Burkina Faso	Chad	Djibouti	Mali	Mauritania	Niger	Nigeria	Senegal
Accumulated precipitation	=	-16mm/dec.	-21mm/dec.	=	=	=	=	=
Intra-annual standard deviation of accumulated precipitation	=	=	=	=	=	+1.26mm/dec.	=	=
Inter-annual standard deviation of accumulated precipitation	=	-7.85mm/dec.	-7.39mm/dec.	-1.63mm/dec.	=	-2.18mm/dec.	-13.67mm/dec.	+3.10mm/dec.
Accumulated reference evapotranspiration	=	+423mm/dec.	+451mm/dec.	+477mm/dec.	+496mm/dec.	+411mm/dec.	=	+814mm/dec.
Climatic water balance	=	-439mm/dec.	-472mm/dec.	-484mm/dec.	-496mm/dec.	-406mm/dec.	=	-823mm/dec.
3-month accumulated SPEI	=	-0.0005/dec.	-0.0004/dec.	-0.0005/dec.	-0.0005/dec.	-0.0004/dec.	-0.0002/dec.	-0.0007/dec.
6-month accumulated SPEI	=	-0.0006/dec.	-0.0005/dec.	-0.0007/dec.	-0.0007/dec.	-0.0005/dec.	-0.0002/dec.	-0.0008/dec.
12-month accumulated SPEI	=	-0.0009/dec.	-0.0007/dec.	-0.0009/dec.	-0.0010/dec.	-0.0008/dec.	-0.0002/dec.	-0.0010/dec.

3. Mean and maximum temperature are both expected to increase for all countries in the SURAGGWA, under both the RCP4.5 and RCP 8.5 scenarios, confirming the trend observed in historical data. While some states are expected to experience stable or increased annual accumulated precipitation levels, projected trends expect an increase in intra-annual precipitation variability in the participating countries under the RCP8.5 scenario. This result seems to point to a change in the precipitation distribution, with fewer but more intense rain episodes.[19] Furthermore, a significant increase in water demand is expected in the next 40 years, due to the high and increasing reference evapotranspiration, leading to consistently low and decreasing climatic water balance under both scenarios. The consistent decline in SPEI across all states also points to an increase in the frequency and intensity of hydrological, agricultural, and geological[20] droughts.

4. The above-mentioned climate trends [21] tend to aggravate the negative impacts of poor agro-silvo-pastoral management practices and pressure on natural resources caused by population growth and other factors. Climate change aggravates land degradation and desertification caused by human mismanagement, posing a serious threat to crops, forests, rangelands, agriculture and livestock-dependent communities. Climate change is expected to have a significant influence on the ecology and distribution of tropical ecosystems, though the magnitude, rate and direction of these changes are uncertain.[22] The Sahel is one of the ecosystems that is the most sensitive to climate change and climate variability worldwide. With rising temperatures and increased frequency and intensity of droughts, wetlands and riverine systems are increasingly at risk of being disrupted and altered, with structural changes in plant and animal populations. Increased temperatures and droughts can also impact succession in forest systems while concurrently increasing the risk of invasive species, all of which affect ecosystems.

Table 2 - Projected climate trends under RCP4.5 scenario for water stress indices (2020-2060)[23]

[20] Geological droughts are based on the Annual SPEI and used to measure the state of water content in the aquifer.

[21] The current climate and historical and projected trends are further detailed in Annex 2, pp. 9-53.

22 T. M. Shanahan et al., “CO_2 and Fire Influence Tropical Ecosystem Stability in Response to Climate Change,” Nat. Publ. Gr., no. July, pp. 1–8, 2016, doi: 10.1038/srep29587.

[23] World Bank Climate Change Knowledge Portal (WBCKP) https://climateknowledgeportal.worldbank.org/ These numbers are estimations not definitive values.

Projected trends under the RCP 4.5 scenario								
Index	Burkina Faso	Chad	Djibouti	Mali	Mauritania	Niger	Nigeria	Senegal
Accumulated precipitation	=	=	=	− *-8mm/dec.*	=	=	=	− *-16mm/dec.*
Intra-annual standard deviation of accumulated precipitation	=	=	=	=	=	=	=	− *-1.66mm/dec.*
Inter-annual standard deviation of accumulated precipitation	=	+ *+2.13mm/dec.*	+ *+3.92mm/dec.*	=	− *-2.55mm/dec.*	− *-1.26mm/dec.*	+ *+3.66mm/dec.*	− *-7.19mm/dec.*
Accumulated reference evapotranspiration	+ *+265mm/dec.*	+ *+380mm/dec.*	=	+ *+569mm/dec.*	+ *+607mm/dec.*	+ *+403mm/dec.*	+ *+289mm/dec.*	+ *+720mm/dec.*
Climatic water balance	=	− *-369mm/dec.*	=	− *-576mm/dec.*	− *-602mm/dec.*	− *-400mm/dec.*	− *-268mm/dec.*	− *-730mm/dec.*
3-month accumulated SPEI	− *-0.0002/dec.*	− *-0.0004/dec.*	− *-0.0002/dec.*	− *-0.0005/dec.*	− *-0.0006/dec.*	− *-0.0004/dec.*	− *-0.0002/dec.*	− *-0.0005/dec.*
6-month accumulated SPEI	− *-0.0002/dec.*	− *-0.0005/dec.*	− *-0.0002/dec.*	− *-0.0006/dec.*	− *-0.0007/dec.*	− *-0.0005/dec.*	− *-0.0002/dec.*	− *-0.0006/dec.*
12-month accumulated SPEI	− *-0.0002/dec.*	− *-0.0006/dec.*	− *-0.0003/dec.*	− *-0.0007/dec.*	− *-0.0009/dec.*	− *-0.0007/dec.*	− *-0.0002/dec.*	− *-0.0007/dec.*

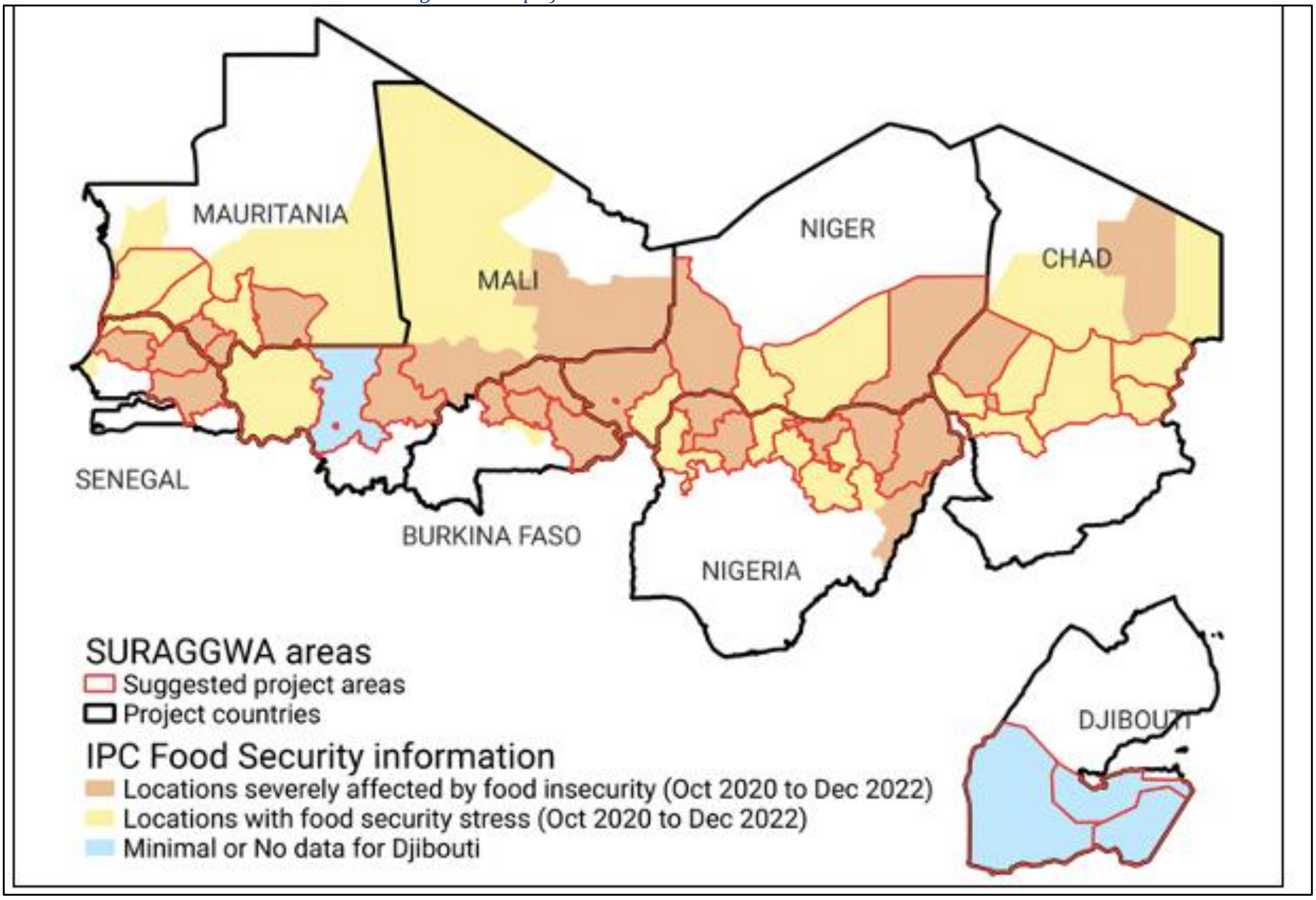
5. The Sahel is highly exposed to climate change, yet impacts vary across different regions. The Sahel will gradually become hotter, with some areas experiencing increased, but more erratic rainfall. Extreme weather events, including droughts and floods, are expected to intensify in this context. Niger and Mauritania, for instance are among the top 10 countries with the highest share of the populations affected by natural hazard-related disasters between 2000 and 2019 (standardized to population size).[24] Crop production is forecasted to drop by at least 20 per cent in the region, which will reduce food availability and farmers' incomes while increasing competition over diminishing natural capital, leading to greater instability and migration, especially of rural youth.[25] The climate models produced using the IFAD Climate Adaptation in Rural Development Assessment Tool[26] indicated that the production of the main crops in the targeted countries will be severely affected by future climate change: average millet production is predicted to decrease by 10 per cent, groundnut by 11 per cent, and rice by 8 per cent over the next 20 years. The tool shows that at median risk setting, the yields of a large range of 17 crops that were analyzed is likely to go down over the next 10 years. In Burkina Faso, the yields of ground nut and sugar beet will be the most affected, in Chad, wheat, soy and ground nuts will be most affected. In Mali, wheat yields, ground nut and maize will be the most affected over the next 10 years. In Mauritania, ground nuts, millet, wheat and sugar beet will be most affected. In Niger, wheat yields are expected to go down significantly every year followed by soy and groundnuts. In Nigeria, wheat is also expected to be impacted significantly. In Senegal there is expected to be a decrease in all major crops except cassava which is expected to increase every year (Annex 2 gives country level analysis). These yield reductions are expected to have negative impacts on 50 million people[27] in the region, including smallholders who already experience high levels of food insecurity and poverty.[28] According to the Ecological Threat Report 2021, climate change also has far-reaching implications for national and regional economic, political and social stability and security in the GGW area, which will increasingly transcend the capacity of each country to manage these issues alone.[29]

Topography, Demographics, Economy

24 CRED & UNDRR, "Human Cost of Disasters. An Overview of the Last 20 Years 2000–2019," 2020.
25 Cordaid (2020), Saving lives and reducing conflict in the Sahel region, Relief Web, https://reliefweb.int/report/burkina-faso/saving-lives-and-reducing-conflictsahel-
Region
26 Information on IFAD's Climate Adaptation in Rural Development (CARD) Assessment Tool is available from https://www.ifad.org/en/web/knowledge/-.
/publication/climate-adaptation-in-rural-development-card-assessment-tool
27 Marcus Arcanjo (2019), Risk and Resilience: Climate change and Instability in the Sahel, Climate Institute, 24 October, https://climate.org/risk-and-resilienceclimate-
change-and-instability-in-the-sahel/
28 FAO (2016), Strengthening resilience to food and nutrition insecurity in the Sahel and Western Africa: Good Practices Booklet, https://www.fao.org/3/i6263en/I6263EN.pdf
29 Institute for Economics & Peace (2021). Ecological Threat Report 2021: Understanding Ecological Threats, Resilience and Peace, Sydney, October,
http://visionofhumanity.org/resources

6. The topography of the Sahel is mostly flat. Altitudes mostly range between 200 and 400 m in elevation, while most of Senegal and the coastal areas of Mauritania lie at sea level. Some isolated mountain areas with an elevation between 1,000 and 2,000 m can be found in northern Niger, northern and eastern Chad. The Sahel is a semi-arid region separating the Sahara Desert to the north and tropical savannas to the south. It comprises several, quite diverse agro-ecological zones, which divide the Sahel into latitudinal sections, moving from a mainly hyper-arid climate in the north to a semi-arid climate in the south. In general, water resources in the Sahel are distributed unequally both in space and time. At national level, some countries, such as Nigeria, have abundant water resources, while others, such as Burkina Faso, have to deal constantly with water scarcity.[30]

Figure 1- Map of SURAGGWA Area and Selected countries



7. The population in the 8 countries where land restoration activities will take place under SURAGGWA, was estimated to be 330 million people[31], with 116 million in the Great Green Wall Area. Nigeria is the most populous country with 217 million inhabitants[32] or 66% of the total population in the 8 countries. Despite high mortality rates of children under five[33], the Sahel is home to some of the fastest growing societies in the world with high population growth rates between 2.6% and 3.8%.[34] Based on these trends, it is predicted that the population will have more than doubled by 2050.[35] Furthermore, the population in the Sahel is one of the youngest in the world, since 64.5% of the population are below the age of 25.[36]

30 Climate Risk Profile. Sahel. UNHCR.
31 https://data.worldbank.org/indicator/NY.GDP.MKTP.KD.ZG?locations=BF
32 https://https://www.worldometers.info/population/africa/western-africa/
33 The World Bank, "Mortality Rate, under-5," 2019. https:// databank.worldbank.org/reports. Sep. 20, 2021).
34 United Nations, "UNISS- United Nations Integrated Strategy for the Sahel," 2019.
35 The World Bank, 2020. https://www.worldbank.org/en/news/opinion/2020/12/16/the-world-bank-can-onlyaccomplish-its-mission-of-ending-extreme-poverty-in-africaby- prioritizing-the-sahel-region (accessed Sep. 20, 2021).
36 United Nations, "The Sahel: Land of Opportunities,"United Nations Africa Renewal, 2021. https://www.un.org/africarenewal/sahel (accessed Sep. 20, 2021).

8. The economic growth rate varies widely within the region and GDP per capita ranges from USD 414 in Niger to 2,787 in Djibouti.[37] In the Sahel, agriculture is the most important sector and provides livelihoods for the majority of the population. The agriculture sector's[38] contribution to Gross Domestic Product (GDP) varies from 2.8% in Djibouti to 52.7% in Chad. Apart from a recent general economic recession due to the COVID-19 pandemic, the region is marked by steady economic growth. The overall economic growth rate in the Sahel is higher than the continental average and is based on the export of primary goods.[39] The region offers great potential considering the abundant natural resources, such as oil in Chad and Nigeria, and great capacity for renewable energy.[40] Despite this potential, people in the Sahel are affected by multiple interrelated crises. According to the World Bank Poverty Headcount Ratio, large parts of the Sahelian populations live below 1.90 US dollars a day. Overall, nearly 50% of the population of the region live in extreme poverty and as a consequence, most countries in the Sahel rank among the lowest on the Human Development Index.[41]

Table 3-Basic Country Profile[42]

Table 3 Basic country profile								
Country	Pop 2020 in GGW area	GDP 2021 (growth %)	National GDP per cap (USD nominal)	agric % of GDP	% pop below nat'l poverty line	HDI Rank	HDI Index	GDI value Index
Burkina Faso	7.786.382	6,9	926	31,00	40	185	0,45	0,90
Chad	3.216.041	1,2	885	52,70	88	190	0,39	0,77
Djibouti	893.043	4,3	2.787	2,80	70	171	0,51	0,51
Mali	5.102.568	3,1	927	36,19	78	186	0,43	0,89
Mauritania	3.936.520	2,3	1.723	14,90	59	158	0,56	0,89
Niger	21.461.744	1,4	414	36,91	41	189	0,40	0,84
Nigeria	71.590.554	3,6	2.432	21,96	40	163	0,54	0,86
Senegal	2.773.096	6,1	1.428	16,90	67	170	0,51	0,87

Source: A range of sources including World Bank, WFP, FAO, etc.

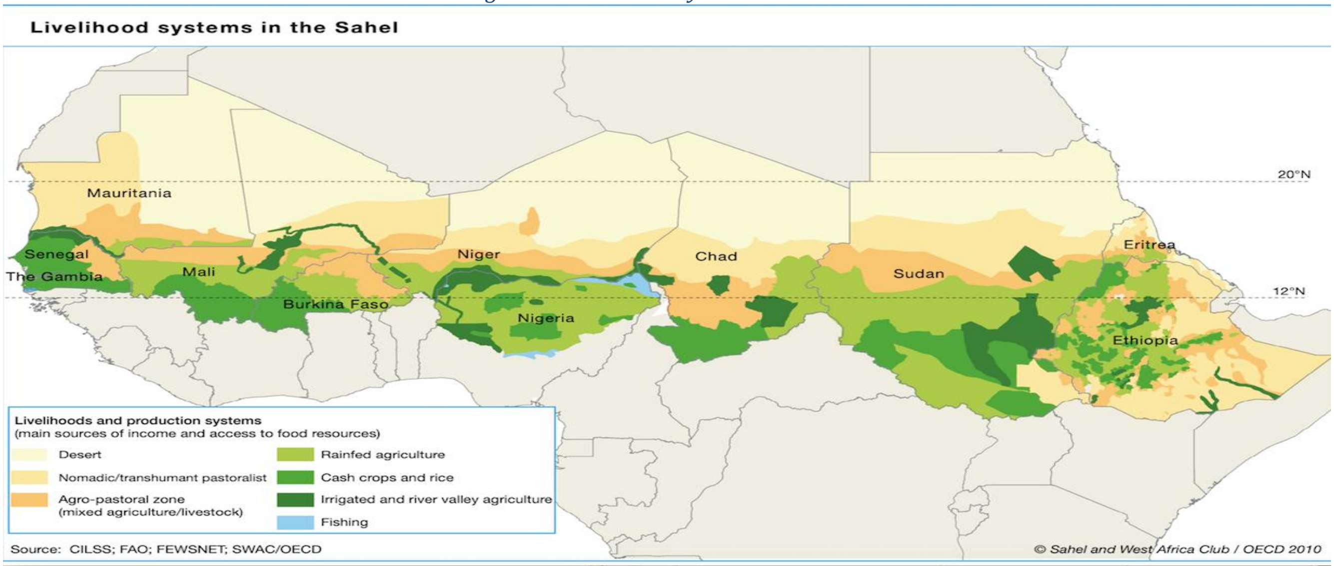
Vulnerability of the Livelihoods and Ecosystems to Climate Change

9. The livelihoods and ecosystems in the Sahel are extremely vulnerable to climate change. Agriculture, the dominant economic activity in the Sahel, is constrained by limited water availability, with soil moisture from precipitation often being the only available source of water.[43] Agriculture is further restricted by up to 50% of rainfall evaporating before crops can make use of it.[44] Dry spells and droughts are the primary reason for crop failures, especially when they occur during essential phases of crop growth.[45] Agricultural droughts is represented by the 6 month SPEI; the overall value is decreasing, highlighting an increase in the intensity and/or frequency of agricultural droughts (Annex 2). Use of irrigation facilities to bridge these gaps is rare due to insufficient surface run-off and thus insufficient water for extraction from wells or rivers, making irrigation impractical in large areas of the Sahel.[46] This makes agricultural production and subsequently food security in the Sahel highly dependent on rainfall patterns and thus vulnerable to climate change.[47] The northern Sahel is drier than its southern counterpart due to its proximity to the Saharan biomes, consisting mostly of marginal lands. It is thus dominated by transhumant pastoralism and some farming of drought-resistant sorghum and millet crops.[48]

37 https://www.google.com/search?q=Income+level+of+Burkino+Faso&sxsrfz
38 According to FAO's definition, agriculture includes the sub-sectors crop growing, livestock raising, fisheries and forestry.
39 A. Day and J. Caus, "Conflict Prevention in the Sahel: Emerging Practice across the UN," United Nations Univ., 2019.
40 A. Day and J. Caus, "Conflict Prevention in the Sahel: Emerging Practice across the UN," United Nations Univ., 2019.
41 UNDP, "Human Development Data Center," 2020. Accessed: Sep. 20, 2021. [Online]. Available: http://hdr.undp.org/en/data.
42 The table includes all 11 countries included in the SURAGGWA programme for all three components.
43 S. Doso Jnr, "Land degradation and agriculture in the Sahel of Africa: causes, impacts and recommendations," J. Agric. Sci. Appl., vol. 3, pp. 67–73, 2014, doi: DOI:10.14511/ jasa.2014.030303.
44 J. Rockstr.m and M. Falkenmark, "Agriculture: Increase water harvesting in Africa," Nat. 2015 5197543, vol. 519, no. 7543, pp. 283–285, Mar. 2015, doi: 10.1038/519283a.
45 J. Rockstr.m and M. Falkenmark, "Agriculture: Increase water harvesting in Africa," Nat. 2015 5197543, vol. 519, no. 7543, pp. 283–285, Mar. 2015, doi: 10.1038/519283a.
46 S. Doso Jnr, "Land degradation and agriculture in the Sahel of Africa: causes, impacts and recommendations," J. Agric. Sci. Appl., vol. 3, pp. 67–73, 2014, doi: DOI:10.14511/ jasa.2014.030303.
47 K. Sissoko, H. van Keulen, J. Verhagen, V. Tekken, and A. Battaglini, "Agriculture, livelihoods and climate change in the West African Sahel," Reg. Environ. Chang., vol. 11, no. S1, pp. 119–125, Mar. 2011, doi: 10.1007/s10113-010-0164-y.
48 S. Doso Jnr, "Land degradation and agriculture in the Sahel of Africa: causes, impacts and recommendations," J. Agric. Sci. Appl., vol. 3, pp. 67–73, 2014, doi: DOI:10.14511/ jasa.2014.030303.

10. Potential for agricultural expansion and intensification is low. Agriculture in the southern Sahel is more diversified, including subsistence crops such as groundnuts, cassava, cow peas and cash crops such as cotton, maize and sugar cane as well as extensively used forest and parkland areas comprised of natural vegetation. Soil quality is a prominent yield-limiting factor in the South Sahel where rainfall is higher. The soil has low availability of Nitrogen and Phosphorous and poor water-holding capacity.[49] As yields have stagnated in the past 30 years with only momentary periods of growth,[50] the needs of a rapidly growing urban population and an increased market orientation of agriculture have necessitated rapid expansion of croplands to keep pace with demand, leading to more marginal areas being put under production and unsustainable agricultural practices such as shortening of fallow periods, further degrading,. soils via soil-mining caused by over-cultivation.[51] Studies which have tried to isolate the impact of climate events such as previous droughts and trends from anthropogenic factors have found it difficult to separate the two.[52] Overgrazing and land clearing to promote growth of more palatable grasses caused by high livestock densities and the expansive needs of a rapidly growing population are common issues leading to land and soil degradation, as a quarter of the population in the northern Sahel is engaged in animal husbandry and livestock herds are an integral part of local cultural practices, determining social status and livelihoods.[53] As a result, more than 40% of the Sahelian population are food insecure[54] with low potential for agricultural intensification or adoption of adaptation strategies due to low incomes, low education levels and a lack of market infrastructure.[55]

Figure 2 - Livelihood Systems in the Sahel



11. In the absence of proposed interventions, the vulnerability of the area to both the anthropogenic pressures and climate risks will continue to grow. Rain-fed agriculture, in particular, is vulnerable to climate change. Repeated cycles of droughts and floods make it increasingly harder for the local population to sustain subsistence agricultural practices. Extreme weather events can lead to widespread crop failure and a reliance on food assistance programs. Additionally, the impact of climate change is straining the relationship of herders and pastoralists and thus also ethnic relations. For centuries, pastoralists have crossed the Sahel following seasonal patterns, which allowed them to feed their herds. The scarcity of water, pasture and fertile soil force people to migrate. Such displacement can lead to conflicts over land and resources between herders and farmers, which in turn further fuel displacement dynamics. Climate variability

49 S. Doso Jnr, "Land degradation and agriculture in tSahel of Africa: causes, impacts and recommendations," J. Agric. Sci. Appl., vol. 3, pp. 67–73, 2014, doi: DOI:10.14511/ jasa.2014.030303.
50 V. Bjornlund, H. Bjornlund, and A. F. Van Rooyen, "Why agricultural production in sub-Saharan Africa remains low compared to the rest of the world – a historical perspective," Int. J. Water Resour. Dev., vol. 36, no. sup1, pp. S20–S53, Oct. 2020, doi: 10.1080/07900627.2020.1739512.
51 K. Sissoko, H. van Keulen, J. Verhagen, V. Tekken, and A. Battaglini, "Agriculture, livelihoods and climate change in the West African Sahel," Reg. Environ. Chang., vol. 11, no. S1, pp. 119–125, Mar. 2011, doi: 10.1007/s10113-010-0164-y.
52 Environmental and Anthropogenic Degradation of Vegetation in the Sahel from 1982 to 2006. https://www.mdpi.com/2072-4292/8/11/948
53 Ibid.
54 Ibid.
55 V. Bjornlund, H. Bjornlund, and A. F. Van Rooyen, "Why agricultural production in sub-Saharan Africa remains low compared to the rest of the world – a historical perspective," Int. J. Water Resour. Dev., vol. 36, no. sup1, pp. S20–S53, Oct. 2020, doi: 10.1080/07900627.2020.1739512.

strongly interacts with other conflict drivers and thus can be difficult to distinguish. However, climate impacts represent a key factor with regards to displacement.[56]

Table 4. Summary of observed and projected climate hazards in the eight Sahelian countries and their impacts on the livelihoods of rural, agro-pastoralist communities.

Climate hazards (statistically significant changes)	Observed climate impacts	Future climate impacts
Temperature: Observed and projected increasing average temperatures. Observed increase in extreme heat events. Projected increasing extreme heat events across the region.	**Production and harvest:** Loss of productivity in crops and pastures due to higher evapotranspiration. Loss of agro-silvo-pastoral production systems due to repeated droughts.	**Production and harvest:** Future yield projections of IFAD CARD show decreased productivity in all rain-fed annual dryland crops. Pasture yields are also reduced, A projected increase in the number of consecutive extreme heat days in the Sahel is expected to cause substantial heat stress on livestock, with declines in milk, poultry, and egg yields.
Precipitation – Drought: High interannual rainfall variability, increasing anomalies in distribution of rainfall over the cropping season.	**Production and harvest:** Drought and moisture stress: major impacts on annual crops and on some fodder crops. Reduced annual growth rate. Reduced soil moisture, soil compaction. Water scarcity and quality degradation through higher concentration of contaminants. Loss of productivity in arable land and pasture.	
Precipitation – Heavy rainfall and flooding: Observed and projected increasing heavy rainfall events.	**Production and harvest:** Heavy rainfall events can cause major crop damage, especially in the most productive valley bottoms, as flood waters gather from surrounding degraded lands, where water infiltration capacity has been considerably reduced.	Higher temperatures and dry spells will further increase water demand of rainfed crops.

12. Investments in improving land and natural resource management get to the heart of what many communities view as both their most pressing human security concerns, and the factors that contribute to persistent conflict and competition. The programme's investments in land restoration and support to related smallholder value chains will thus contribute not just to climate change mitigation and adaptation, but also to peace-building through community-based approaches with a territorial lens.[57]

Related Programme and Project Interventions

13. There are a large number of programmes and projects designed to address the issues in the Sahel. To address these increasingly complex and interrelated challenges and support the transition towards climate resilient, low emission agriculture, the Great Green Wall Initiative (GGWI) was launched in 2007 by the African Union. The GGWI's initial objectives were to address land degradation, climate change adaptation and mitigation, and protect biodiversity and forests. Under the GGWI, environmental aspects and natural capital have been integrated into the development agenda and a multi-stakeholder dialogue has been established to ensure country ownership. It has also created opportunities for the deployment of scaled-up investments based on successful experiences on the ground. During the One Planet Summit in January 2021, the launch of the Great Green Wall Accelerator was announced to help meet the financial requirements of the programme. The Accelerator will be coordinated through the Pan Africa Agency for the Great Green Wall (PA-GGW) with support from UNCCD. It aims to facilitate collaboration among donors and stakeholders involved in the Great Green Wall Initiative and help all actors to better coordinate, monitor, and measure the impact of their actions. While funds have been committed, they have not been provided in a timely manner[58] and for those provided disbursement has been slow. To date, multilateral and bilateral organizations have raised more than USD 19 billion for this initiative.[59]

56 Internal Displacement Monitoring Centre, "2020 Global Report on Internal Displacement," Geneva, 2020. [Online]. Available: https://www.internal-displacement.org/ publications/2020-global-report-on-internal-displacement.

57 FAO is actively involved in UN-wide initiatives combining climate change and peace-building efforts, see e.g. the 2023 thematic review on climate security and peace building, https://www.un.org/peacebuilding/sites/www.un.org.peacebuilding/files/documents/climate_security_tr_web_final_april10.pdf

58 Green Wall Accelerator. United Nations Convention to Combat Desertification.

59 Green Wall Accelerator. United Nations Convention to Combat Desertification.

14. A host of multilateral development agencies and bilateral donors also have a long list of projects that are being implemented in the GGW Area. SURAGGWA will build on FAO's long-lasting support to Sahelian countries to achieve the goals of the GGW, including its Action Against Desertification (AAD) project funded by the European Union and Turkey; and to scale-up successful practices from the GEF/World Bank, African Development Bank (Programme for Integrated Development and Adaptation to Climate Change in the Niger Basin) and FAO's Operation Acacia supported by Italy for sustainable gum Arabic production and food security. A 2022 evaluation of AAD by FAO showed that the project was relevant and contributed to improving the productivity of agro-silvo-pastoral landscapes and the capacity to plan land restoration and manage forest and land resources.[60] Livelihood improvements were found in all countries, with a concrete positive incidence on household income, food security, crops and milk production, and community interactions. The project increased awareness and informed and supported policy makers developing intervention strategies that address desertification, land degradation and drought (D/LDD). SURAGAWA will also deploy the lessons learned from GEF projects to upscale across the landscape, and will seek co-finance to ensure complementary across the Sahel landscape.

15. FAO's contributions and involvements in GGW through the GCF, include the current SURAGGWA project, the FAO-GCF GAMS project in Sudan (2021) and four Readiness projects developed and implemented respectively in Burkina Faso (2020), Chad (2022), Niger (2022) and Senegal (2021). The SURAGGWA programme has been designed to ensure coordination and complementarity with other GGW interventions, the collection, exchange and application of lessons learnt in order to ensure a harmonized, replicable approach across the GGW region, which will increase the overall impact of this GCF transboundary programme and encourage collaboration among the countries in the region. As of October 2021, a range of projects are being implemented across the GGW countries by 17 different Accredited Entities. In response to demands of the countries in the region, the GEF has invested over US$800 million in grants and leveraged an additional US$6 billion from national governments, development partners and other multi-lateral donors for projects in the Sahel region. This financing has helped GGW member countries to boost their efforts to promote practices for improving crop and livestock productivity, restoring degraded parklands and strengthening resilience and adapting to climate change. Building on the GEF's experience with programmes using an integrated landscape approach, the GEF 7 "Harnessing the GGWI for a Sustainable and Resilient Sahel" project is currently being implemented by UNEP to engage with all GGW stakeholders in fostering dialogue with countries and fleshing out a longer-term vision. Some key initiatives are identified in the Table below.

Table 5 - List of Key Projects and Programmes

Initiative	Implementing entity	Achievements / Objectives
Large-scale Assessment of Land Degradation to guide future investment in SLM in the GGW countries (2019-2024)	UNEP Co-financing USD 12,171,000 GEF Project Grant USD 1,045,890	• Assess available tools and methodologies for scientific measurement of the ecological impacts of land degradation and SLM practices in support of future investment decisions in the GGW region.
The IFAD-led GEF Integrated Approach Pilot on Food Security (2017-2022)	IFAD US$900 million diverse sources.	• Five GGW countries (Burkina Faso, Ethiopia, Niger, Nigeria and Senegal (out of 12 countries) Projects to ensure the long-term sustainability and resilience of production systems, particularly in drylands. Institutional and policy strengthening
BRIDGES - Boosting Restoration, Income, Development, Generating Ecosystem Services (2017-2020)	FAO Turkey USD 3.6 million	• Restoration of 5,000 hectares of dryland forests and landscapes. • Reinforcing value chains of non-timber forest products. • Building GGW information and monitoring systems. • Compiling, managing and sharing knowledge and good practices, promoting communications and visibility
Action Against Desertification Program (2014-2019)	FAO EUR 19 930 479,	• Capacity-building for rural communities, government and NGO partners to create an enabling environment, large-scale restoration and SLM of degraded agro-silvo-pastoral lands, landscapes and forests. • Dissemination of good SLM practices. • Creation of income-generating activities (IGA) and employment opportunities in rural areas, through the sustainable production, processing and marketing of agricultural products and forest goods and services.

[60] https://openknowledge.fao.org/server/api/core/bitstreams/4f085418-811f-4ef3-a96f-99bf033f17a7/content

		• Farmer field schools and knowledge exchanges about the causes and the best ways to combat and prevent desertification
Closing the Gaps in the Great Green Wall: Linking Sectors and Stakeholders for Increased Synergy and Scaling-up (2016- 2019)	IUCN Co-financing USD 12,035,943 GEF Grant USD 1,726,400	• Improve the involvement of civil society organizations and vulnerable groups in the GGW Initiative and the national GGW agencies.
The Sahel and West Africa Programme in support of the GGW (2013–2019)	World Bank GEF USD 80.4 million LDCF $20.4 million SCCF $4.6 million from the Special World Bank investments $1.8 billion in cofinancing in 12 countries.	• 12 sustainable land and water management country projects • Training for farmers on SLM practices and improved agricultural technologies One regional project on coordination, M&E and knowledge management
Local Environmental Coalition for a Green Union (2014-2018)	UNCCD EUR 7 million.	• Micro-projects implemented in 23 communities in five countries (Mali, Niger, Burkina Faso, Chad, and Senegal) Regional level capacity building activities on SLM and innovative financing complemented the projects on the ground.
Inclusive Green Financing Initiative (IGREENFIN I and II): Greening Agricultural Banks & the Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall (GGW) countries	GCF, EUR 177.4 million (Phase I) Phase II at CN stage	• Build and scale up the resilience and adaptive capacity of 378,600 smallholder farmers organized around 2500 farmers' organizations (FOs), cooperatives and 1500 micro, small and medium-sized enterprises (MSMEs) by removing barriers to farmers' access to financial and non-financial services that support the adoption of best climate change adaptation and mitigation practices and solutions.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

16. The Sahel's Dryland ecosystems consist of forests, woodlands, wetlands, grasslands and parklands which include arable lands and silvo-pastoral areas with scattered trees. The GGW area hosts a large diversity of plant and animal species that help sustain human livelihoods and enhance their resilience, but a considerable part of it has been degraded and continues to lose its potential for providing the many ecosystem services that the communities and biodiversity conservation depend on. Land restoration generates major climate change adaptation benefits, through the range of ecosystem services (provisioning, regulating, cultural and supporting) they provide. These include important non-timber forest product (NTFP) commodities that generate additional sources of income, whether seasonally or in poor agricultural years, thus offering a safety-net to sustain livelihoods and enhancing resilience to climate variability and climate change.[61] These ecosystems play a key role in adaptation when they are properly managed, restored and reduce pressure on vulnerable natural resources (water, soils, wood and non-wood products, fodder and pastoral land). Land restorative activities increase soil fertility, preserve soil water holding capacity, reduce run-off, protect land and agricultural production from increasingly intense floods and droughts, increase infiltration of rainwater, and regenerate grasses for livestock. At the same time, restoration of degraded dryland ecosystems generates considerable climate change mitigation benefits from increased soil organic matter, as well as from below and above ground biomass.

17. The Sahelian region is experiencing the full impact of climate change, which has dramatic consequences on the eco-systems and people's livelihoods. Temperatures in the region are rising 1.5 times faster than in the rest of the world.[62] In the Sahel, droughts are becoming more and more intense. Climate change is also causing an increased frequency of heavy rains and violent thunderstorms. However, the land is too dry to absorb the sudden rush of water leading to destructive floods and numerous flooding episodes such as the ones observed in Mali and Niger in 2019. These events are degrading the natural resources that are essential to the agro-silvo-pastoral livelihood systems that underpin the economy in much of the area. Land degradation has adverse impacts on agricultural productivity, food insecurity, and rural livelihoods, and the connections to other environmental problems. The degradation of ecosystems and consequent depletion of vegetation and biodiversity has also impacted the availability of water and exposed the soils to erosion. The cycle of droughts and floods has serious consequences for the populations who rely mainly on agriculture and livestock. Insufficient rainfall means that crops fail or have low yields, while livestock struggle to find drinking water and forage which are their main source of feed. This has aggravated food and nutrition insecurity and compromised the sustainability of livelihoods. The Inter-governmental Panel on Climate Change (IPCC) predicts that agricultural yields will fall by 20% per decade by the end of the 21st century in some areas of the Sahel. The competition for the use of the scarce natural resources in the region has been aggravated due to climate change and has become a source of growing conflict between farmers and herders. In some areas, the scarcity of natural resources further aggravates conflict and is leading to migration.[63] The governance mechanisms are struggling to deal with the many challenges that confront these fragile environments.

Key Barriers and Constraints

18. **Constraints Faced by Smallholder Farmers and Herders:** Livelihoods in the Sahel countries heavily depend on the soil, water, and vegetation resources. The state of these resources has been steadily deteriorating in many areas as a result of expanding human settlement and demand for more food and fuelwood. Frequent droughts accompanied by unplanned, unsustainable, and poorly managed use of land and water (surface and groundwater) have, along with natural variability, caused the drying up of national and transboundary rivers and lakes, while wind erosion has removed valuable top soil. As a result, the natural vegetation of most of the Sahel

[61] Sacande, M. & Parfondry, M., 2018. Non-timber forest products: from restoration to income generation. Rome, FAO. 40 pp. License: CC BY-NC-SA 3.0 IGO. https://www.fao.org/documents/card/fr/c/CA2428EN/
[62] The Sahel in the Midst of Climate Change. OCHA Services. March, 2020
[63] T Julie Mayans. The Sahel at the Heart of Climate Change Issues. Food Security And Livelihood Adviser in Solidarités International

has been dramatically altered and the ecosystem degraded. Threatened by climate change, displacement and conflict, communities in the Sahel face threats to their traditional livelihoods. While some of the threats are posed by state policies and legislation which lead to agriculture encroachment and marginalization of pastoral communities, others are due to the changing rainfall patterns. In a region highly dependent upon subsistence agriculture and pastoralist livelihoods, climate variability and environmental degradation have made traditional livelihoods difficult to sustain. Consequently, migration, livelihood adaptation, social unrest, and political instability emerge from the ecological challenges the Sahel is facing. The most vulnerable households have limited ability to adapt to climate risks and rehabilitate degraded lands or adapt to climate factors that have eroded the productivity of their lands. The common property nature of the eco-systems leads to lack of clarity of use and tenure rights leading to lack of investment and overuse. Poor communities in the Sahel do not have the incentives or adequate resources to invest in rehabilitating the ecosystems which could provide them invaluable services in terms of water, food, medicine, forage and livelihood security or on their own small plots of land to increase their productivity and incomes. Local communities have limited access to seeds, inputs, mechanized equipment, processing inputs, adaptive technical capacity or financial resources. There is limited interaction with markets and aggregation through associations is too weak to capitalize on opportunities for enhancing and diversifying incomes. Gender inequalities within the Sahel pose a very real challenge for adaptation and resilience strategies.[64] While women play an important role in natural resource management, they have limited knowledge, decision-making authority or capital to increase their incomes or well-being.

Table 6 - Key Barriers and Underlying Factors

Main barriers	Underlying Factors
Lack of incentives for local communities to manage natural resources sustainably	• Lack of formally recognized use rights over communal lands leading to their overuse. • Increased pressures on dryland ecosystems due to increase in human and livestock populations, aggravated by climate change, and by the increased inter-community conflicts[65] the combination of these two may cause. • Abandonment or loss of ancestral techniques, resulting from the above
Rural communities have limited organizational, managerial, and technical capacity to engage in land restoration and associated NTFP value chains	• Rural communities in the Sahel tend to have low levels of income and literacy/numeracy, high rates of poverty and limited opportunities for employment and income generation. • There is a lack of skills of farmers including both men and women to adapt to climate risks by using more adaptive crop varieties and production methods, including soil and crop management practices that would increase their resilience.
Women have limited knowledge, decision-making power and capital to enable them to invest in land restoration and associated NTFP value chains	• Patriarchal norms that have led to low levels of participation in decision-making, limited asset ownership, low levels of education attainment, poor health status and limited economic participation and decision-making. • Insufficient consideration of gender in rural development efforts.
Poor marketing conditions and value-chain linkages for small-scale NTFP producers	• Weak bargaining and negotiating power of small producers and collectors due to low volume and limited capacity to aggregate production. • Limited opportunities for the formal private sector to invest in rural areas of the Sahel due to fragmented and low production, large number of small farmers with weak organization capacity to aggregate their produce and provide opportunities for economies of scale or reduction of transaction costs for the private sector.
Lack of adequate financial products and services for smallholder NTFP producers and SMEs	• Low risk appetite for lending to sectors with high risk and limited mechanisms for reducing risks especially in view of growing climate threats and crop failure and losses.

[64] McOmber, C. (2020), "Women and climate change in the Sahel", *West African Papers*, No. 27, OECD Publishing, Paris, https://doi.org/10.1787/e31c77ad-en.
[65] Inter-community conflicts are one of the main risks to the success of the programme, so conflict assessment, prevention and management are a key element of programme design and implementation mechanisms chosen, especially for Component 1, land restoration (see also Annex 6, Environmental and Social Management Framework, especially its Appendix 9, "Conflict assessment, prevention and management".

	• Fragmented nature of agriculture and NTFP production with high transaction costs due to the large number of unorganized smallholders. • Lack of appreciation of the market potential to innovate for these special areas which have are perceived to yield low returns.
Weak institutional capacity of GGW institutions to monitor, evaluate, scale up and mobilize funding for land restoration	• Lack of mainstreaming land degradation issues into national policies and frameworks: land restoration tends to be framed as an "environmental protection" activity, rather than an economic activity that can support climate-resilient rural development. • Technical monitoring capacity using advanced tools is a relatively new field for which technical expertise is missing in the public sector.
Absence of clear policies and limited national/regional capacity to engage with carbon finance	• Public sector institutions are structured to work in silos, on their own narrow mandates, whereas land use carbon finance requires a cross-sectoral approach • Analytical and advocacy capacity for making the case for land restoration as a climate change mitigation investment is limited in key sector agencies

Source: Review of literature and discussions during Stakeholder consultations.

19. **Constraints faced by the Private sector**: The formal private sector has generally been missing from these degraded landscapes because of the limited demand for inputs and technologies that they can supply and the fragmentary and scattered nature of products that they can collect and market. Apart from a few large commercial forestry or agribusiness/food operations, most conventional restoration projects generate returns that are too low to attract private investors. There is currently lack of private sector rental markets for scaling up innovative restoration techniques. However, where there is potential for organizing communities and enhancing demand for land restoration and rehabilitation of the eco-systems, private sector has been known to establish well-functioning rental markets for soil conservation and water savings techniques. SURAGGWA plans to facilitate the participation of the private sector through encouraging rental markets that can provide technology for up scaling the restoration techniques and seed supply that will be required for sustainable land management techniques. Private sector is reluctant to market local products if they are unable to secure the volumes in sufficient quantity that can make the enterprise economically feasible for them. By organizing communities to collect and aggregating local products to a scale that becomes attractive for the private sector, SURAGGWA can play a pivotal role in marketing a range of non-timber forest products with a potential for generating foreign exchange, employment and incomes for women and youth.

20. **Constraints Faced by Public Sector**: An overarching issue appearing in most GGW countries is weak governance and ability to deal with environmental change, which is a main barrier to implementation. Institutional challenges include the lack of high-level political support for the environmental policy agenda from the governments of the GGW member states and weak organizational structures and processes for the implementation of environmental projects, development initiatives or programmes. GGW institutions have weak ability to finance, implement, monitor, evaluate or scale up successful experiences implemented by development agencies. Even where national governments in the GGW have put in place mechanisms for decentralization and coordination they have not adequately resourced the institutions with staff or funds. The lack of managed knowledge, information sharing and coordination mechanisms at the national and regional level leads to insufficient coordination and collaboration between GGW countries, project developers and donors. This is especially important for taking advantage of lessons learned, innovations and success stories, whose replication and scaling up could accelerate and ensure an efficient expansion of the GGWI's activities. Another key barrier in the public sector is the lack of a system to monitor and report on GGW activities on the ground. M&E expertise is absent in general, which hampers the establishment of proper M&E systems at the project and national levels and across the GGW Initiative as a whole. It also results in shortcomings in documentation and hinders the sharing of lessons learned, which is key to avoiding negative developments and capitalize positive results achieved by the projects under the GGW.

21. The objective of the current multi-country programme titled ***Scaling-Up Resilience in Africa's Great Green Wall*** **(SURAGGWA)** is to promote a major paradigm shift by removing barriers that vulnerable households who depend on sylvo-pastoralism, agro-forestry or pastoralism face in becoming more resilient to major climate change

impacts, while ***mitigating emissions*** in restored landscapes. More specifically, the programme seeks to address the barriers identified above in the following manner:

Table 7 - Main Barriers and SURAGGWA Interventions

Main barriers	SURAGGWA Intervention
Lack of incentives for local communities to manage natural resources sustainably	A consolidated approach that organizes and strengthens local communities to make investments in land restoration and use natural resources sustainably, and that creates effective mechanisms for reducing and managing inter-community conflicts where necessary.[66]
Rural communities have limited capacity to engage in land restoration and NTFP value chains	All programme investments in participatory land restoration and NTFP value chain have a strong capacity-building focus
Women have limited skills, voice, decision-making power and capital	Gender issues are mainstreamed in all programme activities
Poor marketing conditions and value-chain linkages for small-scale NTFP producers	Through improving the capacity of NTFP producers groups, the programme aims to reposition these groups in the NTFP markets
Lack of adequate financial products and services for smallholder NTFP producers and SMEs	By facilitating direct interaction between NTFP producer groups and financial services providers, and by working in synergy with IFAD iGREENFIN, the programme will ensure better service provision for rural smallholder producers.
Weak institutional capacity of GGW institutions	Introduction of tried and tested, user-friendly land restoration monitoring technologies will help GGW institutions make the case for restoration and improve coordination.
Absence of clear policies and limited national/regional capacity to engage with carbon finance	Collaboration with both national and regional climate finance institutions will help to mainstream land use carbon in policy frameworks and financing mechanisms.

22. While there have been various efforts launched in the GGW area in the past by a host of development agencies which have promoted sustainable land use, these efforts have been limited in scope due to the limited incentives to engage local communities or the private sector. The innovative aspect of the SURAGGWA approach is to provide strong incentives for communities to benefit from Non-Timber Forest Products for additional income sources as well as a source of food and medicine that are available in these areas and the opportunities to grow fodder for their animals, promoting livelihood diversification. In addition, SURAGGWA proposes a unique approach by emphasizing the inclusion of the private sector through providing equipment to MSME and encouraging the establishment of rental markets for equipment needed for large scale land restoration, supply of seed and the services needed along the NTFP value chains. A critical element of the SURAGGWA approach is providing strong incentives for the participation of local communities in managing the degraded common lands through secure use rights that will give them an opportunity to use the fodder and NTFPs grown on these lands, development of degraded farmlands to increase their production and the engagement with the private sector to facilitate their links for input supply and links to the markets. These elements were missing in earlier projects such as the Action Against Desertification (AAD). SURAGGWA will undertake the following specific strategies in the GGWA;

- Increase sequestration and reduce emissions of GHG through sustainable land-use management and large-scale restoration of dryland forests and agro-silvo-pastoral systems, (mitigation) and restore degraded farmlands and landscapes (adaptation);
- Enhance resilience of the livelihoods of vulnerable communities to the impacts of a changing climate on productive landscapes, by forming non timber forestry products low-emission value chains;
- Strengthen the regional and national GGW institutions and disseminate best practices, early climate warning information and lessons learnt from and up scaling restoration experiences.

[66] Inter-community conflicts are one of the main risks to the success of the programme, so conflict assessment, prevention and management are a key element of programme design and implementation mechanisms chosen, especially for Component 1, land restoration (see also Annex 6, Environmental and Social Management Framework, especially its Appendix 9, "Conflict assessment, prevention and management".

23. The Theory of Change (TOC) of SURAGGWA is that the participation of local communities along with strengthened technical and institutional capacities are *sine qua non* conditions for successful adaptation and mitigation efforts in the Drylands. The design of the SURAGGWA programme builds on the experience of the FAO-implemented Action Against Desertification Project (AAD) which has demonstrated that local communities can be motivated to invest their labour and resources (including local knowledge about appropriate species and ways to ensure their regeneration) to achieve sustainable land restoration. The approach promoted by the programme, including the use of mechanical ploughs and animal traction, accompanied by training in a range of land rehabilitation, water infiltration and soil improvement techniques, including the production and collection of restoration seeds and seedlings will contribute to long-term uptake of such practices. The AAD experience has also demonstrated that communities have a strong vested interest in preserving and restoring communal and forest lands by growing forage crops and providing access to non-timber forest products (NTFP) that poor communities can use to enhance their livelihoods through direct consumption and through income generation. Assisting communities with rehabilitating farmlands can also build resilience to climate change and provide vulnerable households with more sustainable livelihoods. The private sector (including local and international firms and farmer organizations) can play an important role by helping to build the value chains for input supply and product marketing both locally and regionally. The theory also contends that with targeted support to the Pan-African and national institutions, they can play a more effective role in coordination, resource mobilization and monitoring and evaluation.

24. The Theory of change of SURAGGWA is based on the following postulate; **IF** the capacity of smallholder rural communities and their organizations is enhanced to engage in restoration of degraded community and family lands and improve market linkages and access to financial services, along with strengthened GGW institutional capacities, **THEN** rural livelihoods and ecosystems will become more resilient to climate change, while reducing CO_2 emissions **BECAUSE** of diversified, climate-smart livelihood options resulting from agro-ecology/agroforestry practices, sustainable production and marketing of NTFPs, along with increased capacity of GGW institutions and providing long-term incentives to rural communities to engage in restoration and sustainable management of restored lands, thus meeting both climate change adaptation and mitigation objectives over the long-term.

25. The concerted impact of the investments that are being proposed under the SURAGGWA programme will shift the development pathways towards a low emission and climate resilient direction because it will enable local governments and communities to become aware of the potential of the investments to protect their natural environment as a valuable asset that can become a sustainable source of livelihoods, generating income and employment. The pathways being designed under SURAGGWA form the core of its TOC and are outlined below:

26. Pathways *to Eco-System and Landscape Restoration*: Local governments and communities do not have the resources to make upfront investments in the vast agro-silvo-pastoral lands which are highly degraded due to a combination of climate change impacts and poor land use practices. The farming and herding communities are scattered and not well organized to develop and sustainably use the common lands. There is limited financial and technical capacity of local communities to participate in the restoration of degraded range lands and little incentive to do so without clarification of the use rights and security of tenure. The smallholders are also not able to use their individual farm plots effectively due to lack of water, agricultural inputs and adaptation techniques in the face of the harsh climate factors and limited resources. However, the pathway to rehabilitation of the natural resource base is premised on the willingness of the communities to undertake this task and the availability of technology and technical solutions which can assist in restoration of the lands at scale.

27. The experience of AAD and a range of other investments in large scale land restoration has shown that the local communities are willing to invest in restoring degraded lands and can be organized to develop and use them on a sustainable basis. Furthermore, the local authorities can work with local communities in a participatory manner to identify areas that need rehabilitation and grant clearly established use rights. This requires access of the local authorities and communities to modern equipment for mechanized ploughing on the vast lands, training in techniques which are able to harvest the infrequent and unpredictable rainfall and implement effective land conservation, soil management and afforestation techniques. The communities can also be organized to collect the seeds of appropriate species that can preserve or enhance the biodiversity of the eco-systems and engage in seed and seedling production as productive income generating activities. These investments reduce desertification, preserve the eco-systems by scaling-up successful restoration practices with well-adapted native

species for increased provision of ecosystem services, including biodiversity conservation. In addition to these climate change adaptation benefits, the regeneration of these areas can make a significant contribution to carbon sequestration, reducing GHG emissions. The links of these communities to the growing carbon markets can provide an additional incentive to increase the biomass in the area through direct payments for carbon off-sets. It will be important to build these links over time and SURAGGWA, working in synergy with other investments such as the GCF-funded iGREENFIN initiative, is can help to strengthen the institutional arrangements necessary for the purpose.

28. *Pathways to Enhancing Livelihoods:* The development of degraded lands can provide an important source for enhancing local livelihoods. The increased utility of the degraded lands after rehabilitation provides a real incentive to local communities to manage the lands and sustain the investments. The experience from AAD shows that local communities are involved with the collection of fodder grasses and Non-Timber Forest Products (NTFP)[67] from these lands which can be an important source of income. The most immediate benefit that communities derive through the protection of these lands is the generation of grasses which are an important source of fodder for livestock. Livestock is an important element of livelihoods in the Sahel and the increased availability of grasses and tree fodder can help to enhance productivity, yields and improve animal health. The SURAGGWA approach builds on the experience that improving non-timber forest product value chains provides a key incentive for local communities to develop these lands. The Theory of change also assumes that the private sector is interested in the marketing of NTFPs in domestic, regional and export markets as there is an unmet and growing demand for these products in the cosmetic, pharmaceutical and food industries. The private sector can play an important role in marketing the range of NTFP provided there is sufficient volume produced with the attributes demanded by the market. Mutually beneficial partnerships can be forged between local communities and the private sector for the collection and production of gum arabic, Balanites oil, honey, Baobab, Jujube, Neem, etc. A proactive way to enhance value-addition for NTFP is through the innovation of new products, increase in the efficiency of transactions within the value chain, and ensuring a fairer distribution of the value added amongst the different actors along the chain.[68] Thus, SURAGWWA intends to give local communities and the private sector a stake in the production, processing and marketing of NTFPs. The investments in land rehabilitation and value chains of the NTFP can generate employment, provide livelihoods and improve food security and nutrition of the communities and give them a long-term incentive for sustainable management of these eco-systems. The rehabilitation of the degraded farm plots will also enable households to supplement their diets through nutrition gardens in their homestead.

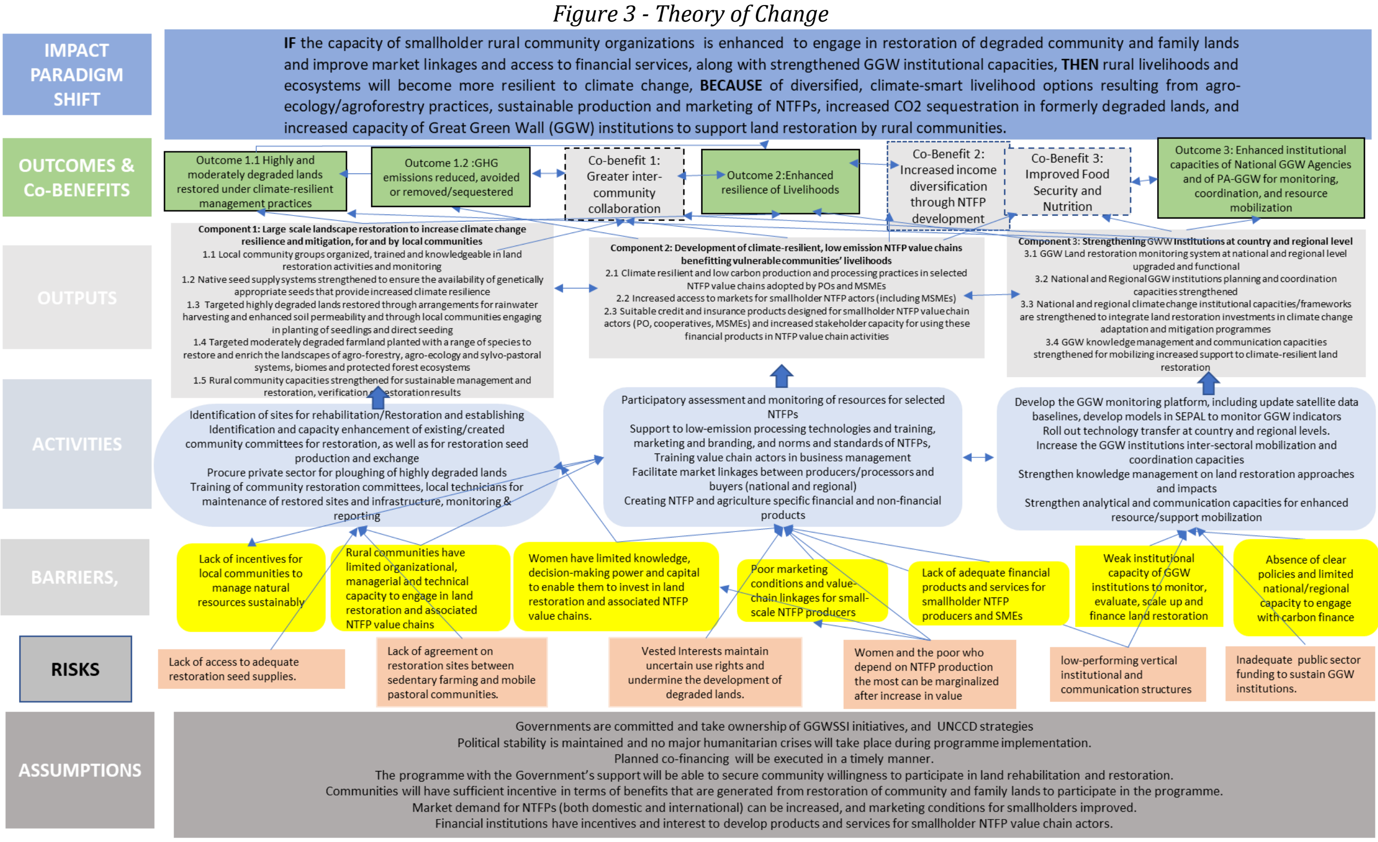
29. *Pathway to Accessing Financial Services:* The SURAGGWA programme recognizes that smallholders have limited opportunities to access formal credit due to the low presence of formal financial institutions in the targeted areas, high cost of outreach in remote areas and the perceived and real risks associated with lending to smallholders and non-timber forest product (NTFP) value chains. Individuals and SMEs often do not possess the documentation required to open bank accounts or access credit. They lack collateral or other mechanisms to guarantee loans. The programme, working in close collaboration with IFAD's GCF-funded iGREENFIN initiative, will reduce these barriers and thereby enhance access to finance by strengthening the capacity of financial institutions as well as the capacity of farmers and firms to access finance. In turn, the private sector can access funds from financiers interested in financing NTFP value chains that have the potential to enhance rural livelihoods and landscapes, bringing economic, social and environmental benefits. Domestic and international investors have expressed interest in engaging or scaling up their engagement in NTFP value chains such as gum arabic, Balanites, baobab and neem in the Sahel. Some investors also expressed interest in providing funding for climate finance, food production and accelerating nature-based solutions for ecosystem restoration. The investors already present in the Sahel support a range of private companies that specialize in the production and manufacture of NTFP products.[69] The programme will help to lay the groundwork for private companies to attract such financing, for example, through strengthening standards and easing access to equipment.

30. *Pathways to Scaling-Up Large Scale Climate Action*: The ToC of the programme recognizes that the enormous task of building the Great Green Wall across the Sahel on a vast area of more than 100 million hectares requires close coordination, considerable finances, effective techniques, dissemination of the lessons, tools and

[67] The NTFPS that are being targeted by SURAGGWA are detailed in the Component 2 description. Details are provided in Annex 2.
[68] Participative Innovation Platforms (PIP) for Upgrading NTFP Value Chains in East Africa. September 2020.
[69] Participative Innovation Platforms (PIP) for Upgrading NTFP Value Chains in East Africa. September 2020.

knowledge and strong monitoring tools. Changes in land use, especially communal lands, and reforms of government structures that enable greater decentralization and local control of natural resources can be significant enablers of change. SURAGGWA is based on the premise that successful experiences can be scaled up through increased coordination among the GGW institutions, strengthening of the monitoring capacity, updating the satellite database and use of innovative models to monitor GGW indicators. With such improved coordination and strengthening, the Great Green Wall institutions will be able to formulate coherent proposals, attract funding and support implementation of larger scale climate action investments, as well as creating knowledge structures for sustainable land management, and improving international awareness of the challenges of land degradation and drought in the Sahel for higher impact and reinforced collaboration.

Figure 3 - Theory of Change



31. The large majority of the eight SURAGGWA countries, while highly diverse, do share a considerable number of important projected climate change impacts, as documented in table 2 above: large increases in the accumulated reference evapotranspiration, sizable reductions in the climatic water balance, and in the 3-, 6- and 12-month accumulated Standardized Precipitation Evapotranspiration Indices (SPEI). Details of the specific climate change impacts on each of the eight SURAGGWA countries are discussed in Annex 2, Feasibility Study. The following table provides an overview of the climate hazards and barriers that will be faced by most rural land users in the Great Green Wall zone of the eight SURAGGWA programme countries, the proposed SURAGGWA investments that will help them face these hazards and barriers, and the climate impacts and vulnerabilities addressed by the programme.

Table 8 Climate change hazard/barriers; proposed interventions; impacts/vulnerabilities addressed by the programme

Climate Hazards/ Barriers	**Proposed Interventions**	**Impacts/vulnerabilities addressed**
Increased temperature	• Promotion of climate-resilient land restoration and management methods on highly and moderately degraded land (Component 1):	• Decrease in surface area of productive agro-silvo-pastoral systems

	▪ Restoration methods on moderately degraded lands include agroforestry to reduce heat stress of annual crops ▪ Use of climate resilient seeds/seedlings	• Heat stress leading to decrease in productivity of crops/pasture
Increased drought and heavy rainfall/flooding events & high interannual rainfall variability	• Use of machine and animal plowing in land preparation activities for restoration, to increase water infiltration (Component 1) ▪ Use of climate-resilient seeds/seedlings ▪ Integrated production systems (agroecology) • Organize communities for sustainable landscape management, including controlled use of restored silvo-pastoral systems	• Decrease in productivity due to drought and moisture stress • Reduced water availability for crops and livestock • Decrease in productivity and loss/damage due to floods • Water runoff, soil erosion
Limited capacity to access NTFP markets and financial institutions	• Capacity building for smallholder producers and MSMEs in NTFP value chain development (Component 2) • Capacity strengthening through technology & knowledge dissemination (Components 1-3) • Improved access to financial institutions (Component 2)	• Limited knowledge and financial capacity to adapt to the changing climate among agro-pastoralists and the institutions supporting them

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Outcome number	**GCF Mitigation Results Area (MRA 1-4)**				**GCF Adaptation Results Area (ARA 1-4)**			
	MRA 1 Energy generation and access	**MRA 2** Low-emission transport	**MRA 3** Building, cities, industries, appliances	**MRA 4** Forestry and land use	**ARA 1** Most vulnerable people and communities	**ARA 2** Health, well-being, food and water security	**ARA 3** Infrastructure and built environment	**ARA 4** Ecosystems and ecosystem services
Outcome 1.1: Highly and moderately degraded lands restored under climate-resilient management practices	☐	☐	☐	☒	☐	☐	☐	☒
Outcome 1.2: GHG emissions reduced, avoided, or removed/sequestered	☐	☐	☐	☒	☐	☐	☐	☒
Outcome 2: Enhanced resilience of livelihoods	☐	☐	☐	☐	☒	☐	☐	☐
Outcome 3: Enhanced institutional capacities of National GGW Agencies and of PA-GGW for monitoring, coordination,	☐	☐	☐	☒	☒	☐	☐	☒

and resource mobilization								

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Co-benefit 1: Greater inter-community collaboration	☒	☒	☒	☒	☒	☐
Co-Benefit 2: Increased income diversification through NTFP development.	☐	☐	☒	☐	☒	☒
Co-benefit 3: Improved food security and nutrition	☐	☒	☒	☒	☒	☐

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

32. Sahel's dryland ecosystems consist of forests, woodlands, wetlands, grasslands and parklands that include farm/arable lands and agro-ecology areas with scattered trees. The area hosts a vast biodiversity of plant and animal species that help sustain human livelihoods and enhance their resilience. These lands are fast degrading and losing their potential for providing the range of services and the protection they offer to the environment and communities dependent upon them. Sahel's dryland ecosystems (forests, woodlands, wetlands, grasslands and parklands with scattered trees) host a vast biodiversity of plant and animal species that help sustain human livelihoods and enhance their resilience. Restoration of degraded dryland ecosystems generates large mitigation benefits from increased soil organic matter, as well as from below and above ground biomass. At the same time, land restoration generates major climate change adaptation benefits, through the range of ecosystem services (provisioning, regulating, cultural and supporting) they provide. These include important non-timber forest product (NTFP) commodities that generate additional sources of income, whether seasonally or in poor agricultural years, thus offering a safety-net to sustain livelihoods and enhancing resilience to climate variability and climate change.[70] These ecosystems play a key role in adaptation when they are properly managed, restored and reduce pressure on vulnerable natural resources (water, soils, wood and non-wood products, fodder and pastoral land). Land restorative activities increase soil fertility, preserve soil water holding capacity, reduce run-off, protect land and agricultural production from increasingly intense floods and droughts, increase infiltration of rainwater, regenerate grasses for livestock.

33. SURAGGWA will consist of three components designed to help make rural livelihoods more resilient to climate change and sequester carbon, as well as enhance GGW institutional capacities to ensure sustainability of restoration results through monitoring, coordination and knowledge management, by addressing the key barriers identified in the Theory of Change (B.2.b) and realize the outcomes identified above (B.2.b). This will be undertaken through the implementation of:

- **Component 1: Large scale landscape restoration to increase climate change resilience and mitigation, for and by local communities** which will address the key issues associated with the degradation of the natural resource base through the restoration of agro-silvo-pastoral landscapes;
- **Component 2: Development of climate-resilient, low emission NTFP value chains benefitting vulnerable communities' livelihoods** which will address the challenges rural communities face in accessing markets and financial services; and
- **Component 3: Strengthening GGW Institutions at country and regional level,** which will focus on institutional strengthening of the Great Green Wall institutions at country and regional level to improve institutional coordination, results monitoring, resource mobilization and knowledge management and sharing. The set of interventions included in the multi-country programme were selected based on their potential to bring

[70] Sacande, M. & Parfondry, M., 2018. Non-timber forest products: from restoration to income generation. Rome, FAO. 40 pp. License: CC BY-NC-SA 3.0 IGO. https://www.fao.org/documents/card/fr/c/CA2428EN/

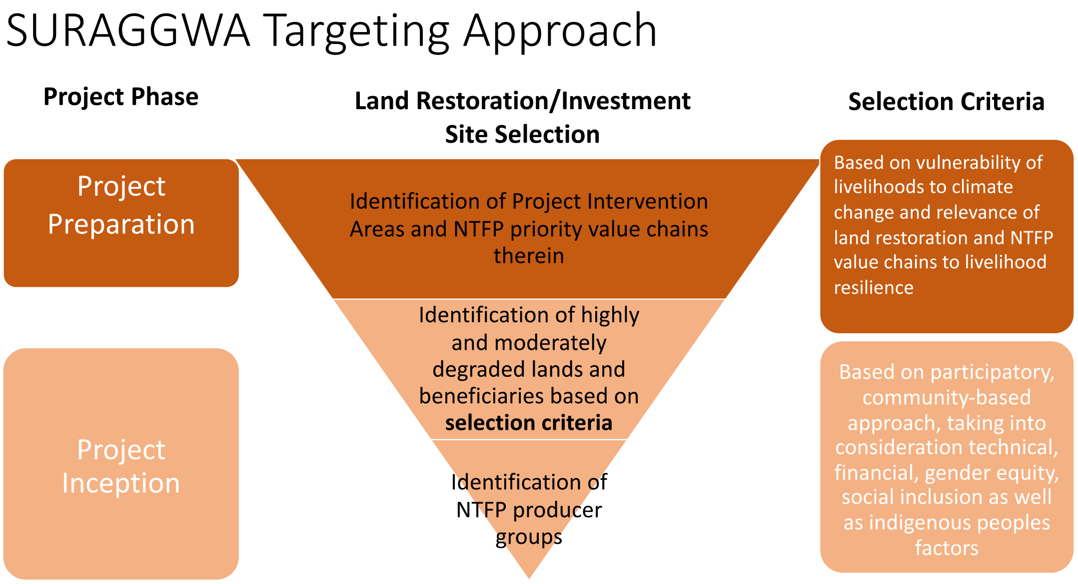
about a paradigm shift through providing critical support in a sustainable manner.[71] The trade-offs or the opportunity cost associated with some of the interventions and the rationale for the specific set of interventions chosen have been detailed in Annex 2: Feasibility Study.

34. Additionally, the set of interventions across the land restoration and NTFP value-chain development components will leverage synergies with a targeting approach that builds upon the identification of programme intervention areas based upon the identification of degraded lands through the use of existing baselines through Africa Open DEAL, and a mapping of priority areas including through remote-sensing and drones, to ensure consistency with the climate rationale along the Great Green Wall Initiative geographical areas (please see Annex 2: Feasibility study). This exercise will also include the identification of priority NTFP value chains with existing production in the identified programme intervention areas (please refer to Table 9 for NTFP priority value chains), during the programme preparation phase, after which specific investment sites, including both highly degraded communal lands and moderately degraded farmlands, will be identified following the specific selection criteria as outlined in Table 7 and 8), through a participatory, community-based approach taking into consideration gender equality, social inclusion and consideration of indigenous peoples' rights. These criteria have been identified and validated during programme preparation, with the understanding that the participatory approach that will be used for restoration site selection might provide additional selection criteria to allow for local realities. Based upon the identified restoration sites, the specific restoration methods and species selection for highly degraded lands and specific NTFP production at each moderately degraded site, in addition to the identification and selection of existing NTFP producer groups will follow (refer to Table 10 for Component 2 selection criteria).

35. SURAGGWA uses a strategic approach for beneficiary selection that prioritizes communities most vulnerable to land degradation, those dependent on natural resources for their livelihoods, and groups with a demonstrated commitment to sustainable land management. The selection process is focussed on the availability of degraded land to be restored in the villages, the motivation and commitment of community members to take part in restoration activities, including in-kind contributions such as land and labour; the non-existence of unresolved land issues and/or inter-village disputes; and preferably the pre-existence of community-based structures and organizations. Furthermore, the programme will engage smallholder farmers, pastoralists, and local cooperatives who are directly affected by desertification and have the potential to contribute to long-term forest restoration. Additionally, preference is given to communities that have experience with agroforestry, soil and water conservation, or traditional land stewardship practices, as their involvement can enhance programme success. Gender and social inclusion is also be key criteria, ensuring that women, youth, and marginalized groups have equitable access to programme benefits and decision-making roles. Furthermore, engagement with local authorities, customary landowners, and community-based organizations will help ensure that beneficiaries have secure land tenure, reducing the risk of disputes and ensuring sustained participation. By prioritizing beneficiaries who align with ecological and social sustainability goals, the programme can maximize its environmental impact and long-term viability.

Figure 4 – SURAGGWA Targeting Approach and Criteria

[71] Please refer to Sacande, M., Parfondry, M. and C. Cicatiello (2020) "Restoration in Action Against Desertification: A Manual for Large-Scale Restoration to Support Rural Communities' Resilience in Africa's Great Green Wall.



36. The identification of highly and moderately degraded lands and beneficiaries will use the criteria laid out in Table 7 and Table 8, below. A sub-set of beneficiaries engaging in restoration activities in moderately degraded farmlands will also benefit from NTFP value chain activities (assuming a 30% overlap) under Component 2. The specific selection criteria for beneficiaries under Component 2 activities are provided in Table 9. These beneficiaries will be supported, in a participatory and voluntary manner, to develop their skills under the priority NTFP value chains, as laid out in Table 10.

Table 4- Selection Criteria for the Selected Sites for Highly Degraded Lands[72]

Type	***Criteria***	***Means of Verification***
Climate	The site is impacted by climate events and risks based on a physical examination of the site, climate data, local weather station data or verbal records by village elders.	Photographs, record of observations and species and note for the record on the Site file with names and dates of interviews collected attested by local leaders and M&E team.
Physical	There is at least a minimum parcel of 300 hectares per restoration site selected.	A rough map will be provided and recorded in the programme records.
Technical	A site visit by technical experts indicates that the selected site can benefit from the menu of activities included for land rehabilitation and restoration.	Technical site report with photographs, names of species, impact potential and identification of potential interventions prepared in consultation with local community experts and elders.
Population	There are at least 40 minimum households who have access to use rights on the selected site.	A record of HHs will be maintained in programme records.
Social	The community members agree to use an existing organization or form a new one for the purpose of programme activities and nominate 2-3 members to provide coordination and facilitation for programme activities. There are no reported conflicts in the area and no rights of community members, whether pastoral or settled, will be infringed by the restoration activities.	A written document signed by 2-3 community elders in the presence of a majority of the households who have use rights in an open meeting organized by the local Government agency with photographs maintained of the meeting and a resolution signed by the local leaders.
Gender	The local community members agree in a meeting in which a majority of the users of the communal site are present to promote the interests of women through the programme and nominate 2 to 3	Photographs maintained of the meeting and a resolution signed by the women leaders.

72 For further details on the exact restoration methodology approach please refer to: https://openknowledge.fao.org/items/467cc151-490b-48ca-a1a6-4290ec774376

	women members who will promote those interests and ensure the communication and participation of women members.	
Logistical	A clear agreement from the community to proceed further with terms of reference signed between the community, local authorities and the programme.	A signed Terms of Partnership.

Table 5 - Selection Criteria for the Selected Sites for Moderately Degraded Lands[73]

Type	*Criteria*	*Means of Verification*
Climate	The site is impacted by climate events and risks based on a physical examination of the site, climate data, local weather station data or verbal records by village elders.	Photographs, record of observations and species and note for the record on the Site file with names and dates of interviews collected attested by local leaders and M&E team.
Physical	There is at least a minimum parcel of 500 hectares of moderately degraded land per restoration site selected.	A rough map will be provided and recorded in the programme records.
Technical	A site visit by technical experts indicates that the selected site can benefit from the menu of activities included for land rehabilitation and restoration.	Technical site report with photographs, names of specifies, impact potential and identification of potential interventions prepared in consultation with local community experts and elders.
Population	There are at least 125 minimum households who have access to use rights on the selected site.	A record of HHs will be maintained in programme records.
Social	There is relatively equal distribution of land with an average of around 4 has per HHs. Where HHs own more than this they can use their skills and awareness and new techniques to restore more than 4 HHs per HH but programme inputs will be based on a share of 4 HH per HH. The community members agree to use an existing mechanism for organizing into groups or form new groups and nominate 2-3 members to provide coordination and facilitation for programme activities. There are no reported conflicts in the area and the rights of no community members, pastoral or settled will be infringed by the restoration and rehabilitation activities.	A written document signed by 2-3 community elders in the presence of a majority of the households who own land or have use rights in an open meetings organized by the local Government agency with photographs maintained of the meeting and a resolution signed by the local leaders with a listing of land rights.
Gender	The local community members agree in a meeting in which a majority of the users of the communal site are present to promote the interests of women through the programme and nominate 2 to 3 women members who will promote those interests and ensure the communication and participation of women members through separate women's groups.	Photographs maintained of the meeting and a resolution signed by the women leaders and a group leader in areas where separate groups are formed for women.
Logistical	A clear agreement from the community to proceed further with a terms of reference signed between the community, local authorities and the programme.	A signed Terms of Partnership.

37. <u>Selection criteria of the beneficiaries for component 2:</u> A number of criteria has been elaborated to guide the programme implementation staff in selecting the beneficiaries of component 2 interventions: producers, collectors, processors and distributors of non-timber forest products (NTFP) and fodder - referred to collectively as NTFP value chain actors. These criteria consider socio-economic, technical and organizational dimensions/aspects. Applying the criteria will require some flexibility and adaptation, taking into account countries' contexts and specificities. For example, ensuring that women have equal access to NTFP value chain opportunities (see criterion 6 below) will require different culturally appropriate methods in the eight countries. Access to programe support and interventions may be subject to some additional conditions that would be elaborated in the different beneficiary countries according to their specificities.

Table 9 - Selection criteria of beneficiaries

	Socio-economic criteria	**Technical and organizational criteria**
	1. *Dependence on forest and/or tree resources*: Beneficiaries should be those who are dependent	1. *Capacity and skills*: Beneficiaries should have the minimum

[73] For further details on the exact restoration methodology approach please refer to: https://openknowledge.fao.org/items/467cc151-490b-48ca-a1a6-4290ec774376

Producers/collectors/ processors/distributors	on forest and/or tree resources for their livelihoods. Involvement in restoration of degraded is considered an advantage; 2. *Availability of the natural resources/raw materials*: Beneficiaries should have access to NTFP/fodder resources required for the selected value chains (see Table x), and have an interest in sustainable management of these resources; 3. *Market demand/requirement:* Beneficiaries should be ***able and willing*** to meet the product requirements/quality standards for the NTFP products, whether for the market or for direct consumption, improving their own food and nutrition security. This requires an understanding of the targeted end users (including domestic and international markets) and their preferences, as well as the ability to produce products that meet the required quality standards; 4. *Social and environmental sustainability:* Beneficiaries should be committed to social and environmental sustainability in the production and marketing of the NTFP products. This involves practices that promote conservation and restoration of biodiversity and ecosystem services, and equitable sharing of benefits, among others; 5. *Gender equity and social inclusion*: Beneficiaries should include both men and women, including youth, with a focus on ensuring that women and youth have equal access to NTFP value chain opportunities and benefits (without discriminating against socially disadvantaged or vulnerable groups); 6. *Economic viability*: Beneficiaries should be able to generate income and improve their livelihoods through their participation in the NTFP value chains. This requires a comprehensive business case of the costs, benefits, and risks associated with the selected value chains and the ability to manage these effectively. 7. *Commitment*: The beneficiaries should commit themselves to the principles and requirements of the programme (to avoid/minimize opportunists). Programme staff in every beneficiary country should find a suitable way of engaging the beneficiaries.	capacity and skills required to participate in the selected value chains (such as harvesting, processing, packaging and marketing of the products, etc.); This also includes some willingness to improve their capacity and skills; 2. *Organizational and functional requirements:* Beneficiaries should be part of an organized group or be ready to be organized in a producer group/ cooperative with a shared understanding and vision; The organized groups could be formal or informal and where existing, will be reinforced. They should be ready to report transparently to their members. 3. *Access and benefits sharing:* Beneficiaries should be ready to equitably share benefits among group members, and to share knowledge about sustainable use and processing of NTFP resulting from their participation in the programme.

Table 10 - List of priority NTFPs by country

Burkina Faso	Chad	Djibouti	Mali
Balanites (*Balanites aegyptiaca*), **Baobab** (*Adansonia digitata*), **Gum arabic** (*Acacia* spp), African locust bean (*Parkia biglobosa*), Jujube (*Ziziphus mauritiana*) Tamarind (*Tamarindus indica*), Shea (*Vitellaria paradoxa*), Neem (*Azadirachta indica*), **Fodder,** **Honey**, Moringa (*Moringa oleifera*)	**Balanites** (*Balanites aegyptiaca*), **Gum arabic** (*Acacia* spp), Jujube (*Ziziphus mauritiana*) **Fodder,** **Honey**, Moringa (*Moringa oleifera*)	Gum arabic (*Acacia* spp), Jujube (*Ziziphus mauritiana*), Frankincense (*Boswellia* spp), Date palm (*Phoenix dactylifera*), **Fodder,** **Honey**, Moringa (*Moringa oleifera*)	**Balanites** (*Balanites aegyptiaca*), **Baobab** (*Adansonia digitata*), **Gum arabic** (*Acacia* spp), Ronier (*Borassus aethiopum*), **Fodder,** **Honey**

Mauritania	Niger	Nigeria	Senegal
Balanites (*Balanites aegyptiaca*), **Baobab** (*Adansonia digitata*), **Gum arabic** (*Acacia spp.*) Jujube (*Ziziphus mauritiana*) Doum palm (*Hyphaene thebaica*), *Neocarya macrophylla* **Fodder,** **Honey,** Moringa (*Moringa oleifera*)	**Balanites** (*Balanites aegyptiaca),* **Baobab** (*Adansonia digitata*), **Gum arabic** (*Acacia spp*.) Jujube (*Ziziphus mauritiana*) Doum palm (*Hyphaene thebaica*) *Nymphea lotus* *Sporobolus robustus,* **Fodder** **Honey,** Moringa (*Moringa oleifera*)	**Balanites** (*Balanites aegyptiaca*), **Baobab** (*Adansonia digitata*), **Gum arabic** (*Acacia* spp.), Neem (*Azadirachta indica*), **Fodder,** **Honey**,	**Balanites** (*Balanites aegyptiaca*), **Baobab** (*Adansonia digitata*), Jujube (*Ziziphus mauritiana*) **Fodder,** **Gum arabic** (*Acacia* spp), **Honey**, Moringa (*Moringa oleifera*)

Source: Results from country consultation workshops and exploitation of secondary data

Component 1: Large-scale landscape restoration to increase climate change resilience and mitigation, for and by local communities[74].

Outcome 1.1: Highly and moderately degraded lands restored under climate-resilient management practices
Outcome 1.2: GHG emissions reduced, avoided or removed/sequestered

Output 1.1: Local community groups organized, trained and knowledgeable in land restoration activities and monitoring.

38. In each programme area, specific climate data analysis and GIS mapping will be undertaken to identify preliminary sites for restoration. The programme will then work with communities to prioritize and select highly degraded common lands and moderately degraded farmlands according to the specific criteria set out in Tables 7 and 8. Following the Action Against Desertification manual[75], these selected communities will be asked to help elaborate and agree to a set of terms of partnership that will: (i) identify and agree on the communal agro-silvo-pastoral lands which will be restored (>50 ha plots for mechanized interventions), on land on which use rights are well recognized and agreed among the local communities and local governments, which is then formalized through locally appropriate means; (ii) organize village management committees for land maintenance; and (iii) identify village level technicians whose capacity will be developed. Communities will then take responsibility for (iv) providing labour for planting seeds and seedlings and (v) maintaining the restored plots. The management committees, one per village, will oversee the seeding/planting, monitoring and maintenance activities, and manage the benefits and proceeds generated from the interventions on common lands. The programme will strongly encourage the participation of women and youth in the committee and in its leadership positions, given their frequently key role in the establishment and maintenance of restored areas.

Activity 1.1.1: Identify potential sites for restoration and to inform resource base for value chain priorities based on analysis of climate data, GIS maps and potential for impact with local authorities and community

39. The purpose of this activity is to assist communities and local authorities to work together on identifying specific restoration areas, and their potential linkages with NTFP value chain development. The project will generate the necessary evidence base for prioritization of Great Green Wall (GGW) restoration areas by undertaking climate data analysis and by generating GIS Maps that will be used for consultations with local community groups and authorities under Activity 1.1.2 defined priorities for restoration and for NTFP value chains (sub-activity 1.1.1.1).

Activity 1.1.2: Identify and mobilize communities for participatory selection of the specific sites, ensuring inclusion of women.

[74] Outputs 1.1 through 1.5 under this component do not have grant making or cash transfers elements to the concerned local communities.
[75] https://doi.org/10.4060/ca6932en

40. This activity will start with consultations of the local communities having expressed an interest in restoration and in NTFP value chain development (sub-activity 1.1.2.1), identified by activity 1.1.1 above, Subsequently, selected NGOs and service providers (SP) will facilitate the mobilization of local households into community groups (CG) and provide technical assistance to help the CG identify potential restoration sites and take stock of legal and customary tenure rights applying to these sites, with technical backstopping provided by FAO (sub-activity 1.1.2.2). Community restoration committees – including separate community women's groups (CWG) where necessary – will be established for the restoration of highly degraded common lands (sub-activity 1.1.2.3). For the restoration of moderately degraded individual/family farmland, producer organizations will be established (sub-activity 1.1.2.4). To facilitate and promote the role of women in restoration, regional CWG workshops and national consultations will be held to support the development of gender and safeguards related documents (sub-activity 1.1.2.5).

41. The process for engagement of local communities will build on best practices used by the FAO Land Tenure and related projects in the region, facilitating national and local stakeholder processes to ensure a responsible governance of tenure and engaging groups with statutory as well as groups with customary (formal/informal) tenure and land-use rights. In so doing, the process will contribute towards securing women's tenure rights and access to land and natural resources, as identified in the Technical *Guide on the Integration of the Voluntary Guidelines* on the Responsible Governance of Tenure into the Implementation of the United Nations Convention to Combat Desertification and Land Degradation Neutrality.[76]

Activity 1.1.3: Train programme's restoration teams and community management organizations

42. The programme will follow a "training of trainers" approach, first at national level, and then moving down to community level, where village technicians will be identified among the membership of the community-level Restoration Management Committees (RMC) and then trained to support their communities. This tool for strengthening the capacities of community members in land restoration techniques has been successfully applied in Sahelian land restoration efforts, including AAD, by FAO and by other organizations..

43. In order to implement this activity, the programme will select candidates for becoming field trainers at national level, from government and NGO technicians and other Service Providerion, and train them on Farmer Field Schools (FFS) and soil conservation, protection and land restoration techniques (sub-activity 1.1.3.1): Once trained, these trainers will be deployed to provide community-level trainings through the FFS model, focusing on soil conservation and protection and land restoration techniques (sub-activity 1.1.3.2).

Output 1.2. Native seed supply systems strengthened to ensure the availability of genetically appropriate seeds that provide increased climate resilience.

44. Good physiological quality (capacity for germination and propagation) and quantity of seeds and seedlings of well-adapted and resilient native species are a fundamental element in dryland restoration and rehabilitation interventions in the Great Green Wall countries. There is a significant shortage of good seeds from wild native species and a shortage of seed for agriculture crops adapted to the local growing conditions. The previous FAO project, AAD initiated a large-scale effort in Niger, Mali, Burkina Faso and Nigeria to produce restoration seeds. This effort will be scaled up to involve reliable supplies through collaboration with national forest seed centers, forestry research institutes and rural communities' associations. Communities will be trained and organized for the collection and conditioning of local seeds. In addition, selected communities will be supported to establish plant nurseries. It is expected that native seed (germplasm) supply systems will be set up to ensure supply of around 2,000 tons of restoration seeds with high germination rates to seed and plant in the selected restoration sites. Communities will be encouraged to develop small private business for gathering wild native seeds and establishing and caring for native species nurseries, to be used in their village and farm plots and beyond.

76 https://www.fao.org/publications/card/en/c/CB9656EN/.

Activity 1.2.1: Identify and train community technicians on good quality and quantities of restoration seeds and native seed supply

45. The purpose of this activity is to organize community members involved in seed collection and storage, and build their capacity. First, community technicians will be trained by national research institutions (sub-activity 1.2.1.1). Subsequently, the programme will provide technical assistance for the formation of community-level seed supply groups (sub-activity 1.2.1.2). NB The trainings will be cascaded down from community technicians to community members under the next activity, 1.2.2.

Activity 1.2.2: Identify, organize and train community members involved in seed and seedling production for land restoration activities.

46. This activity is essential for scaling up local communities' capacity for land restoration at affordable cost. It involves a two-stage training efforts, first the training of restoration seed producer groups in basic seed and seedling production (sub-activity 1.2.2.1) and then the training of communities and equipping them with seed collection, treatment and storage kits to prepare seeds for restoration by seeding and planting. (sub-activity 1.2.2.2).

Activity 1.2.3: Establish, equip and operate community nurseries for seedling production and dissemination

47. Many restoration activities will be implemented through direct seeding. But where the context is suitable, especially in terms of the availability of water sources, community nurseries can make an important contribution to efforts to restore degraded land as well as improve food security, through the association of vegetable gardens. The programme will establish and equip community groups to operate nurseries and vegetable gardens, and to collect and store restoration seeds (sub-activity 1.2.3.1). Subsequently, community groups and village technicians will be trained to sustainably manage these nurseries and use the seed collection equipment and storage facilities effectively (sub-activity 1.2.3.2). Finally, the programme will create a network to facilitate the dissemination and exchange of restoration seeds from established nurseries to programme restoration sites (sub-activity 1.2.3.3).

Activity 1.2.4: Local communities' seed supply of native seeds is integrated into national seed system.

48. The purpose of this activity is to connect the local community nurseries (that have been established and supported under activity 1.2.3) to the national seed centres. The national seed centres have the capacities for identifying, preserving and further disseminating good quality seeds and can thus strengthen the national restoration seed supply, and fill in shortfalls locally and nationally. This will not only help to improve the performance of other GGW restoration efforts, but also generate income for community seed producer groups.

49. The programme will create a system for the identification of restoration seeds for the national seed supply (sub-activity 1.2.4.1) and support the aggregation and dissemination of restoration seeds at and from national seed centres (sub-activity 1.2.4.2).

Output 1.3: Targeted highly degraded lands restored through arrangements for rainwater harvesting and enhanced soil permeability and through local communities engaging in planting of seedlings and direct seeding.

50. On highly degraded lands, where natural regeneration is blocked by the poor condition of the eroded soil and lack of soil seedbank, the programme will carry out large scale mechanized land preparation to increase soil permeability and promote rainwater infiltration, allowing soil moisture retention for up to two months beyond the rainy season, enabling seeds and/or seedlings to grow and establish in the dry and harsh conditions in the Sahel. Building up biodiversity is the backbone for the adaptation practices to strengthen resilience of local communities and landscapes in such extreme conditions, while restored areas will also act as important carbon sinks.

51. However, due to the advanced level of degradation of those lands, experiences drawn from AAD have shown that heavy mechanized interventions are needed to ensure positive results, as manual preparation methods are less efficient and have often failed to provide impactful results. In addition, average costs of such mechanized interventions (including land preparation, seeding/planting) are found to be cheaper than the all-manual methods.

Barren lands are ploughed using tractors and Delfino ploughs, which work by creating a pattern system of micro-basins and underground storage compartments, mimicking the traditional half-moons applied in the Sahel. The technique allows rainwater infiltration of up to 1,000 litres per half-moon. This land preparation method allows better soil permeability and efficient infiltration and storage of rainwater, needed for plant growth. The pattern of the micro-basins reduces the amount of evaporation as well as the amount of soil erosion. The increased amount of infiltrated rainwater between the plough lines improves soil fertility and plant production. FAO experience has shown that with appropriate planting and seeding of well-adapted native species, considerable re-greening and vegetation cover results within three rainy seasons.

Activity 1.3.1: Highly degraded land restoration plans prepared and implemented through mechanized, animal, and manual traction techniques.

52. As is obvious from the above, this activity requires a significant investment from the project (financial, for mechanized ploughing) and from the local communities (labour, for non-mechanized soil preparation, and seeding and planting of considerable areas of highly degraded lands. Therefore, it requires prior agreement on site-specific restoration plans drafted in collaboration with the community restoration committees (sub-activity 1.3.1.1) formed under Output 1.1. Following agreement on the site-specific restoration plans, private sector service providers – with an existing national presence and with access to suitable equipment – will be mobilized to assist communities with mechanized deep-ploughing of agreed restoration sites (sub-activity 1.3.1.2). These larger-scale land preparation activities will take place in the dry season, ensuring that the restoration sites are ready for rainwater harvesting, seeding and planting as the rainy season sets in.

Activity 1.3.2: Sowing and planting in prepared sites with communities/villages

53. Following the completion of site preparation activities, local communities will engage in direct sowing of seeds from prioritized native grass fodder and woody species and in planting of seedlings coming from village nurseries. (sub-activity 1.3.2.1). The seeds for these restoration operations will be generated through activities under Output 1.2, above.

Activity 1.3.3: Training of community members in monitoring & maintenance

54. The monitoring and maintenance of seeded and planted sites is essential for obtaining optimum restoration results. Therefore, local communities will be trained for management (including protection through physical means or social fencing) and surveillance of restored plots (sub-activity 1.3.3.1). They will also be supported for the maintenance of the restored sites during two years following planting/seeding (sub-activity 1.3.3.2).

Output 1.4: Targeted moderately degraded lands planted with a range of species to restore and enrich the landscapes of agro-forestry, agro-ecology and silvo-pastoral systems, biomes and protected forest ecosystems.

55. In other less degraded areas with some surviving plant species, other sylvo-pastoral, agro-ecology and agro-forestry techniques will be used to enrich productivity on households' plots (i.e. up to 3 ha per household), improving their crop and/or fodder production. The programme will use agroecology principles and techniques which is fundamentally different from any other approach to sustainable development. Based on bottom-up and territorial processes, it helps solve local problems through context-specific solutions. Agroecological innovations are based on the joint production of knowledge, combining science with the traditional, concrete and local knowledge of producers. By strengthening their autonomy and their capacity to adapt, agroecology empowers producers and populations to be key actors of change.

56. In sub-Saharan Africa numerous case studies prove that agroecology can contribute to food and nutrition security while restoring resources, ecosystem services, and biodiversity (Annex 2). These studies also show that agroecology can play an important role in social cohesion, resilience building, and climate change adaptation. SURAGGWA will consolidate the production base and economic stability of the agroecology sector by facilitating the interaction between community groups and private operators developing promising services such as farmer advice, mechanization, production and marketing of seeds and organic inputs, mechanization, etc…

Activity 1.4.1 Sub-national moderately degraded land restoration plans prepared and implemented

57. Since this activity concerns the restoration of moderately degraded land, soil preparation will rely mainly rely on animal traction and manual labour. Beneficiary households will be supported with key information to prepare costed plans for the restoration of their prioritized sites (sub-activity 1.4.1.1). Local communities will also receive technical guidance and support for the land preparation and other restoration activities (sub-activity 1.4.1.2).

Output 1.5. Rural community capacities strengthened for sustainable management and restoration, and verification of restoration results.

58. To ensure greater sustainability of programme actions and not depend on continuous programme support, the village restoration committees will reinforce pre-existing village organisations already established through other projects/activities and/or will nominate two to four community technicians with gender and youth balance. Village technicians will provide support and follow-up to the restoration and maintenance with programme technical support as well as monitor progress of a combination of activities. The capacity of the Village technicians will be strengthened for large scale restoration of highly degraded lands, as well as for soil and water conservation and revegetation techniques for moderately degraded lands*.* They will then be expected to track field performance and maintenance of the village plots. The programme will encourage information/knowledge exchange among these technicians establishing a field technicians' network using mobile telephones.

59. The village technicians will also monitor, ground-truth and report on progress of the plots, which will be fed into the programme's reporting system linked to the national and regional GGW M&E system. Aerial and satellite information analyzed by national GGW institutions will corroborate progress made and provide a mechanism for independent verification of results reported by communities. During the first two years, the programme will provide direct support to them after which they will be expected to largely operate independently, with minimum supervision but close coordination and acknowledgement by the GGW-NAs and other national institutions involved in restoration monitoring. Those performing well will be encouraged through awards and recognition.

Activity 1.5.1: Village technicians trained and equipped in managing restoration areas and verifying restoration results, in a participatory manner

60. Village technicians will be nominated by the communities, and confirmed in collaboration with the programme's technical assistants, ensuring a balanced representation of gender and youth. Village technicians will then be trained on data collection tools and technologies, as well as participatory approaches (sub-activity 1.5.1.1). Village technicians will also be provided with relevant equipment and tools to facilitate the management of restoration activities and the verification of results (sub-activity 1.5.1.2). These village technicians will in turn be responsible for training communities on restoration and maintenance activities.

Figure 5 - Restoring natural capital at scale in the Sahel

Example of FAO's *Action Against Desertification* operations in Burkina Faso - 2016-2018, where barren land is ploughed using the Delfino Plough (that mimic the traditional "half-moon" technique dug by hand) for larger scales, improved soil permeability and rainwater harvesting of up to 1000 liters per half-moon. With appropriate planting of well-adapted native species by local communities (right species in the right place, at the right time), considerable re-greening and vegetation cover resulted within three rainy seasons.

Component 2: Development of climate-resilient, low emission non-timber forest product (NTFP) value chains benefiting vulnerable communities' livelihoods

Outcome 2: Enhanced resilience of livelihoods

61. This component aims at improving resilience of the livelihoods of the local agro-silvo-pastoral communities and smallholder NTFP collectors, processors and sellers in the Great Green Wall zones in the selected Sahel countries. This will be achieved by supporting the development of non-timber forest product (NTFP) and fodder value chains. This component will address the main constraints faced by the actors along the NTFP and fodder value chains (VC). These constraints include: (i) fragmented and small-scale production; (ii) limited capacity to add value; (iii) lack of information on the volume of production of the different resources[77]; (iv) inadequate equipment and technology for processing; (v) lack of access to finance; (vi) insufficient marketing and branding and promotion of the product attributes; (vii) weak management capacities; (viii) limited attention to and awareness about norms and standards to export to regional or international markets, among others. Annex 2 provides details about the potential of some of the key NTFP products that are produced in the region. Women account for more than 70% of the workforce responsible for harvesting, informal production and local distribution of many of these products from the Sahelian tree crops. Hence, increasing productivity and efficiency within these value chains has the potential to ensure food security and enhance incomes for women.

62. This component is organized to deliver three main outputs: (i) **output 2.1: Technical and organizational capacities of stakeholders strengthened in the selected NTFP-VC; (ii) Output 2.2: Improved market inclusion for NTFP stakeholders; (iii) Output 2.3: Enhanced access of smallholders and SMSs to financial services.** The implementation of these interventions under this component will be coordinated with those of component 1 and 3 given their complementarity and for synergies and more impacts. Specific coordination and synergies aspects between the components will be described in the different interventions of the components. The results from the interventions will be assessed against the following output indicators: (i) Number of value chain organizations (including producer, processor, distributor organisations) established and operational ; (ii) Increase in real gross sales generated by business plans in execution; (iii) Value of NTFP produce marketed by value chain actors

[77] Lack of baseline data on the number of trees, production levels and export volumes and values from GGW countries makes it difficult to accurately quantify the market potential of the prioritized value chains (WEF, 2022)

disaggregated by value chain segment (USD equivalent in market value); (iv) Number of financial products, tailored for agriculture or NTFP value chains, developed and made accessible to programme beneficiaries along the value chains, including MSMEs.

63. Under component 2, the planned country level activities will be **supported by two categories of regional-level activities:** (i) activities to provide effective and efficient support to programme activities that are similar across all countries: e.g. harmonized approaches to capacity development for NTFP value chain analysis (including resource assessment and market research) and the elaboration of NTFP business plans, as well as the capitalizing and sharing of lessons learned from experience and partnership building for the promotion of the NTFP in the region; and (ii) activities to address structural constraints affecting NTFP value chains in all SURAGGWA countries, e.g. product standards and regulations and information required for regional and international export markets, an issue that many NTFP processors and distributors struggle with. To implement these regional-level activities, the regional Programme Management Unit (PMU) will collaborate closely with the country implementation units (CIU) and work in partnership with regional institutions such as the Pan African Agency for the Great Green Wall (PAGGW), the Network of Farmers Organizations and Agricultural Producers of West Africa (ROPPA), the Economic Community of West African States (ECOWAS) and its specialized agencies (including ARAA, its agency for agriculture and food), and the private sector institutions including national[78] and international ones such as World Economic Forum, etc.

64. The policy, regulatory and enabling environment for business in the Sahel is complex and if not adequately tackled could hinder the success of commercially viable land restoration projects in agricultural value chains. National governments, together with international organizations and development finance institutions, have an important role in contributing to ensuring a conducive/enabling environment where businesses thrive and local communities participate in, and benefit from them. This entails actions to improve governance through the strengthening of mechanisms for consultation and participation, as well as to facilitate SME access to information, finance and international markets. The SURAGGWA multi-country programme can contribute to the efforts of these other partners to improve the enabling environment for NTFP value chain development in the region by bringing key actors together to exchange lessons learned and by promoting harmonized NTFP standards across the region. SURAGGWA multi-country programme will also intervene in the information, sensitization and dissemination of norms, standards, regulations and instruments pertaining the exchanges/commercialization of NTFP. These regional sub-activities will be implemented in synergy with capacity development interventions planned under component 3.

65. For the implementation of this component, the programme will adopt a multi-tiered approach and work both at the village level and the market level. At the village level, the programme will adopt a community-based approach for the inclusion of the small producers and collectors which is likely to involve women as the main beneficiaries. At the market level, the programme will work with value chain actors which could be individuals, SMEs, firms engaged in processing and marketing of the NTFPs in national, regional or international markets. A list of priority NTFP has been identified based on published reports and interactions at the country level. These include Balanites oil, Gum Arabic, Baobab powder and leaves, fodder and honey. The final list will be selected based on discussions with local communities, key value chain actors and review of additional information during implementation.

Output 2.1: Climate resilient and low carbon production and processing practices in selected NTFP value chains adopted by Producer Organizations and MSMEs.

66. The programme's interventions will target formal as well as informal organizations in the sector. A profile of some of the actors engaged in the main value chains such as Shea Butter, African Baobab, Balanites, Moringa, etc. is given in Annex 2.

Activity 2.1.1: Train and provide TA to POs and MSMEs in selected NTFP value chains for enhanced organizational and managerial capacities (registration, structuring according to OHADA, training on management, administration etc.)

[78] Examples of private sector entities in the beneficiary countries in the table below

67. Under Activity 2.1.1, the programme will support the NTFP value chain actors such as individual collectors/producers, processors and distributors, cooperatives and associations, SME, formal or informal, to become more professional in their structure and operations. This will be done inpartnership with local NGOs and institutions. It will assist the NTFP value chain actors to improve their coordination and resolve common problems to achieve enhanced efficiency and profitability.

68. The programme will also support and facilitate the establishment and operationalization of platforms of stakeholders for collaboration in the NTFP. The establishment of the platform will be conducted in phases. For the first phase, the efforts will be focused on the establishment and operationalisation of the platform at national and local levels and the second phase will be organised to facilitate the interaction among players across the SURAGGWA programme countries. The objective of this activity is to assist the value chain actors from different layers (production/collection, processing, and distribution) to create a platform that will improve the coordination and the governance and the value chains for efficiency and profitability. The programme will encourage, sensitize, facilitate and support interaction of the various actors through regular meetings. The concerned actors will be asked for contribution in the organization/financing of their meetings in order to stimulate ownership of the process and its sustainability (financial and/or in-kind contributions are expected from the beneficiaries for the organisation of their meetings).

69. Well-functioning platforms can facilitate access to affordable packaging materials and equipment for NTFP, access to tailored finance/credit, access to affordable technologies for the processing of NTFP (e.g. Balanites oil extraction). The platforms will also offer opportunities to share professional information and negotiate business growth. At regional level, the programme will support and facilitate an inter-country platform of NTFP actors for exchange of information, experiences, lessons learned and business opportunities. Even though the interventions of component 2 focused on the existing resources, the establishment of these platforms will take into consideration the actors from component 1 when relevant, in order to take into consideration the actors from the newly restored areas of the programme.

70. Initially, the programme will deliver technical assistance to existing Producer Organizations (PO) and Micro- Small and Medium Enterprises (MSME) to increase their administrative and management capacities (sub-activity 2.1.1.1). It will also provide technical assistance to new PO and MSME, and train them in administrative and management capacities (sub-activity 2.1.1.2).

71. The programme will also deploy technical assistance to develop legal texts and tools, and support the formal registration – and other administrative processes of the bodies following the Organization for the harmonization of Business Law in Africa (OHADA) rules and principles – of both existing PO and MSME and the new ones formed with the support of the programme (sub-activity 2.1.1.3). Finally, the programme will support and facilitate the establishment and operationalization of platforms of NTFP actors to enhance collaboration in the value chain and promote the creation of professional associations (sub-activity 2.1.1.4).

Activity 2.1.2: Train and coach local POs and MSMEs in sustainable production and collection practices to enhance NTFP quality and availability

72. Under Activity 2.1.2, trainings will be conducted for enhancing the capacity of local producers/collectors (men, women and youth) to improve the quality of the products and sustainable management of natural resources. The main constraints highlighted by the collectors and processors during the field consultations were related to the collection, transport and storage of the raw materials. This is the case for gum arabic, Balanites, baobab and jujube fruits, etc. The objective of this activity is to improve the mechanism for production/collection/harvesting, handling, transport and storage of the NTFP. The capacity building of producers/collectors will combine theoretical and practical aspects and will contribute to enhance the quality of the products and the efficiency of the different operations.

73. Considering the diversity of the sector/value chains, the training modules will depend on the priority products per country and the specific technical aspects to improve their collection, storage, processing, etc. This is the case for, among others, fodder and honey, products that are very specific and will require tailored approaches and tools.

During the implementation phase, specific attention will be paid to support for youth, women and other marginalized groups.

74. Specific training modules and resources will be developed and/or updated, based on the needs of the NTFP VC actors (sub-activity 2.1.2.1). These new and updated training materials will then be used to provide technical assistance (training and coaching) for PO and MSME. This training will be essential for the sustainable management of NTFP resources, and for the improvement of the quality of primary products and reduction of losses.

Activity 2.1.3: Train and coach POs and MSMEs to improve the processing practices (including packaging and labelling) of the selected products

75. Under activity 2.1.3, technical training will be conducted to improve the processing and packaging and labelling of the selected products: Balanites oil, Moringa powder, gum arabic, honey, jujube, baobab, etc. This will help PO and MSME to improve the efficiency of their operations as well as the quality of the final products, and hence their profits and revenue. The content of the training will include the techniques of processing, hygiene, food safety, food quality and standards, packaging, storage and conservation, etc. The practical training will be delivered in collaboration with specialized research institutes and/or private actors that are equipped for that.

76. The implementation of this activity will be coordinated with access to new technologies that result out of partnership with institutions on emission-lowering technologies and equipment to enhance NTFP processing at scale (See Activity 2.2.5) The findings from market analysis investigations will be used to design and tailor the training modules to help meet the market demand.

77. NTFP processing training modules and resources will be developed and updated, based on the needs of the VC actors (sub-activity 2.1.3.1). Using these modules and resources technical assistance (training and coaching) will be provided for PO and MSME (sub-activity 2.1.3.2).

Output 2.2: Increased access to markets for smallholder NTFP actors (including micro- and small enterprises)

78. Products derived from GGW tree crops that deliver ecological and socioeconomic benefits show great potential to compete in the personal care and superfoods[79] markets. The main objective of this output is to improve the market value of the NTFP and the incomes of the actors engaged in the sector, especially for smallholders/MSME including women and youth. It will provide market-based solutions that address the socioeconomic reality of the people living in the Great Green Wall area, as well as the latter's ecological/environmental vulnerability to climate change. In this context, the successful engagement of the private sector in commercially viable, socially and ecologically responsible restoration efforts constitutes a major opportunity to deliver restoration at scale, while providing additional livelihood opportunities to the communities. This output is composed of a series of interventions that aim at removing constraints and barriers that limit the marketing and commercialization of the NTFP at local, national, regional and international levels. The following are the main activities envisaged under this output.

Activity 2.2.1: Identify NTFPs with market and sustainable production potential that can be integrated in land restoration efforts funded under Component 1

79. One of the main challenges for the promotion of the NTFP is related to lack of proper information of the potential of the various NTFP (Balanites, Jujube, Baobab, Tamarindus, Shea, etc.). There is a deficiency of robust baseline data on the number of trees, production levels and export volumes of the NTFP in the GGW countries. This intervention will capitalize on the current efforts to (i) conduct an assessment of the potential of natural resources for the NTFP and (ii) setting up of a mechanism for the monitoring of the NTFP resources.

80. This activity will promote and support a participatory assessment and monitoring of the resources for the selected NTFP.[80]. It will be implemented in collaboration with component 1 and 3 and will adopt a participatory approach that will strengthen the capacities of the national institutions for ownership and sustainability. This intervention will

[79] There are many non-timber forest products (NTFPs) that are considered to be "superfoods" due to their high nutritional value and health benefits. Some of these superfoods in the NTFP sector include: Baobab, Moringa leaves, honey, etc.
[80] Refer to the table of the list of priority NTFP

contribute to the improvement of knowledge and data of the NTFP resources, contribute to improved decisions by producers and buyers based on credible information while supporting the sustainable management of the resources. The interventions related to this activity will be planned and implemented in collaboration and complementarity with component 1 since this will involve the actual available resources as well as the areas and resources that will derive from the land restoration (component 1). Since this activity is more than relevant to all the beneficiary countries, it will be implemented also in complementarity and synergies with component 3 focused capacity building.

81. This intervention will contribute to the identification of priority NTFP products that could be marketed with sustainable increase in production to ensure availability of resources for the market as well as ensuring sustainable management of the natural resource base, including restored areas. This will be through the regional, specialized NGO Tree Aid that has a well-known approach to ensure economic viability and environmental sustainability of these activities.The activity will comprise the assessment of the potential of restoration areas and the surrounding landscape for the sustainable production of NTFP (sub-activity 2.2.1.1) and the integration of a NTFP resource management module in Component 1 land restoration efforts (sub-activity 2.2.1.2).

Activity 2.2.2. Support to marketing and branding of the NTFP (promotional activities, nutritional value pamphlets of NTFP, organization of trade fairs, tv shows, etc.)

82. Activity 2.2.2 will provide support to marketing and branding of the NTFP (promotional activities, nutritional value pamphlet, organization of trade fairs, TV shows, etc.): Except for gum arabic and shea butter, most of the NTFP of the GGW countries have limited competitiveness, despite their ecological and socioeconomic benefits that show greater potential to compete nationally as well as internationally in the domains of personal care and the superfoods markets, among others. Lack of professional marketing and branding of the products by SMEs engaged in the NTFP sector is among the reasons for this setback. This activity aims to support the NTFP value chain actors, especially MSME, in the promotion and branding of different products. These interventions will complement the activities under output 2.1 which is dealing with capacity building of the processors.

83. The following specific interventions will be conducted as part of the marketing promotion: (i) Support to the preparation of a master plan or guide for marketing and promotional activities including digital tools (sub-activity 2.2.2.1); (ii) Support to the implementation of the master plan including TV shows, radio programmes, social networks/medias[81] on the nutritional properties and other characteristics of the NTFP (sub-activity 2.2.2.2). To ensure success, the CIU will collaborate with the SME, research institutions and the private sector including mobile phone operators in the preparation of the TV/Radio programmes and messages that will impact on all the actors engaged in the NTFP business including the consumers and the buyers; (iii) Support to the organization and/or participation in trade fairs. The programme will support SME in the preparation and attendance in trade fairs (sub-activity 2.2.2.3); and (iv) Direct coaching of MSMEs in the branding of their specific NTFP products (sub-activity 2.2.2.4). This intervention will provide tailored support to the MSME/value chains actors. The programme will collaborate with public and private institutions in the implementation of this intervention. Specialized private organizations have been identified in the participating countries with strong capacity and experience in supporting MSME in the marketing, branding as well as commercialization of their products. It is also envisaged to collaborate with leading celebrities in the promotional activities of the NTFP.

Activity 2.2.3: Support Producer Organizations (PO) and Micro- Small and Medium Enterprises (MSMEs) in business plan development and management

84. Under this activity, the programme will conduct trainings of value chain actors in business management: Targeted trainings on market requirements, financial literacy, farms/business and risk management and sustainable use of resources, among other topics, will enable producers and SME to establish stable and profitable partnership with NTFP buyers, and improve their productive capacity. Specific criteria will be considered to attract and promote female-led businesses/actors. The experts that will deliver the training will support the value chain specialists

[81] Some good experiences have been capitalized in Senegal in the promotion, branding and marketing of NTFP with Soreetul group (Soreetul (sooretul.com)). https://shop.soreetul.com/en

working on the programme at the national level. Synergies and complementarity will be sought with the other activities of the programme such as the training on financial literacy and access to finance.

85. The training sessions will cover a series of thematic topics that will derive from a rapid situation analysis. First, information and awareness-raising efforts will be undertaken to inform value chain actors on the opportunities offered by the programme and the conditions and modalities to access the available support. Different channels will be used depending on the type of audience and their location as well as their areas of intervention. The programme will adopt a *''demand-driven''* approach and offer the service on a *''first come, first served''* basis. Hence, the interested OP/MSME will express their interest to be considered for the training and will be selected based on specific criteria. Subsequently, the selected value chain actors will receive a training from technically strong service providers that will help them elaborate their business plans.

86. The training of the NTFP VC actors will be followed by their coaching to assist them in the correct implementation of the acquired skills and of the business plans that they drafted during the training sessions. Coaching for the implementation of OP/MSME business plans consists of practical support for key actions, such as facilitating access to funds from financial institutions and/or other sources. The mechanism for the financing of the business plans of the value chain actors is detailed in outcome 2.3. The coaching sessions will be carried out with specific indicators that will be commonly defined in collaboration with the business owners and will help to assess its effectiveness and efficiency the support. These indicators will be related to the training topics. These interventions will be implemented by service providers through contractual arrangements with specialized organizations or individual experts/coaches[82].

87. The following interventions will be implemented: (i) Conduct trainings of PO and MSME engaged in NTFP on business plan development and management skills, through the use of standardized tools such as RuralInvest (sub-activity 2.2.3.1); (ii) Technical assistance and coaching for PO and MSME in the preparation and implementation of their business plans (sub-activity 2.2.3.2). (iii) Coaching of SMEs:

Activity 2.2.4: Enhance access to national, regional and international markets for NTFP, including through the setting up of norms and standards to improve the marketing of the NTFP;

88. Under Activity 2.2.4, the programme will enhance access to national, regional and international markets for NTFP: Lack of information on production levels and export volumes and values of NTFP in the GGW countries makes it difficult to quantify accurately the market potential of the prioritized value chains. Except for shea and gum arabic, few market studies have been conducted at the regional/international level. Nonetheless, distributors and off-takers, primarily in the food and cosmetics industries, are expanding their product and ingredient range to include Sahelian high-potential tree crops such as shea, Balanites, baobab, etc. that fit the needs of fast-growing and high value markets. The programme will carry out the following interventions: (i) Conduct market investigations and value chain assessments for national, regional and international opportunities for the NTFP; (ii) sharing the findings from the investigations; (iii) facilitate markets linkages between producers and buyers through roundtables.

89. The programme will facilitate discussions and negotiation between the producers/processors and the buyers (off-takers). These discussions and negotiation could lead to commitments and/or contractsl including with some institutional buyers such as WFP for their school feeding programmes[83]. This option and opportunity has been discussed with WFP during the consultations workshop in Niamey – Niger. (iv) information and/or training of the SME on market requirements pertaining to international trade. This information will also influence and help the SME to reshape their plans and objectives in terms of targeted markets. These interventions will help in overcoming market-entry barriers to international markets arising from complex standards and regulations. The successful engagement of NTFP actors in the private sector in commercially viable, socially and ecologically responsible

[82] The following elements should be considered in terms of criteria in the process of selection of the business coaches: Experience, attitude, willing to share, expertise in the field/domain, Accessibility, connections and networks, expectations, a love of teaching, old you accountable + the presence in the field/proximity considering the security issues.

[83] Many NTFP are of interest and being considered for school feeding programme directly as a specific meal/menu or incorporated in other meals as complements: Honey, edible leaves from forest such as Moringa, Baobab leaves and pulpe, fruits and nuts, etc. Many of NTFP are used in the fabrication/processing of jams, juice, biscuits, and fruit salads. There is a good example from Senegal with Lionceau that makes varieties of baby foods using NTFP as ingredients (Baobab, Tamarindus, Moringa, etc.): www.le-lionceau.com/

restoration efforts constitutes a major opportunity to deliver restoration at scale while providing additional livelihood opportunities.

90. The objective of this intervention is to assist the beneficiary countries through their specialized institutions in the elaboration, validation and diffusion of norms and standards for Non-Timber Forest Products (NTFP) that will guide the production and commercialization of the concerned products as well as protect the consumer. The NTFP sector is characterized by a lack of clear and consistent regulation in terms of norms and standards governing the production and commercialization of the different products and/or their dissemination among the value chain actors. Some GGW countries are more advanced in terms of setting up norms and standards for the production and commercialization of NTFP nationally and internationally. The prime example is Burkina Faso, which possesses norms and standards for more than 90% of its NTFP. In some countries, the norms and standards exist, but the national institutions and authorities fail to inform and train the concerned actors of their existence and application. The programme will also intervene in the information/communication of the norms as well as for the training of the actors and the control of their application in the field.

91. The programme will also collaborate with research institutions in order to identify properties of the various NTFP (anti-inflammatory, diuretic and digestive properties, fight against diabetes and obesity, prebiotic properties, gluten replacement, low calorific value, etc.) where it is not yet available. The following specific interventions will be implemented: (i) Support to the elaboration/update of norms and standards for the NTFP. For this sub-activity, the programme will provide technical and financial support to the national institutions for the elaboration and/or for the updating of norms and standards (sub-activity 2.2.4.1) l; (ii) Information, sensitization and/or training of the NTFP value chain actors on the norms and standards (sub-activity 2.2.4.2). Considering that some countries are much more advanced than others in this area, the programme will also facilitate exchange of experience and lessons among the beneficiary countries, through regional workshops (sub-activity 2.2.4.3). These regional workshops will also share the results of programme-funded market investigation and value chain assessments for national, regional and international NTFP opportunities (sub-activity 2.2.4.4). Significant contributions from the regional entities (ECOWAS, ROPPA, ARAA, etc.) will be expected and key lessons learned from the agriculture sector will be capitalized. The implementation of this intervention will also cover the areas that will be restored under component 1 of the programme.

92. The programme will facilitate market linkages between NTFP producers and buyers through information sessions and B2B roundtables for business facilitation (sub-activity 2.2.4.5), addressing another key constraint to increased profitability of good-quality, sustainably produced NTFP. Country-level seminars will be organized on market requirements in international trade, targeted to MSME with export potential (sub-activity 2.2.4.6). Finally, a regional workshop will be organized to share experience gained and lessons learned among the participating countries of the SURAGGWA multi-country programme (sub-activity 2.2.4.7). This will be a periodic and rotated (among the beneficiary countries) activity that will gather the main stakeholders in the implementation of the programme including the direct actors (producers/collectors, processors, distributors, etc.), services providers, Government and non-Government institutions' actors, other programme implementation partners, private sector's actors, regional institutions, etc. Given the nature of this activity, its implementation will be coordinated with the other elements of the multi-country programme particularly component 3 as above-mentioned.

Activity 2.2.5: Identify, train and equip POs and MSMEs with emission-lowering technologies and equipment to enhance NTFP processing

93. Under Activity 2.2.5, partnerships will be established with institutions working on emission-lowering technologies and equipment to enhance NTFP processing. For most of the NTFP, technologically advanced manufacturing processes and significant R&D investments are required for enhancing the efficiency of the process and enhancing the quality of the products to achieve competitiveness. Consequently, the programme will collaborate with research institutions that are specialized in the research and manufacturing of food processing equipment and more specifically in NTFPs. The Country Implementation Units will assess the different available opportunities. The intervention will consider new technologies as well as existing ones. For the existing ones, the programme will intervene in supporting their identification, selection and dissemination. The programme could bear the costs related to the import (if coming from abroad), demonstration and promotion/dissemination of the new technologies/equipment. The programme will offer some financial and technical contributions to assist and

participate in the efforts being made in the research of technological solutions for the processing of NTFPs such as the production of Balanites oil that is very laborious, etc. The mechanism for providing these financial and technical contributions to partner institutions will be defined in consultations with the stakeholders and based on the situational analysis and assessment that will be conducted at the start-up of the programme. The results of these initial investigations will guide the selection process of the partners. (See Annex 2).

94. Key interventions to be conducted for implementing this activity are: (i) identifying emission-lowering technologies (RET) and equipment to enhance NTFP processing (sub-activity 2.2.5.1); and (ii) deploy identified equipment through capacity building agreements with local PO and MSME (sub-activity 2.2.5.2).

Output 2.3 Suitable credit and insurance products designed for smallholder NTFP value chain actors (PO, cooperatives, MSMEs) and increased stakeholder capacity for using these financial products in their NTFP value chain activities

95. The output aims to ease the specific barriers around access to finance for smallholder farmers, producer organizations (PO), cooperatives and associations and micro, small and medium enterprises (MSME) to engage in and scale up income generating activities from agricultural value chains, particularly of Non-Timber Forest Products (NTFPs). These barriers include low physical and digital financial institution penetration in the programme's target areas, low information, lack of collateral, limited products and services tailored to NTFPs and low financial literacy and familiarity with documentation requirements (Annex 2 describes the key constraints to access to finance for the region as well as by country). The output will harness synergies with other access to finance interventions financed by the GCF, notably IGREENFIN that will provide about USD 282 million in financing (through credit lines (senior loans) or technical assistance) for local financial institutions in the Great Green Wall through its Green Finance Facility, thereby reducing credit risk for the institutions themselves. Annex 2 provides the breakdown of IGREENFIN financing, by country and time period. IGREENFIN aims to build and scale the resilience and adaptive capacity of producer organizations, cooperatives and micro, small and medium-sized enterprises (MSMEs) by removing key barriers to farmers' access to financial and non-financial services that support the adoption of best climate change adaptation and mitigation practices and solutions (Annex 2 details the access to finance interventions proposed by iGREENFIN and how the SURAGGWA output will complement those). In particular, the programme's interventions aim at increasing potential borrowers' eligibility to access credit through local financial institutions receiving funding from the IGREENFIN Green Finance Facility (*parallel co-financing*), thereby increasing credit uptake, and enhancing the performance of credit once disbursed through lowering the risk of default. The proposed activities not only capitalize on the technical assistance provided under IGREENFIN to national agricultural banks such as Banque Agricole du Senegal, BNDA Mali and Banque Agricole de Burkina Faso to strengthen their capacity to disburse credit as well as the regulatory environment, but they also further expand them to financial institutions such as commercial banks, microfinance institutions and rural finance players. Further, the activities under Output 2.3 expand the limited reach of IGREENFIN in countries with less developed financial sectors or no GCF nationally accredited entities such as Djibouti.

96. Farmer and other producer organizations and associations face a range of supply and demand side constraints in accessing finance. The main demand side constraints include (i) lack of information about bank products and processes; (ii) long production cycles for NTFP products such as Balanites oil; (iii) borrowers' inability to fulfil documentation requirements to open bank accounts and access loans, and (iv) limited guarantors or property titles, especially for women and youth, to serve as collateral. Financial institutions identified supply side barriers to providing credit to smallholders or MSMEs. These include (i) low bank branch penetration in rural areas, (ii) absence of long-term credit products suitable for the Sahel in several countries, (iii) lack of awareness and training of personnel from financial institutions in agricultural value chains in several countries (Nigeria, Burkina Faso, Niger, Djibouti, Chad, Mauritania), (iv) security risks (in Mali, Northern Nigeria and Burkina Faso), (v) high default risk and absence of agricultural insurance compounding this (particularly in non-agricultural banks in Mali and Nigeria) and (vi) high risk of lending to the agriculture sector and low appetite for lending to the sector.

Activity 2.3.1: Train and sensitize value chain actors on financial institutions' offerings, enhance financial literacy and financial record keeping

97. Activities under Output 2.3 will tackle access to finance constraints through demand and supply-side interventions that will come together to address the barriers in a holistic manner. From the *demand side*, *first*, the programme will map the existing, relevant financial products and services available in the GGW regions, by country for the agriculture and forestry sector (sub-activity 2.3.1.1). Using the mapping thus generated, the programme will familiarize value chain actors (e.g., smallholders, farmers organizations, MSMEs) with the suite of financial services

and options already available to them through local financial institutions (sub-activity 2.3.1.2). *Second*, It will also enhance their understanding of basic financial concepts and practices around record keeping through trainings on financial literacy and financial record keeping, increasing their eligibility to access credit (sub-activity 2.3.1.3). The iGREENFIN capacity development activities do not include any intervention targeting financial literacy and this remains an important gap that the SURAGGWA programme can fill. *Third*, it will link potential borrowers with financial institutions through the business management framework (*linked to output 2.2 above*).

Activity 2.3.2 Technical assistance provided to develop five new financial products (of which at least one insurance product) tailored to agriculture and NTFP value chains with financial institutions, including those collaborating with iGREENFIN.

98. This activity aims to support the enabling environment for access to financial products and services. Through the design of context and region-specific financial products, the activity will make available suitable instruments for channeling short- and long-term finance to support value chain activities. These activities will be undertaken ensuring inclusion of financial institutions already collaborating with iGREENFIN to accelerate access to credit and finance by targeted beneficiaries in relevant value chains. The programme will support the enabling environment through helping design five tailored financial products, one in agricultural insurance and two each in credit and savings.[84] One of the credit products will be a digital-only product to enhance the reach of financial services in remote areas with low physical bank branch penetration.

99. Before designing any new financial products, the programme will assess the existing financial products related to NTFP offered by financial institutions in GGW countries, and their geographic coverage and NTFP actor uptake (sub-activity 2.3.2.1). On the basis of this assessment, it will design two new credit products, one aimed at working capital and one at investment in agriculture and NTFP, and share these products with national and regional financial institutions (sub-activity 2.3.2.2). Subsequently, the programme will test and disseminate these two new credit products, in collaboration with national and regional financial institutions (sub-activity 2.3.2.3).

100. The programme will design one agricultural insurance product (indexed) for the drylands context, with customizable features that can be tailored by country, and share this insurance product with national and regional financial institutions (sub-activity 2.3.2.4). Finally, the programme will develop two new savings products, one digital and one conventional, for building collateral for farmers, and share this with national and regional financial institutions (sub-activity 2.3.2.5).

Activity 2.3.3: Train staff (loan officers and others) from financial institutions, including those collaborating with iGREENFIN, on how to assess agricultural and NTFP value chain risks and climate-related risks (national and regional level)

101. From the *supply side*, *first*, considering the limited lending to NTFP value chains, the programme will train and sensitize loan officers in financial institutions (national agricultural banks, commercial banks, microfinance institutions and rural finance actors such as cooperatives, including those collaborating with iGREENFIN) about the profitability of NTFP value chains, showcasing successful experiences and prospects, as well as on agricultural finance. Two in-person trainings of loan officers will be provided, with at least 80 beneficiaries (sub-activity 2.3.3.1). *Second*, it will train loan officers and other personnel who disburse credit in understanding, assessing and managing agricultural risks and climate-related risks in the agriculture sector with a focus on NTFPs. Four trainings of loan officers will be organized in virtual mode (sub-activity 2.3.3.2). Many of these trainings will also be organized at a regional level, through regional networks such as ECOWAS and AFRACA.

Activity 2.3.4: Facilitate linkages between last-mile providers of finance (producer organizations, village savings groups, microfinance networks) and financial institutions, including those participating in iGREENFIN.

102. In many cases, large financial institutions may not have an extensive physical presence in rural areas. This activity aims to facilitate linkages between financial institutions at all levels, including the providers of credits at the last mile such as village savings groups and producer organizations. The programme will facilitate linkages between last-mile providers of finance such as village savings groups, microfinance networks and producer organizations with national or regional financial institutions such as commercial banks or agricultural development banks. The country-level trainings under Activity 2.3.3 will serve as the forum for these linkages.

103. The programme will map providers of last mile finance in rural GGW regions, by country (sub-activity 2.3.4.1). This data will then be used to facilitate connections between mapped providers of finance and national/regional financial institutions, including those participating in iGREENFIN (sub-activity 2.3.4.2).

[84] Currently, the availability of funds to local financial institutions also remains a hurdle. However, through its Green Financial Facility, iGREENFIN aims to overcome this challenge, increasing local banks' funding and thereby ability to disburse credit to the end users, and thus the present programme does not aim to duplicate this intervention.

Activity 2.3.5. Pilot access to credit through digital financial solutions in at least one country and facilitate scale up, if successful

104. This activity aims to increase access to finance, particularly in rural and remote areas where long distances to physical bank branches prevent smallholder farmers from opening accounts or accessing credit. The programme will collaborate with UNCDF to rollout a digital credit solution in one country, accompanied by e-advisory services that inform farmers about weather conditions to assist them in making evidence-based planting decisions and lower the risk of default. This activity will be co-financed by UNCDF and a national agricultural development bank. FAO will enter into partnership agreement with UNCDF and BNDA for them to develop and roll out digital financial credit products. UNCDF and BNDA will roll out the products at their own costs

105. The programme will test and finalize tailored a digital credit solution with financial institution and fintech in one country (sub-activity 2.3.5.1). It will then roll out this digital credit solution accompanied by an e-weather advisory solution in the country (sub-activity 2.3.5.2). Finally, a seminar will be organized to discuss performance of the product and explore potential for scaling up (sub-activity 2.3.5.3).

Activity 2.3.6. Improve knowledge management and exchanges to increase adoption of best practices across local financial institutions

106. Finally, the programme will also facilitate knowledge and peer-to-peer exchanges between financial institutions at the regional level. A potential entry point for these regional knowledge sharing events and trainings is to utilize the existing pan-African network of the African Rural and Agricultural Credit Association (AFRACA). Their member organizations include agricultural development banks, commercial banks, microfinance institutions and producer organizations. They are present in five of the eight SURAGGWA countries. The programme will organize regional seminars for financial institutions to share experiences with NTFP value chains in the GGW region (sub-activity 2.4.6.1).

Component 3: Strengthening the Great Green Wall institutions at country and regional level

Outcome 3: Enhanced institutional capacities of National GGW Agencies and of PA-GGW for monitoring, coordination, and resource mobilization

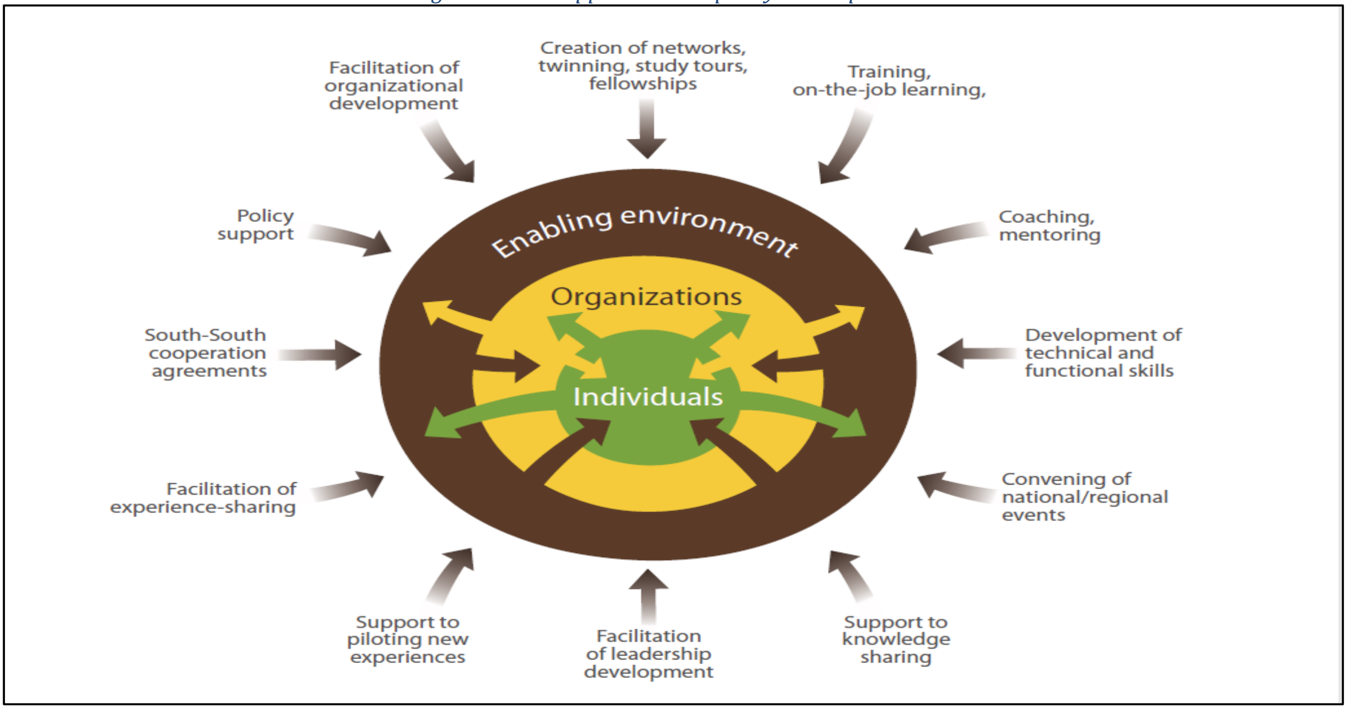
107. Building capable and effective institutions is pivotal to achieving the very ambitious goals assigned to the GGW initiative while being critical to the sustainability of the programme's results, and to enable the GGW Initiative to effectively address the challenge of scaling up. The main target and beneficiaries of the capacity development actions conducted by SURAGGWA are the institutions officially created and mandated to spearhead the GGW initiative at the national and regional levels, namely: (i) the national GGW agencies in the 8 participating countries (and 11 countries specifically for capacity building in ecological monitoring); (ii) the Pan-African GGW Agency.

108. While some gains have already been made in building these institutions through the efforts of the member states and the impetus of the African Union, capacity remains insufficient as pointed out in several recent analyses. SURAGGWA intends to contribute to this endeavor, while aligning its interventions with the needs expressed by the beneficiary institutions, and by complementing the efforts deployed by all other partners, and in particular the effort to build a GCF sponsored Regional Support Platform. SURAGGWA will therefore intervene in four targeted and strategic areas that are fundamental to the successful handling of the countries' (and regional) cycle of GGW investment. This includes: (i) monitoring and reporting capacities, which will be the major focus of SURAGGWA's institutional strengthening (Output 3.1); (ii) planning and coordination capacities which are closely linked to the previous point (Output 3.2); (iii) resource mobilization capacities, with a particular emphasis on carbon finance (Output 3.3); and finally (iv) knowledge management and communication capacities. (Output 3.4).

109. The intervention strategy of the component is strongly based on three principles: first, working jointly at the regional and national levels on a common set of capacity development areas of critical importance for all the Sahel GGW countries. This will allow economies of scale, synergies, and to facilitate convergence towards common and interoperable tools and practices between countries and at the regional level as for example for land restoration monitoring. This will also allow for the definition of common indicators and the promotion of cross-country learning. The second principle is that of adaptability to the realities and specific needs of each of the member countries of the initiative. It is clear that each country has a varying level of capacity that must be taken into account. Also, the specific contexts mean that the paths taken by each country to build certain capacities or to develop their GGW institutions mandates may be different. The SURAGGWA programme will adapt to these realities and propose a

menu of capacity development activities from which countries can select more specific options. Third, capacity is conceived in a holistic manner by implementing FAO's proven corporate approach to capacity development. This implies that action takes place at three levels in a coordinated and complementary manner as illustrated in the following diagram.

Figure 6 - FAO Approach to Capacity Developmentm



110. The component aims to make a significant contribution to developing the capacity of GGW institutions, in coherence and synergy with the efforts of other partners, notably the Accelerator Initiative, and in close complementarity with capacity development activities financed by the GCF, in particular through the IGREENFIN programme. The establishment of the institutional architecture for the GGW was undertaken as a result of the strong political commitment at the country and African Union level. The GGW impetus led to the creation of GGW NAs in all member countries included in the initiative. However, the level of institutional embedding as well as the degree of structuring of the GGW NAs varies from one country to another. In identifying the areas of institutional support, SURAGGWA will complement the work of several on-going initiatives such as the Action Against Desertification programme, the Accelerator Initiative, as well as the Regional Support Platform initiative.

111. The following diagram illustrates the anchoring of SURAGGWA's efforts in the regional support platform initiative, showing the alignment between the main areas of intervention of the three Components and the major thrusts of the accelerator initiative.

Figure 7 - SURAGGWA Anchorage in Accelerator

Component 1 (Regional programme Pillar 2)	Component 2 (Regional programme Pillar 1)	Component 3 (Regional programme Pillar 5)
• Large scale landscape restoration to increase climate change resilience and mitigation, for and by local communities.	***• Financing and building-capacities for bio-based businesses and community enterprises***	***• Strengthening the GGW institutions, to better support the implementation and monitoring of impactful climate action.***

112. The choice of thematic areas for these contributions is based on four elements: (i) on the experience gained from the AAD project; (ii) areas of expertise where FAO has a recognized technical edge and comparative strength at the international level (iii) Needs gleaned through consultations with GGW institutions and NDAs and (iv) capacity development activities already planned by other GGW partners (including UNCCD, UNEP, UNDP and IFAD).

113. Finally, the component provides capacity building elements that are innovative on at least two levels. Firstly, the capacity development actions targeted on monitoring and evaluation will mobilize the most innovative digital tools (while taking care to build a system that is inexpensive and therefore sustainably embeddable). Secondly, the actions in favour of access to carbon finance are also inherently innovative, particularly with regard to the capacity to materialize the opportunities related to the implementation of Article 6, which remains at an extremely embryonic stage in the sub-region despite its importance and potential.

Output 3.1: GGW Land restoration monitoring system at national and regional level upgraded and functional

114. Monitoring and evaluation as well as reporting are a weak element in all these countries, even though some of them have made progress in this regard. The current tools being used are not very efficient and provide little quantitative or qualitative data. The reporting processes in place are ad hoc and very project specific and generally do not last beyond the projects' completion. Furthermore, the reporting arrangements are partial and do not always adequately assess and evaluate the progress of the GGW initiative. This output aims to address the dearth of land-use information for decision makers and sectoral planners in GGW countries. The activities and sub-activities under the component will help build expertise and capacity to use innovative and cost-effective land use assessment tools, especially FAO's Collect Earth tools. SURAGGWA will contribute to reducing duplication and ensuring consistency with other monitoring and reporting efforts on biophysical and socio-economic outcomes from GGW interventions. Reporting outcomes of the GGW accurately, consistently and through standardized processes is critical and by reporting to high-level platforms including the UN General Assembly will raise the profile of the Great Green Wall actions. .

The approach taken by the multi-country programme will be to build on the achievements of the AAD project, which has developed technology-based approaches that are cutting-edge and efficient, and at the same time are locally appropriate and proven in terms of existing capacity. The agencies and other GGWI stakeholders expect that the monitoring capacity will be built in a partnership manner, building on existing institutions and capacities in each country and at the regional level. This means concretely and typically that for each country, a partner institution of the Great Green Wall National Agency (GGW-NA) with expertise in monitoring technologies will be identified for participation. SURAGGWA will carry out coordinated capacity development actions with the GGW-NAs as well as with the national and regional monitoring partner organizations. The GGW-NAs and partner organizations will be trained to (i) use the SURAGGWA monitoring and evaluation system; (ii) develop their capacity as trainers to enable these institutions to play a leading role in supporting the adoption of the system or some of its basic elements by as many GGW land restoration interventions and projects as possible.

115. Building on previous work under AAD, and in close collaboration with PA-GGW, SURAGGWA will contribute to the M&E module of the regional GGW platform. The formulation work of SURAGGWA has also allowed the identification of a set of common GGW initiative indicators initiated within the framework of the Accelerator with the support of the UNCCD. The framework will be used as a basis for the definition and implementation of tools. The framework has been the subject of a multi-country consultation, its use will facilitate the work of harmonization of indicators and interoperability of monitoring systems. What is developed through SURAGGWA will feed the Framework for Ecosystem Restoration Monitoring (FERM) and therefore showcase the contributions of GGW to the UN Decade on Ecosystem Restoration. The system will include geospatial data.

116. FAO has already collected and established biophysical baselines at the country level and for GGW restorable lands (see EarthMap.org) and trained country experts in the use of these tools. The information will continue to be an open source available to interested public, private and civil society actors, and will help to assess progress, coordinate sustainable development and climate action efforts in the region using a climate-based GIS. Baselines will be updated and enhanced through the different FAO OpenForis tools, in particular through the development of user-friendly GGW geospatial modules to enable mapping GGW indicators, using cloud computing, remote sensing and machine learning, integrating socio-economic and biophysical information. The System for Earth Observation, data access, processing and analysis for land monitoring (SEPAL) modules will enable generating up-to-date wall-to-wall information from collected sample data using Collect Earth and for monitoring specific indicators and remote sensing indices, land suitability, and zoning in support of GGW activities, with linkages to ABC Map and the IPCC GHG software. The resulting wall-to-wall information will be completed with socio-economic and biophysical field data collection in 2024 and 2027. It will entail the use of a data collection software package (KoboToolBox) that

combines (1) geo-referenced monitoring of restoration sites as linked to Component 1 activities and (2) household and market level monitoring of critical socio-economic indicators as linked to both Component 1 and Component 2 activities which will also be geo-tagged.

117. Under the programme, training and orientation will also be provided on tools for estimating GHG impacts per unit of land, expressed in tCO_2-e per ha and per year[85], including the Africa Open DEAL methodology and the *EX-ACT tool (Ex-Ante Carbon Balance Tool)* – with the latter having been used to quantify the carbon impact of SURAGGWA (see Annex 22). In addition to these remote sensing applications, SURAGGWA will also provide training for the analysis of soil carbon, which is an essential component of the CO2 results of SURAGGWA and other restoration programmes in the Sahel, but is rarely monitored systematically. For the monitoring of the climate change adapation, SURAGGWA will strengthen the capacity of GGW actors to use theThe *Tool for Agroecology Performance Evaluation (TAPE)*[86] which will aim to provide evidence to policymakers and other stakeholders on how agroecology can contribute to more sustainable food and agricultural systems. TAPE will be used in collaboration with the DyTAES network[87] and also enable cross-sectoral and cross-ministerial cooperation. It will contribute to the empowerment of producers through self-diagnosis and assessment of their system's transition level and performance. Assistance on this tool will be on a tailored basis, depending on country interest.

Activity 3.1.1: Development, testing and deployment of National GGW land restoration monitoring tools and system

118. The programme will analyze existing monitoring mechanisms and tools used throughout the PAGGW and each of the countries – including the overall architecture and existing institutions for data collection and processing – and identify needs at national and regional level (sub-activity 3.1.1.1). Subsequently, GGW monitoring modules and smartphone apps will be developed and tested (sub-activity 3.1.1.2). Finally, the programme will equip and train NAGGW for national GGW monoitoring. Equipment will include computers, laptop, printers, tablets and smartphones (sub-activity 3.1.1.3).

Activity 3.1.2: Transfer knowledge and skills for the use of the GGW Land restoration monitoring system by national and regional authorities.

119. The programme, working closely together with national and regional GGW institutions, will: (i) compile manuals, document procedures (protocols), develop field guides on using the GGW platform (sub-activity 3.1.2.1) Subsequently, monitoring tools will be deployed and documentation produced under activity 3.1.1 will be shared (sub-activity 3.1.2.2). Finally, the programme will provide hands-on trainings and piloting of GGW land restoration monitoring tools (sub-activity 3.1.2.3).

Activity 3.1.3: Development and deployment of regional multi-stakeholder Monitoring platform

120. The next step in enhancing and “de-projectizing” restoration monitoring systems in the GGW zone is to develop and deploy a regional multi-stakeholder monitoring platform. This will be essential for a more transparent sharing of restoration results and lessons learned from different approaches applied in different biophysical and socioeconomic contexts among national and regional stakeholders

121. The first intervention under this activitiy is to design, develop, test and deploy the GGW platform and all its modules, including not just remote sensing applications but also a soil carbon analysis tool[88], ensuring inter-operability between Regional platform and monitoring app(sub-activity 3.1.3.1). Subsequently, the programme will set up a multi-layer Geographic Information System (GIS) of the 8 SURAGGWA countries and their GGW areas, and populate it with regional layers from various sources (sub-activity 3.1.3.2). This product will be translated in at least two languages, including English and French (sub-activity 3.1.3.3). Once the GIS is set up and translated, the programme will roll out, test and operationalize it through technology transfer at country and regional levels (sub-acti vity 3.1.3.4).

[85] FAO Ex-ANTE Carbon-Balance Tool both
[86] FAO TAPE Tool
[87] DyTAES network for the agroecological transition
[88] SURAGGWA will strengthen capacity for soil carbon analysis in all 8 programme countries, ensuring a harmonized regional approach to this key issue.

B

Activity 3.1.4 Building the capacity of NAGGW and PAGGW to establish databases of ongoing restoration projects and programmes contributing to national and regional GGW results

122. With the regional land restoration monitoring platform having been developed and tested, and rolled out to the 8 GGW national agencies, and the PA-GGW and other key national and regional actors, the next step is to populate the platform with data from ongoing restoration projects and programmes that are contributing to national and regional GGW results.

123. The first intervention under this activity is the creation of a stocktaking repository – in Excel, or another easily sharable format – of identified projects and programmes for each country (sub-activity 3.1.4.1). Once this is completed, the programme, in close collaboration with PA-GGW and the 8 NAGGW, will provide technical assistance for building databases at country level (sub-activity 3.1.4.2).

Activity 3.1.5: Prepare regulatory national frameworks for the surveillance of all GGW labelled interventions in all GGW countries

124. With restoration monitoring systems and restoration projects/programmes databases operational, it will be important to agree on national regulatory frameworks for the surveillance of GGW-labelled interventions in GGW countries, and on a division of labour among NAGGW and other restoration actors for applying this framework. This framework, and the division of labour for applying it may vary somewhat from one country to another, so it will be important to involve key restoration actors in a multi-stakeholder process to formulate and validate the respective frameworks in each of the 8 countries.

125. The programme will support the drafting and multi-stakeholder validation of a regulatory framework document for land restoration monitoring, as well as the publication of the documents, in each of the 8 countries (sub-activity 3.1.5.1).

Activity 3.1.6: Develop operational partnerships between the NAGGWs and PAGGW and scientific and technical institutions in the region (such as universities, research institutes, CSEs, Agrhymet, ACMAD) on issues of ecological monitoring and adaptation to climate change

126. To ensure the technical quality and the operational utility of the land restoration monitoring systems established under activities 3.1.1 to 3.1.5, the development of operational partnerships between national and regional GGW actors on the one hand, and scientific and technical institutions on the other will be essential.

127. The programme will train selected staff from scientific and technical institutions in GGW land restoration monitoring system, methodologies and protocols (sub-activity 3.1.6.1). It will also promote the signing of Memoranda of Understanding (MoU) and agreement on joint working modalities between NAGGW/PAGGW and itechnical institutions identified and trained under the programme (sub-activity 3.1.6.2).

Activity 3.1.7: Strengthen the capacities of national stakeholders in the use of the relevant tools developed by FAO (for example, Collect Earth and Africa OpenDeal database), and other partners such as the Ecological Monitoring Centre (EMC), the Sahara and Sahel Observatory (OSS) and CILSS.

128. As highlighted above, FAO – working in partnerships with Google as well as research institutions – has developed many restoration monitoring tools that are highly adapted to Africa's difficult IT context, with many data layers available for free and with all results calculations done "in the cloud", rather than blocking local computers for days. Training relevant national and regional actors in the use of these tools will improve the performance and reduce the cost of and the effort required for restoration monitoring.

129. The programme will develop and deploy tools and training modules, and provide hands-on training courses to key restoration actors (sub-activity 3.1.7.1). It will also organize one training per year for 15 land restoration montiroing specialists from NAGGGW and partner institutions (sub-activity 3.1.7.2).

130. In conclusion, it is important to note that within the framework of output 3.1, the said "regional" activities are activities carried out for the benefit of the SURAGGWA countries and involve the implementation of common and interoperable mechanisms and tools whose future management and development are under the responsibility of the PA-GGW (which will itself receive support from SURAGGWA to ensure this role). The Joint National and

Regional activities address the institutional strengthening needed at the individual country and regional level and may involve joint learning for efficiency purposes and cross-fertilization but do not include or require setting up specific mechanisms at the regional level.

Output 3.2: National and Regional GGW institutions planning and coordination capacities strengthened

Activity 3.2.1: Establish National GGW Coalitions to promote coherent coordination and planning at country level

131. Through this output, the programme will strengthen the capacity of GGW institutions at the national and regional levels to address the challenge of planning and coordinating ongoing land and landscape restoration initiatives that are recognized as an integral part of the GGW dynamic. The GGWI has fostered a great deal of interest in all the countries that have joined it, both from civil society and from private sector actors, NGOs, foundations, development partners, government agencies and their programmes. This has resulted in a multiplication of land restoration interventions claiming to be aligned with or contributing to the GGWI. Although the situation varies from one country to another, the difficulties of coordination are strongly felt by all GGW NAs.

132. While this problem is linked partly to the broader issue of aid coordination, SURAGGWA can provide systemic support to NAGGW to improve their planning and coordination performance. The output will contribute to the objective of meeting this challenge by implementing a coherent set of capacity development actions for improved coordination and planning. This will result in well-defined benefits in the form of well-functioning and inclusive multi-stakeholder consultation, coordination and planning processes and frameworks, which will improve the flow of information on the initiatives taken and promote the coordination of interventions. These regular exchanges will progressively allow all the actors to better identify their own contribution to the GGWI and also to define together more precisely the basic principles of intervention to which they commit themselves by being part of or joining the Initiative.

133. The planning exercises initiated by the GGW-NAs are often unreliable in terms of forecasts and poorly followed up by the initiatives implementing land restoration projects. The activities under this output will contribute to the objective of meeting this challenge by implementing a coherent set of capacity building activities. This will result in tangible benefits in the form of well-functioning multi-stakeholder and inclusive consultation, coordination and planning processes frameworks, which will improve the flow of information on the initiatives taken (a basis for better planning) and promote the coordination of interventions.

134. In order to generate a larger impact, SURAGGWA will align itself with the recent initiative of setting up national coalitions in each of the GGWI countries.[89] These coalitions will serve as a foundation upon which planning coordination activities can be facilitated and conducted in each country. These activities will allow the GGW-NAs to conduct technical exchanges aimed at coordinating the coverage and geographic targeting strategy of GGWI interventions. and define together more precisely the basic principles of intervention to which they commit themselves by being part of or joining the initiative.

135. The programme will establish a baseline for this result by undertakeing a critical analysis of the existing consultation and coordination frameworks, from the local to the national level, in areas related to the GGW and recommend solutions for improving coordination mechanisms for GGW actors (sub-activity 3.2.1.1). With this baseline in place, the programme will design and support the establishment of country-specific GGW National Coalitions, including a 5-year strategy and action plan for their further development (sub-activity 3.2.1.2). It will also provide technical assistance for the initial functioning of the GGW National Coalitions in the 8 countries (sub-activity 3.2.1.3).

Activity 3.2.2: Prepare and issue planning and regulatory frameworks for the coordination of all GGW aligned interventions (projects, programmes, activities, etc.)

[89] https://csopanel.org/briefing-note-on-the-great-green-wall-accelerator/

136. With the land restoration monitoring frameworks in place and key tools deployed as well as capacity for their use strengthened through the preceding activities, the key elements are in place for making the planning efforts of the NAGGW and their national coalitions more effective.

137. The programme will hold a workshop at regional level with the PAGGW and targeted NAGGW to discuss and prepare the generic framework (sub-activity 3.2.2.1). Subsequently, technical assistance will be provided to draft the regulatory framework in each of the targeted countries and at regional level (sub-activity 3.2.2.2). the

138. Within the framework of output 3.1, the said “regional” activities are activities carried out for the benefit of the SURAGGWA countries and involving the implementation of common and interoperable mechanisms and tools whose future management and development are under the responsibility of the PA-GGW (which will itself receive support from SURAGGWA to ensure these roles). The Joint National and Regional activities address the institutional strengthening needed at the individual country and regional level and may involve joint learning for efficiency purpose and cross-fertilization but do not include or require setting up specific mechanisms at the regional level.

Output 3.3: National and regional climate change institutional capacities/frameworks strengthened to integrate land restoration investments in climate change adaptation and mitigation programmes

139. This output aims at building the capacities of the institutional actors of the Great Green Wall (GGW) to mobilize the financial resources necessary to achieve the very ambitious objectives set. The output complements activities included in other SURAGGWA components to strengthen GGW resource mobilization capacities. Under component 1, the more cost-effective restoration methods used in combination with geo-referencing of restoration sites will enable actors to demonstrate climate change mitigation and adaptation impact and efficiency – thus remedying the lack of visibility that has prevented development partners from committing sufficient funds. Under component 2, the use of smallholder value chain support to engage local communities to commit land and labour to restoration efforts, is not only an important resource mobilization strategy, but also a key element of the theory of change of the SURAGGWA multi-country programme as a whole.

140. A number of resource mobilization strategies have already been prepared in the past at the continental and country levels.[1] These initiatives have been unable to raise funds at the scale required and funds are becoming harder to generate as traditional donors have also reduced their development budgets given the pressures they face in the aftermath of COVID-19 and the impact of the Ukraine war. The resource mobilization capacity development efforts to be undertaken by the SURAGGWA multi-country programme, will be focusing on national agencies for the Great Green Wall and Ministries of Finance and therefor be tailored to each specific national context. However, for efficiency purposes the effort will be coordinated at the regional level (through the SURAGGWA Regional Coordination Unit). As needed and suitable, training and tools will be delivered in a coordinated manner through orientation sessions organized by groups of countries. Support will be tailored to help countries to put in place their own mechanisms and arrangements for attracting climate and other innovative finance through private sector carbon developers, social impact investors, fintech operators and financial service providers, among others. The support will be narrowly focused on two very specific aspects of resource mobilization, each of which will constitute areas of innovation for the GGW initiative: (i) access to carbon finance, which is a major challenge and opportunity given the need for adaptation and mitigation; (ii) support for the accreditation efforts of national GGW agencies to climate development funds.

141. In relation to Carbon Finance, the output aims to harness opportunities and strengthen capacities in GGW countries to access carbon finance as a potential source of additional revenue from their land restoration activities that incentivizes scaling up. The Africa Carbon Markets Initiative identifies “forestry and land use” as one of the types of projects capable of generating significant carbon credits.[90] In the drylands, generating the quantity of sequestration that could become viable to attract international financing requires large tracts of land and a large number of participating producers. International financing opportunities exist through the VCM or emerging transactions under article 6 of the Paris Agreement, also linked to compliance markets in buyer countries. In addition, opportunities for carbon finance in and from the region are emerging related to policy commitments to create domestic compliance carbon markets or carbon taxes with potential offset mechanisms.

[90] Roadmap Report, p.17.

142. The main barriers to access *international* carbon markets include scarce experience in the Sahelian countries with carbon finance in agriculture and forestry, fragmented carbon market activities in the region with a limited number of developers and existing projects, inadequate information around opportunities and projects, and a lack of tailored methodologies appropriate for the drylands context.[91] At policy and regulatory level, insecure tenure, carbon rights and challenging business climates are key factors holding back investments. When it comes to *domestic* carbon finance opportunities, the barriers to incentivize restoration relate to the lack of existing models in the GGW countries and capacity gaps in integrating agriculture and forestry into domestic carbon market or tax and offset schemes.

143. The programme will work in coordination and partnership with national and regional initiatives that strengthen country capacities for Article 6, voluntary carbon markets and the creation of domestic carbon tax or carbon markets (Nigeria) across sectors. It will collaborate with these initiatives to harness specific opportunities and strengthen capacities for carbon finance for agriculture and forestry, with a particular focus on restoration. The carbon finance support under the programme aims to unlock financing for additional sequestration beyond those financed directly through the GCF grant and hence enhance country NDC achievements and further ambition. The output will address relevant constraints within the sectoral and geographical scope of the programme, through two categories of interventions: strengthening countries' capacities to access finance from international carbon market opportunities and supporting the creation of carbon finance opportunities for restoration as part of developing national and regional carbon markets.

Activity 3.3.1: Strengthen public- and private sector understanding of and capacity to engage with carbon markets for climate change adaptation and mitigation through land restoration

144. Under the set of activities related to Activity 3.3.1, the programme will facilitate access to international carbon finance for restoration through three sub-activities. Under sub-activity 3.3.1.1, it will assess and enhance carbon finance options, relevant to the specific regional (Sahel) and landscape (dryland) context. Specifically, it will assess the carbon sequestration potential of planned restoration interventions under the programme and identify promising locations and approaches to pursue carbon finance opportunities. It will also take stock of the region's on-going experiences with carbon finance, with a focus on agriculture and forestry (both at the country and regional level) and assess lessons and implications for choice of options. SURAGGWA will assess relevant carbon finance methodologies suitable to GGW restoration already established and used in carbon finance transactions as well as additional methodologies currently under development. As needed, it will complement these by developing additional methodologies for high integrity restoration credits to leverage carbon market opportunities. Finally, it will create and maintain a dynamic database on demand and differentiated buyers and their respective requirements (for example, landscape/regional, system/technology specific). Currently, the SURAGGWA countries do not have a consolidated overview of relevant carbon market opportunities, and specific prospects. Information is partial, dispersed and outdated. This sub-activity would fill this gap and ensure new developments in this dynamic field are captured. The programme will also work with and build the capacity of national or regional organizations to maintain and update this overview and database at regular intervals beyond its implementation period, complementing expected future global information resources, including under the UNFCCC, as appropriate.

145. Under sub-activity 3.3.1.2, the programme will strengthen countries' understanding of and capacity to pursue carbon finance for landscape restoration in the GGW through regional training, knowledge sharing and peer to peer exchanges, in collaboration with regional carbon market initiatives such as the West Africa Alliance on Carbon Markets and the Africa Carbon Markets Initiative. It will provide training to national and subnational policymakers on carbon finance opportunities for restoration based upon assessment of sequestration potential and existing and potential methodologies in the Voluntary Carbon Market and Article 6. It will also facilitate learning exchanges between countries in the region, linked to existing initiatives and organizations (see Annex 2).

Activity 3.3.2: Pilot the establishment of domestic carbon accounting framework that integrates agriculture and forestry in Nigeria, and identify additional countries for potential replication[92].

[91] (ACMI Roadmap Report, p. 18-22).

[92] The institutions mentioned under this Activity will not receive any GCF fund disbursements. Rather, the activities focus on creating networks that will share knowledge generated by SURAGGWA on methodologies and tools for preparing National Carbon Accounting Frameworks.

146. During the Hand-in-Hand Forum organized by FAO in October 2022, several countries expressed interest in engaging with programs that connect smallholder farmers to international carbon markets. In response to country demand, the programme will examine the feasibility and weigh the pros and cons of various schemes with an existing or planned presence in the Sahelian region for specific countries or sub-national settings. For instance, new initiatives and companies such as Climate Asset Management, Agroforestry CRUs for the Organic Restoration of Nature (ACORN) and Earthbanc that target bridging the gap between carbon finance and small-scale producers have expressed interest or are in the process of engaging in carbon finance in the region. Annex 2 provides an overview of these carbon finance investors / opportunities, including the technical specifications underlying their business models.

147. Annex 2 discusses the carbon finance landscape in Nigeria and highlights its feasibility as an entry point for the programme's support given the country's strong political enagement, institutional framework and potential to serve as a model for other Sahelian countries in developing domestic carbon pricing and compliance markets. The mandate of the recently formed Nigerian Council on Climate Change[93] includes setting up the policies and regulatory framework for Nigeria's carbon market or tax.[94] Carbon credits from land use can potentially be marketed domestically to offset tax or compliance targets levied on other sectors, as is already practiced in some countries.

148. The integration of the agriculture and forestry sectors in Nigeria's carbon market policies and regulatory framework would help create the incentives for private investments to generate carbon credits for restoration to meet domestic demand. The activity will include providing technical support to the Nigerian National Council on Climate Change on the inclusion of agriculture and forestry in their planned domestic carbon tax or emissions trading scheme. It will also help engage different stakeholders from the public, private and civil society sectors to feed into how agriculture and forestry are included in the carbon market framework. The programme will also support validating and/or developing methodologies and mechanisms to enable carbon finance flows to GGW restoration in line with the overall domestic market setup.

149. Understanding and meeting the requirements for concrete international voluntary carbon market opportunities or evolving Article 6 financing opportunities linked to their specific technical potentials and policy contexts will be essential for GGW countries to scale up their land restoration efforts in the medium to long term. This activity will enhance countries' capacity to engage with carbon finance opportunities for GGW restoration. This will be achieved through three sub-activities. The programme will provide this support in Nigeria as the pioneering country in the region on domestic compliance markets. The Nigerian carbon accounting framework (including CO2 markets) is currently being formulated by its National Council on Climate Change. This activity would be developed and documented to provide a template that will then be used to support other programme countries.

150. Under sub-activity 3.3.2.1, the programme will assist Nigeria to integrate agriculture and forestry sectors in the planned national carbon accounting system as well as in domestic carbon markets or carbon tax regulations (and offsets). This will be achieved by examining the requirements and features of the relevant emerging schemes, building on a general assessment of the potential benefits for land restoration from the programme in the Great Green Wall area. It will assess their technical feasibility and identify technical and institutional gaps (e.g., land tenure and carbon rights, Monitoring, Reporting and Verification (MRV), institutional capacity in public institutions, local private sector and civil society partners) in Nigeria.

151. Under sub-activity 3.3.2.2, Technical Assistance will be provided to Nigeria's National Council on Climate Change (NCCC), and other relevant institutions, on addressing specific issues in national carbon accounting systems and domestic carbon markets, e.g., carbon tax and offsets or other options.

152. Under sub-activity 3.3.2.3, the programme will use the lessons learned from the process, approach and lessons from integrating the agriculture and forestry sectors carbon market policies in Nigeria to support other programme countries in exploring the potential of introducing similar carbon accounting frameworks in their respective countries.

[93] In its Climate Change Act, 2021, the government of Nigeria lays out the mandate for the National Council on Climate Change. This mandate includes exploring a carbon tax or an Emissions Trading System. Source: https://faolex.fao.org/docs/pdf/NIG208055.pdf
[94] The policy options for the Council include the direct inclusion of agriculture and forestry in a cap and trade system, or a regulatory regime in which domestic credits from agriculture and forestry can be bought as (partial) offsets by companies in sectors that have a compliance target within the domestic carbon market (for example, Colombia).

The programme will provide information and capacity development support to policymakers and local developers at the country level, in partnership with the West Africa Alliance on Carbon Markets, which has confirmed that it views land use as a potential growth area but that it does not have the requisite technical knowledge to engage fully with the sector, expressed its interest in working alongside the SURAGGWA programme. This is very much aligned with the vision on the integration of agriculture and forestry in the regional carbon market, as expressed by the Economic Commission of West African States (ECOWAS) Regional Climate Strategy and Action Plan (2022-2030).[95]

Output 3.4 GGW knowledge management and communication capacities strengthened for mobilizing increased support to climate-resilient land restoration

153. Knowledge management is an essential function on which the GGWI has not performed particularly well to date. The need is clearly expressed and recognized by many actors for improvement of this function, in particular in a context where there is a strong political will to scale up, which implies being able to build on successful models. This point was very clearly underlined by the *Great Green Wall Implementation Status and Way Ahead to 2030 report,*[96] and then taken up as a constraint to be addressed as a priority in the framework of the Accelerator initiative. It is expected that IFAD (with GCF co-financing) will take the lead through the IGREENFIN project, by strengthening capacities at the country and regional levels. The results of the lessons learning exercises sponsored by SURAGGWA will be shared through the regional KM support platform. This output will capitalize on the substantial technical expertise of FAO in analyzing and presenting technical papers and policy briefs.

154. Although highly visible in the international arena, the GGW initiative suffers from several communication deficits. Firstly, there is a profound deficit in technical communication within the scientific events organized by the practitioners working on land restoration issues in drylands. The second communication gap concerns the production of clear and useful messages on its actions and principles of intervention at the local level. Indeed, the success of the GGW initiative relies to a large extent on the adherence of the communities to the approach, as much as the actions to be carried out, (restoration works) and then the maintenance and use of the investments. It is therefore crucial that the GGW NAs have effective and efficient communication mechanisms that have wide outreach. In particular, the messages and modes of communication will have to be adapted to target audiences such as women and youth, as they play key roles and are the primary beneficiaries of the services and products of landscape restoration. The programme will finance initial investments in the preparation of messages and communication media and will help to establish sustainable working relationships between the GGW Nas and field communication actors (rural radio, social networks). In terms of scientific communication, the support provided will allow the PA-GGW and the GGW Nas to play a leading role in the organization of scientific events and in particular the African Drylands Week.
Table B.3.1 in Annex 4 presents a detailed mapping of SURAGGWA activities by country, and intervention under GCF financing.

Activity 3.4.1: Develop innovative and robust methods for evaluating/demonstrating resilience benefits of climate change investments in land restoration and NTFP value chains in GGW

155. Consolidation of lessons-learned, proof of concept of land restoration models and approaches under SURAGGWA and other relevant GGW projects, to inform climate change adaptation fundraising activities. The data and information collected through programme activities will serve as the basis for these studies and will include the development of specific methodologies through technical assistance, working in conjunction with the programme team and national and regional GGW authorities.

156. This activity aims to ensure that robust knowledge products, including analytical and impact studies as well as lessons. learned in terms of approaches and results is documented and disseminated, including through the participation to high visibility events including the African Union's biennial Drylands Week showcase. It is anticipated that the capitalization work could focus on themes such as: conflict sensitive approaches, youth employment, improvement of women's income, use of VGGTs, etc.-

[95]https://www.climatestrategy.ecowas.int/images/documentation/ECOWAS%20Regional%20Climate%20Strategy_adopted%20june%202022.pdf
[96] Page 31 : ": there are no proper and managed knowledge/information sharing and coordination mechanisms at the national and regional levels,….."

157. The following interventions will be undertaken as part of this activity. First, Methods developed, tested and evaluated, including biophysical and socio-economic resilience outcomes (sub-activity 3.4.1.1). Then, technical assistance will be provided for application of innovative and robust assessment methods to relevant institutions (sub-activity 3.4.1.2). On the basis of these interventions, a regional workshop on lessons learned and experience sharing workshop on sustainable land management and adaptation to climate change approaches, for both restoration and economic benefits (sub-activity 3.4.1.3).

158. The programme will also provide technical assistance for reporting on conflict-sensitive land restoration activities (sub-activity 3.4.1.4), for a report on Youth and women inclusion in restoration and NTFP value chain development activities (sub-activity 3.4.1.5), and for a report on Land tenure and VGGTs and restoration activities (sub-activity 3.4.1.6).

Activity 3.4.2 Communication, visibility and dissemination of knowledge

159. Regarding Knowledge Management (KM), SURAGGWA will complement the efforts deployed by IGREENFIN by focusing on lessons learning on the specific experience of the SURAGGWA programme. The results of the capitalization work will be shared through the regional KM support platform. The programme will fund participation in the different events organized by iGREENFIN, the African Union (Drylands Week) and other relevant fora (sub-activity 3.4.2.1).

Activity 3.4.3: Train communication specialists from GGW's national structures to implement their communication plan and develop tools adapted to different target groups, particularly women and youth.

160. The KM actions carried out at the SURAGGWA level will allow the relevant actors of the programme to participate in the exchanges and events animated by the KM platform (contribution to 'community of practice', etc.) In terms of communication capacity building, the activities carried out by SURAGGWA will aim at equipping the communication units of the NAGGW, in connection with the partner administrations that act on the ground close to the field. The programme will finance initial investments in the preparation of messages and communication materials and will help establish sustainable working relationships between the NAGGW and field communication actors (rural radio, social networks).

161. The programme will fund participation in both francophone and anglophone training courses (sub-activity 3.4.3.1).

Activity 3.4.4: Develop, in relation to the different target groups identified, adapted supports and modes of information, such as: the use of social networks (Facebook, Twitter, etc.) and teleconferences, the organization of meetings; workshops, the production of films, leaflets, brochures, etc.).

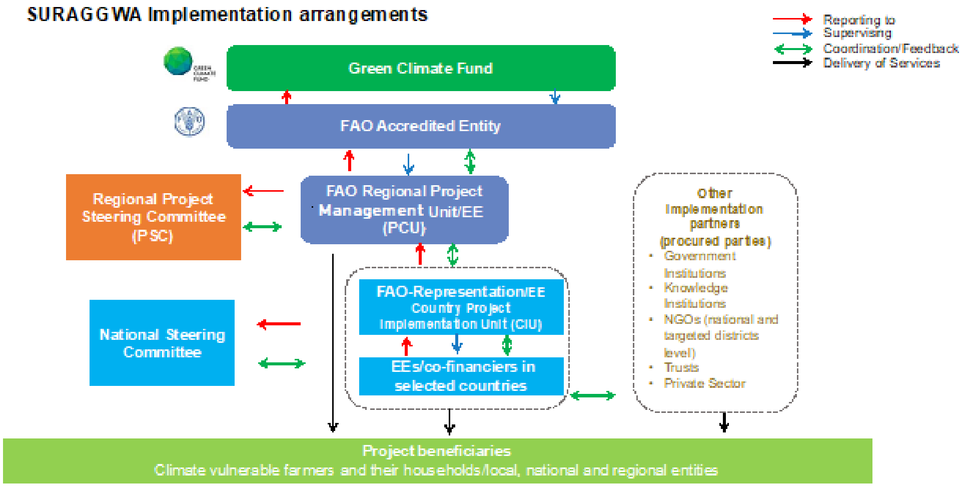
162. The following interventions will be implemented under this activity. A communication tools design workshop will be organized in each of the 8 countries (sub-activity 3.4.4.1) and communication materials will be prepared and deployed to targeted populations – with communication campaigns launched and their impacts tracked in each of the 8 countries (sub-activity 3.4.4.2).

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

163. The SURAGGWA design envisages working closely with a range of regional and international organizations supporting the implementation of the Great Green Wall initiative (GGW) as well as the National Governments in each of the selected countries for the programme. The Pan African Agency of the GGW (PAGGW) created in 2010 has been given the responsibility for the coordination and monitoring of the implementation of the GGW and for the mobilization of necessary resources in relation with the African Union (AU) and the Member States. They all share a common aim focused on the restoration of degraded land, economic development, increasing adaptation and resilience to climate change and the fight against food insecurity and migration. At the national level, Member States have created National GGW Agencies or focal points to supervise and coordinate the implementation of national GGW priority actions. The PA-GGW is expected to play a key role in coordinating country-level

commitments to advancing the objectives of the GGW and is currently being supported by the UNCCD to implement the Great Green Wall Accelerator and by other UN agencies. The implementation of SURAGGWA will build on the synergies with the GGW Accelerator and closely coordinate and complement various efforts with a range of other initiatives working in the GGW area.

Figure 8 - Implementation Arrangements



Project Governance

164. **A Regional Programme Steering Committee (PSC)** will be set up to provide general oversight and strategic guidance in the successful implementation and ensure that the multi-country programme is aligned and well-coordinated with the priorities and policies of the GGW. The PSC will identify how best to capitalize on the synergies with other projects and programmes and draw on lessons learned and how best to upscale them in the GGW initiative. The key responsibility of the PSC will be to identify constraints and opportunities for the multi-country programme, including ensuring the co-financing flow from the counterparts and if need be, enhance resource mobilization opportunities and make recommendations to the Programme Management Unit (PMU) on how to overcome constraints and benefit from opportunities, in order to ensure complementarities and visibility of the programme and the GGW global with other initiatives. It will support the mobilization of additional resources for scaling up the programme and for identifying synergies and leveraging with other major international initiatives and binding commitments and for highlighting the priorities of the area, the catalytic performances, lessons learned and identifying opportunities for collaboration.

165. The first Programme Steering Committee will take place in the Pan African Agency of the Great Green Wall (PA-GGW). Subsequent PSC will take place either virtually, physically on a rotational basis in Dakar and Nouakchott, or in hybrid format. PSC members will consist of representatives of the African Union Commission, as Chair of PSC, assisted by the PA-GGW and two vice-chairs from donor partners and key regional implementation partners, GGW-NAs (and/or representatives of National Steering Committees or GGW coalitions) and FAO. The members will be nominated on a rotational basis according to the three major components of the programme and the GGW global vision. The PSC will invite to its sessions key donor and development agencies working in the Sahel on the GGW initiative and the Southern African Development Community (SADC-GGW) as observers, as well as the World Bank, GEF, AfDB, EU, UN system, etc. to draw on their experience. Private impact investors, Direct Access Entities, financial institutions, research institutions and private sector representatives will be invited as observers as and when appropriate, based on their specific potential contribution to the successful implementation of the multi-country programme. PA-GGW will serve as secretariat for the PSC meetings and it will be responsible for building synergies with current and planned projects co-financing the GGW. The tasks of the Secretariat, in collaboration with the Regional Programme Management Unit (PMU) (described below) will include: (i) Preparing and sending out Invitation Letters at least one month prior to the opening of the PSC Meeting; (ii) production and dissemination of background documents for the meetings, and (iii) production and dissemination of meetings reports.

166. SURAGGWA will also establish **National Steering Committees (NSC)** in each of the 8 target countries where activities will be implemented on the ground. This will be an inter-ministerial steering committee as part of the GGW national coalitions, which will include the GGW-NAs and the NDA as the Co-Chairs, relevant implementing partners, research institutions, technical experts and representatives from the FAO country offices. Private sector partners and representatives from the beneficiary communities will be invited to the NSC as observers and for advice and consultation as and when required. A notification would be issued to establish the programme and national SC's during the inception phase and will be a disbursement condition for each country. Staff of the CIUs will serve as secretariat for the NSC meetings and will prepare background information, reports and organize these meetings.

Accredited Entities

167. FAO will serve as the Accredited Entity (AE) for the programme. The multi-country programme supervision function will remain independent of the Executing Entity functions performed by FAO (see Executing Entities section below). The segregation of responsibilities within FAO will ensure that the Organization can independently and effectively perform the AE functions. As such, FAO will be responsible for the overall management of the project, including (i) all aspects of project appraisal; (ii) administrative, financial and technical oversight and supervision throughout project implementation; (iii) ensuring funds are effectively managed to deliver results and achieve objectives; (iv) ensuring the quality of project monitoring, as well as the timeliness and quality of reporting to the GCF; and (v) project closure and evaluation. FAO will ensure these responsibilities in accordance with the detailed provisions outlined in the Accreditation Master Agreement (AMA) between FAO and GCF.

168. FAO's role as AE will be attributed to relevant Offices and divisions at FAO headquarter (HQ), in Rome Italy and the FAO Regional Office for Africa (RAF), Sub-Regional Offices, and FAO Country Offices in target eight countries.

169. To fulfil the AE functions, the Project Task Force (PTF) will be established by FAO in line with FAO project cycle guidelines and will be composed by the Budget Holder (BH), Lead Technical Officer (LTO), Funding Liaison Officer (FLO), HQ Technical Officer and other technical officers, as appropriate. Members of the PTF will perform the necessary supervision and oversight functions, including supervision and backstopping missions during the entire implementation period, as required. The PTF will be accountable for the quality of programme documentation and implementation throughout the programme cycle and will actively work to manage the agreed results of the programme and ensure appropriate use of resources.

Executing Entities

170. The programme will be executed by FAO and the Government of eight countries as Executing Entities. Activities funded by GCF proceeds will be executed by FAO and the Government of Chad, Djibouti, Mauritania and Senegal while activities co-funded by the Governments and FAO will be directly executed by each co-financier. The specific arrangements in each country were closely coordinated with the NDAs and a list of Executing Entities is outlined in the Table 11 below.

Table 6.a Key Institutions Responsible for SURAGGWA implementation

Level	EE
Regional	FAO Regional Office for Africa (based in Ghana), FAO Sub-Regional Office for West Africa (based in Senegal)
Burkina Faso	FAO Burkina Faso Representation Government of Burkina Faso, acting through Ministry of Environment, Water and Sanitation
Chad	FAO-Chad Representation Government of Chad, acting through Ministry of Environment, Fisheries and Sustainable Development
Djibouti	FAO-Djibouti Representation Government of Djibouti, acting through Ministry of Environment and Sustainable Development
Mali	FAO-Mali Representation Government of Mali, acting through Ministry of Environment and Sustainable Development
Mauritania	FAO-Mauritania Representation Government of Mauritania, acting through Ministry of Environment & Sustainable Development

Niger	FAO-Niger Representation Government of Niger, acting through Ministry of Hydraulics, Sanitation and Environment
Nigeria	FAO-Nigeria Representation Government of Nigeria, acting through Federal Ministry of Environment
Senegal	FAO-Senegal Representation Government of Senegal, acting through Ministry of Environment and Sustainable Development

Table 7.b Project sub-activities per EE

Sub-activity	Executing Entity (EE)									Funding source
	Burkina Faso	Chad	Djibouti	Mali	Mauritania	Niger	Nigeria	Senegal	Regional	
1.1.1.1	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
1.1.2.1	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
1.1.2.2	FAO	FAO	FAO	FAO	FAO	FAO	FAO	Gov	-	GCF
1.1.2.3	FAO	Gov	Gov	FAO	Gov	FAO	FAO	Gov	-	GCF
1.1.2.4	FAO	Gov	Gov	FAO	Gov	FAO	FAO	Gov	-	GCF
	-	-	-	-	-	-	-	Gov	-	Gov-Senegal
1.1.2.5	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	GCF
1.1.3.1	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
1.1.3.2	FAO	Gov	Gov	FAO	Gov	FAO	FAO	Gov	-	GCF
1.2.1.1	FAO	Gov	Gov	FAO	Gov	FAO	FAO	Gov	-	GCF
1.2.1.2	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
1.2.2.1	FAO	Gov	Gov	FAO	Gov	FAO	FAO	Gov	-	GCF
1.2.2.2	FAO	FAO	FAO	FAO	Gov	FAO	FAO	Gov	-	GCF
1.2.3.1	FAO	FAO	FAO	FAO	Gov	FAO	FAO	Gov	-	GCF
	-	-	-	-	-	-	-	Gov	-	Gov-Senegal
1.2.3.2	FAO	Gov	Gov	FAO	Gov	FAO	FAO	Gov	-	GCF
1.2.3.3	FAO	FAO	FAO	FAO	Gov	FAO	FAO	Gov	-	GCF
1.2.4.1	FAO	FAO	FAO	FAO	FAO	FAO	FAO	Gov	-	GCF
1.2.4.2	FAO	FAO	FAO	FAO	FAO	FAO	FAO	Gov	-	GCF
1.3.1.1	FAO	Gov	Gov	FAO	Gov	FAO	FAO	Gov	-	GCF
1.3.1.2	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
	-	-	-	FAO	FAO, Gov	FAO	Gov	Gov	-	FAO, Gov-Mauritania, Gov-Nigeria, Gov-Senegal
1.3.2.1	FAO	FAO	FAO	FAO	Gov	FAO	FAO	Gov	-	GCF
1.3.3.1	FAO	FAO	FAO	FAO	Gov	FAO	FAO	Gov	-	GCF
1.3.3.2	FAO	FAO	FAO	FAO	Gov	FAO	FAO	Gov	-	GCF
1.4.1.1	FAO	Gov	Gov	FAO	FAO	FAO	FAO	Gov	-	GCF
1.4.1.2	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
	-	-	-	FAO	FAO, Gov	FAO	Gov	FAO, Gov	-	FAO, Gov-Mauritania, Gov-Nigeria, Gov-Senegal
1.5.1.1	FAO	Gov	Gov	FAO	Gov	FAO	FAO	Gov	-	GCF
1.5.1.2	FAO	FAO	FAO	FAO	Gov	FAO	FAO	Gov	-	GCF
	-	-	-	-	Gov	-	-	-	-	Gov-Mauritania
2.1.1.1	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	GCF
2.1.1.2	FAO	Gov	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.1.1.3	FAO	Gov	Gov	FAO	FAO	FAO	FAO	FAO	-	GCF
2.1.1.4	FAO	Gov	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.1.2.1	FAO	Gov	Gov	FAO	FAO	FAO	FAO	FAO	-	GCF
2.1.2.2	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.1.3.1	FAO	Gov	Gov	FAO	FAO	FAO	FAO	FAO	-	GCF
2.1.3.2	FAO	Gov	Gov	FAO	FAO	FAO	FAO	FAO	-	GCF
2.2.1.1	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	GCF
2.2.1.2	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF

2.2.2.1	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.2.2.2	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.2.2.3	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.2.2.4	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.2.3.1	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.2.3.2	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.2.4.1	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.2.4.2	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.2.4.3	-	-	-	-	-	-	-	-	FAO	GCF
2.2.4.4	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.2.4.5	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.2.4.6	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.2.4.7	-	-	-	-	-	-	-	-	-	GCF
2.2.5.1	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	GCF
2.2.5.2	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
	-	-	-	-	-	-	Gov	FAO, Gov	-	FAO, Gov-Nigeria, Gov-Senegal
2.3.1.1	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.3.1.2	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.3.1.3	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.3.2.1	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.3.2.2	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.3.2.3	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.3.2.4	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.3.2.5	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.3.3.1	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	GCF
2.3.3.2	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.3.4.1	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.3.4.2	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
2.3.5.1	-	-	-	-	-	-	-	-	FAO	GCF
2.3.5.2	-	-	-	-	-	-	-	-	FAO	GCF
2.3.5.3	-	-	-	-	-	-	-	-	FAO	GCF
2.3.5.4	-	-	-	-	-	-	-	-	FAO	GCF
2.3.6.1	-	-	-	-	-	-	-	-	FAO	GCF
3.1.1.1	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	GCF
	Gov	Gov	Gov	Gov	Gov	Gov	Gov	Gov	-	Each government
3.1.1.2	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
3.1.1.3	FAO	Gov	Gov	FAO	Gov	FAO	FAO	Gov	-	GCF
3.1.2.1	-	-	-	-	-	-	-	-	FAO	GCF
3.1.2.2	-	-	-	-	-	-	-	-	FAO	GCF
3.1.2.3	-	-	-	-	-	-	-	-	FAO	GCF
3.1.3.1	-	-	-	-	-	-	-	-	FAO	GCF
3.1.3.2	-	-	-	-	-	-	-	-	FAO	GCF
3.1.3.3	-	-	-	-	-	-	-	-	FAO	GCF
3.1.3.4	-	-	-	-	-	-	-	-	FAO	GCF
3.1.4.1	FAO	Gov	Gov	FAO	Gov	FAO	FAO	Gov	-	GCF
3.1.4.2	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
3.1.5.1	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
3.1.6.1	-	-	-	-	-	-	-	-	FAO	GCF
3.1.6.2	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
3.1.7.1	-	-	-	-	-	-	-	-	-	GCF
3.1.7.2	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
3.2.1.1	FAO	Gov	Gov	FAO	Gov	FAO	FAO	Gov	FAO	GCF
	Gov	Gov	Gov	Gov	Gov	Gov	Gov	Gov	-	Each government
3.2.1.2	FAO	Gov	Gov	FAO	Gov	FAO	FAO	Gov	-	GCF
3.2.1.3	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	GCF

3.2.2.1	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
3.2.2.2	-	-	-	-	-	-	-	-	FAO	GCF
3.3.1.1	-	-	-	-	-	-	-	-	FAO	GCF
3.3.1.2	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	GCF
3.3.2.1	-	-	-	-	-	-	-	-	FAO	GCF
3.3.2.2	-	-	-	-	-	-	-	-	FAO	GCF
3.3.2.3	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	GCF
3.4.1.1	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
3.4.1.2	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
3.4.1.3	-	-	-	-	-	-	-	-	FAO	GCF
3.4.1.4	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
3.4.1.5	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
3.4.1.6	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
3.4.2.1	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
3.4.3.1	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	-	GCF
3.4.4.1	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	GCF
3.4.4.2	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	FAO	GCF

171. The regional **Programme Management Unit (PMU)** will be responsible to facilitate the management of the overall multi-country programme. The PMU will be composed of a coordination team, based in the FAO-Sub-Regional Office for Africa in Senegal. In addition, a Liaison Officer will be based in the PA-GGW in Nouakchott, who will closely coordinate with the PA-GGW HQ offices located in Mauritania and support the Secretariat. The liaison based in PA-GGW Mauritania will enable building synergies with current and planned projects co-financing the SURAGGWA and better coordination with the Accelerator. The PMU will support County Implementation Units (CIUs) efforts to implement activities at country level, as well as implement regional-level activities. It will also monitor progress against planned results at both country- and regional level, and ensure the overall coherence of all programme activities. Furthermore, the PMU will provide overall management of the programme, in collaboration with the PSC, as well as technical backstopping, and ensure timely implementation, reporting and trouble-shooting.

172. The actual implementation of the programme will be undertaken at the country level through **Country Implementation Units (CIUs)** which will be housed at the FAO country offices (in the case of Mauritania and Senegal, the CIU will be housed in the National Agency for the Great Green Wall). A National Programme Coordinator will be appointed by FAO, in consultation with the national counterpart institutions. Each CIU will also recruit a land restoration specialist, who will serve as CIU Coordinator, a national value chains and rural finance specialist, an ecological and monitoring and evaluation specialist, and a social safeguards and gender specialist, as well an administrative, procurement and finance staff.

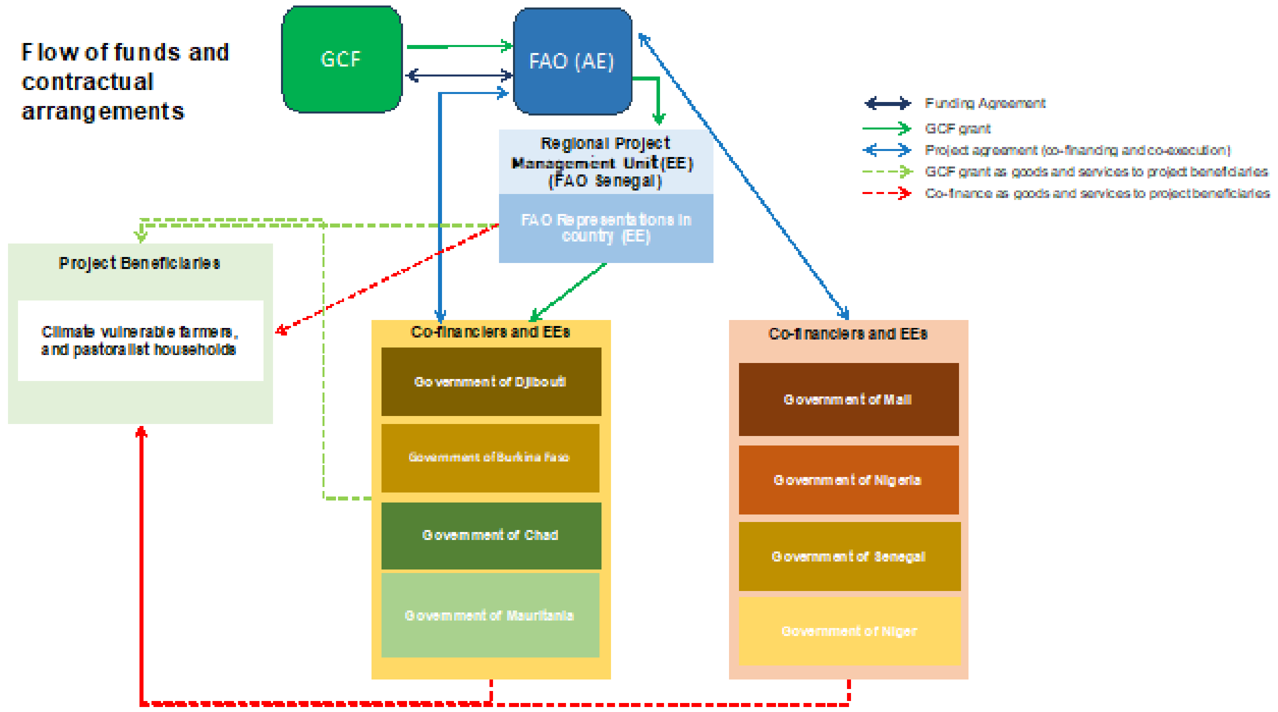
173. **Direct implementation** on the ground will be undertaken at country level, through a combination of modalities that will use the well-established capacity of FAO and government co-Executing Entities in the respective host countries. FAO, line Ministries, service providers, technical partners would be part of Country Implementation Units that will assume responsibility for the implementation of the country level activities. FAO and NDAs will interact with other sectoral ministries, the national GGW agency, regional research institutes, GCF direct accredited entities (where available), service providers and NGOs who would act as implementing partners or service providers for on-the-ground activities under the overall technical support and coordination of the CIU.

Flow of funds and legal arrangements

174. In line with the programme implementation arrangements outlined above, the Funded Activity Agreement (FAA) will be signed between FAO and GCF and GCF proceeds received by FAO in its capacity as Accredited Entity will flow to the co-Executing Entities, namely the Government of Chad, Djibouti, Mauritania and Senegal for the execution of programme activities funded by GCF proceeds.

175. The Project Agreement between FAO and the Government of the eight countries which also include relevant clauses related to co-financing (and execution of the cofinancing) from the Government (Subsidiary Agreement).

Figure 9 - Flow of funds and contractual arrangements



176. The co-financiers (the eight Governments) will be responsible for reporting to the AE in accordance with the detailed provisions outlined in the GCF policies as well as AMA, Funded Activity Agreement (FAA) between FAO and GCF and the Project Agreements to be signed between each government's co-financier and FAO in its capacity of AE, on co-financing activities execution, and the allocated co-financing amount. Each co-financier will be responsible for executing and managing their co-financing funds under the coordination of the Programme Management Unit (PMU) and through the Programme Steering Committee. There will be no contractual agreements to be entered into between the AE and beneficiaries in the countries and FAO will not make cash transfers directly to programme beneficiaries.

177. **FAO**, a specialized technical agency of the United Nations, is well placed to undertake the planned activities given its potential for addressing climate change adaptation and mitigation, reducing poverty and boosting resilient livelihoods. FAO is a key technical partner of the African Union and its Member States, which have devised a clear and detailed approach and methodology for solutions in the Sahel. FAO has implemented a large number of climate adaptation and mitigation projects worldwide and is currently supporting regional technical cooperation programme and projects on NDCs and AFR100. FAO will continue to play a key role of convener and facilitator, bringing countries and partners and sharing experience for scaling up its successful results. In the interest of effective realization of the GGW vision, from 2010 to 2013, with funding from the European Union, FAO supported the African Union Commission and AU member states as they prepared national climate strategies and action plans, including NAP, NAMAs and NDCs, developed a *Regional Harmonized Strategy for GGW*, created a community of practice, and mobilised partnerships. A major outcome was that all country action plans include large-scale restoration of agro-silvo-pastoral systems, which combine arable farming, agro-ecology, livestock and tree-based production, as one of the highest priorities, which SURAGGWA will up-scale. FAO has designed a comprehensive approach for large-scale dryland restoration for GGW backed by the Global guidelines for the restoration of degraded forests and landscapes. FAO will build on its ongoing partnerships - African Union Commission, PA-GGW, UNCCD, EU, AfDB, UNEP, IFIs (WB, IFAD, AFD, BADEA), regional research and other organizations and NGOs – to ensure proper coordination, experience exchange, avoid duplication and ensure impact in the Sahel. FAO's vast and unique technical and organizational qualities and experience are well suited to manage this large multi-country programme. Additionally, FAO has leveraged co-financing through various financial instruments with sources of funds including from GEF, the World Bank, and Global Affaires Canada, reflecting the ongoing partnership and collaboration towards achieving the GGW restoration targets.

Other Project Partners

178. The partners, including mainly the CBO, women and youth groups, local service providers, will be engaged during project implementation as a service provider in accordance with relevant FAO Manual Sections. The AE will conduct procurement based on its FAO Manual Sections as per reaccreditation while the co-EEs will conduct procurement following their own procurement policy and rules.

179. SURAGGWA will use both private sector and Direct Access Entities (DAEs) as service providers. In accordance with FAO Manual Section 501, FAO will procure services from public and private sector entities such as Centre de Suivi Écologique (CSE) and La Banque Agricole both in Senegal for renting equipment, purchase of restoration seeds, capacity building and other restoration and income generation (NTFPs) activities. In relation to the DAEs, a list of the DAEs in the programme countries was examined to assess how they might be involved in the programme. DAEs including CSE and La Banque Agricole will also benefit from capacity building of this programme.

180. **For Access to Finance**: The programme will work directly to strengthen the capacity and promote innovation of at least one entity that is accredited with the GCF: Banque Agricole du Senegal is a Senegalese national entity based in Senegal that focuses on the provision of financial products and services to the country, including the project's areas of intervention. Additionally, the Banque Nationale de Developpement Agricole (BNDA), will provide USD 10 million in parallel financing. The project will include capacity development activities for the banks' loan officers on specific areas such as evaluating and assessing agricultural risk and developing new financial products. The project – in collaboration with UNCDF – will collaborate with the Banque Agricole du Senegal to roll out a digital financial credit product along with e-advisory related to crops/weather to the bank's clients. The Country Implementation Units (CIUs) in each of the countries will cover the functions related to the national activities around access to finance

181. **Carbon Finance:** In implementing its carbon finance activities, the project will target relevant staff from national governments and their partners, regional organizations or networks working on carbon markets, in particular the West Africa Alliance on Carbon Markets, ECOWAS, the Africa Carbon Markets Initiative, and the pan-African agency for the Great Green Wall. In particular in Nigeria, the project will work in close collaboration with the Nigerian National Council on Climate Change. For capacity development activities such as trainings or workshops, the programme will target networks of the West Africa Alliance on Carbon Markets as well as the Pan-African Agency of the Great Green Wall. The programme will to provide technical assistance and capacity development to national agencies to ensure that national GGW agencies and their partners are better equipped to engage with carbon markets for climate change adaptation and mitigation.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

182. GCF funding is requested for the GGWI because of the large unmet needs of the programme. The recent UNCCD report on the status of the GGWI has clearly stated that if the current level of funding and pace of implementation continue, the GGWI's objectives and all the co-benefits of this flagship initiative will not be met.[97] According to this report, it is estimated that a minimum investment of US$33 billion from national and international sources will be required to support countries in achieving the GGW objectives by 2030. With the shift in focus of the GGWI from tree planting to more comprehensive rural development, including improving the production systems of smallholder farmers and pastoralist communities to increase food security, create jobs and boost the local economy through sustainable land management, countries are now tackling systemic challenges in an integrated and holistic manner to ensure long-term sustainability in the entire Sahel region. Stronger support, coordination and engagement by development partners, international financing institutions and the private sector are critical for harnessing emerging technologies and innovations and promoting policy options that unlock market opportunities and capital for scaling up interventions. Financial needs for impact at scale are estimated to average between US$ 40-130 million per country annually. More resources will thus have to be mobilized from a range of domestic and international sources. GCF funding is critical not only in terms of the direct support it can provide but the potential for leveraging other resources.

[97] September 2021. UNCCD.

183. GCF funding is justified because the programme is located in very vulnerable and fragile areas of the selected countries which are deeply impacted by climate risks and climate change. The encroachment of the desert, the increase of extreme weather events, the variability of rainfall and the harsh climate conditions, in combination with sometimes poor land use practices, are eroding the productivity of the fragile lands and ecosystems. Of the eight countries, 5-6 are among the poorest in the world, as measured by the UN Human Development and climate resilience indices. The land tenure pattern in the areas being targeted includes both common property resources and small parcels of farmlands owned by small producers. There is little incentive to make long-term investments in the common lands over which the pastoral and farming communities have use rights. The individual households which own the small plots of land are too poor to make any investments for which there is no immediate return. The upfront costs of rehabilitating the land are high and the benefits are diffused among a large number of households who are not organized to benefit from them. The smallholder farmers have limited access to financial services and their risk profile is such that few, if any, financial institutions have the risk appetite to loan to them for adapting their agriculture production systems to climate change. The private sector has little incentive to invest in areas where there is no security of tenure, production is scattered and volumes and quality do not justify investments with the high associated risks and no immediate mechanisms to internalize the externalities generated from the investment. The public sector has limited resources and other urgent priorities which take precedence. These countries are highly indebted LDCs and do not have sufficient funds for upscaling interventions that have the potential of providing vast mitigation benefits from the restoration of agro-silvo-pastoral landscapes and ecosystems, as well as improving the resilience and livelihoods of the poor communities depending upon these areas.

184. GCF funding is being requested for SURAGGWA to ensure the success of the GGW initiative by providing significant impetus to its mitigation and adaptation mandate and highlight its significance in Africa. GCF support is imperative for the GGW given that it is an African-led flagship programme demonstrating how to harness the power of nature to provide policy solutions to multiple and complex environmental threats, such as land degradation, desertification, drought, climate change, biodiversity loss, poverty and food insecurity, simultaneously. The GGW has inspired many African countries which are now associated with it and its work is contributing to the implementation of the post-2020 global biodiversity framework. GGW is among the iconic global campaigns targeted for completion during the Decade of Ecosystem Restoration ending in 2030. While the Great Green Wall countries have attracted financing from a host of other donors and have been doing so for the last two decades, the financial needs are significant. New funding of USD 14.32 billion was announced in the One Planet Summit for Biodiversity co-organized by France, the United Nations and World Bank in January 2021. However, not all the resources promised are delivered and there is some uncertainty about the funding that will eventually be made available. The country Governments are also identifying innovative ways to raise funding such as through the establishment of a biodiversity Fund for contributing a portion of the resources that may result from the cancellation of national debts. In a post COVID context, Sahelian countries are struggling with budgets and funding and firm commitment and support is required. While the funding requested from GCF is relatively small in relation to overall GGW financing needs, it will contribute significantly to demonstrating the interventions that can help upscale land rehabilitation and building climate resilience of vulnerable communities threatened by climate change.

185. The SURAGGWA multi-country programme will ssupport smallholders to enable them to undertake land restoration activities on highly degraded common lands and moderately degraded farmlands. For highly degraded common lands, this investment cost will include the cost of deep ploughing through the use of both mechanized, animal-drawn and manual means, the cost of land contouring for preparing water catchments, assistance with recuperating soil health, provision of seeds and seedlings and the cost of capacity building and training in the use of traditional and modern techniques of land restoration. Programme financing is expected to cover only the initial costs of restoring degraded lands and the surrounding ecosystems. Thereafter, the management of the restored lands will be the responsibility of the local communities (for common property lands) and farm households. The consultations with the local communities prior to the planning of restoration activities will ensure their priority grass, shrub and tree species will be included and will generate early economic returns, e.g. from grasses that can be used as animal fodder. As the trees grow in the managed restoration areas, the production of non-timber forest products will provide sufficient economic incentives for local communities to continue maintaining and using these lands on a sustainable basis.

186. SURAGGWA will also provide funds to hire service providers to organize the women and men who rely on non-timber forest products and help them facilitate access to markets and build links with the private sector to earn

additional income from a range of promising products such as Balanites, baobab, neem, jujube, forage crops, etc. The programme will finance the cost of establishing and organizing commodity groups, basic equipment and infrastructure for storage, sorting and processing to aggregate the produce and add value and attract private sector partnerships. The private sector and the commodity groups will be expected to finance the subsequent costs once the systems are set up and the market links are established. It is expected that the private sector partners will provide the additional financing needs to reach scale for any of the commodity groups as both working capital or investment capital through value chain financing arrangements through their own funds or through access to formal financial institutions.

187. The GCF grant is essential as the Government does not have the resources to cover these investment costs due to the limited public sector capacity to generate funds and the high level of country indebtedness that inhibits further borrowing. The private sector has limited incentive to invest in these areas and there has been limited incentive for them to invest without providing the local communities and smallholder farmers with initial support in helping to aggregate the local produce, increase volumes and quantity of non-timber forest products to make the enterprise viable for them.

188. GCF concessionality is further justified based on the significant potential for SURAGGWA to generate climate change mitigation as well as adaptation benefits. There is significant potential for carbon sequestration due to the rehabilitation of the degraded lands in the Sahel and helping smallholder producers adapt to the climate threats by diversifying their livelihoods on a sustainable basis through investments in sustainable land management practices. The level of concessionality requested for the countries under SURAGGWA is justified based on their economic and environmental vulnerability, the high level of country indebtedness and the low level of readiness to adapt to climate impacts (See section D.4). The programme is also expected to make a significant contribution to GCF's goal of carbon sequestration and investment in vulnerable hotspots at imminent risk. Component 3 is designed to strengthen the regional and country-level architecture of the GGW and facilitate information exchange for upscaling and replicating the positive experience from successful projects of this nature. The investment in this component is justified based on the high value added of capacity building and institutional strengthening in supporting land restoration activities by local communities and smallholder farmers.

B.6. Exit strategy (max. 500 words, approximately 1 page)

189. The multi-country programme has developed a clear exit strategy for its investments. ***Under component 1***, SURAGGWA will make just one-time investments in the restoration of both highly degraded common lands and moderately degraded smallholder farmlands. The programme will monitor and support the investments for the first two years and thereafter, the users and owners of the land are expected to continue maintaining them on their own due to better defined use rights and the additional incentives of using the grasses for forage for their animals and non-timber forest products as a source of additional income. The programme is designed to *strengthen existing groups or establish new ones* to undertake the land rehabilitation and value chain development activities. The programme will work with community groups build local capacity and support village technicians who will provide support and follow-up for the restoration activities and assist in the monitoring and maintenance of the investments on common lands. Generally, people who put themselves forward as village volunteers are activists willing to provide support and assistance on an on-going basis. They also have the capacity for creating supportive networks and working with NGOs to leverage additional support for their communities. Previous experience with AAD shows that communities prioritize rewarding the Village Technicians from the first benefits or incomes coming from restoration activities and interventions. This provides additional incentives to them to continue their support. Communities under SURAGGWA will be encouraged to follow the same pattern. The programme will also *establish community seed and seedling production for land restoration activities*. These groups are expected to sustain themselves through the sale of seed and saplings to local communities. The programme will *strengthen the rental markets for deep ploughing equipment*, training local technicians in restoration techniques and providing a model of community organization and with the capacity and access to seeds and sapling from local suppliers to expand the restoration activities on their own either individually or collectively.

Table 8 - Exit Strategy Component 1 Land Restoration

Type of Investment	*Programme Investments*	*Exit Strategy*	*Maintenance*
Organization of Community Groups	Development of technical networks in villages.	Organize and train community groups and work with them for two years.	The community groups are expected to maintain the restored common lands and farm lands.
Village Technicians	Build capacity.	Provide them initial training and monitoring support for first two years.	The VT are expected to continue their work on their own through a combination of incentives and activism and support from NGOs and local social networks.
Rehabilitation of common property degraded lands	Assistance with deep ploughing, land contouring, seed and seedlings, labour and water catchments.	Once the land is deep ploughed the seeds of woody and herbaceous native species are sown directly, and inoculated seedlings planted. The programme will provide only one time assistance in the first year of the programme cycle in the selected area and thereafter only supervision and monitoring will be undertaken for the first two years.	The ploughed lands species are very resilient and work well in degraded land, providing vegetation cover and improving the productivity of previously barren lands. After the initial assistance, communities are expected to maintain these lands due to the potential for additional source of income from fodder and non-timber forest products and better defined use rights.
Enhancing productivity on farm lands.	Subsidizing initial investment costs for improved protection against wind and soil erosion and increasing supply of water for improved productivity.	The programme will only provide initial investment costs for farm lands.	Individual households will have added incentive to protect and maintain the lands on which productivity has been enhanced.
Provision of mechanized equipment including Delfino and other deep ploughs.	The programme will encourage private rental markets to provide the mechanized deep ploughs.	The equipment is supplied based on a contractual relationship.	The private sector will operate and maintain the equipment and machinery.
Production of seeds and seedlings.	Mobilize community seed production groups, establish links with private sector and strengthen the capacity of public sector institutions for production of high-quality seed and quality control.	The community groups will be supported for the initial 2 to 3 years through provision of initial inputs, training and linking with public and private sector. The public sector will be given key training and equipment for the purpose and the private sector will be assisted only with establishing links between the producers and those communities in need of the seed and seedlings.	It is expected that 70% of the seed groups will continue to produce the seed and the public sector institutions will use their own budgets to continue to produce and monitor the production of high quality seeds and seedlings. The private sector is expected to tailor its production and distribution based on market conditions and supply and demand factors.
Capacity building for enhanced use of both traditional and modern	A one-time training will be provided to selected community members, staff	The programme will train master trainers in the	The curriculum of the training institutions will be modified to

techniques of restoration and enhancing bio-diversity.	from research and forestry centres in the selected countries as well as modification in the curriculum for researchers, extension staff and community members.		incorporate some of the traditional and technical knowledge so that the techniques are imparted to others trained in these institutions.

190. The exit strategy under component 2 dealing with non-timber forest products *hinges on the continued interest of smallholder producers in NTFP production and marketing and in their engagement with the private sector*. Local communities who rely on the collection of the non-timber forest products will be better linked with the private sector along the value chain. These private sector actors are involved in aggregating the collection of the products, transporting, processing and trading them in the domestic and international markets. There is considerable private sector interest in a range of NTFPs, however, their engagement has been limited due to lack of community organizations and groups that can help to aggregate produce and assure them of a certain minimum volume according to an agreed schedule. During the formulation stage, *private sector entities have shown interest in negotiating purchase agreements*, and utilizing their own funds to up-grade the processing and marketing of the produce. SURAGGWA will organize women and men engaged in collection and facilitate their interaction with the private sector. The exit strategy is premised on previous experience which indicates that once mutually beneficial contractual relationships are established between community groups collecting the NTFPs and private sector interested in purchasing them, these relationships sustain on the strength of the benefits generated for both. The project programme will promote the participation of women and youth as members and leaders of the restoration and VC committees. The project programme will provide the organizational and technical capacity for aggregating the produce and enhancing the quality through improved collection, storage and processing techniques and equipment. Incentives for both parties are expected to increase over time with increased volumes and quality of production.

191. It is also expected that SURAGGWA will facilitate links to enable the value chain actors to access finance from a range of commercial financial institutions. The Output on Access to Finance (Output 2.3) will address key constraints to finance, particularly around the uptake and efficacy of credit. The activities ensure the outputs sustainability beyond the project life cycle through building the capacities of financial institutions and potential users of finance. The facilitation of initial linkages between financial institutions and the providers of credit in rural areas such as village savings associations and farmer associations will expand the reach of financial services in previously unbanked areas. The development of innovative products and channels such as digital financial solutions will enhance the potential reach of financial services going forward. Finally, the creation of a tailored product on savings will provide a vehicle for smallholder farmers, organizations or MSMEs to build their collateral for future credit, even beyond the project life cycle. This increased access will enable those with the potential to grow to use the funds to expand their market. Several Direct Access Entities such as La Banque Agricole have applied for accreditation with GCF and in the future they could use the GCF Facility to expand their own operations to contribute further to the objectives of the GGWI as well.

Table 9 - Exit Strategy Component 2 Non-Timber Forest Product Value Chains

Type of Investment	***Programme Investments***	***Exit Strategy***	***Maintenance***
Organization of Collectors of NTFPs.	Assistance with formation of NTFP groups.	Initial support and monitoring in formation for first two years.	Community Groups will meet on their own.
Training in collection, storage and simple processing techniques.	Training	One time training in identified techniques.	Monitoring and supervisions will be undertaken by SURAGGWA during the first three years and final year.
Grant for equipment for collection, storage and simple processing.	Equipment and Facilities	Capital Cost	Community Groups will maintain the equipment and facilities.

B

Facilitation of Interaction with Private Sector	Market Linkages	Initial contacts with private sector.	The private sector is expected to renew contracts with communities.
Access to Finance	Introduce value chain actors to financial service providers. Building the capacities of financial institutions and potential users of finance Development of innovative products and channels such as digital financial solutions.	Initial introduction and facilitation. Tailored training programmes for specific duration. Uptake by the clients.	The potential clients and financial Institutions are expected to maintain the relationship based on demand and supply factors. Enhanced capacity will enable the development of business plans by clients. Direct Access Entities build independent relationship with GCF.
Carbon Finance	Strengthen countries' readiness to engage with emerging carbon market opportunities and access new, growing and continued sources of financing.	Building productive partnerships and in-country capacity.	Through embedding technical working groups and networks.

192. Through its activities strengthening access to finance and equipment and strengthen standards, SURAGGWA will facilitate links to enable value chain actors such as off-takers, suppliers and larger producers to access finance from a range of commercial financial institutions in the medium and longer term. This value chain financing could serve as an additional, substantial source of financing even beyond the SURAGGWA and iGREENFIN project durations. Annex 2, chapter VI describes the feasibility of Value Chain Financing in the region.

193. The Output on Access to Finance (Output 2.3) will address key constraints to finance, particularly around the uptake and efficacy of credit. The activities ensure the outputs sustainability beyond the project life cycle through building the capacities of financial institutions and potential users of finance. The facilitation of initial linkages between financial institutions and the providers of credit in rural areas such as village savings associations and farmer associations will expand the reach of financial services in previously unbanked areas. The development of innovative products and channels such as digital financial solutions will enhance the potential reach of financial services going forward. Finally, the creation of a tailored product on savings will provide a vehicle for smallholder farmers, organizations or MSMEs to build their collateral for future credit, even beyond the project life cycle. This increased access will enable those with the potential to grow to use the funds to expand their market. Several Direct Access Entities such as Banque Agricole du Senegal have applied for accreditation with GCF and in the future they could use the GCF Facility to expand their own operations to contribute further to the objectives of the GGWI as well.

194. The interventions designed under Component 3 are focused on strengthening the capacity at the country and regional level to galvanize further support for the GGW Initiative. SURAGGWA will seek to strengthen existing institutions and mechanisms in the individual countries, and at the regional level and will not create new or parallel structures. Similarly, existing coordination and governance frameworks will be used to the fullest extent possible, while making improvements where appropriate. The work of strengthening monitoring services will be done in a manner that is consistent with national and international standards. This will contribute to long-term sustainability by ensuring that the functions of the GGW agencies remain relevant.

195. The development of strong capacity for monitoring and evaluation is designed to demonstrate the success of the initiative based on strong and credible evidence and generate additional funding sources. The component will promote and integrate cost-effective technologies and tools for strengthened land restoration monitoring. The national and regional GGW institutions will subsequently be able to generate information of significant value to their decisions makers and their development partners at a reasonable cost. The Component will create an enabling

environment for the GGW-National Agencies (GGW-NAs) to provide accurate land restoration monitoring services which will provide the foundation for uptake of the generated information in decision-making. The observed value of improved datasets and regular reporting of positive outcomes will reinforce support for the ongoing maintenance of observation tools and services including human resources. The GGW-NAs will be asked to formally commit to maintaining some of the core functions that will be introduced by SURAGGWA after its completion. Some Direct Access Entities like the Centre de Suivi Écologique (CSE) in Senegal whose core activities include environmental monitoring, natural resources management and conducting environmental impact assessments are also expected to build partnerships to develop climate change projects in the areas of environment, agriculture and livestock.

196. Capacities will be built to ensure even larger replicability of proposed actions and ensure feasible projects are prepared and funded. Through this component, the programme will ensure strong ownership and coordination by the main regional actors, who can play a key role in the coordination of the GGW interventions and demonstrate that they have a sustainable model that can be replicated in other areas. FAO would actively support such national aspirations as part of the GGW umbrella framework, which the PA-GGW, having received more capacity building and hands on experience, could help to further develop. Several development agencies and national governments have agreed to provide additional funds based on the success of some on-going projects such as the AAD.

197. The phasing out of technical assistance and the financing of selected recurrent costs necessary for the functioning of the supported functions will be gradual. Technical assistance will be mobilized intensively at the beginning of the programme and then withdrawn gradually as the institutions gain in competence. Similarly, the assumption of certain recurrent operating costs of certain services that SURAGGWA seeks to strengthen, such as monitoring land degradation/restoration, will be programmed on a declining scale and will be completed prior to the end of the tprogramme. This phase-out will be accompanied by assistance in finding alternative sources of funding, including country budget contributions.

198. The output will address the barriers to countries' access to carbon finance for restoration through participation in domestic and international carbon markets. Through strengthening capacity, disseminating knowledge and supporting selected agreements between stakeholders, the programme supports countries' readiness to engage with emerging carbon market opportunities and access new, growing and continued sources of financing. By partnering with sub-regional and regional initiatives on carbon markets, the programme can both draw upon existing carbon market expertise and complement this with expertise in agroforestry and restoration. Embedding support in evolving technical working groups and networks within relevant existing initiatives will lead to greater efficiency and sustainability. Overall, the output will also contribute to the Programme's programme's economic sustainability beyond its implementation period since the stream of carbon revenues will contribute to smallholder farmer incomes beyond its life cycle, paving the way for different sources of financing that can sustain existing restoration and support upscaling.

Table 10 - Exit Strategy Component 3 Strengthening the Great Green Wall Institutions at country and regional level

Type of Investment	***Programme Investments***	***Exit Strategy***	***Maintenance***
Monitoring & Evaluation Capacity	Technical Assistance. Introduction of modern technologies and tools.	Withdraw after the TA is completed and new tools and technologies have been introduced.	Government will operate and maintenance the system.
Planning and Coordination.	Technical Assistance for improving the system and building capacity.	Withdraw after the TA is completed and new mechanisms and systems are introduced.	Existing structures and systems will be strengthened.
Carbon Finance	Strengthening capacity, disseminating knowledge and supporting selected agreements between stakeholders	Withdraw after the TA is completed and capacity is strengthened.	Embedding support in evolving technical working groups and networks within relevant existing initiatives.

			Stream of carbon revenues to smallholder farmer incomes can sustain existing restoration and support upscaling.

C. FINANCING INFORMATION

C.1. Total financing

(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency	
	150.0			million USD ($)	
GCF financial instrument	**Amount**	**Tenor**	**Grace period**	**Pricing**	
(i) Senior loans	Enter amount	Enter years	Enter years	Enter %	
(ii) Subordinated loans	Enter amount	Enter years	Enter years	Enter %	
(iii) Equity	Enter amount			Enter % equity return	
(iv) Guarantees	Enter amount	Enter years			
(v) Reimbursable grants	Enter amount				
(vi) Grants	150.0				
(vii) Results-based payments	Enter amount				

(b) Co-financing information	Total amount			Currency		
	72.0			million USD ($)		
Name of institution	**Financial instrument**	**Amount**	**Currency**	**Tenor & grace**	**Pricing**	**Seniority**
Government[98]	In kind	4.0	million USD ($)	Enter years Enter years	Enter%	Options
Government[99]	Grant	48.2	million USD ($)	Enter years Enter years	Enter%	Options
FAO[100]	Grant	19.8	million USD ($)	Enter years Enter years	Enter%	Options
Click here to enter text.		Enter amount	million USD ($)	Enter years Enter years	Enter%	Options
	Options		million USD ($)	Enter years Enter years	Enter%	Options
	Options		million USD ($)	Enter years Enter years	Enter%	Options

(c) Total financing (c) = (a)+(b)	Amount	Currency
	222.0	million USD ($)
(d) Other financing arrangements and contributions (max. 250	*Please explain if any of the financing parties including the AE would benefit from any type of guarantee (e.g. sovereign guarantee, MIGA guarantee).* *Please also explain other contributions such as in-kind contributions including tax exemptions and contributions of assets.*	

[98] All the eight Countries (Detailed in the annex 4`- Country costs Summary table)
[99] Government of Mauritania acting through NAGGW (16 million USD), Government of Nigeria - Agro-Climatic Resilience in Semi-Arid Landscapes (ACReSAL) Project Through Federal Ministry of Environment (20,0 million USD), Government of Senegal through Global Affairs Canada(12.2 million USD)
[100]Mali, Mauritania, Niger, Senegal through GEF financing Respectively 4.9 million USD, 6 million USD ,5.8 million USD and 3.1 million USD),

words, approximately 0.5 page)	*Please also include parallel financing associated with this project or programme (refer to the co-financing policy).*

C.2. Financing by component

Please provide an estimate of the total cost per component and output as outlined in section B.3. above and disaggregate by source of financing. More than one co-financing institution can fund a single component or output. Provide the summarised cost estimates in the table below and the detailed budget plan as annex 4.

Component	Output	Indicative cost	GCF financing		Co-financing		
		Amount USD ($)	Amount USD ($)	Financial Instrument	Amount USD ($)	Financial Instrument	Name of co-financiers
Component 1: Large scale landscape restoration to increase climate change resilience and mitigation, for and by local communities.	Output 1.1: Local community groups organized, trained and knowledgeable in land restoration activities and monitoring	13,164,167	12,752,167	Grant	412,000	Cash	Government of Senegal
	Output 1.2. Native seed supply systems strengthened to ensure the availability of genetically appropriate seeds that provide increased climate resilience.	47,088,991	45,587,221	Grant	1,501,770	Cash	Government of Senegal
	Output 1.3: Targeted highly degraded lands restored through arrangements for rainwater harvesting and enhanced soil permeability and through local communities engaging in planting of seedlings and direct seeding.	51,554,333	16,767,809	Grant	2,472,000	Cash	FAO
					8,000,000	Cash	Government of Mauritania
					2,700,000	Cash	FAO
					4,404,593	Cash	FAO
					12,709,931	Cash	Government of Nigeria
					4,500,000	Cash	Government of Senegal
	Output 1.4: Targeted moderately degraded lands planted with a range of species to restore and enrich the landscapes of agro-forestry, agro-ecology and sylvo-pastoral systems, biomes and protected forest ecosystems.	25,406,957	5,362,475	Grant	2,143,446	Cash	FAO
					4,000,000	Cash	Government of Mauritania
					2,450,000	Cash	FAO
					1,095,822	Cash	FAO
					4,090,922	Cash	Government of Nigeria
					5,500,000	Cash	Government of Senegal
					764,292	Cash	FAO
	Output 1.5: Rural community capacities strengthened for sustainable management and restoration, and verification of restoration results	5,629,000	1,629,000	Grant	4,000,000	Cash	Government of Mauritania
Component 2: Development of climate-resilient, low emission non-timber forest product (NTFP) value chains benefiting vulnerable communities' livelihoods	Output 2.1. Climate resilient and low carbon production and processing practices in selected NTFP value chains adopted by Producer Organizations and MSMEs.	13,815,950	13,815,950	Grant			
	Output 2.2: Increased access to markets for smallholder NTFP actors (including	19,587,157	15,424,357	Grant	1,980,000	Cash	Government of Nigeria
					238,000	Cash	Government of Senegal
					1,944,800	Cash	FAO

	micro- and small enterprises)						
	Output 2.3 Suitable credit and insurance products designed for smallholder NTFP value chain actors (PO, cooperatives, MSMEs) and increased stakeholder capacity for using these financial products in their NTFP value chain activities	3,087,000	3,087,000	Grant			
Component 3: Strengthening the Great Green Wall institutions at country and regional level	Output 3.1: GGW Land restoration monitoring system at national and regional level upgraded and functional	7,961,500	7,412,500	Grant	76,500	In-kind	Government of Burkina Faso
					103,500	In-kind	Government of Chad
					58,500	In-kind	Government of Djibouti
					40,500	In-kind	Government of Mali
					76,500	In-kind	Government of Mauritania
					40,500	In-kind	Government of Niger
					76,500	In-kind	Government of Nigeria
					76,500	In-kind	Government of Senegal
	Output 3.2: National and Regional GGW institutions planning and coordination capacities strengthened	5,469,000	4,920,000	Grant	76,500	In-kind	Government of Burkina Faso
					103,500	In-kind	Government of Chad
					58,500	In-kind	Government of Djibouti
					40,500	In-kind	Government of Mali
					76,500	In-kind	Government of Mauritania
					40,500	In-kind	Government of Niger
					76,500	In-kind	Government of Nigeria
					76,500	In-kind	Government of Senegal
	Output 3.3: National and regional climate change institutional capacities/frameworks are strengthened to integrate land restoration investments in climate change adaptation and mitigation programmes	2,636,000	2,636,000	Grant			
	Output 3.4. GGW knowledge management and communication capacities strengthened for mobilizing increased support to climate-resilient land restoration	2,549,000	2,549,000	Grant			
Contingencies		4,252,272	4,252,272	Grant			

Monitoring & Evaluation		8,760,750	6,304,250		200,000	In-kind	Government of Burkina Faso
					212,500	In-kind	Government of Chad
					175,000	In-kind	Government of Djibouti
					150,000	In-kind	Government of Mali
					144,500	In-kind	Government of Mauritania
					168,000	In-kind	Government of Niger
					280,000	In-kind	Government of Nigeria
					235,500	In-kind	Government of Senegal
					891,000	Cash	FAO
Project management and implementation		11,028,635	7,500,000	Grant	110,000	In-kind	Government of Burkina Faso
					103,500	In-kind	Government of Chad
					87,000	In-kind	Government of Djibouti
					242,918	Cash	FAO
					125,000	In-kind	Government of Mali
					221,000	In-kind	Government of Mauritania
					300,000	Cash	FAO
					235,000	In-kind	Government of Niger
					295,000	In-kind	Government of Nigeria
					1,294,217	Cash	Government of Nigeria
					400,000	Cash	FAO
					115,000	In-kind	Government of Senegal
Indicative total cost (USD)		221,990,712	150,000,000				71,990,712

This table should match the one presented in the term sheet and be consistent with information presented in other annexes including the detailed budget plan and implementation timetable.

In case of a multi-country/region programme, specify indicative requested GCF funding amount for each country in annex 17, if available.

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?	Yes ☒ No ☐
C.3.2. Does GCF funding finance technology development/transfer?	Yes ☒ No ☐

199. The programme will support both capacity building and technology transfer. Capacity building activities will include building the technical expertise of community members in rehabilitation of degraded rangelands, farm lands and enhancing their capacity for adaptation to climate change through training them in water retention and soil conservation techniques, collection and production of seed and sapling of both local plant varieties, forage crops, appropriate shrubs and bushes for soil conservation and regeneration, wild berries and NTFPs. Farmers will also be given training in how to adapt their farming practices in the face of change in precipitation and temperatures by varying the planting time and using specifies which are better adapted and deploying farming techniques that will help to increase resilience. It is expected that USD 10 million will be used from the GCF funding amount for capacity building at the

community level. In addition, there is a strong element of capacity building of regional and national GGW institutions for improving their systems of monitoring and evaluation, coordination, planning and resource mobilization through carbon markets.

200. The programme is expected to introduce new technologies for rehabilitation of land in the area such as deep ploughing and contour ploughing which mitigates the impacts of floods, storms, and landslides on the crops by reducing soil erosion up to 50 percent, controlling runoff water, increasing moisture infiltration and retention and thus enhancing soil quality and composition.[101] The programme will encourage the use of private sector firms that will use their equipment for contour ploughing and making contour beds and establishing a sustainable rental market for the scaling up of this technology. It is expected that around USD 33 million of GCF funds will be used to rent farm equipment from private sector service providers and that this will help to establish a viable rental market for farm equipment.

201. There is also expected to be transfer of technology with regards to monitoring and evaluation tools to measure the impact of the programme activities as a result of increase of vegetative cover, carbon sequestration and impact on GHG emissions through the OpenForis toolkit developed by FAO. The FAO has developed a Framework for Ecosystem Restoration Monitoring (FERM) and an associated geospatial platform to improve data access, transparency, and ensure actions to meet restoration commitments are guided by the best available science. The national governments will be trained in the use of the FERM platform which uses FAO's corporate Hand-In-Hand geospatial architecture. Continuous improvement of these systems will support a science-based restoration movement through transparent and fit-for-purpose monitoring in support to the United Nations Decade on Ecosystem Restoration. Users are able to interrogate and interact with key geospatial information related to the biophysical, and the socio-economic dimension for their ecosystem of interest. The FERM platform also has functionality for uploading national and sub-national data, enabling integration of geospatial data locally, nationally, regionally, and globally. The platform includes functionality for creating compelling restoration impact stories, based on user specific geospatial data for a defined area of interest.

202. For the more advanced user, FAO will provide and train users in the integration with FAO's geospatial processing platform, SEPAL, to analyse and regularly update the programme's geospatial information with near real time imagery. The platform also allows restoration stakeholders and national entities to share their information on restoration progress at different scales through the FERM Registry which will be interoperable with other restoration monitoring platforms, enabling transparency in restoration monitoring and reporting. Under the programme, training and orientation will also be provided on the *EX-ACT (Ex-Ante Carbon Balance Tool)* to estimate GHG impacts per unit of land, expressed in tCO_2-e per ha per year[102] and its equivalent in terms of economic returns; (ii) *Tool for Agroecology Performance Evaluation (TAPE)*[103] which will aim to provide evidence to policymakers and other stakeholders on how agroecology can contribute to more sustainable food and agricultural systems. TAPE will be used in collaboration with the DyTAES network[104] and also enable cross-sectoral and cross-ministerial cooperation. It will contribute to the empowerment of producers through self-diagnosis and assessment of their system's transition level and performance. Assistance on this tool will be on an tailored basis, depending on country interest.

[101] TECA - Technologies and Practices for Small Agricultural Producers. FAO. December, 2020
[102] FAO Ex-ANTE Carbon-Balance Tool
[103] FAO TAPE Tool
[104] DyTAES network for the agroecological transition

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's Initial Investment Framework.

D.1. Impact potential (max. 500 words, approximately 1 page)

203. SURAGGWA is designed to contribute to the achievement of GCF's objectives of promoting a paradigm shift towards low emission and climate resilient development pathways to reduce greenhouse gas emissions and to help vulnerable communities in the Sahel adapt to the impacts of climate change. Preliminary estimates indicate that land restoration efforts funded by SURAGGWA will sequester 65.1 million tons CO_2e over the 20-year lifespan of the programme (Annex 22)[105]. The main investment area which is expected to contribute to creating carbon sinks is the restoration of 1.273 million hectares of highly and moderately degraded lands. The carbon sequestration is estimated from increased tree canopy cover due to the restoration activities in eight Sahel countries covered by the programme. The analyses are mainly based on the results of recent research on global tree restoration potential. The emissions reductions and removals will be measured using the Collect Earth methodology to convert measured canopy cover increase into carbon stocks change (Annex 22).

204. The multi-country programme is expected to directly benefit 1,881,002 million people and indirectly benefit an estimated at 3.861.925 million. The main beneficiaries of the programme will be from the restoration of degraded lands. The targeted communities are essentially agro-pastoral in nature and use an agro-silvo-pastoral system that provides ecosystem services, which constitute a nature-based solution, acting as a barrier against climate change and its impacts such as desertification, and floods, reducing run-off, increasing infiltration of rainwater, regenerating grasses for livestock and making land fertile for crop production.

205. The targeted communities own around 172,000 of large ruminant and close to 580,000 of small ruminants, based on the assumption that participating house-holds own livestock in the same proportion as in the country on average with each HHs owning the average number of small ruminants and cattle among livestock owners. These are low yielding animals with poor health status and poor diets. It is expected that with increased availability of fodder and forage crops and diversification of livelihoods through value chain investments in non-timber forest products, the silvo-pastoral communities may invest in higher productivity animals and reduce their low yielding livestock. The increased production of fodder and tree browse for livestock is a key adaptation requirement for pastoralists throughout the Sahel. The restoration of moderately and heavily degraded lands in key locations will increase the resilience of farm and pastoralist households facing a future with higher temperatures and increased moisture stress. The combined impact of heathier livestock and decrease in herd size will also have an appreciable impact on methane emissions.

206. The programme will also introduce improved and new climate-resilient livelihood options such as through increased participation in Non-Timber Forest Products (NTFP) and increased employment in the NTFP value chain, as well as through the stabilization of crop yields on moderately degraded farmlands with the application of agro-ecological approaches. These aspects will be particularly important for women who are the majority participants in the collection and sale of these forest products. Food security is expected to improve both directly and indirectly, as the main immediate impact of increased income for poor communities is on food consumption, The direct impact on food security will be through increased yields, as a result of better adapted crop varieties, improved livestock productivity through access to increased forage crops and through the impact on incomes and employment as a result of increased participation in NTFP value chains. Water security will improve as a result of ecosystem restoration and retention of water in soils and its immediate impact on food production. The programme is also designed to introduce innovations in the restoration of degraded lands which will strengthen climate change resilience through selection of more resilient plant species, creating buffers to wind and soil erosion.

207. The programme is expected to have a significant impact on women. About 2,860,313 beneficiaries of the programme or 51% are expected to be women (Annex 23). Of these, 937,040 are expected to be direct women beneficiaries and 1,923,273 are expected to be indirect beneficiaries. The programme will increase opportunities for livelihood improvement and provide concrete benefits to men and women from smallholder farmers and pastoralists. The programme will enhance women's income and employment opportunities through its investments in both land rehabilitation and NTFP in which mostly women are engaged. The programme has identified a Gender Action Plan (Annex 8) which indicates clearly how women's participation will be encouraged and the specific targets and budgets allocated for women in each of the output areas. The programme will adopt a gender transformative approach which requires identifying structural and normative barriers preventing women's access.

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[105] This was evaluated using FAO's Ex-ACT tool. The net result of the SURAGGWA programme is slightly lower at 65.9 million tCO_2e, as it takes into account a slight increase in livestock emissions due to the fodder production contribution of the restored land, see Annex 22, carbon impact potential assessment.

208. The programme is aligned with the following results for 2024-2027 or the Updated Strategic Plan: i) Support for developing countries with the restoration of 1.273 million hectares ii) Support for developing countries that results in 1,544,770 million beneficiaries adopting innovations that strengthen climate-resilience through sustainable restoration and agricultural practices, securing livelihoods while reconfiguring food systems.

GCF SURAGGWA proposal USP-2 Targets table

Strategic Priority	**Description**	**Contribution**
Country Programming	More than 100 developing countries directly supported to advance NDCs, NAPs or LTS	Eight developing countries assisted to implement their NDCs, NAPs and LTSs
Direct Access	Doubling the number of DAEs with approved GCF funding proposals through strengthened climate programming capacity and increasing the allocation of GCF resources through DAEs.	Capacity building of DAEs participating directly in programme implementation (CSE Senegal, La Banque Agricole du Senegal) and potential future DAEs, such as the National agencies for the Great Green Wall, improving their land restoration coordination, monitoring and reporting capacities.
Food	190 to 280 million beneficiaries adopting low-emission climate-resilient agricultural and fisheries	1,063,950 million people improve food production through adoption of climate-resilient agro-silvo-pastoral land use systems, improving sustainability of food cropping through techniques like agroforestry, and increasing production of tree foods and livestock fodder.
Ecosystems	120 to 190 million hectares of terrestrial and marine areas conserved, restored or brought under sustainable management	Restoration of 1.273 million hectares of degraded land
Locally-led Adaptation	40-70 approved proposals for adaptation projects, including for locally-led adaptation action.	Improved local capacity for attracting investment in adaptation projects, through clearly demonstrated mitigation and local livelihoods benefits from smallholder value chains
Innovation and Market Creation	900 to 1500 local private sector early stage ventures and MSMEs provided with broad-based seed and early-stage capital for innovative climate solutions, business models and technologies	Strengthening the capacities and provide TA for cooperatives and 1, 795 MSMEs, improving the enabling environment for private sector investments in land restoration, through improved capacity for demonstrating carbon benefits of climate change adaptation investments and through national and regional policy dialogue. Additionally, the strengthening of NTFP value chain market actors and linkages.
Greening Finance	90 to 180 national and regional financial institutions supported to access GCF resources, and other green finance, particularly for MSMEs	Supporting 15 local financial institutions to access GCF resources through financing of beneficiary business plans and strengthening capacities for future green financing solutions.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

209. There are several key elements included in the SURAGGWA approach which are expected to catalyse impact beyond a one-off programme investment. In this respect, SURAGGWA is expected to be transformative by *building capacities of local communities* in understanding how to implement land rehabilitation investments in highly degraded common lands and moderately degraded farm lands. The local government agencies are expected to provide a supportive framework which will guarantee local communities use rights thereby encouraging them to make investments in these lands.Furthermore, the programme will *facilitate the development of private sector rental markets* that can be used by local communities collectively or individually. The establishment of the rental markets will also lead to the development of associated ancillary services such as repair and maintenance services for the deep ploughing equipment and tractors, etc. The establishment of community groups to collect local seed varieties and the setting of private nurseries

for saplings will also encourage plantation. The successful engagement of the private sector in commercially viable and ecologically responsible restoration efforts will constitute a major opportunity to deliver restoration at scale while providing additional livelihood opportunities to the local communities.

210. The programme has the potential to transform the livelihoods of approximately 259.837 smallholder farmer and pastoralist households in the programme area through investing in climate change-resilient land management, smallholder organizational capacity building, improving market access for smallholder non-timber forest products (NTFP) and brokering more equitable relationships with private sector buyers and financial institutions. The non-timber forest product value chains that will be supported in the programme area play an important role in rural smallholder production systems in other Sahelian countries and in other parts of the programme countries. The world market demand for certain widely traded NTFPs, such as shea butter, baobab, and gum arabic is increasing, especially the latter, following its recent recognition as a dietary fibre by the US FDA. The "return to nature" product development path followed by an increasing number of food and personal care multinationals and the increased consumer interest in so-called superfoods, bodes well for the future market potential of NTFPs occurring in the programme area, such as desert date (*Balanites aegyptiaca*) and baobab (*Adansonia digitata*). Furthermore, the improved quality and quantity of smallholder NTFP produce will facilitate more equitable relationships with product off-takers and financial service providers – which will in turn will open the door for similar benefits accruing to other rural households in the SURAGGWA programme area and beyond. By linking financial rewards to land management decisions favouring climate change adaptation outcomes, SURAGGWA will help rural smallholder households and communities in some of the most vulnerable environments in the world to greatly increase the resilience of their livelihood systems. The increased prices obtained by smallholder NTFP producer organizations will create the financial incentives for them to continue to invest in the management of lands restored with SURAGGWA support and in restoration of additional degraded lands. These fundamental transformations will continue to have an impact long after the programme ends.

211. SURAGGWA will attempt to scale up the approach of the Global Environment Facility (GEF), which has played a leading role in influencing a shift from the initial vision of a tree planting venture to one focused on integrated management of natural resources for improving livelihoods and landscapes. There are already indications of the scaling up of the approach of GEF through similar investments. In response to demands of the countries in the region, the GEF has invested over US$800 million in grants and leveraged an additional US$6 billion from national governments, development partners and other multi-lateral donors for projects in the Sahel region. This financing has helped GGW member countries to boost their efforts to promote practices for improving crop and livestock productivity, restoring degraded parklands and strengthening resilience and adapting to climate change. Building on the GEF's experience with programmes using an integrated landscape approach, the GEF 7 "Harnessing the GGWI for a Sustainable and Resilient Sahel" project is currently being implemented by UNEP to engage with all GGW stakeholders in fostering dialogue with countries and fleshing out a longer-term vision. It will serve as the vehicle to design a programme with potential to mobilize larger investments in GEF-8. The total amount of GEF-8 resources that will be dedicated to support the GGW will be determined upon completion of the replenishment process and discussion with beneficiary countries on the use of their System for Transparent Allocation of Resources (STAR) allocations.

212. There are new opportunities for financing which are emerging from innovative and non-traditional sources. There is a growing market dealing with carbon markets and carbon trading in the voluntary market. Carbon emissions reductions and removals programmes such as the Forest Carbon Partnership Facility (FCPF) have mainly focused on countries with a humid climate. Drylands however, which cover two thirds of the African continent, have considerable carbon sequestration potential, as trees can be added to most degraded farming landscapes with negative opportunity cost to the land user, i.e. they have a positive impact on existing crops and pasture, if the right tree species are used and recommended densities respected. The restoration methods employed by the SURAGGWA programme are highly cost-effective, relying mainly on self-interested local community efforts and on proven, low-cost technologies such as direct seeding and managed natural regeneration. In combination with cutting-edge, transparent monitoring techniques to document the programme's carbon emission reductions and removals results, this will help participating countries to make a much stronger case for investment in land restoration as part of both domestic and international climate investments announced in their NDCs. Lessons learned from experience in the SURAGGWA programme area – especially those concerning cost-effective restoration methods and accurate, transparent methods for monitoring land restoration and carbon mitigation results – will be shared with other Sahelian countries, and with other parts of the programme countries, as well as with interested Financial and Technical Partners. The innovative mechanisms for mobilizing carbon finance for smallholder land restoration that SURAGGWA will

develop and test in some programme countries, including the integration of land use carbon in Nigeria's domestic emissions trading scheme, have enormous potential for scaling up and replication.

D.3. Sustainable development (max. 500 words, approximately 1 page)

213. The programme is expected to address several of the Sustainable Development Goals (SDGs) namely the SDG 1 which is aimed at eradicating poverty, SDG 2 which commits to zero hunger, gender equality (SDG 5), decent work and economic growth (SDG 8); Climate action (SDG 13); Life on land (SDG 15), and Partnerships for the Goals (SDG 17). Experience from many countries in the Sahel, where investments are being made in soil and water conservation (SWCM), show that these investments result in improved agricultural productivity and food security, economic security, improvement in groundwater tables, tree regeneration, and increase in biodiversity. These investments also have an impact in reducing migration and poverty in the Sahel, especially in areas with a larger proportion of farmers and herders.[106]

214. **Environmental benefits[107]:** The programme is designed to help secure several environmental benefits through its investments in participatory land restoration and subsequent sustainable management of natural resources through its protection of drylands against erosion and desertification and increase the potential for carbon sequestration through increasing the grass and tree cover on degraded lands. It is expected that CO2 emissions will be reduced during the programme's lifecycle due to land restoration by local communities, through both agroforestry on moderately degraded lands and higher-cost interventions on highly degraded lands. For the latter, the programme will rehabilitate the selected sites through deep ploughs which creates micro-catchments through the Vallerani system. This improves water infiltration and water retention, while loosening the soil and improving nutrient access. Tree and grass seedlings are trapped in micro-catchments which helps to build the natural vegetation cover with native trees and grass species, providing habitat for pollinators and other animals into the area, increasing biodiversity. [108] The programme will introduce agriculture practices and technologies that reduce soil erosion and soil fertility loss, enhance soil moisture retention and maintain soil temperatures suitable for crop production. In Niger, farmers have developed innovative ways to regenerate and multiply valuable trees whose roots already lay under their land. Farmers in the area, also introduce 'farmer-managed natural regeneration' (FMNR) first pioneered by outside actors but spread rapidly by farmers once they observed its success.[109] Sahelian farmers, driven to desperation by the great droughts of the early 1970s and the 1980s, have ingeniously modified traditional agroforestry, water and soil management practices to restore the fertility of their land.

215. **Social co-benefits including food security:** About a quarter of the 100 million people who live in the Sahel rely on pastoralist livelihoods, 70 per cent of whom are living in poverty.[110] Climate change has made this livelihood strategy difficult, as access to water and foraging resources have greatly reduced with the increase in warming and dry weather. This, coupled with social vulnerabilities including political, ecological and social, present challenges for climate adaptation for pastoralists.[111] The herders and farmers in the area complete for a meagre resource base, which in periods of droughts, growing desertification and soil erosion is further intensified. Conflicts intensify when livestock stray into fields with standing crops especially when feed is scarce. The social and water conservation investments that will be promoted by the programme are expected to *improve food security and crop/pasture yields while conserving the natural vegetation and increasing the resilience of both pastoralists and farmers' livelihoods.* Rehabilitation of degraded pastures will include working with communities to identify and demarcate community pasture areas and the direct seeding of naturalized tree, shrub and herbaceous species. Clarifying use rights and allowing people to grow and use the common lands for alternative feed sources, can reduce conflicts. This is likely to reduce the social tensions

[106] Regreening the Sahel: A quiet agroecological evolution. OCHA. November 2020.
[107] The environmental benefits of the programme are an inherent part of the climate change outcomes.
[108] Rehabilitation of the Sahel (AGED) – Burkina Faso.
[109] Regreening the Sahel: A quiet agroecological evolution. OCHA. November 2020.
[110] Women and Climate Change in the Sahel. OECD. March, 2020.
[111] Women and Climate Change in the Sahel. OECD. March, 2020.

that currently beset many of the Sahel countries threatened by climate change.[112] Several countries have clarified the land tenure arrangements as a result, for example, a Rural Land Tenure Law (N°034-2009/AN) enabled legal recognition of rights legitimized by customary rules and practices and helped to define local land charters in Burkina Faso to develop and implement appropriate controls of pasture use.[113] Furthermore, the Sahel region has struggled with food insecurity and severe issues of malnutrition which have been made worse by conflict and climate change. The region has among the highest rates of childhood malnutrition and stunting in the world.[114] Food security and nutrition benefits also accrue through the impact on diet and nutrition that results from increased incomes from selling better fed livestock, improved production of dairy, increased access to collecting food from the wild, increased incomes from selling forage and non-timber forest products[115].

216. **Economic co-benefits: smallholder income diversification.** It is expected that the programme will lead to a range of economic benefits that will enhance the incomes of poor smallholder farmers, and thus provide the underlying basis for making the investments sustainable. The programme is expected to increase productivity of lands through increased production of forage and fodder crops, increased productivity of livestock, increased production of crops on degraded farmlands and increased production of non-timber forest products such as Balanites, gum arabic, baobab leaves, neem, honey and sale of seeds and seedlings, etc. The GGW implementation status report indicated sizable income generation since 2007 across all GGWI countries. The programme is designed to strengthen selected NTFP value chains identified by local communities and the private sector in each of the sites. The market linkages will help identify the quality and volume of each of the NTFP and assist with better collection, storage, transport and processing techniques to enhance the value of the production. The experience of on-going projects has also demonstrated substantial income in some cases from collecting and selling forage from restored land (the Niger and Senegal). Improved beekeeping and honey production also offer a sustainable source of income. Well-fed livestock increases the selling price and thus generate revenues for the household. In addition, some grass species growing on the rehabilitated sites allow some income generating activities like *Cassia tora* for the production of Sekko and *Eragrostis tremula* to make brooms. A range of new jobs will be generated during the rehabilitation of the lands, during crop production and the collection, sale, transport and processing of NTFPs. The programme is also designed to encourage the private sector to establish rental markets for farm equipment and machinery for land rehabilitation and its upscaling. The activities are expected to generate backward and forward linkages that will itself generate additional employment for drivers, transporters, mechanics, processors and other value chain actors.

217. **Gender-sensitive development benefit:** Gender inequity within the Sahel is particularly pronounced. Women have to assume the responsibility for key tasks within the household such as food and water, fuelwood, fodder, care of livestock, vegetable gardening, etc. This responsibility has become harder to assume due to extreme weather events and changing climate which impacts water, food and feed availability. The programme will increase opportunities for livelihood improvement and provide concrete benefits to men and women from smallholder farmers and pastoralists. The programme will enhance women's income and employment opportunities through its investments in both land rehabilitation and NTFP in which mostly women are engaged. The programme has identified a Gender Action Plan (Annex 8) which indicates clearly how women's participation will be encouraged and the specific targets and budgets allocated for women in each of the output areas. Specific actions will be developed to strengthen the technical and managerial capacities of women engaged with livestock, crops and NTFPs. Gender-disaggregated data will be assessed against the appropriate indicator to measure women's participation, livelihoods, employment and incomes. However, the project intends to adopt a gender transformative approach in the manner in which it builds resilience for the agriculture and pastoralist livelihoods in the Sahel. Such an approach is likely to generate sustained benefits for women beyond those that are achieved during the project life-cycle by helping to change some of the deep-seated patriarchal norms and the traditional perceptions held about girls and women

[112] Inter-community conflicts are one of the main risks to the success of the programme, so conflict assessment, prevention and management are a key element of programme design and implementation mechanisms chosen, especially for Component 1, land restoration (see also Annex 6, Environmental and Social Management Framework, especially its Appendix 9, "Conflict assessment, prevention and management".

[113] Rehabilitation of the Sahel (AGED) – Burkina Faso.

[114] Women and Climate Change in the Sahel. OECD. March, 2020.

[115] 10.1016/j.ecolecon.2024.108311

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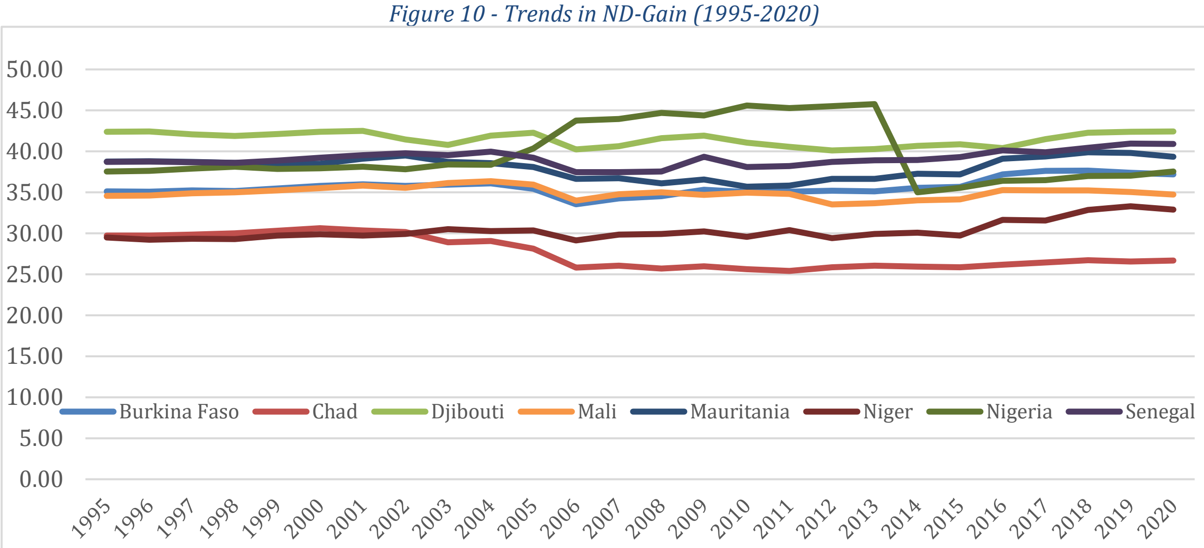
in the region. Key to gender transformation, will be listening to and responding to the needs of women in the local context. Women's knowledge systems and preferences for communication will be considered when constructing a system of information sharing that would be appropriate for women given the special context and constraints within which they operate.

D.4. Needs of recipient (max. 500 words, approximately 1 page)

218. The Sahel countries, and in particular the GGW area are highly vulnerable in both economic and environmental terms. The Notre Dame-Global Adaptation Index (ND-GAIN)[116] shows a country's current vulnerability[117] to climate disruptions and assesses a country's readiness by considering economic governance and social readiness.[118] An assessment of the ND-GAIN rankings for the SURAGGWA countries shows a high degree of vulnerability and lack of readiness. Five of the countries in the project are in the bottom 7% of the countries in terms of their overall rank and all the GGWI countries are in the bottom 21% of the 191 countries assessed. The chronic nature of the vulnerability and lack of readiness is indicated by the more or less static lines and little movement in their scores over the last 25 years as indicated by the figure below.

Figure 10 - Trends in ND-Gain (1995-2020)



219. Participating countries in this programme are highly vulnerable to climate threats and risks such as drought, rising temperatures and variability and unpredictability in rainfall which can cause disease in both plants and animals, locust infestations, etc. Their exposure and vulnerability to climate hazards is projected to increase in the future. Climate projections suggest that the Sahel will be hotter and drier with more frequent extreme events. In most scenarios, temperature rise will accelerate, meaning that the continent on average could be between 2°C and 6°C warmer by the end of this century (ACMAD, 2020). Climate impacts have had very adverse negative impacts on the fragile landscapes in this vulnerable area. The desert has been encroaching on the lands, biodiversity has been reducing and soil erosion has been impacting the productivity of the land. The reduction in productivity directly impacts food security, incomes, employment and it increases rural poverty. The figure below outlines the Vulnerability and readiness index of the Sahel countries. In the climate vulnerability index and readiness to improve resilience – the eight countries rank between 131-180 out of 181 countries.

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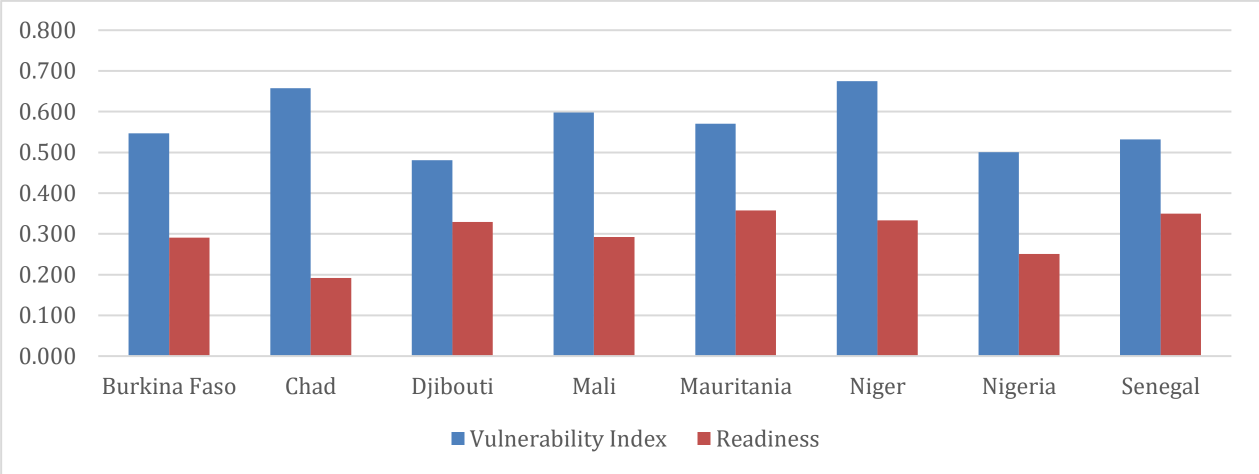
[116] ND Gain Index. University of Notre Dame. The Index brings together over 74 variables to form 45 core indicators to measure vulnerability and readiness.
[117] The ND-GAIN's framework breaks the measure of vulnerability into exposure, sensitivity and adaptive capacity.
[118] Economic Readiness: The investment climate that facilitates mobilizing capitals from private sector. Governance Readiness: The stability of the society and institutional arrangements that contribute to the investment risks.

Figure 11 - Vulnerability and Readiness in SURAGGWA Countries



220. The low state of economic and social development of the Sahel countries is outlined in Table 14 below. These countries have variable growth rates, low GDP per capita, very high rates of poverty, low human development and gender development indices and most have high reliance on the agriculture sector. For this reason, the majority of the Great Green Wall countries have prioritized the Agriculture, Forestry and Land Use (AFOLU) sector in their NDCs, both in terms of climate change adaptation and mitigation. It would be difficult to find a region with greater need for climate change resilience. The impacts of climate change (Section B.1) have been devastating to the region, causing increased poverty (20 million food insecure), migration (1 million displaced), inter-ethnic group conflict over depleting natural resources, and loss of ecosystems, vegetation and biodiversity.

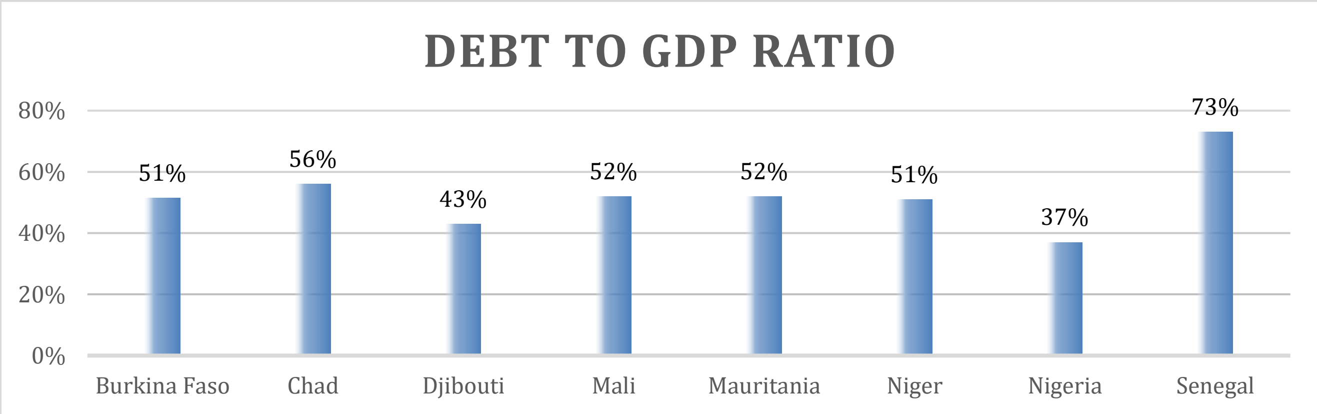
Table 11 - Economic and Social Development Profile of the Sahel Countries

	GDP Growth 2021[119]	Income GDP per capita USD (nominal)[120]	Agriculture Contribution to GDP	(%) population below the National Poverty Line	HDI Rank	HDI Index	GDI Value Index
Burkina Faso	6.90%	926	31%	40%	185	0.449	0.903
Chad	-1.20%	885	53%	88%	190	0.394	0.77
Djibouti	4.30%	2787	3%	70%	171	0.509	0.509
Mali	3.10%	927	36%	78%	186	0.428	0.887
Mauritania	2.30%	1723	15%	59%	158	0.556	0.89
Niger	1.40%	414	37%	41%	189	0.4	0.835
Nigeria	3.60%	2432	21.96%	40%	163	0.535	0.863
Senegal	6.10%	1,428	16.90%	67%	170	0.511	0.874

221. In the NDCs of the eight countries where SURAGGWA will be implemented on the ground, the calculated costs of adaptation and mitigation vary from USD 7 bn to 34 bn (excluding Nigeria which covers much larger non-dryland territory). These countries have limited capacity to allocate funds to the GGW area and have limited capacity to borrow from the public sector or private markets. While many African states took the opportunity to access international debt markets for the first time through the use of Eurobonds between 2007 and 2020, most of the Sahel countries were not among them. There are many different debt situations to be found among African countries. But general concerns about rising debt levels became more audible in 2019, even before the pandemic shock when debt alarms amplified.[121] Overall, the debt sustainability profile in

Africa is low and the debt to GDP ratios are very high and exceed 50% in most cases with rates as high as 73% in Senegal and 56% in Chad. There is considerable uncertainty about how much medium-term debt sustainability has been eroded as a result of the disruptions caused by COVID-19. What is clear, however, is that buffers have been eroded and another subsequent crisis would be much harder to fight unless there are efforts to reduce debt risk and shift to better ways of accessing resources.

Figure 12 - Debt to GDP Ratio in the Sahel



222. The Sahel countries face a host of barriers in mitigation and adaptation to climate threats. While the technologies and methodologies exist to upscale appropriate interventions, the principal barriers include insufficient access to financial resources, lack of institutional capacity, limited engagement of the private sector and limited mechanisms for community mobilization to invest in common property resources, lack of access to funds for investments on private land and lack of access to markets. The Sahel's forests, woodlands, wetlands, and grasslands are important ecosystems that host vast biodiversity of plant and animal species which help sustain livelihoods and their resilience. They provide a range of ecosystem services including important NTFP commodities that generate additional sources of income in poor agricultural years, and enhance resilience to climate variability and climate change; they often provide the extra income necessary, a safety-net to sustain livelihoods.[122] These ecosystems play a key role in adaptation when they are properly managed and restored. The ecosystems also reduce excess pressure on vulnerable natural resources (water, soils, wood and non-wood products, fodder and pastoral land). These restorative activities increase soil fertility, preserve water quality and water cycles, reduce run-off, protect land and agricultural production from increasingly intense floods and droughts, increase percolation of rainwater, regenerate grasses for livestock. There are large mitigation benefits realized from restoration of ecosystems due to the generation of below and above ground biomass.

Table 12 - Country profiles and international commitments

Countries	National Administrative area	National Restoration potential	Nationally Determined Contribution	GGW Restoration potential	AFR100 Restoration commitments
	Mha	*Mha*	2030 GHG targets	*Mha*	*Mha*
Total	**624.64**	**275.83**	*% reduction*	**123.73**	**25.6**
Burkina Faso	27.42	9.07	11.60	4.85	5.00

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[119] https://data.worldbank.org/indicator/NY.GDP.MKTP.KD.ZG?locations=BF
[120] b:
https://www.google.com/search?q=Income+level+of+Burkino+Faso&sxsrf=ALiCzsYkUiOEPz1iG99A2uwWazDEs0BiKQ%3A1666938406450&ei=JnZbY4qQG8SN9u8PoZ-h4A8&ved=0ahUKEwiKu9C8pYL7AhXEhv0HHaFPCPwQ4dUDCA8&uact=5&oq=Income+level+of+Burkino+Faso&gs_lcp=Cgdnd3Mtd2l6EAMyBwghEKABEAoyDAghEBYQHhAPEAoQHTIKCCEQFhAeEAoQHTIKCCEQFhAeEAoQHTIKCCEQFhAeEAoQHToKCAAQRxDWBBCwAzoFCAAQgAQ6BggAEBYQHjoFCCEQoAE6CAghEBYQHhAdOgQIIRAVOgoIIRAWEB4QDxAdSgQITRgBSgQIQRgASgQIRhgAUJwFWNIvYJUyaAFwAXgAgAGTA4gB4BuSAQcyLTguMy4xmAEAoAEByAEIwAEB&sclient=gws-wiz

[121] Africa's Hard Won Market Access. December, 2021. IMF.
[122] Sacande, M. & Parfondry, M., 2018. Non-timber forest products: from restoration to income generation. Rome, FAO. 40 pp. License: CC BY-NC-SA 3.0 IGO. https://www.fao.org/documents/card/fr/c/CA2428EN/

Chad	128.4	64.58	53.11	22.16	1.4
Djibouti	2.32	2.11	20.0	1.06	n/a
Mali	124.68	54.73	134.03	24.68	10
Mauritania	103.07	38.34	-	19.10	n/a
Niger	126.7	61.22	25.0	41.23	3.2
Nigeria	92.38	38.43	16.0	8.52	4.0
Senegal	19.67	7.35	25.0	2.13	2.0

D.5. Country ownership (max. 500 words, approximately 1 page)

223. The Sahel has suffered from long periods of drought as a result of which the African Union developed and approved the Great Green Wall for the Sahara and the Sahel Initiative in 2007, with the assistance of a host of partners including FAO. Regional and national level GGW institutions were set up and each country nominated an NDA and developed their strategies for both mitigation and adaptation. At the regional level, several initiatives were established to demonstrate the seriousness regarding protection of the Sahel. In 2010 an African Agency of the Great Green Wall was established to monitor, track progress and ensure a more coordinated support to the existing GGW member states, structures and institutions. In 2014, G5 Sahel (G5S) was established to provide an institutional framework for coordination of regional cooperation in development policies and security matters in West Africa namely for Burkina Faso, Chad, Mali, Mauritania, and Niger. New impetus has been added to these efforts by the international development community. In addition, there are a range of projects which have been initiated for strengthening institutions, ensuring readiness and implementation on the ground. There is strong country ownership and an abiding commitment to achieving the targets that have been identified in the relevant climate strategies in each of the Sahel countries. This commitment is driven by the realization that climate change has to be addressed as a priority as it poses a significant threat to the region. The most recent pledge was made in 2021, through the creation of the Great Green Wall Accelerator announced at the One Planet Summit giving new impetus to the initiative.

224. All the countries in the Sahel have ratified the UN Convention on Biological Diversity (CBD), the Convention to Combat Desertification (CCD), the Framework Convention on Climate Change (UNFCCC) and the Kyoto Agreement. The countries have also signed and ratified the Paris agreement on climate change. By ratifying the UNFCCC, these countries have committed to implementing measures to adapt to climate change. In addition to the reports and commitments submitted to the UNFCCC, the Sahel countries have prepared national climate change-related strategies, policies and actions: A brief synopsis of these strategies is provided in Table 161313 below.

Table 13 - Climate Change Strategies and Policies at the Country Level

Country	Climate Change Strategies & Policy Documents
Burkina Faso	Burkina Faso: has prepared a number of strategies and proposed actions: Strategic Framework for Investment in Sustainable Land Management (SFI-SLM); Strategy for Accelerated Growth and Sustainable Development (SAGSD); National Adaptation Plan (NAP); NAMA Framework (2008); National Action Programme for Adaptation for Climate Change (NAPA) (2007); The National Strategy for implementing the Climate Change Convention (2001)
Chad	Country Resilience Priorities (PRP) AGIR CHAD (2020); National Investment Plan for the Rural Sector (PNISR) (2014 - 2020); National Adaptation Programme of Action for Climate Change(NAPA);
Djibouti	Djibouti submitted its Second National Communication in 2014 and its Nationally-Determined Contributions to the UNFCCC in 2016. These strategies also work in parallel with Djibouti's Vision 2035. To increase its adaptive capacity to projected impacts from climate change, Djibouti is committed to the increased use of renewable energy, capacity building and is in the process of developing a National Strategy for a Green Economy
Mali	National Policy on Climate Change (2008); National Climate Change Strategy (NCCS) (2011)126;
Mauritania	prepared one of the first National Action Programme for Adaptation (NAPA) (2004) submitted to the UNFCCC; National Environmental Action Plan (PANE);
Niger	National Action Programmes for Adaptation (NAPAs); Strategic Framework for Sustainable Land Management (SF-SLM);
Nigeria	The National Agricultural Resilience Framework (NARF 2014)127; National Adaptation Strategy and Plan of Action for Climate Change Nigeria (NASPA-CCN)128; Nigeria Climate Change Policy Response and Strategy;

Senegal.	Senegal's Nationally Determined Contribution (NDC), adopted in 2015, is part of the forward-looking "Plan Sénégal Émergent (PSE)" vision, its strategy and development plans, as well as sectoral programs for sustainable management of its natural and environmental resources.

225. The Sahel countries are low emission countries. Nevertheless, each has made a commitment to lower emissions further taking into consideration their business-as-usual scenarios. At the end of 2015, all the Sahel countries submitted an Intended Nationally Determined Contribution (INDC) to UNFCCC. They subsequently submitted NDCs and most have recently updated them in which they stressed their vulnerability to climate change, especially for food security, and while committing to do their fair share for emissions, highlighted adaptation as the priority and included adaptation actions. While the proposed NDC adaptation actions reflect each country's priorities, there are some actions which are common to the countries given that they face similar climate threats. These include the need for large scale restoration of degraded lands; afforestation/reforestation/agroforestry and improved pastures; climate-smart or conservation agriculture including improved seeds, crop diversification; rotational grazing for livestock, enhancing productivity through improved husbandry, feed and animal management practices that can enhance productivity and assist in reducing methane emissions from livestock; water harvesting and efficiency; and early warning systems.

Table 14 - Countries NDC and Climate Change targets relevant to SURAGGWA

	Mitigation	Adaptation Priorities
Burkina Faso (2021-25)	Reduce GHG emissions by 31,682.3 Gg CO2eq by 2030, i.e. 29.42% compared to the Business as Usual scenario.	Agriculture, water management, and land use: • Restore and maintain land fertility of 1.575 million ha of cropland; • Restore 1.125 million ha of degraded land for pasture and forest; • 10,000 tons of fodder collected and stored each year; • 30,000 ha of stream banks protected; • Compost from biodigestors fertilize 750,000 ha.
Chad October 2021	Cumulative reduction of GHG emissions by 2030 to 88,350 kt CO2eq (unconditional and conditional measures) with an overall mitigation target of 19.3% compared to the baseline scenario.	Agriculture, livestock and fisheries: • Develop intensive and diverse cultivation; • Use improved inputs, (organic fertilizers including composts, adapted plant varieties); • Agroforestry; • Land and water conservation; • Common grazing zones, creating and popularizing fodder banks, crossbreeding of animal species; • Development of enclosed fish farming areas.
Djibouti 2015	The Republic of Djibouti is committed to reducing its GHG emissions by 2030 by 40%, i.e. nearly 2Mt CO2e.	Six priority areas: • Ensuring water access; • Promotion of best practices in the agriculture, forestry, fishery and tourism sectors; • Reduce vulnerability to the effects of climate change for the most exposed social, economic or geographic sectors; • Protect and enhance ecosystems and maintain the services they provide • Ensure the development of sustainable and resilient cities; and • Ensure resilience and sustainability of the country's key infrastructure.
Mali (29/9/15)	Commits to reducing emissions by 29% for agriculture, 31% for energy and 21% for forests and land use, each by 2030, and in comparison, to a BAU scenario. This is an average reduction of 27%. This is conditional upon international support, although around 40% of this can be met unconditionally. Includes a section on adaptation, though only for the period 2015-2020.	Agriculture: • 92,000 ha under climate smart agriculture and sustainable land management; • Improve livestock rotation over grazelands to reduce farmer-livestock conflict over 400,000 ha; • Improved crop and livestock varieties; • Small scale agricultural development, including fruit trees for reforestation, and vegetation cover and erosion prevention (post 2020); Land use and forestry: • Anti-desertification and protection of 9 million ha; • Reforestation of 325,000 ha. Water and water supply: • Rainwater harvesting and storage to ensure universal potable water access; • 75,000 rural households have drinking water from drinking water systems and water collection structures; • Watershed management (post-2020) • Wastewater treatment (post-2020)

Mauritania (2021-2030)	Committed to reducing approximately 33.56 million tonnes Eq-CO2, i.e. 22.3% during the period 2020-2030.	Agriculture and land management: • Aerial seeding of degraded land (10,000 ha per year) to promote regeneration of the natural environment; • Restoration of natural pastures (deferred grazing and rangeland management); • Exploration of aquifers (drilling) Fisheries and aquaculture: • Promotion of fish-farming and responsible fishing on Lake Foum Gleita; Water and water management: • Rehabilitation and integrated management of sustainable wetlands against the effects of climate change; • Drinking water supply systems in rural areas equipped with solar energy; Climate risk management: • Protecting cities of Nouakchott and Nouadhibou against risks marine emersion and silting;
Senegal 2020	Senegal has revised its targets and is now committing to a relative reduction in greenhouse gas emissions by 5% unconditional by 2025 and 7% by 2030, compared to the BUA. This reduction may be increased to 23% and 29% respectively, to horizons 2025 and 2030, if Senegal benefits from the support of the international community. Overall emissions projections until 2025 : 32 648 Gg CO2 2030 : 37 761 Gg CO2 NDC (Unconditional objective of achieving GHG emission reductions with national capacity) 2025 : - 1 632.4 Gg C02 equivalent 2030 : - 2 643.27 Gg C02 equivalent NDC + (Conditional objective of achieving GHG emission reduction the support of the international community) 2025 : - 7 509.04 Gg C02 equivalent 2030 : - 10 950.69 Gg C02 equivalent	Agriculture (through PRACAS2 2019-2023) : NDC : • 99,621 ha of agricultural land under assisted natural regeneration (ANR) • 4,500 ha under compost, by 2030 • Make available organic manure and improved compost with biogas production • NDC + : • 28,500 ha of irrigated rice to an Intensive Rice Cultivation System (IRCS) reducing both water use and methane emissions. • 498,105 ha for ANR and 14,400 ha for compost. Forestry (through “Lettre de Politique de l’environnement”) NDC : • Increase annually the reforested/restored areas by about 1,297 ha of mangrove and 21,000 ha of various plantations;o Reduce the burned areas due to late fires by 5% and those due to controlled fires by 10% compared to 2015. NDC+ : • Secure 500,000 ha of forests, • Reforest and restore 4,000 ha/year of mangroves, • Carry out 500,000 ha of various plantations • Reduce the area burned by bushfires by 90% by the fifth year of implementation of the management plans. NB: These efforts will reduce the deforestation rate by 25%, from 40,000 ha/year in 2010 to 30,000 ha/year in 2030.
Niger 2021	the AFAT sector: Unconditional Reductions: 4.50% (BAU 2025) and 12.57% (BAU 2030) and Conditional Reductions: 14.60% (BAU-2025) and 22.75% (BAU 2030) - the Energy sector: Unconditional Reductions: 11.20% (BAU-2025) and 10.60% (BAU2030) and Conditional Reductions: 48% (BAU-2025) and 45% (BAU-2030).	Agriculture and sustainable land management: • Restoration of agricultural/forestry/pastoral lands: 1,030 000 ha.; • Assisted natural regeneration: 1,100,000 ha.; • Fixation of dunes: 550,000 ha.; • Management of natural forests: 2,220,000 ha.; • Hedgerows: 145,000 km.; • Planting of multiuse species: 750,000 ha.; • Planting of Moringa oleifera: 125 000 ha.; • Seeding of roadways: 304,500 ha.; • Private forestry: 75,000 ha.
Nigeria 2021	20% below BAU by 2030 and 47% conditional on international support. Or 100MTCo2100 MTCO2 eq below2018 levels	Agriculture: • Adopt improved agricultural systems for both crops and livestock (e.g. diversify livestock and improve range management); • Increase access to drought resistant crops and livestock feeds; • adopt better soil management practices; • provide early warning/meteorological forecasts and related information); • Implement strategies for improved resource management (e.g., water efficiency of irrigation systems; increase rainwater & sustainable ground water harvesting); • Increase planting of native vegetation cover & promotion of re-greening efforts; • Focus on agricultural impacts in the savanna zones, particularly

		the Sahel, the areas that are likely to be most affected by the impacts of climate change.

226. There is strong ownership in the Sahel for SURAGGWA. At the project inception and formulation stage, there was close participation through regional and country level workshops, individual interviews, and extensive stakeholder consultations (Annex 7). These meetings were used to identify overall target areas, specific activities, financing requirements, local contributions and set targets for each country, using well-specified criteria including capacity and country priorities. The countries have demonstrated their commitment further by identifying opportunities for seeking international climate finance for the project, establishing partnerships and institutional arrangements and focal points for implementation. Some of the countries are also making efforts for additional technical assistance and funding for the establishment of a Green Financing Facility that will provide concessional loans and climate smart credit risk tools in order to enhance access to credit, contribute to de-risking the sector and foster greater knowledge on climate finance and adaptation across the selected countries. These measures are designed to directly help overcome the key barriers that countries face in rural finance and enable them to engage in green financing.

227. The design of component 3 is the outcome of a two-year on-going consultation process with the selected beneficiary countries as well as regional and international partners. In particular, the preparation of the concept note and the Full Funding Proposal was based on a series of workshops that allowed participants to express their needs and to align future activities with existing programmatic frameworks at the national and regional levels. At the end of the design phase, all countries and the GGW PA identified monitoring of land degradation/restoration, coordination and planning, communication, resource mobilization and knowledge management as the priorities in which SURAGGWA should invest to help strengthen their Great Green Wall institutions. These priorities are also directly consistent with those expressed in the GGW strategies prepared by the countries in 2011-2012. The second round of consultations that led to this proposal allowed for a second round of engagement to identify more specific areas of work - among the themes listed above - where SURAGGWA would add the most value, taking into account the capacity building initiatives of other Development Partners.

D.6. Efficiency and effectiveness (max`. 500 words, approximately 1 page)

228. **Financial Structure**: The financial structure of the programme is based on cost estimates of land rehabilitation on both highly degraded and moderately degraded farm land as well as the other costs associated with capacity building for enhancing the resilience of farming and pastoralist communities and institutional strengthening for enhanced coordination, implementation and monitoring and evaluation. The initial costs were based on FAO's field experience gleaned from on-going projects such as the Action Against Desertification and confirmed during field visits to several of the Sahelian countries. The targets of the project are based on these cost estimates which will be incurred on addressing some of the key constraints faced by the vulnerable communities in the proposed sites. The programme has been designed to provide only a one-time input for rehabilitating the degraded lands and thereafter, the farmers and pastoralists will sustain the restored areas, because the production of forage and NTFPs from the restored lands will give them incentives to do so. Once the technical, organizational and commercial capacities of the farmers and pastoralists have been strengthened, their ability to adopt more resilient farming practices and invest in restoration of additional land will ensure further scaling up of project results.

229. **Private Sector Orientation**: In order to make the investments viable, there is a strong role envisaged for the private sector in the programme. SURAGGWA will encourage the development of a vibrant farm equipment rental market for hiring deep ploughs to implement the Vallerani System (VS).[123] This is a low cost, high productivity system with a high degree of efficiency which has demonstrated significant results in a wide range of countries where it has been implemented. SURAGGWA has modified the earlier approach of procuring Delfino ploughs in favour of placing greater reliance on the private sector to develop the local rental market for a more sustainable approach to land restoration. In addition, the value chain activities for NTFPs

[123] A method of working arid and semi-arid land to bring life back and restore degraded soils.

will also rely on establishing effective market linkages so that the private sector is able to provide sustainable access to markets.

230. **Cost-Benefit Analysis of the Mitigation and Adaptation Outcomes:** GCF is being requested to provide total financing of USD 150 million for SURAGGWA. The programme is expected to benefit around 1.881.002 people directly at a per capita cost of USD 80 which is above the average cost of GCF projects reported at USD 17 per capita, though within the range of per costs of FAO implemented GCF projects.[124] The per capita cost within GCF and other projects generally vary considerably depending upon the level of support provided to the beneficiaries. Projects and programmes that which focus only on capacity building, awareness and information dissemination are much lower in cost while those which provide concrete support for tangible investments are naturally much higher in terms of the intensity of support. SURAGGWA provides a combination of soft and hard support. The unit cost per CO_2 sequestered from agroforestry can vary widely depending on a number of factors such as the specific location, tree species, management practices, and monitoring and verification requirements. However, here are some general comparisons of the unit costs per ton CO_2 sequestered from agroforestry compared to other types of carbon removal which are presented in Table 15 below. Based on preliminary estimates, it is expected that the project will help to sequester 65.1 million tons of CO_2 equivalent at a unit cost of USD 3.41 per metric ton. This is 28% below the average cost of USD 4.75 per mt CO_2 equivalent reported in the GCF dashboard.[125] Agroforestry is generally considered to be a relatively low-cost option for carbon sequestration compared to other types of carbon removal activities. However, the costs can vary widely depending on the specific circumstances and practices used. This SURAGGWA unit cost of USD 3.41 per tCO_2e[126] also compares very favourably with the current estimates of the social cost of carbon, which the World Bank puts at between USD 55 and USD 108 per ton CO_2e for 2024, increasing in the future as climate change impacts intensify. Some observers argue that these figures do not yet include all of the widely recognized and accepted scientific and economic impacts of climate change and are therefore far lower than the true costs of carbon pollution.[127]

Table 15 - Comparative Cost of Carbon Sequestration

Type of Activity	Unit Cost per tonne of CO_2 (USD)
Afforestation and reforestation:	10 to 200
soil carbon sequestration	3 to 200
Direct air capture	100 to 1000
Blue Carbon (seagrass beds and mangroves)	10 to 100
Agro-Forestry	5 to 50

231. **Results of the Economic and Financial Analysis.** The Economic and Financial Analysis (EFA) of the proposed programme is based on a Cost-Benefit Analysis (CBA) applied to a range of typical community land and farmland restoration models in each of the eight countries (Reforestation/Afforestation using a mix of native tree, shrub and grass species and Agroforestry integrating native tree species in cropping systems) as well as to the improvement of smallholder NTFP value chains such as gum, Balanites, Moringa, jujube, baobab, etc.) by promoting better production, product conditioning and agro-processing, storage and marketing.

232. The result of the financial analysis (from the – private – point of view of smallholder beneficiaries) confirms the high potential of land restoration activities for the participating rural households, producing a return of 1.52 USD on average for each USD invested, with outcomes ranging from 1.1 USD to 3.19 USD. The adoption of sustainable land management practices integrating the cultivation of fodder and other crops generating fast returns would allow a substantial increase in household income (up to 65%) over the first 3 years of the investment, depending on the level of adoption.

[124] The per capita calculation uses the committed funding of USD 11.4 billion and the 666 million beneficiaries reported at the end of November 2022 by GCF in its dashboard.
[125] The unit cost uses the committed funding of USD 11.4 billion and the 2.4 bn CO_2 reported at the end of November 2022 by GCF in its dashboard.
[126] Calculated as the total programme cost (250 million) divided by the sequestration result of 65.9 million tCO2e.
[127] The true cost of carbon pollution. Environmental Defence Fund. November 25, 2022.

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233. **Economically speaking, SURAGGWA is a moderately viable programme, generating a net present value (NPV) of USD 154.2 million and an economic internal rate of return (ERR) of 19.2percent** (on a total budget of USD 220.0 million, USD150 million of which would be funded by GCF), even without valuing any of the environmental benefits. The vulnerability to climate change of the rural communities involved, in combination with their extremely limited access to financial services and the high level of indebtedness of the eight SURAGGWA countries, however, fully justify the mobilization of grant funding from the GCF.

234. The full economic potential of the project, when the projected GHG mitigation are valued appropriately, is much higher. **Using more conservative prices, at current market price (31 USD per tCO2-eq), SURAGGWA would generate a net present value (NPV) of US$ 754.5 million and an economic internal rate of return (ERR) of 57.5 percent over a 20-year period**. These results further emphasize the fact that the public benefits that would be generated by the programme largely outweigh the private benefits created. An analysis was undertaken of the returns on some of the key NTFP which show high potential returns (Annex 3).

235. **Industry Best Practices:** SURAGGWA has been patterned on the best practices of the Action Against Desertification project in its pilot and scaled up versions. Lessons have been learnt from these projects over the last decade, regarding the use of traditional knowledge (restoration is based on the half-moon soil preparation technique); primary emphasis on developing community buy-in, management skills, and benefits from restoration as a requisite for adoption and sustainability. A few adjustments have been made in the previous design as a more cost-effective, efficient and sustainable option. The first of these is the decision not to purchase any heavy equipment by the project but to encourage the use of rented equipment that will help build private sector growth and conserve the project r sources for expanding the project area. The second adjustment is the focus on seed and seedling production to ensure that adequate seed is available for forage crops and for threatened local species and varieties that can build resilience to climate change.

236. **Leverage and co-financing:** By demonstrating effective approaches for greening the Sahel and helping the vulnerable communities to build resilience, SURAGGWA can mobilize considerable resources that are potentially available from a host of development partners committed to providing funds for climate change mitigation and adaptation. FAO has also established a strong reputation in this field and has the potential to attract funds from a host of other agencies such as the World Bank, EU, ADB, AFDB, IFAD, GEF, etc and a range of bilateral donors who have strong confidence in FAO's technical capacity.

237. **Mobilization of private and commercial funds:** It is expected that SURAGGWA will result in *mobilizing both private and commercial capital* for the sustainable development of the Sahel. When pastoralists and farmers are given a real incentive to preserveingthe capital investments made by the project, they are also expected to provide their labour and own resources in the development of the land as it is the eco-systems on which they depend. Similarly, *enterprises dealing with NTFP value chains are expected to make investments in a range of equipment, machinery, facilities that will grow their business and increase their incomes*. These small enterprises are expected to invest their own capital as well as borrow from the commercial financial institutions. There are several projects which have also put in place arrangements for enhanced access to financial services such as the IFAD supported IGREEFIN project, that SURAGGWA will collaborate with closely. Programmes like SURAGGWA provide *increased confidence to financial institutions of the viability of NTFPs as a viable business opportunity and send a strong signal to the commercial financial sector* to direct their resources to this sector, important for livelihoods, income, employment and business growth and climate mitigation and resilience. On-going projects in several countries, including lessons used in the formulation of a GCF co-financed project on gum arabic in Sudan, show an increase in smallholder investment in restoration once producers receive better prices for NTFP products.

E. LOGICAL FRAMEWORK

E.1. Project/Programme Focus

☒ Reduced emissions (mitigation)
☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project will contribute (Description)
	Description	Rating		
Scale	It is estimated that only 4 million hectares of the total potential restorable area of 123 million hectares in the GGW countries has been rehabilitated so far. There is limited public sector financing and little incentive for local communities to rehabilitate degraded communal resources, low technical and financial ability to protect and develop farm lands to increase crop and livestock yields. There are insecure tenure rights and lack of a model for protecting and restoring degraded land in a highly vulnerable area threatened by climate risks and other conflicts.	Low	Local communities develop an abiding interest in rehabilitating and protecting degraded communal lands as it offers them an opportunity for gathering forage crops for their livestock, collecting non-timber forest products as an additional source of income, from which women benefit in particular. In addition, farmers have the technical knowledge to adapt to climate risks and are able to procure inputs and technologies for improved management and productivity of their farm lands. Local communities are interested in scaling up the rehabilitation in collaborative arrangements with local authorities which ensure use rights. There is a vibrant private sector that is able to provide equipment on rent for deep ploughing and seeds and seedlings for plantation are available locally. The local communities and private enterprises are able to expand their linkages for sale of a range of NTFP products with access to finance from the commercial sector and impact investors. Public sector is able to demonstrate the impact of the investments through credible evidence and coordinate and plan in participation with l research institutions, development partners and DAEs.	The project aims to restore 133 238hectares of highly degradedlands and 1.140 million hectares of moderately degraded lands and develop key value chains for building resilience of local communities and give them a stake in the management of the communal lands. The project aims at demonstrating a model of land rehabilitation, ecosystem restoration and value chain development in collaboration with local communities, the private sector and national governments. A recent evaluation of AAD showed that at the rehabilitation sites, vegetation had been successfully established, land had been rehabilitated, and the density of trees and shrubs had increased dramatically.
Replicability	Replicability of the models that have been tested and tried is currently very low due to lack of financing, weak regulatory, institutional and technical capacities, weak market linkages and limited engagement with the private sector which can potentially play a key role by providing	Low	It is expected that strengthened national governments will be able to provide supportive regulatory frameworks that can better define use and tenure rights on common property resources to encourage sustainable use and ecosystem development. The technical capacity building of small farmers and herders as well as capacity for seed and seedling production will enable replication of the innovative techniques and	The key outputs expected from the replication will be increase in the area rehabilitated and protected, increase in water retention, reduction in desertification and soil erosion, increase in capacity for building climate resilience in farming practices. The replication is expected in areas adjacent to the selected sites through local exchange and information dissemination that is expected to be promoted through local social

	finance, equipment, inputs and access to markets.		climate resilient practices by the local communities. The rental markets for farm equipment will be available to encourage replication by others and engagement with the private sector will help to replicate and encourage the production of NTFPs, provide access to finance.	networks and tribal and clan affiliations and propagation by the private sector, which is always seeking to expand its markets. The expected outcomes of the replication will be increased carbon sequestration through increased plantations, enhanced climate resilience through adapting farming practices to changes in weather leading to reduction in crop losses, increase in yields and increase in incomes through increased production and sale of NTFPs.
Sustainability	Some aspects of the existing models have not been very sustainable in the past because of insecure tenure and use rights, limited community engagement due to the low level of benefits realized, limited engagement of the private sector in terms of providing an active rental market for renting of equipment, provision of seeds and other key inputs and the low level of production of local non-timber forest products which could not sustain the interest of the local communities or the private sector at the end of the project life.	Low	Local communities are expected to be interested in sustaining the rehabilitated lands due to the benefits they provide such as increased water for their soils, protection against desertification and soil erosion and increased forage crops for their livestock. The adoption of the new techniques and on-farm practices is expected to be sustained due to the increased benefits that the practices provide of increased resilience against climate change, reduction in crop losses and increase in yields. The sustainability of the investments in NTFPs is expected due to the mutually beneficial linkages established between the local communities and private sector which enables them to access the markets and increase their livelihoods and incomes. The national Governments are expected to enhance their capacity for monitoring and demonstrating the credible evidence that the approach is successful, enhance capacity for coordination, planning, resource mobilization and forging effective partnerships.	It is expected that the rehabilitation of 133 238 hectares of degraded common lands will be sustained by local communities due to well established and clear use rights and the benefits that are realised from sustaining the vegetation cover on these lands. In addition, the increased production potential on 1.140 million hectares of farm lands will be sustained by the farming families who own these lands mainly because the practices that are imparted provide them benefits of reduction in losses and increase in yields and enable them to withstand any negative impacts of climate change. Similarly, the 250,000 households who are engaged in a host of value chains related to seed production and NTFPs are expected to sustain their production and market linkages because of the increase in incomes and diversification of their livelihoods.

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final[128]	
MRA4 Forestry and land use	Core 1: GHG emissions reduced, avoided or removed/sequestered	EX-Ante Carbon-balance Tool (EX-ACT)		7.27 Mn TCO2 eq mitigated	25.1 Mn tonnes CO_2 eq mitigated Over 10 years	The project is initiated on time and the overall implementation proceeds as planned.

[128] The final target means the target at the end of programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

					65 Mn TCO2eq over 20 years	The growth and survival rates of the species are not disrupted by undue climate factors. Planned co-financing will be executed in a timely manner The programme with the Governments' support will be able to secure community willingness to participate in land rehabilitation activities Communities will have sufficient incentives in terms of benefits that are generated from restoring of community and family lands to participate in the programme Political stability is maintained and no major humanitarian crises will take place during programme implementation
ARA1 Most vulnerable people and communities	Core 2: Direct and indirect beneficiaries reached	Annual progress reports; mid-term and final evaluation Remote-sensing and GIS data collection, that will be included in Annual Performance Reporting, will be utilized to ensure ecosystem based monitoring of spillover effects	0	Direct: 658,351 [129] of which women: 327,964	Direct: 1,881,002 Of which women: 937,040	. Planned co-financing will be executed in a timely manner The programme with the Governments' support will be able to secure community willingness to participate in land rehabilitation activities Communities will have sufficient

[129] The total number of direct beneficiaries is calculated based upon the assumption that each restoration site for highly degraded lands will be of 300 ha, with 1 community group per site and assuming each community group is composed of 40 households. For moderately degraded lands, the assumption is that on average each farm plot is 4 HA, with 20 households organized in community groups. There is an assumption of 5% overlap between the direct beneficiaries across highly degraded restoration sites and moderately degraded farm plots, and a 30% overlap between component 1 and component 2 beneficiares. Please refer to Annex 23 for further details.

		of land restoration activities. Biennial Outcome Assessment (geo-referenced)		Indirect: 1,351,674 Of which women: 673,146	Indirect: 3,861,925 Of which women: 1,923,273	incentives in terms of benefits that are generated from restoring of community and family lands to participate in the programme Political stability is maintained and no major humanitarian crises will take place during programme implementation The indirect beneficiary numbers have been calculated based on conservative assumptions relating to knowledge transfer between households. During programme implementation these numbers will monitored and updated according to the ecological monitoring tools in the programme M&E approach. Furthermore, indirect beneficiaries are derived from land restoration activities under Component 1 Assumptions: please refer to Annex 23 excel template for assumptions related to population beneficiary calculations.
	Supplementary 2.1: Beneficiaries	Annual perform ace reports; mid-	0			A: Market demand for NTFPs (both domestic and

	(female/male) adopting improved and/or new climate-resilient livelihood options[130]	term and final evaluation Biennial Outcome Assessment (geo-referenced)		168,116	480,330	international) can be increased, and marketing conditions for smallholders improved. Assuming that 75% of the direct beneficiaries of value chain activities under component 2 adopt improved or new climate resiient livelihood options through Component 2 activities
	Supplementary indicator 2.5:Beneficiaries (female/male) adopting innovations that strengthen climate change resilience[131]	Annual perform ace reports; mid-term and final evaluation Biennial Outcome Assessment (geo-referenced)	0	540,670	1,544,770	Communities will have sufficient incentives in terms of benefits that are generated from restoring of community and family lands to participate in the programme Assuming that 75% of all direct beneficiaries from land restoration activities under component 1 adopt an innovation that strengthens climate change resilience
ARA4 Ecosystems and ecosystem services	Core 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice	Annual perform ace reports; mid-term and final evaluation. Remote-sensing and GIS data collection, that will be included in Annual Performance Reporting, will be utilized to ensure	0	961,928 ha	1,273,751 ha	Climatic conditions will allow the establishment and growth of the enriched and restored landscapes. Restored area management committees continue to operate effectively.

[130] Climate-resilient livelihood options refer to the stabilisation of crop yields through the adoption of agro-ecological approaches and other specific activities to restore moderately degraded farmlands.; increased adoption of production and processing within NTFP value chains in beneficiary households; increased diversification of NTFP production as a result of programme activities. Please also refer to crop vulnerability as per CARD analysis.

[131] Innovations are included as part of all programme components. Under component 1 innovations include the combination of mechanised ploughing together with direct seeding, thus reducing overall costs and reliance on nurseries; focus on native species and mxing with grass specific and the application of science and technological innovations in relation to species selection, seed handling technologies, as well as community consultations for the selection of species. Under component 2, innovations include the uptake of technologies that improve efficiency in the processing of NTFPs, as well as innovations relating to financial access.

		ecosystem based monitoring of spillover effects of land restoration activities. Biennial Outcome Assessment (geo-referenced)[132]				

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)					
Core Indicator	**Baseline context (description)**	**Rating for current state (baseline)**	**Target scenario (description)**	**How the project will contribute**	**Coverage**
Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner	Little capacities and regulatory mandates of NAGGW to consistently identify most effective land restoration activities and tracking of investments and results against GGW Initiative targets. Additionally, many countries have weak to no standards and certifications in the NTFP value chains, which can be strengthened in terms of processing and marketing for domestic and export markets.	low	*NAGGW have strengthened mandates to request specific information from projects directly relating to GGW Initiative, with a core set of indicators (both biophysical and socio-economic, geo-referenced) that projects in the GGW area report against.* *Relevant national agencies have strengthened capacities to access carbon finance for restoration with at least two countries engaging in concrete opportunities. At least one country appraises specific options to incorporate restoration in its evolving regulatory framework to develop domestic carbon markets/taxes/offsets.* *In the context of NTFP value chain development, countries will have strengthened existence of certification and regulatory schemes to strengthen*	The project will support the update and revision of frameworks relating to NAGGW mandates and responsibilities to facilitate the monitoring and reporting of GGW-related projects consistently across all investment sources. The project will strengthen national capacities to a) incorporate restoration in domestic carbon markets or tax/offset systems and b) engage with carbon finance opportunities (voluntary carbon markets and Article 6). There will also be in relation to standards and certifications within the NTFP value chains targeted under SURAGGWA.The programme will support the formulation of norms, standards and certification in relation to NTFPs where they do not exist. This will build on the experiences of other participating countries, as well as	Multi-countries

[132] The biennial outcome assessments combine ground-truthing with remote sensing information on land restoration status.

			production and marketing.	specific studies undertaken that will inform the formulation of these frameworks.	
Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation	Little systematic use of mechanized ploughs for large-scale land restoration, as well as GIS and remote-sensing technologies for monitoring and reporting of results on land restoration efforts. Furthermore, there is weak innovation and technologies adopted in value-added processing activities, equipment and technologies in NTFP value chains, as well as in financial products that contribute to climate change adaptation and mitigation (eg. Carbon credit markets)	low	*Rural communities will have access to suitable technologies for large-scale land restoration of communal lands, as well as access to profitable technologies and equipment to strengthen resilience and improve options for access to markets. Additionally, NAGGW and other agencies will be able to systematically monitor progress against GGW indicator framework through monitoring and evaluation system (including remote-sensing and GIS) deployed as part of project.* *In the context of NTFP value chains, actors will have access and know-how in relation to necessary equipment and technologies for fit-for-purpose and efficient processing[133] and marketing of NTFP products.*	*The project activities will boost adoption of multiple technologies in order to boost land restoration activates of highly degraded lands through mechanized ploughing, as well as through the adoption of remote-sensing GIS technologies and approaches for cost-effective monitoring.* *Under the NTFP value chain development activities, value chain actors will be equipped with fit-for-purpose equipment to boost output, and will be provided with the necessary technical training in the use of such equipment.*	Multi-countries
Core indicator 7: Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level	While NTFP markets exist in all 8 countries of intervention, the capacities of actors across the value chains from production to marketing require systematic support to improve entrepreneurial activities	low	*Rural communities, public actors and private sector will have stronger capacities and greater access to markets, credit, etc.. to adopt suitable and market-driven adaptation options through NTFP value chain development.*	Under Outcome 2, programme activities will support market linkages and entrepreneurial activities for facilitating a post-project financial exit strategy. This will include among other the promotion of dialogue and engagement between producer groups and private sector entrepreneurs (including for market distribution,	Multi-countries

[133] Many actors across the value chain do have access to equipment, however, it is inefficient or not fit-for-purpose, with some value chains characterized by manual processing or incorrect machinery that leads to high loss of product and decreases overall output. Please refer to Annex 2 for further details.

	and linkages to markets.			domestic/international) to further expand the sector's output as well as value addition. In addition, the project will collaborate with financial institutions to stimulate further investment in the NTFP sub-sector, through exploring options such as value chain financing.	

E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Outcome 1.1: Highly and moderately degraded lands restored under climate-resilient management practices	Core 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice	EX-Ante Carbon-balance Tool (EX-ACT), country-level carbon accounting, supported by remote sensing imaging. Secondary source of information through Africa OpenDeal	0	961,928 ha	1,273,751 ha	A: Planned co-financing will be executed in a timely manner A: The programme with the Governments' support will be able to secure community willingness to participate in land rehabilitation activities A: Communities will have sufficient incentives in terms of benefits that are generated from restoring of community and family lands to participate in the programme A: Political stability is maintained and no major humanitarian crises will take place during programme implementation A: No major forest fire or natural disaster occurs that would destroy large areas of land .

						A: Population growth and migratory patterns remain unchanged; A: Climatic conditions will allow the establishment and growth of the enriched and restored landscapes. A: Countries continue to implement their climate laws and programs according to plans A: Restored area management committees continue to operate effectively. Target number of hectares is based hectares to be brought under sustainable restoration and management approaches under Output 1.3, Output 1.4 and Output 1.5.
Outcome 1.2: GHG emissions reduced, avoided or removed/sequestered	Core 1: GHG emissions reduced, avoided or removed/sequestered	EX-Ante Carbon-balance Tool (EX-ACT), country-level carbon accounting, supported by remote sensing imaging. Secondary source of information through Africa OpenDeal	0	5,653,860 tCO2 eq	22,413,008 tCO2 eq by programme end (2036) 65,094,801 tCO2 eq over programme lifetime	A: Planned co-financing will be executed in a timely manner A: The programme with the Governments' support will be able to secure community willingness to participate in land rehabilitation activities A: Communities will have sufficient incentives in terms of benefits that are generated from restoring of community and family lands to participate in the programme A: Political stability is maintained and no major humanitarian crises will take place during programme implementation

						A: External shocks do not impact survival rates of the species, and are not disrupted by undue climate factors. A: Population growth and migratory patterns remain unchanged; A: No wholesale of land occurs and land tenure regimes remain stable A: Countries continue to implement their climate laws and programs according to plans Data sources: Baseline emissions from AfricaOpenDEAL AFOLU emissions reductions estimates were obtained using EXACT (see Annex 22). The carbon sequestration figures calculated to programme lifetime of 20 years (taking into account country-specific programme implementation length + balance of years of capitalization depending on country-specific implementation plan).
Output 1.1: Local community groups organized, trained and knowledgeable in land restoration activities and monitoring	1.1.1 Number of community restoration management committees trained and knowledgeable in restoration practices (disaggregated by % women participation, % youth participation. % women in management position, % youth in management position)	Annual Performance Reports, Quarterly activity monitoring Source: Records of regular meetings of restoration management committees Biennial Outcome Assessment (KAP Survey to be undertaken as part of Biennial Outcome Assessment)	0	11,015 community groups/organizations	14,700 community groups/organizations Target 50% participation of women	Moderately degraded farmlands will have 40 households on average per community organization covering 4 ha per household. For highly degraded lands, it is assumed that one community group will cover 300 ha on average. Communities agreeing on providing land and labor and management/administration time and skills to restore communities' lands.

						Functionality of the restoration committees is defined by success of social fencing and sustainable management after initial restoration activities are undertaken. Communities agree to give more prominent roles to women and youth in community groups. This assumes that specific and distinct Community groups will be built around existing community groupings and where they do not exist they will be organized around highly degraded lands and moderately degraded lands.
Output 1.2: Native seed supply systems have been strengthened to ensure the availability of genetically appropriate seeds that provide increased climate resilience.	1.2.1 Number of community groups actively collecting, storing and disseminating native restoration species. 1.2.2 Share of women participating in seed supply systems/networks	Quarterly activity/community monitoring and reporting Biennial Outcome Assessment (geo-referenced survey) Source: records of rural community organizations on sales and/or distribution of restoration seeds by source of seed (informal/community -based or formal seed supply system) and by variety released.	0	735 restoration seed producer groups At least 50% participation of women	735 restoration seed producer groups At least 50% participation of women	A: Communities agreeing on organizing (CBO) and providing labor and management/administration time and skills for quality biodiverse collections and handling restoration seeds of useful native species. A: Existing tree, shrub and grass stock of genetically appropriate native species suitable for seed production can supply the restoration of the targeted ha by the programme
Output 1.3: Targeted highly degraded lands restored through arrangements for rainwater harvesting and enhanced soil permeability and through local communities	1.3.1 No. hectares of highly degraded land restored and/or under restoration 1.3.1.a. Share of women that have noticed time savings as a result of land	Quarterly activity monitoring through remote-sensing Ground-truthing including qualitative assessment of restored ha Annual Progress Reports	0	110,109 ha 20% women report time savings	133,238 ha 50% women report time savings	A: Planned co-financing will be executed in a timely manner A: The programme with the Governments' support will be able to secure

engaging in planting of seedlings and direct seeding**.**	preparation/restoration practices	Source: Remote-sensing to be implemented through Component 3; qualititative assessments/ground-truthing carried out during field missions for monitoring				community willingness to participate in land rehabilitation activities A: Communities will have sufficient incentives in terms of benefits that are generated from restoring of community and family lands to participate in the programme A: Political stability is maintained and no major humanitarian crises will take place during programme implementation A: Communities are able to mobilize sufficient resources for the labor required for rainwater harvesting and enhanced soil permeability. A: Sufficient seedlings available to supply the area targeted (linked to success of activities under Output 1.2) A: Manual labor is available, where needed A: Mechanised ploughs (Delfino) available for large scale land preparation in countries
	1.3.2 Percent Increase in regreening, NDVI[134] and vegetation cover in restored areas	Baseline, Mid-term and Final NDVI Assessment (qualitative assessment) Annual Programme Progress reports Source: earthmap/satellite imagery/drone data/KoboToolBox geo-referenced data extraction Ground-truthed through Biennial Outcome Assessment (to include qualitative NDVI assessment)	TBD (at initial baseline in PY1)	5%	20%	

134 Investigation of NDVI (Normalized Difference Vegetation Index) values in cropland and non-cropland areas within the GGW areas of Senegal and Chad, and with increase during 10-year period (2012-2022) shows that NDVI increase typically range from 10-20%. The calculation was done in EarthMap.org with Landsat data at 30m resolution. For context, NDVI is a reliable proxy for land restoration and productivity (Chen et al, 2021; Birtwistle et al, 2016; Ruijsch et al, 2022; Helman et al, 2014; Sun et al, 2011; Meroni et al, 2017).

Output 1.4: Targeted moderately degraded lands planted with a range of species to restore and enrich the landscapes of agro-forestry, agro-ecology and sylvo-pastoral systems, biomes and protected forest	1.4.1 No. of ha of moderately degraded lands restored and/or under restoration 1.4.1.a. Share of women that have noticed time savings as a result of land preparation/land restoration practices	Quarterly activity monitoring through remote-sensing Ground-truthing including qualitative assessment of restored ha Annual Progress Reports Source: Remote-sensing to be implemented through Component 3; qualitative assessments/ground truthing carried out during field missions for monitoring	0	851,818ha	1.140 million ha 50% women report time savings.	A: Planned co-financing will be executed in a timely manner A: The programme with the Governments' support will be able to secure community willingness to participate in land rehabilitation activities A: Communities will have sufficient incentives in terms of benefits that are generated from restoring of community and family lands to participate in the programme A: Political stability is maintained and no major humanitarian crises will take place during programme implementation A: That the existing tree stock of genetically appropriate NTFP species suitable for seed production exists to supply the restoration of the targeted ha by the programme. A: Communities and households agree on part taking and planting recommended species. A: Communities and households protect the plots and continuously look after seedlings of the planted species.
	1.4.2 Percent increase in NDVI and vegetaion cover in restored areas	Biennial Outcome Assessment (geo-referenced and to include qualitative NDVI assessment) Annual Programme Progress reports Source: earthmap/satellite imagery Ground-truthed through Biennial Outcome Assessment (to include qualitative NDVI assessment)	0	5%	10%	
Output 1.5. Rural community capacities strengthened for sustainable management and restoration, and	1.5.1 No. of local restoration technicians whose technical capacities[135] have been strengthened for	Annual Progress Reports, quarterly activity monitoring Biennial Outcome Assessment[136]	0	279 20% overall participation of women in trainings	279 20% overall participation of women in trainings	A: Community sensitization activities, and related capacity building activities are fully adopted by the community

[135] Reference to capacity development outcome surveys mentioned previously.
[136] Reference to capacity development outcome surveys mentioned previously.

verification of restoration results	land restoration and monitoring (disaggregated by % women participation, % youth participation)					organization, with a well-defined and sustainable exit strategy. A: Assuming 2 village technicians will be trained to support 5 Community Groups (or 10 community groups on moderately degraded lands). A: Community activities, and related capacity building activities are fully adopted by the community organization, with a well-defined and sustainable exit strategy.
Outcome 2.1. Enhanced resilience of livelihoods	Supplementary 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options[137]	Annual progress reports and final evaluation Biennial Outcome Assessment (geo-referenced)	0	168,116	480,330	A: Communities will have sufficient incentives in terms of benefits that are generated from restoring of community and family lands to participate in the programme A: Market demand for NTFPs (both domestic and international) can be increased, and marketing conditions for smallholders improved. Assuming that 75% of all direct beneficiaries from land restoration activities under component 1 adopt an innovation that strengthens climate change resilience

[137] Climate-resilient livelihood options refer to the stabilisation of NTFP yields towards the increased commercialisation of NTFPs through the adoption of agro-ecological approaches and other specific activities to restore moderately degraded farmlands under Component 1, as well as ; increased adoption of production and processing within NTFP value chains in beneficiary households under Component 2. ;

Output 2.1: Climate resilient and low carbon production and processing practices in selected NTFP value chains adopted by Producer Organizations and MSMEs.	2.1.1 Number of people participating in trainings and coaching provided by the programme (disaggregated by type of training, gender, age)	Project monitoring and community reporting Source: Business plan execution of value chain organisations	0	900	1,795 40% women-led and women participation	A: 30 persons per value chain organization on average trained per session; that value chain activities will concern 5 market region/zone per country; and that each region will be covered by 30 value chain organizations. A: 50% or organizations will be established and operational by mid-term.
Output 2.2: Increased access to markets for smallholder NTFP actors (including micro- and small enterprises)	2.2.1. Increase in real gross sales among programme beneficiaries in targeted NTFP value chains after implementation of programme activities (Percentage; disaggregated by women-led, youth-led, value chain)	Beneficiary business planning/accounting books records (Reporting on Business plan execution of value chain organizations)	0	10%	20% 40% increase in real gross sales for women-led businesses	A: programme activities increase efficiency in production, thereby increase total output combined with increased producer prices expected through market access (both domestic and international) will lead to an increase in quality and quantity of production, thus increasing gross sales. Furthermore, there is great potential in NTFP markets internationally[138]
	2.2.2. Number of commercial purchase agreements signed.	Trade fair results/pledges and beneficiary business planning/accounting books and records	0	At least two purchase agreements per NTFP value chains per country	At least two purchase agreements per NTFP value chains per country	
Output 2.3 Suitable credit and insurance products designed for smallholder NTFP value chain actors (PO, cooperatives, MSMEs) and increased stakeholder capacity for using these financial products in their NTFP value chain activities	2.3.1 No. of financial products tailored for agriculture or NTFP value chains, developed and made accessible to programme beneficiaries (disaggregated by type of products/services: Credt, savings, digital credit, insurance; disaggregated by women/men)	Programme activity monitoring, community reporting Biennial Outcome Assessment (geo-referenced) Source: Review of documents/products /services offered from financial institutions, as well as information collected during survey regarding any real increases in credits/services availed by project beneficiaries	0	16	24 20% share of women value chain actors who have access to financing and non-financial services and products	A: Financial institutions have incentives and interest to develop products and services for smallholder NTFP value chain actors. A: Banks and potential users have demand for these financial products; currently few financial instruments specific to the NTFP value chains exist (with the exception of Senegal and Mali, where some instruments do exist already).

[138] There is a lack of robust baseline data on the number of NTFP species, production levels and export volumes and values from GGW countries. However, market studies on gum Arabic and shea provide some insights into the economic potential of strengthening processing and marketing through increasing productivity, quality and processing and many of the value chains identified in SURAGGWA have market potential in international markets for cosmetics, personal care and for other uses. Please refer here for more information: https://www.weforum.org/reports/the-untapped-potential-of-great-green-wall-value-chains-an-action-agenda-to-scale-restoration-in-the-sahel/

						Assuming that within the project cycle, a financial product on agricultural insurance will be developed/implemented to lower the risk associated with financing NTFPs. Assumption that through strengthening of NTFP value chains there will be increased value chain financing opportunities from private sector actors and thus may serve as an additional source of access to finance. Assuming 2 new credit products and 1 agricultural insurance product developed per country

Outcome 3.1 Enhanced institutional capacities of National GGW Agencies and of Pan African GGW for monitoring, coordination, and resource mobilization	Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner	Annual Performance Reports Biennial Outcome Survey Publications of knowledge products, analyses and reports Resource mobilization pledges from programme-support events	0	NA-GGW and PA-GGW agencies and institutions are actively engaging in capacity development activities and starting-up their hands-on use of the tools, methodologies and coordination and resource mobilization mechanisms promoted by the programme, with the contribution of their own resources (in-kind and financial).	NAGGW have strengthened mandates to request specific information from projects directly relating to GGW Initiative, with a core set of indicators (both biophysical and socio-economic, geo-referenced) that projects in the GGW area report against. ddRelevant national agencies have strengthened to attract diversified resources, including capacities to access carbon finance for restoration with at least two countries engaging in concrete opportunities. At least one country appraises specific options to incorporate restoration in its evolving regulatory framework to develop domestic carbon markets/taxes/ offsets.	A: Governments are committed and take ownership of GGWSSI initiatives, and UNCCD strategies A: Governments are committed to the development of land degradation / restoration monitoring services A: GGW Stakeholders and partners are willing to adopt new Monitoring system including indicators, governance, institutional and regulatory mechanisms A: National and regional GGW institutions can recruit and retain relevant staff with required qualifications and skills

Output 3.1: : GGW Land restoration monitoring system at national and regional level upgraded and functional	3.1.1 Number of GGW agency and other ecological monitoring agency staff trained and using the new monitoring and reporting e system	Training reports and Annual Performance Reports Biennial Outcome Assessment Source: document review of GGW contributions to national reporting to national and countries international commitments; i.e. NDCs, SDGs	0	300	480	A: Governments are committed to the development of land degradation / restoration monitoring services A: GGW Stakeholders and partners are willing to adopt new Monitoring system including indicators, governance, institutional and regulatory mechanisms A: National and regional GGW institutions can recruit and retain M&E experts with required qualifications and skills A: Skills building training does not result in high staff turnover.
Output 3.2: Regional and National GGW Institutions planning and coordination capacities strengthened	3.2.1. No. of GGW planning and coordination meetings undertaken as part of programme meetings undertaken.	Annual Performance Reports Minutes/records of GGW coalitions meetings and action-plans	0	All programme countries have GGW planning and coordination strategies, documents and frameworks drafted	All programme countries have regular GGW coordination and planning meetings, as per their specific country strategies.	There is one coalition per country, and one at the regional level around the PA-GGW It is expected that by mid-term all activities would have been delivered and thus changes in organizational outcomes should be measurable by then.

Output 3.3: National and regional climate change institutional capacities/frameworks are strengthened to integrate land restoration investments in climate change adaptation and mitigation programmes.	3.3.1 No. of knowledge exchange, and peer-to-peer events undertaken, informed by relevant knowledge products on carbon markets for climate change adaptation and mitigation funding	Annual Performance Reports Biennial Outcome Survey Source: Document review of Reports on climate finance proposals submitted and new investment/funding acquired Existence of increase and effective knowledge and tools on carbon sequestration to access carbon markets and finance Reports on Ecosystem services acquired or enhanced; baseline; mid-term and final impact evaluation	1	14	34	This assumes that all programme activities relating to trainings and events to enhance communications strategies and tools have been delivered and adopted by all National Agencies and the Pan-African Agency of the Great Green Wall. The targets are calculated assuming that 4 organizations will be targeted in each country (covering Min. of Environment, Agriculture, any specialized climate change unit, relevant staff from NAGGW). At regional level targeting the Pan-African Agency. Assuming 40% completion at mid-term.
Output 3.4: GGW knowledge management and communication capacities strengthened for mobilizing increased support to climate-resilient land restoration	3.4.1 No. of regional and national knowledge products and communications highlighting SURAGGWA results and lessons-learned (including GGW Umbrella Programme events, press, radio shows, workshops etc…) 3.4.1.a. Share of women involved in GGW knowledge management and communications development	Mid-Term and Final Evaluation Biennial Outcome Assessment Source: document review on Adoption of and reporting accurate data/information on large scale restoration best practices applied	0	88 20% women involvement	176 20% women involvement	This includes participation of SURAGGWA in events of other partner agencies and stakeholders, as well as any workshops and communications campaigns relating to results, achievements and visibility of land restoration and monitoring and reporting activities under SURAGGWA by NAGGW and other agencies/partner. This does not include any internal SURAGGWA event that is meant for planning or other purposes. This assumes participation of SURAGGWA in iGREENFIN, and other events; validation and dissemination workshops and

						other events, and including consultation to be organized under African Union Drylands Week.

Project/programme co-benefit indicators						
Co-benefit	Indicator	MoV	Baseline	Mid-Term Target	Final target	Assumptions
Co-benefit 1: Greater inter-community collaboration	The number of diverse communities able to resolve conflict through collaborative measures/mechanisms	Annual progress reporting (including through ESG system) Biennial Outcome Assessment (geo-referenced)	Sporadic inter-community conflicts reported on a regular basis.	Increase in inter-community collaboration mechanisms	Increase in peaceful resolution of conflicts through strengthened inter-community collaboration	A: Political stability is maintained and no major humanitarian crises will take place during programme implementation A: There is relative calm and stability in the region and the overall insecurity and tensions in the region do not spill over to the target sites. The agro-ecological approaches and choice of species under Component 1 activities takes into consideration the needs of transhumant pastoral communities, thus increasing dialogue among different communities interested in common lands and activities to resolve conflicts that might arise.
Co-benefit 2: Increased income diversification towards climate-resilient sources	Share of income from climate-resilient sources [139]	Biennial outcome assessment; baseline; mid-term and final impact evaluation	Baseline to be determined at inception.	5% of beneficiary households report an increase in income from climate-resilient sources	10% of beneficiary households report an increase income from climate-resilient sources	A: Land restoration and NTFP value chain activities are economically attractive to local

[139] Studies show that the AAD approach has a statistically significant impact on the mix of income sources of households engaging in participatory land restoration activities, reducing reliance on the commercialization of agricultural crop products, which are vulnerable to climate variability and shocks, towards livestock-related production (through increased fodder production), and demonstrating an increase in income derived from the sales of NTFPs. Please refer to 10.1016/j.ecolecon.2024.108311 and https://doi.org/10.1016/j.jrurstud.2021.09.021

						community members.
Co-benefit 3: Improved food security	Change in the food security experience scale (FIES methodology[140141])	Biennial outcome assessment; baseline; mid-term and final impact evaluation (FIES indicator module and methodology based on sampling strategy) IPC/CILSS Cadre Harmonise data	Baseline to be determined at inception.	5% of beneficiary households report a decrease in moderate food insecurity.	10% of beneficiary households report a decrease in moderate food insecurity.	Programme interventions will lead to increased income generation through value-chain activities, as well as enhancing productive capacities through harvesting of seeds. Improvement in land productivity is also expected in the long run.

E.6. Project/programme activities and deliverables

Activities (NB. All activities will be implemented across all 8 programme countries, unless otherwise stated)	Description	Sub-activities	Deliverables
Component 1: ***Large-scale landscape restoration to increase climate change resilience and mitigation, for and by local communities.***			
Output 1.1: Local community groups organized, trained and knowledgeable in land restoration activities and monitoring			
Activity 1.1.1: Identify potential sites for restoration and to inform resource base for value chain priorities based on analysis of climate data, GIS maps and potential for impact with local authorities and community	The purpose of this activity is to assist communities and local authorities to identify with the necessary evidence base specific restoration areas, and their potential linkages with NTFP value chain development.	1.1.1.1: Undertake climate data analysis and GIS Mapping of GGW restorable areas, communities in each country for the restoration and priority value chains.	1.1.1.1. One Report that identifies specific restoration sites, including demarcated attested/legalized plots of land to be restored for all areas, jointly prepared by FAO at regional- and country-level and local authorities per country.
Activity 1.1.2: Identify and mobilize communities for participatory selection of the	Organize households into community groups (CG) and participation in the	1.1.2.1: Organize community-level consultations 1.1.2.2: Mobilization of stakeholders for the	1.1.2.1. At least two Community level consultations on restoration practices, maintenance and engagement per year per restoration site per country undertaken

[140] https://www.fao.org/in-action/voices-of-the-hungry/fies/en/: While the Integrated Food Security Phase Classification (IPC) data is used as rationale for site selection, it would be a complex approach to track a smaller set of population. In this case, for simplicity of tracking the co-benefit indicator, we opted for the straightforward Food Insecurity Experience Scale (FIES). FIES's eight questions will be integrated in the data collection app to track the severity of food insecurity over the course of the rpgoramme.

[141] Evidence from previous AAD actions document that the percentage of households that declared to be worried for food availability dropped by 7%, 12% and 13 in Niger, Senegal, and Nigeria, respectively. the share of households that declared to eat less than they should also decreased in all the three countries covered by this study, of − 8% in Niger, − 31% in Senegal and − 9% in Nigeria; and the percentage of households that did not eat for a whole day dropped from 46% to 15% in Senegal and from 69% to 58% in Niger. https://doi.org/10.1016/j.jrurstud.2021.09.021

specific sites, ensuring inclusion of women.	restoration activities, with the facilitation and TA of selected NGOs and SPs, and FAO technical backstopping. The inclusion of women will be assured depending upon the country context, including but not limited to the formation of separate women's groups (CWGs) where deemed relevant.	Identification of the potential sites and stocktaking of legal and tenure demarcations and certifications. 1.1.2.3. Establish in a participatory manner community restoration committees, and CWGs for highly degraded land restoration activities 1.1.2.4. Establish producer organizations for moderately degraded land restoration activities. 1.1.2.5. Regional WCGs workshops and national consultations to discuss gender issues in restoration activities and support development of gender and safeguards related documents	1.1.2.2 Produce mapping of area to be restored, demarcated and certified by communities and households and/or legalized at municipality/decentralised administration level per restoration site. 1.1.2.3. 444 community groups and organizations have signed charter/mandate and membership lists for highly degraded communal land restoration activities. 1.1.2.4. 14 256 producer organizations identified or established around moderately degraded farmlands. 1.1.2.5. One regional workshop report drafted and publicly available, and 8 country level GAPs/Safeguards documents drafted, publicly available, and implemented
Activity 1.1.3: Train programme's restoration teams and community management organizations	Restoration management committees formed under Activity 1.1.1 will engage in capacity building through a cascading of training of trainers at national level, down to the training at community level. This activity aims to strengthen community members' capacities in land restoration techniques	1.1.3.1: Selection and training of trainers on FFS and soil conservation, protection and land restoration techniques (NGOs, technicians and other SPs) of community level technicians Activity 1.1.3.2 – Community-level trainings through FFS model and in soil conservation and protection and land restoration techniques	1.1.3.1. 24 trainers of trainers sessions undertaken on FFS, soil conservation, protection and land restoration techniques (3 per country with 30 participants each) 1.1.3.2. 144 community management organization trainings undertaken through FFS approach on sol conservation, protection and land restoration (assuming an average of 18 trainings per country; 30 participants over 5 days)
Activities	**Description**	**Sub-activities**	**Deliverables**
Output 1.2. Native seed supply systems strengthened to ensure the availability of genetically appropriate seeds that provide increased climate resilience.			
Activity 1.2.1: Identify and train community technicians on good quality and quantities of restoration seeds and native seed supply	Organization and capacity building of community members involved in seed collection and storage . The trainings will be cascaded down from community technicians to community members.	1.2.1.1: Training of community technicians by research institutions 1.2.1.2. Technical assistance for formation of community-level seed supply groups	1.2.1.1. 96 training sessions held on seed and seedling production and propagation 1.2.1.2. 706 seed supply groups formed and receiving technical assistance (link to 1.2.3.1.)
Activity 1.2.2: Identify, organize and train community members involved in seed and seedling	Organization of community groups and identification of producer groups for seed and	1.2.2.1: Training of restoration seed producer groups in seed and seedling production;	1.2.2.1. 735 restoration seed producers (entrpreneurs) trained on seed and seedling production

production for land restoration activities.	seedling production and propagation techniques ;	1.2.2.2: Training and equipping (seed collection and storage facility/kits etc) communities on seeds collection/handling/storage and treatment for restoration seeds and plantings	1.2.2.2. 2,221 people in communities trained and equipped on seed collection and handling
Activity 1.2.3: Establish, equip and operate community nurseries for seedling production and dissemination	Where suitable (e.g. availability of water sources), establish and equip community nurseries for seedling production; organise local communities seed supply through provision of seed collection and storage equipment, as well as associated and create a network for seed exchange	1.2.3.1: Establish, equip community groups to operate nurseries and vegetable gardens, and collect/store restoration seeds 1.2.3.2. Train community groups and village technicians to sustainably manage nurseries and vegetable gardens and properly collect and store restoration seeds. Activity 1.2.3 3: Dissemination of restoration seeds through established nurseries to programme restoration sites.	1.2.3.1 706 nurseries established and equipped to operate nurseries, vegetable gardens and collect/store restoration seeds 1.2.3.2. 2 rounds of training per nursery groups established on sustainable management of nurseries and vegetable gardens (Year 1 and Year 3) 1.2.3.2. 1.270 million ha of moderately and highly degraded lands supplied with restoration seeds
Activity 1.2.4: Local communities' seed supply of native seeds is integrated into national seed system.	Connecting local communities to the national seed centres, who have the capacities of identifying, preserving and further disseminating good quality seeds to strengthen the national restoration seed supply. The associated equipment and training for this activity are under activity 1.2.3. Seed/seedling networks formed to support and fill in shortfalls locally and nationally	1.2.4.1: Identification of restoration seeds for national supply 1.2.4.2. Aggregation and dissemination of restoration seeds at/from national seed centres	1.2.4.1. Catalogue of restoration seeds available (one per country) 1.2.4.2. Availability of identified restoration seeds at national seed centres
Output 1.3: Targeted highly degraded lands restored through arrangements for rainwater harvesting and enhanced soil permeability and through local communities engaging in planting of seedlings and direct seeding.			
Activities	**Description**	**Sub-activities**	Deliverables
Activity 1.3.1: Highly degraded land restoration plans prepared and implemented through mechanized, animal, and manual traction techniques.	Ploughing and scarification of identified degraded lands and sites in collaboration with local technicians and communities	1.3.1.1. Site-specific restoration plans drafted in collaboration with community restoration committees. 1.3.1.2: Land preparation in collaboration with local communities (Deep ploughing)	1.3.1.1. Costed land restoration plans made available per restoration site. 1.3.1.2. 133 000 ha of highly degraded lands prepared for restoration activities.

	Prepared lands at larger scale in the dry season for rainwater harvesting and ready for seeding and planting. Private sector service providers will be identified, with an existing national presence and equipment to support communities in mechanized/deep ploughing techniques.		
Activity 1.3.2: Sowing and planting in prepared sites with communities/villages	Direct sowing of seeds from prioritised native grass fodder and woody species; and planting of seedlings coming from village nurseries Restoration operations with seeded/planted lands that are previously ploughed.	1.3.2.1: Sowing and planting in prepared sites with communities/villages;	1.3.2.1. 133 000 ha of highly degraded lands seeded.
Activity 1.3.3: Training of community members in monitoring & maintenance	Practical training of communities on monitoring vegetation coverage and maintenance of plots Trained community members in tending restored sites and protection /surveillance / management/ social fencing of restored plots.	1.3.3.1: Communities trained on restored lands maintenance 1.3.3.2: Maintenance of planted/seeded sites (2 years maintenances)	1.3.3.1 trainings per year per community group on restored land maintenance 1.3.3.2 Annual land restoration maintenance report per community group per country.
Output 1.4: Targeted moderately degraded lands planted with a range of species to restore and enrich the landscapes of agro-forestry, agro-ecology and sylvo-pastoral systems, biomes and protected forest ecosystems.			
Activity 1.4.1 Sub-national moderately degraded land restoration plans prepared and implemented	Supporting households with key information in the preparation of costed plans; Guiding and supporting the ploughing of identified degraded	1.4.1.1: Households mobilisation for costed plans for land restoration 1.4.1.2: Land preparation in collaboration with local communities and households by mechanized animal and manual traction	1.4.1.1. Regional-level costed moderately degraded land restoration plans made available. 1.4.1.2. 1.140 500 million ha moderately degraded lands prepared and restored.

	lands and sites with households and communities	techniques (mainly households)	
Activities	**Description**	**Sub-activities**	**Deliverables**
Output 1.5: Rural community capacities strengthened for sustainable management and restoration, and verification of restoration results			
Activities	**Description**	**Sub-activities**	**Deliverables**
Activity 1.5.1: Village technicians trained and equipped in managing restoration areas and verifying restoration results, in a participatory manner.	Selection of village technicians responsible for training communities on restoration /management activities. These village technicians will in turn train the community groups with new technical, managerial and administrative knowledge (disaggregated by % women participation, % youth participation)	1.5.1.1: Training of village technicians on data collection tools, technologies and participatory approaches. 1.5.1.2. Village technicians equipped with relevant equipment and tools to facilitate restoration management and verification of results.	1.5.1.1. 258 village technicians trained on data collection across 8 countries 1.5.1.2. Village technicians utilizing procured equipment and data collection methodologies for restoration maintenance.

Component 2: Development of climate-resilient, low emission non-timber forest product (NTFP) value chains benefiting vulnerable communities' livelihoods

Output 2.1. Climate resilient and low carbon production and processing practices in selected NTFP value chains adopted by Producer Organizations and MSMEs.			
Activities	**Description**	**Sub-activities**	**Deliverables**
Activity 2.1.1: Train and provide TA to POs and MSMEs in selected NTFP value chains for enhanced organizational and managerial capacities (registration, structuring according to OHADA, training on management, administration etc.)	The programme will support and strengthen the value chain actors to become more professional in their structure and operations. To assist the value chain actors to improve their coordination and resolve common issues for enhanced efficiency and profitability. It will consist of supporting the actors to organize themselves in term of structure, support the registration and formalization, and strengthen their capacities in terms of management including accounting, administration, leadership, resources management, etc. This will be done in partnership with local NGOs and institutions.	2.1.1.1: Deliver technical assistance to increase administrative and management capacities of existing POs and MSMEs. 2.1.1.2: Deliver technical assistance and facilitation services to establish new POs and MSMEs in NTFP value chain and train them in administrative and management capacities. 2.1.1.3: Deliver technical assistance to develop legal texts, tools and support the registration of POs and MSMEs (existing and those formed through programme activities) 2.1.1.4: Support and facilitate the establishment and operationalization of platforms of actors for collaboration in the NTFP value chain.	2.1.1.1. 1,795 Pos and MSMEs trained and accompanied in business management, marketing food processing and food safety standards. 2.1.1.2. 5260 Pos and MSMEs supported to structure and organize across NTFP value chains 2.1.1.3. 765 Pos and MSMEs supported for registration 2.1.1.4: At least 1 platform operational for at least 2 priority NTFPs per country
Activity 2.1.2: Train and coach local POs and MSMEs in sustainable production and collection practices to enhance NTFP quality and availability	This aims at improving capacities of the actors and the efficiency of the various processes involved in the collection/production of NTFP. This activity will ensure that groups of men and women involved in the primary segment of NTFP value chains (ie. Production and collection of NTFPs) will have the necessary knowledge and skills to ensure that the priority non-timber forest product resources are harvested, collected and managed sustainably and in a way which elevates the quality of the primary product and contribute to reduce losses This will consist of theoretical and practical training and will	2.1.2.1: Develop/update training modules and resources based on the needs of actors 2.1.2.2: Provide technical assistance (training and coaching) for POs and MSMEs	2.1.2.1 Production and collection training modules developed or updated per country 2.1.2.2. 138 training sessions for POs/MSMEs on NTFP production and collection practices (assuming 25-30 persons per training)

	be conducted in collaboration with national specialized institutions and NGOs.		
Activity 2.1.3: Train and coach POs and MSMEs to improve the processing practices (including packaging and labelling) of the selected products	Build the capacities of the actors engaged in the processing, packaging and marketing of the NTFP to improve the efficiency of their operation as well as the quality of the final products and hence their profits and revenue. The content of the training will include the technics of processing, hygiene, food safety, food quality and standards, packaging, storage and conservation, etc. The practical training will be delivered in collaboration with specialized research institutes and/or private actors that are equipped for that	2.1.3.1: Develop/update training modules and resources based on the needs of actors. 2.1.3.2: Provide technical assistance (training and coaching) for POs and MSMEs	2.1.3.1 Processing training modules developed /updated and made available 2.1.3.2 138 training sessions undertaken on processing practices (including packaging and labelling)
Activities	**Description**	**Sub-activities**	**Deliverables**
Output 2.2: Increased access to markets for smallholder NTFP actors (including micro- and small enterprises)			
Activity 2.2.1: Identify NTFPs with market and sustainable production potential that can be integrated in land restoration efforts funded under Component 1	This intervention will contribute to the identification of priority NTFP products that could be marketed with sustainable increase in production to ensure availability of resources for the market as well as ensuring sustainable management of the natural resource base, including restored areas. This will be through the regional, specialized NGO Tree Aid that has a well-known approach to ensure economic viability and environmental sustainability of these activities.	2.2.1.1: Assessment of the potential of restoration areas and the surrounding landscape for the sustainable production of NTFP 2.2.1.2: Integration of NTFP resource management module in Component 1 land restoration efforts	2.2.1.1. NTFP resource assessment detailing development opportunities and challenges for 3-5 products (one report per country) 2.2.1.2. NTFP resource management modules developed and incorporated in land restoration investments (1 per country)
Activity 2.2.2: Support to marketing and branding of the NTFP (promotional activities, nutritional value pamphlets of NTFP, organization of trade fairs, tv show, etc.)	The activity aims at supporting the value chain actors, especially the SME and downstream actors in the promotion and branding of their different products for a better market	2.2.2.1: Support to the preparation of master plan or guide for marketing and promotional activities including digital tools 2.2.2.2: Organization of TV shows and radio diffusions with rural radios	2.2.2.1 One Promotional master plan on NTFP commercialization per country 2.2.2.2. One radio/ TV campaign launched on NTFP

	penetration and improved revenue. Different tools, approaches and materials will be developed to support these promotional activities for the value chain actors. The programme will collaborate with specialized agencies for the implementation of these activities. The coaching part of this intervention consists of providing tailored and close support to the actors in the implementation of their marketing plan with commonly agreed indicators and calendar of activities..	on the nutritional value of the NTFP 2.2.2.3: Support the organization/participation to the trade fairs 2.2.2.4: Coaching and support of the smallholders in the branding of their NTFP products	nutritional value per country 2.2.2.3. 51 trade fairs attended by programme beneficiaries 2.2.2.4 51 branding and marketing coaching sessions undertaken with POs and MSMEs
<u>Activity 2.2.3:</u> Support POs and MSMEs in business plan development and management.	This intervention is designed to enhance the business management skills of the various value chain actors involved with NTFPs. This activity will support the value chain actors in elaborating and implementing their business plans through practical support on technical aspects and access to a range of partners for services. This will also contribute to their professionalization. The programme will work with business development partners.	2.2.3.1: Train POs and MSMEs in the NTFP on business plan development and management skills, including through the use of standardized tools such as RuralInvest. <u>2.2.3.2:</u> Technical assistance to POs and MSMEs in the preparation and implementation of their business plans	2.2.3.1. 90 training sessions for POs and MSMEs in business plan development and management (30 POs/MSMEs per training session). 2.2.3.2. Technical assistance delivered through full-time business coaches, reporting on business plan implementation (one report per country per year of implementation)
<u>Activity 2.2.4:</u> Enhance access to national, regional and international markets for NTFP, including through the setting up of norms and standards to improve the marketing of the NTFP;	This activity will provide evidence-based information on local, regional and international markets for NTFP actors. It will provide specific information on the market demand, the main requirements, the quantities, etc. of the different NTFP that will guide the actors in their business planning development to take advantage of the opportunities.	2.2.4.1: Support to the elaboration/update of norms and standards for the NTFP 2.2.4.2: Information, sensitization and/or training of the NTFP value chains actors on the norms and standards 2.2.4.3: Regional workshop on NTFP standards/norms and commercialization (international trade) International, and market research for NTFP	2.2.4.1. Norms and standards for NTFPs available/updated (one per country) 2.2.4.2. 2 NTFP norms and standards sensitization sessions per country undertaken. 2.2.4.3. One workshop undertaken and report made available. 2.2.4.4. One market assessment per country, including

	The objective of this intervention is to assist the beneficiary countries through their specialized institutions in the elaboration, validation and diffusion of norms and standards that will guide the production and commercialization of NTFPs. All the actors in the NTFP will benefit from this intervention as it will contribute to improve the quality of NTFP products as well as the harmonization of the norms and standards very important for marketing.	2.2.4.4.: Conduct markets investigations/value chains assessment for national, regional and international opportunities for the NTFP; 2.2.4.5: Facilitate markets linkages between producers and buyers through information sessions and B2B roundtables (business facilitation) 2.2.4.6: Country-level seminars targeted to MSMEs with export potential on the markets requirements pertaining the international trade 2.2.4.7.: Regional workshop on lessons learned and experience sharing workshop on NTFP value chain development approaches, for both restoration and economic benefits.	priority NTFP value chains selected. 2.2.4.5. At least two purchase agreements signed per country 2.2.4.6. At least three seminars per country for business linkages 2.2.4.7. One regional workshop undertaken with a focus on NTFP value chain development approaches for restoration and economic benefits, and reports published (publicly available)
Activity 2.2.5: Identify, train and equip POs and MSMEs with emission-lowering technologies and equipment to enhance NTFP processing	The programme will collaborate with research institutions and support them to identify technological solutions for the processing of NTFPs. The process of deployment of processing equipment will be linked to training and capacity building activities. The POs and MSMEs that participate and successfully complete this will be provided with the equipment to ensure sustainability. This intervention will bring private sectot for business and sustainability purposes.	2.2.5.1. Identify emission-lowering technologies (RET) and equipment to enhance NTFP processing. 2.2.5.2. Deploy identified equipment through capacity building agreements with local POs and MSMEs	2.2.5.1. Directory of technologies and equipment made available to the actors. 2.2.5.2. Trained POs and MSMEs have processing equipment
Activities	**Description**	**Sub-activities**	**Deliverables**
Output 2.3 Suitable credit and insurance products designed for smallholder NTFP value chain actors (PO, cooperatives, MSMEs) and increased stakeholder capacity for using these financial products in their NTFP value chain activities			
Activity 2.3.1**:** Train and sensitize value chain actors on financial institutions' offerings, enhance financial literacy and financial record keeping	The activity aims to reduce information barriers to accessing financial products and services, by providing targeted information around relevant products, services and institutions. It also aims to	2.3.1.1: Map the existing, relevant financial products and services available in the GGW regions, by country for the agriculture and forestry sector	2.3.1.1. Eight documents prepared mapping financial services providers and agricultural financial products (1 pereach SURAGGWA country)

	foster an understanding of financial concepts and financial record-keeping, so that smallholders are able to meet service providers' documentation requirements to open bank accounts and access credit.	2.3.1.2: Familiarize value chain actors with the financial options available to them 2.3.1.3: Deliver trainings on financial record requirements of banks for opening a bank account and accessing credit,.	2.3.1.2. Eight short sessions organized on financial options and products within business management module in Activity 2.2 (1 per country) 2.3.1.3. Ten trainings on record-keeping/bank account opening deployed per country
Activity 2.3.2**:** Technical assistance provided to develop five new financial products (of which at least one insurance product) tailored to agriculture and NTFP value chains with financial institutions, including those collaborating with iGREENFIN.	The activity aims to support the enabling environment for access to financial products and services. Through the design of context and region-specific financial products, the activity will make available suitable instruments for channeling short- and long-term finance to support value chain activities. These activities will be undertaken ensuring inclusion of financial institutions already collaborating with iGREENFIN to accelerate access to credit and finance by targeted beneficiaries in relevant value chains.	2.3.2.1: Assess the existing financial products related to NTFPs offered by financial institutions in GGW countries, their coverage 2.3.2.2: Design two new credit products, one aimed at working capital and one at investment in agriculture and NTFPs and share with national and regional financial institutions 2.3.2.3: Test and dissemination of the new credit products, one aimed at working capital and one at investment in agriculture and NTFPs, in collaboration with existing financial institutions 2.3.2.4: Design one agricultural insurance product (indexed) for the drylands context, with customizable features that can be tailored by country, and share with national and regional financial institutions 2.3.2.5: Develop two new savings products, one digital and one conventional for building collateral for farmers and share with national and regional financial institutions	2.3.2.1. Mapping document created of existing products offered by 12 financial institutions in GGW region 2.3.2.2. Two new credit products developed and documentation shared with financial institutions 2.3.2.4. One new agricultural insurance products developed for the drylands context and targeted at smallholders and documentation shared with financial institutions 2.3.2.5. Two new savings products developed, one of which is digital only and documentation shared with financial institutions
Activity 2.3.3**:** Train staff (loan officers and others) from financial institutions, including those collaborating with iGREENFIN, on how to assess agricultural and NTFP value chain risks and climate-related risks (national and regional level)	The activity aims to train and sensitize loan officers in financial institutions (national agricultural banks, commercial banks, microfinance institutions and rural finance actors such as cooperatives) around the profitability of NTFP value chains, showcasing successful	2.3.3.1: Two in-person trainings of loan officers, with at least 80 beneficiaries, 2.3.3.2: Four trainings of loan officers organized in virtual mode	2.3.3.1. Two course curricula and materials developed (one in-person and one e-learning module) 2.3.3..2. Two trainings of loan officers, with at least 80 beneficiaries,

	experiences and prospects as well as on agricultural finance. Specifically, it aims to enhance loan officers grasp and ability to assess risks specific to the agriculture sector in drylands contexts.		organized in-person as a regional activity
Activity 2.3.4: Facilitate linkages between last-mile providers of finance (producer organizations, village savings groups, microfinance networks) and financial institutions, including those participating in iGREENFIN.	In many cases, large financial institutions may not have an extensive physical presence in rural areas. This activity aims to facilitate linkages between financial institutions at all levels, including the providers of credits at the last mile such as village savings groups and producer organizations.	2.3.4.1: Map providers of last-mile finance in rural GGW regions, by country 2.3.4.2: Facilitate connections between mapped providers of finance and national/regional financial institutions, including those participating in iGREENFIN	2.3.4.1. One mapping document prepared on financial service providers in each GGW country 2.3.4.2. Organize 1 session per year for the first two years on facilitating linkages between loan officers and other relevant personnel from different financial institutions at trainings under
Activity 2.3.5. Pilot access to credit through digital financial solutions in at least one country and facilitate scale up, if successful	The activity aims to increase access to finance, particularly in rural and remote areas where long distances to physical bank branches prevent smallholder farmers from opening accounts or accessing credit. The programme will collaborate with UNCDF to rollout of a digital credit solution in one country. It will also provide e-advisory weather services to assist farmers in making evidence-based planting decisions and lower the risk of default. This activity will be co-financed by UNCDF and a national bank.	*2.3.5.1: Test and finalize tailored digital credit solution with financial institution and fintech in one country* *2.3.5.2: Digital credit solution deployed in one country in GGW area* *2.3.5.3: Roll-out digital credit solution and e-weather advisory solution in the country (To farmers, actors)* *2.3.5.4: Organize seminar to discuss performance of the product and potential for scale*	2.3.5.1. One digital credit solution developed and tested with financial institution, in collaboration with UNCDF 2.3.5.2. Digital credit solution and e-advisory service deployed in one country in GGW area 2.3.5.3. One follow-up seminar organized with UNCDF, fintech providers and partner financial institution to present results on take-up, repayment rates and performance of the digital credit product and assess potential for scale-up
Activity 2.3.6. Improve knowledge management and exchanges to increase adoption of best practices across local financial institutions	The programme will facilitate knowledge and peer-to-peer exchanges between financial institutions at the regional level. A potential entry point for these	*2.3.6.1:* Organize regional seminars for financial institutions to share experiences with NTFP value chains in the GGW region	2.3.6.1. Four regional seminars organized for financial institutions

	regional knowledge sharing events and trainings is to utilize the existing pan-African network of the African Rural and Agricultural Credit Association (AFRACA). Their member organizations include agricultural development banks, commercial banks, microfinance institutions and producer organizations. They are present in five of the eight SURAGGWA countries.		

Component 3: Strengthening the Great Green Wall institutions at country and regional level			
Output 3.1: GGW Land restoration monitoring system at national and regional level upgraded and functional			
Activities	**Description**	**Sub-activities**	**Deliverables**
Activity 3.1.1: Development, testing and deployment of National GGW land restoration monitoring tools and system	The implemented activities will provide the GGW-NAs and PA-GGW with an operational system for monitoring land degradation and tracking the restoration efforts undertaken. The system will build on existing systems and skills while allowing for significant upgrades to existing solutions. Activities will combine training in monitoring tools and technologies, organizational strengthening of beneficiary institutions, and targeting activities on the enabling environment that will empower the GGW institutions to carry out their monitoring and reporting mandate. The monitoring systems will include the development of monitoring of vegetation progress at restoration site scale using satellite-based detection of vegetation indices, biomass production, and	3.1.1.1 : Analyze existing mechanisms and tools throughout the PAGGW and each of the countries (overall architecture, existing institutions for data collection and processing) and identify needs. 3.1.1.2: Development and testing of GGW monitoring modules and smartphone app 3.1.1.3: Equip and train for national GGW Monitoring to be deployed in NAGGW (including computers, laptop, printers, tablets, smartphones)	3.1.1.1. Stocktaking report of existing mechanisms and tools at PAGGW and programme supported NAGGW 3.1.1.2. The GGW smartphone App is developed, tested and available to users through Google Play Store repository. 3.1.1.3. GGW's land restoration monitoring tools and modules are tested, deployed and made operational
Activity 3.1.2: Transfer knowledge and skills for the use of the GGW Land restoration monitoring system by national and regional authorities.		3.1.2.1: Compile manuals, document procedures (protocols), develop field guides on using the GGW platform.	3.1.2.1. Suite of methodological guidance drafted and made available to PAGGW and NAGGW, and other relevant local authorities involved in land restoration monitoring.

	tracking of GGW results and evaluation impacts through quasi-experimental satellite evidence and reduction of degraded lands. The monitoring system will allow for the measurement of land degradation around restoration sites (up to 3-4 times the area of any designated restoration site), based upon the successful experiences of the Action Against Desertification project. The monitoring system will be initially piloted through the tracking of SURAGGWA programme results and then transferred to national institutions to allow for tracking of all GGW related projects by the end of the period of implementation.	3.1.2.2: Deployment of tools and sharing of documentation produced under 3.1.1. 3.1.2.3: Conduct Hands-on trainings and piloting of GGW land restoration monitoring tools	3.1.2.2. NAGGW and PAGGW have GGW Land restoration tools and technologies in use. 3.1.2.3. 2 hands-on training and piloting sessions of GGW land restoration monitoring tools of relevant institutions (GGW, research institutes) and on training for regional institutions (including PAGGW, CSE and others)
Activity 3.1.3: Development and deployment a of regional multi-stakeholder Monitoring platform		3.1.3.1: Design, develop, test and deploy the GGW platform and all its modules, ensuring inter-operability between Regional platform and monitoring app. 3.1.3.2: Set up a multi-layer Geographic Information System of the 8 SURAGGWA countries and their GGW areas and populate it with regional layers from various sources 3.1.3.3: Product translated in at least two languages: English and French 3.1.3.4 : Roll out, test and operationalize technology transfer at country and regional levels	3.1.3.1: GGW's land restoration monitoring platform is deployed, tested and made operational for national and regional multi-stakeholder 3.1.3.2: GGW GIS and monitoring operators have regularly twice a year populated restoration data for the 8 SURAGGWA countries and their GGW areas of interventions 3.1.3.3: GGW's land restoration monitoring platform made available and accessible in English and French 3.1.3.4: Platform tested and functional at country and regional levels
Activity 3.1.4: Building the capacity of NAGGW and PAGGW to establish databases of ongoing restoration projects and programmes contributing to national and regional GGW results		3.1.4.1: Stocktaking of ongoing projects and programmes 3.1.4.2. Technical assistance provided for building databases at country level.	3.1.4.1. Stocktaking repository (in excel, or other relevant format) of identified projects and programmes for each country. 3.1.4.2. Database available of ongoing restoration projects compiled and made available
Activity 3.1.5: Prepare regulatory national frameworks for the surveillance of all GGW labelled interventions in all GGW countries		3.1.5.1: Drafting and multi-stakeholder validation of regulatory framework document.	3.1.5.1. Regulatory framework document made available
Activity 3.1.6: Develop operational partnerships between the NAGGWs and PAGGW and scientific and		3.1.6.1: Selected staff from scientific and technical institutions are	3.1.6.1. Trainings undertaken through partnerships (LoA) with OCC, CILSS/AGHRYMET.

technical institutions in the region (such as universities, research institutes, CSEs, Agrhymet, ACMAD) on issues of ecological monitoring and adaptation to climate change		trained in GGW land restoration monitoring system, methodologies and protocols. 3.1.6.2. Promote MoUs and joint working modalities between NAGGW, PAGGW and identified/trained institutions	3.1.6.2. At least one operational partnership per country, and at regional level, operational (eg. MoU signed or staff working jointly on ecological monitoring activities)
Activity 3.1.7.: Strengthen the capacities of national stakeholders in the use of the relevant tools developed by FAO (for example, Collect Earth and Africa OpenDeal database), and other partners such as the Ecological Monitoring Centre (EMC), the Sahara and Sahel Observatory (OSS) and CILSS.		3.1.7.1. Tools and training modules developed and deployed, hand-on training courses 3.1.7.2: One training per year for 15 land restoration monitoring specialists from NAGGW and partner institutions.	3.1.7.1. One training module developed and made available 3.1.7.2. 120 staff trained on relevant tools including Collect Earth, Africa OpenDeal,and tools developed by EMC, OSS and CILSS
Output 3.2: National and Regional GGW institutions planning and coordination capacities strengthened			
Activity 3.2.1: Establish National GGW Coalitions to promote coherent coordination and planning at country-level	The planning exercises initiated by the GGW-NAs are often unreliable in terms of forecasts and poorly followed up by the initiatives implementing land restoration projects. These activities will contribute to the objective of meeting this challenge by implementing a coherent set of capacity building activities. This will result in tangible benefits in the form of well-functioning multi-stakeholder and inclusive consultation, coordination and planning processes frameworks, which will improve the flow of information on the initiatives taken (a basis for better planning) and promote the coordination of interventions. SURAGGWA will simultaneously help pursue the recent initiative of setting up national coalitions in each of the GGWI countries. These coalitions will serve individually in each country as a foundation	3.2.1.1. Undertake a critical analysis of the existing consultation and coordination frameworks, from the local to the national level, in areas related to the GGW and recommend solutions for improving coordination mechanisms for GGW actors (baseline) 3.2.1.2. Design and support establishment of country-specific GGW National Coalition, including 5-year strategy. 3.2.1.3. Technical assistance to the start-up of National Coalition functioning	3.2.1.1. Coordination and planning capacity needs assessment undertaken and made available. 3.2.2.2. One multistakeholder workshop per country undertaken to take note of Capacity Needs Assessment Results and agree on appropriate action for GGW National Coalition establishment, including the development of 5-year strategy and action plan for each National Coalition. 3.2.2.3 GGW National Coalitions are operational and reporting annually on their progress
Activity 3.2.2: Prepare and issue planning and regulatory frameworks for the coordination		3.2.2.1: stakeholder workshop to discuss and	3.2.3.1. One stakeholder workshop at regional level, including PAGGW and

of all GGW aligned interventions (projects, programmes, activities etc.)	upon which planning coordination activities can be facilitated and conducted. These activities will allow the GGWI NAs to conduct technical exchanges aimed at coordinating the coverage and geographic targeting strategy of GGWI interventions and define together more precisely the basic principles of intervention to which they commit themselves by being part of or joining the initiative.	prepare the framework 3.2.2.2. Technical assistance provided to draft regulatory framework at regional and national levels.	targeted NAGGWs held and report made available. 3.2.3.2. Planning and/or regulatory frameworks / mechanisms are developed and issued in all 8 countries
Output 3.3: National and regional climate change institutional capacities/frameworks strengthened to integrate land restoration investments in climate change adaptation and mitigation programmes			
Activities	**Description**	**Sub-activities**	**Deliverables**
Activity 3.3.1: Strengthen public- and private sector understanding of and capacity to engage with carbon markets for climate change adaptation and mitigation through land restoration	The activity will undertake a landscape analysis of existing engagement with carbon markets and identify gaps in regulatory frameworks and capacities to engage with it in the region. This will inform the identification of relevant finance options, capacity needs and required regulatory and institutional changes, and the platforms will provide an avenue to pitch, attract and connect with public and private investors to continue attracting investments to GGW interventions.	3.3.1.1: Assess Region-(Sahel) and Context- (dryland landscape) specific carbon financing options 3.3.1.2: Provide input to private and public investment collaboration platforms (link to 3.3.1.1.)	3.3.1.1. One assessment report on regional options for carbon finance for climate change mitigation and adaptation financing through land restoration. 3.3.1.2. Technical inputs provided for existing platforms to facilitate carbon market engagement by public and/or private actors.
Activity 3.3.2: Pilot the establishment of domestic carbon accounting framework that integrates agriculture and forestry in Nigeria, and identify additional countries for potential replication.	The activity will enhance countries' capacity to engage with carbon finance opportunities for GGW restoration. This will be achieved through two sub-activities. The programme will provide this support in Nigeria as the pioneering country in the region on domestic compliance markets. This activity would be developed and documented to provide a template for similar frameworks in other countries. The Nigerian	3.3.2.1 Integrate agriculture and forestry sectors in the planned national carbon accounting system as well as in domestic carbon markets or carbon tax regulations (and offsets) in Nigeria. 3.3.2.2. Technical Assistance provided to Nigeria's National Council on Climate Change (NCCC), and other relevant	3.3.2.1. Assessment report of potential of agriculture and forestry sector in Nigeria with respect to carbon sequestration and identification of potential issues. 3.3.2.2. Technical assistance provided to staff in relevant national and regional institutions (eg. NCCC, WAACMCF) on GHG estimations, and other relevant methodologies/tools, in AFOLU sectors and on domestic and regional carbon markets .

	carbon accounting framework (including CO2 markets) is currently being formulated by its National Council on Climate Change. In order to advance on the identification of pilot countries, the West African Alliance on Carbon Markets and Climate Finance will also be included in relevant activities.	institutions, on addressing specific issues in national carbon accounting systems and domestic carbon markets, e.g., carbon tax and offsets or other options. 3.3.2.3: Support other programme countries in exploring the potential of similar carbon accounting frameworks, as feasible.	3.3.2.3. Feasibility assessment of other programme countries undertaken to assess readiness (e.g. Senegal) to learn from the Nigerian experience. and disseminate results.
Output 3.4. GGW knowledge management and communication capacities strengthened for mobilizing increased support to climate-resilient land restoration			
Activity 3.4.1: Develop innovative and robust methods for evaluating/demonstrating resilience benefits of climate change investments in land restoration and NTFP value chains in GGW	Consolidation of lessons-learned, proof of concept of land restoration models and approaches under SURAGGWA and other relevant GGW projects, to inform climate change adaptation fundraising activities. The data and information collected through programme activities will serve as the basis for these studies and will include the development of specific methodologies through technical assistance, working in conjunction with the programme team and national and regional GGW authorities. This activity aims to ensure that robust knowledge products, including analytical and impact studies as well as lessons-learned in terms of approaches and results is documented and disseminated, including through the participation to high visibility events including the African Union's biennial Drylands Week showcase.	3.4.1.1. Methods developed, tested and evaluated, including biophysical and socio-economic resilience outcomes. 3.4.1.2. Technical assistance provided for application of innovative and robust assessment methods to relevant institutions. 3.4.1.3.: Regional workshop on lessons learned and experience sharing workshop on sustainable land management and adaptation to climate change approaches, for both restoration and economic benefits 3.4.1.4.Technical assistance for report on Conflict-sensitive land restoration activities 3.4.1.5. Technical assistance for report on Youth and women inclusion in restoration and NTFP value chain	3.4.1.1. One suite of methods and tools for data analysis and evaluation developed, drafted and disseminated. 3.4.1.2. At least one Peer-reviewed journal article published on biophysical and socio-economic resilience impacts and sustainability. 3.4.1.3. One regional workshop report made available (linked to sub-activity 3.4.1.1.) 3.4.1.4. One report promoting integration of lessons-learned from conflict-sensitive land restoration in AFOLU investments drafted and made available 3.4.1.5. One report promoting youth and women inclusion in land restoration in AFOLU investments drafted and made available. 3.4.1.6. One report promoting the adoption of VGGTs and land tenure in AFOLU investments.

	It is anticipated that the capitalization work could focus on themes such as: conflict sensitive approaches, youth employment, improvement of women's income, use of VGGTs, etc.	development activities. 3.4.1.6. Technical assistance for report on Land tenure and VGGTs and restoration activities.	
Activity 3.4.2. Communication, visibility and dissemination of knowledge.	Regarding Knowledge Management, SURAGGWA will complement the efforts deployed by IGREENFIN by focusing on lessons learning on the specific experience of the SURAGGWA programme. The results of the capitalization work will be shared through the regional KM support platform. The KM actions carried out at the SURAGGWA level will allow the relevant actors of the SURAGGWA programme to participate in the exchanges and events animated by the KM platform (contribution to 'community of practice', etc.)	3.4.2.1: Participation in the different events organized by IGREENFIN, African Union (Drylands Week) and other relevant fora	3.4.2.1. Communication materials, presentations highlighting programme lessons learned generated from Activity 3.4.1. made available and visible.
Activity 3.4.3 Train communication specialists from GGW's national structures to implement their communication plan and develop tools adapted to different target groups, particularly women and youth.	In terms of communication capacity building, the activities carried out by SURAGGWA will aim at equipping the communication units of the GMV-NAs, in connection with the partner administrations that act on the ground close to the field. The programme will finance initial investments in the preparation of messages and communication materials and will help establish sustainable working relationships between the GGW NAs and field communication actors (rural radio, social networks).	3.4.3.1: Participation to Francophone and anglophone training	3.4.3.1. 120 communications training sessions undertaken to develop and implement communications strategies/plans
Activity 3.4.4 Develop, in relation to the different target groups identified, adapted supports and modes of information, such as: the use of social networks (Facebook, Twitter, etc.) and teleconferences, the organization of meetings; workshops, the production of films, leaflets, brochures, etc.).		3.4.4.1: Communication tools design workshop undertaken 3.4.4.2. Preparation and deployment of communication materials to targeted populations.	3.4.4.1. One communication tools design workshop per country 3.4.4.2. Communication campaigns launched and tracked in each country.

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

238. **Reporting from FAO to GCF**: Programme-level monitoring and evaluation will be undertaken in compliance with FAO as well as GCF policies. FAO will ensure the existence of a well-designed, operational and effective impact monitoring and reporting system to analyse and quantify the causal and attributable change, the contribution and the overall causal results of the programme. In particular, the monitoring and reporting system will allow for understanding the efficacy, targeting and verifying the assumptions that the programme is making, as well as implementing a knowledge management and learning plan, in close coordination with iGREENFIN, so that elements emerging from the monitoring systems can feedback into the programme implementation and the planning of Outcomes on an annual basis. The M&E Specialist in the Programme Management Unit (PMU) will be responsible for transmitting annual progress reports to GCF Secretariat, as well as results of biennial outcome surveys, and mid-term and final evaluations.

239. **SURAGGWA monitoring and reporting**: In addition to support to GCF projects on knowledge management and innovation, the GCF invited the Food and Agriculture Organization to develop a monitoring and evaluation framework to facilitate consolidated reporting of GCF projects and programmes across the GGW. The monitoring and evaluation framework will provide a multi-scale Monitoring and Evaluation of ecosystem services, low-emissions, climate-resilient pathways. The framework will focus on (i) SURAGGWA Monitoring and Reporting information, applying monitoring and reporting approach drawing on the GEF-7 Sustainable Forest Management Program on Dryland Sustainable Landscapes (DSL), and which will be integrated into the Framework for Ecosystem Restoration Monitoring (FERM) (ii) GGW land restoration monitoring systems development, including baseline information prepared using innovative geospatial tools and technologies, including soil carbon analysis and the use of analytical tools such as Ex-ACT, Africa Open DEAL, Collect Earth, Earthmap, and SEPAL. SURAGGWA will apply the standard procedures established by FAO for monitoring and evaluating projects and programmes. The Programme Management Unit (PMU), which will include an International M&E, gender and safeguards Specialist who will compile and maintain all information, indicators and parameters necessary for the preparation of proramme reports, including quarterly activity monitoring, annual progress reports, which will include progress on carbon sequestration indicators, and biennial outcome surveys (see below for further details), as well as midterm and final evaluations to be submitted to the GCF. Furthermore, the International M&E Specialist will provide capacity building to all country-level M&E specialists and contribute to the development of the GGW Monitoring and Reporting information system and indicator framework, as well as ensuring these are transmitted and discussed by the Regional Steering Committee, that will include representatives from all 8 participating countries, the Pan-African Agency of the GGW, as well as the African Union. The M&E specialists in the Country Implementation Units (CIU) will be responsible for coordinating quarterly activity monitoring and communicating the progress on programme output and outcome-level indicators to the PMU on a regular basis. The monitoring process intends to follow up on the execution of the programme to identify the intermediate milestones achieved in each phase and evaluate its outcomes and fulfilment of proposed targets. Furthermore, Steering Committees will be established at country-level, including the NDA, GGW focal points as well as other relevant Ministries and stakeholders with a view to overseeing annual work planning and budgeting, as well as review of progress against the results indicators on an annual basis. The indicators to be monitored are included in the log frame in section E.

240. **Safeguards and Gender Action Plan reporting**: The programme places particular importance on ensuring the monitoring of differential impacts by sex, age and vulnerability, especially monitoring components relevant to women and more vulnerable populations (including Internally Displaced Populations and Pastoralists). Furthermore, the monitoring system will ensure monitoring of the action plans for gender, internally displaced people and the social and environmental framework to safeguard. This will ensure a comprehensive and holistic monitoring system, besides the quantitative impact monitoring.

241. **SURAGGWA Biennial Outcome Reporting:** Independent consultants or research institutions will be contracted by the Regional PCU to undertake an outcome survey every two years in order to track progress and identify attribution in relation to activities, outputs and outcomes as per the SURAGGWA IRMF. In particular, the biennial outcome assessment will focus on identifying progress in relation to outcome-level and co-benefit indicators, identifying as much as possible contribution of project activities and outputs to outcome indicators and results as per IRMF. The Biennial Outcome Assessment will build upon the data and information in the SURAGGWA MEL System and will be geo-referenced. The Biennial Outcome Assessments will identify critical outcome indicators for closer monitoring and will draw upon the Baseline survey and report and will also inform the interim review and final impact assessments and evaluation. It will include a capacity development module[142] to track adoption and uptake of trainings, practices and knowledge across all components of the project (and including the scorecard methodology for the enabling environment, organisations and at individual level), as well as a qualitative NDVI assessment and beneficiary feedback survey.

[142] KAP Surveys and tools to be utilized

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Please describe financial, technical, operational, macroeconomic/political, money laundering/terrorist financing (ML/TF), sanctions, prohibited practices, and other risks that might prevent the project/programme objectives from being achieved.

Selected Risk Factor 1: Displacement & Insecurity, political instability

Category	Probability	Impact
Other	Medium	High

Description

242. The rise of extremist movements within the region has been a significant cause of violence in the Sahel. Religious extremist groups and militias have gained influence in the Sahel region causing the region to become an epicenter for conflict. The armed conflict in Africa's Central Sahel region has forced more than 2.5 million people to flee their homes in the last decade. Growing terrorist threats and attacks are central to the displacement. The primary driver of this displacement is violence, occurring mainly in Nigeria, Mali and Burkina Faso. Alongside terror threats, deepening environmental challenges are exacerbating the competition for resources and creating further tensions between groups. Violent conflict in the Sahel, compounded by the impacts of climate change, has resulted in widespread displacement and food insecurity. The fragmented security landscape, comprised of Sahelian state forces, international contingencies, and violent extremist organizations has also hindered humanitarian response. This situation has also contributed to political instability in some SURAGGWA countries.

Mitigation

243. In selecting its areas of operation, SURAGGWA will avoid regions where there is violent conflict and displacement due to safety reasons and intervene in areas where the security situation allows the programme to be implemented on the ground. In countries where there has been recent political upheaval, FAO strictly follows high-level UN guidance, including: (i) respecting UN-imposed sanctions; (ii) following UN instructions re interruption of the implementation of "non-critical" activities.[143]

Selected Risk Factor 2: Conflict over meagre resources

Category	Probability	Impact
Technical and Operational	Medium	Medium

Description

244. Conflict in the Sahel is primarily rooted in political and governance-related weaknesses. However, environmental fragility and climate change can exacerbate the conflict. There is overreliance on climate-sensitive, agriculture-based livelihoods which poses additional risks with growth in population as well. The growth in population numbers also contributes to competition over scarce water and land resources and compounds the problems of resource degradation. Combined with climate uncertainty, this results in low yields and high crop losses resulting in increase in food prices and declining food availability and food security. This leads to further competition over the dwindling natural resource base and causes conflicts among the farming and herding communities who rely on livestock. These conflicts can become severe during periods of droughts as communities try and find forage

[143] Currently, none of the 8 SURAGGWA countries are subject to any UN sanctions. The only country with a "UN criticality programme" is Niger, but nearly all emergency, resilience and rural development activities are considered high-priority by the UN, and activities that would be temporarily suspended account for less than 10% of the country's budget for the SURAGGWA programme, i.e. less than 1% of the overall programme budget.

for their animals and often stray into farm lands. It is difficult to work with communities with internal conflicts and disputes and get them on the same platform.[144]
Mitigation
245. The SURAGGWA programme is designed to clarify the arrangements for using the common lands through consultations with local communities. This clarification itself is expected to reduce tensions in the region among the local communities and to contribute to much-needed peace-building efforts. SURAGGWA will initiate its activities by outlining to the communities the benefits of collaboration. In addition, the programme is designed to undertake investments that will help to reduce tensions among communities by providing opportunities for enhancing productivity of the natural resource base. One of the first activities that will be undertaken as part of land rehabilitation under SURAGGWA will be the plantation of forage crops which is extremely important for communities reliant on livestock. The availability of increased forage resources is expected to reduce the tensions among pastoral communities and the conflicts resulting from them, as well as enhance inter-community collaboration. SURAGGWA is expected to improve food security by focusing on rehabilitating degraded lands and also providing local communities alternate sources of income and livelihood through investments in non-timber forest products. In addition, SURAGWWA has been designed with an understanding of customary laws, traditional use rights, coping strategies of the most vulnerable and how these rights can be impacted by changes in the rehabilitation of the communal lands. It is expected that the financial incentives and sensitivity too customary norms and practices will help to reduce any tensions in the area by providing local communities additional and more secure sources of livelihoods.

Selected Risk Factor 3: Programme Governance and Management Arrangements		
Category	Probability	Impact
Governance	Medium	Medium
Description		
246. Experience has shown that in a programme which is being implemented in a wide range of countries, coordination can pose a challenge between the Great Green Wall (GGW) institutions, the programme's Country Implementation Units (CIU) and the range of implementing partners in each of the 8 countries. Often delays can also be caused due to inexperience in proper procurement procedures which can in turn have a significant impact on the implementation schedule and impact the timing of the various activities some of which are very time sensitive as missing one planting season can delay the entire cycle of programme activities by a year or more.		
Mitigation Measure(s)		
247. The programme will adopt a phased approach in which there will be a gradual initiation of activities in the various countries. Those countries which demonstrate a higher degree of readiness will be provided the opportunity of an early start. Those countries which require a longer timeline because of their weak institutional capacity will be provided a longer gestation period with upfront support in putting in place a properly staffed Country Implementation Unit (CIU) with proper training in the contracting and procurement arrangements. FAO will create strong CIUs well versed in procurement methods and provide proper technical assistance to begin the procurement as soon as possible with sufficient time to allow for all steps in the process to be completed. FAO's presence and engagement at the country level will provide oversight, support and technical backstopping. FAO will also establish will establish a Programme Management Unit in its Sub-Regional Office in Senegal to support overall implementation and efficient performance. These measures are expected to minimize the associated risks.		

Selected Risk Factor 4: Sustainability		
Category	Probability	Impact

[144] Inter-community conflicts are one of the main risks to the success of the programme, so conflict assessment, prevention and management are a key element of programme design and implementation mechanisms chosen, especially for Component 1, land restoration (see also Annex 6, Environmental and Social Management Framework, especially its Appendix 9, "Conflict assessment, prevention and management".

Governance	Medium	Medium
Description		
248. The outcomes expected from the Institutional Strengthening component and its sustainability could be affected by the limited capacity of National and Regional GGW institutions to operationalize and sustain the mechanisms and up-graded tools for monitoring and tracking, analysing and sharing lessons and their inability to mobilize resources and upscale the interventions. This could happen due to lack of high level commitment, lack of engagement and appointment of senior and technically qualified experts to operationalize the monitoring system, inadequate budget allocation for the key activities that need to be sustained during and beyond the project life.		
Mitigation		
249. During the programme formulation stage and at various stages during the implementation of the GGW Initiative, national governments have committed to continue investing money in activities that will address D/LDD to develop policies and create the enabling environment to facilitate the setting up of land restoration and reforestation initiatives with elements of income-generating activities. These commitments have ben reiterated and will be formalised during the initiation of SURAGGWA by each participating country at the highest level to deploy the required experts and allocate the funds needed to operate and maintain the arrangements after the programme is completed.		
Selected Risk Factor 5: Elite Capture		
Category	Probability	Impact
Technical and operational	Medium	Medium
Description		
250. Some of the land and sites that the programme will be implemented on is used by farmers, agro-pastoral, and nomadic populations – all of whom engage in traditional land-use arrangements that provide mutual food and livelihood benefits. In these settings, even the most degraded land has value as these are important areas of passage and grazing for livestock, particularly during the rainy season, and are sources of wild plants and wood gathered by women. Sometimes the introduction of management practices to rehabilitate the lands can be a source of exclusion. The use of area enclosures – a land management practice that seeks to restore degraded land by excluding livestock and humans from openly accessing it in the short to medium term – runs the risk of exacerbating vulnerability; and in the absence of good land governance, possibly causing harm. Furthermore, increasing the value of degraded land can appreciably increase its value and lead to capture by elites and displace the local population from the rights they have traditionally enjoyed. The introduction of inter-cropping can sometimes lead to individualized claims on community land which can compromise the use rights of others in the community. Women can in particular be impacted in their capacity as one of the primary users of common lands for gathering non-timber forest products, fuelwood, fodder, etc. One illustration of this was in the case of Nigeria, where land was effectively restored, but where parcels were also sold outside of the community, in areas that lacked good land governance arrangements.		
Mitigation Measure(s)		
251. SURAGWWA has been designed with an understanding of customary laws, traditional use rights, coping strategies of the most vulnerable and how these rights can be impacted by changes in the rehabilitation of the communal lands. The programme will ensure that these arrangements do not compromise the interests of the most vulnerable segments of the local communities, especially women. Thus emphasis will be placed on ensuring that clear, enforceable land-use agreements are in place prior to the initiation of restoration activities and to ensure that these are properly communicated, accepted and respected by all concerned. Arrangements will be made in accordance with FAO's social safeguards to put in place an appropriate monitoring and supervision mechanism and grievance redress mechanism.		
Selected Risk Factor 6: Survival Rates of Vegetation		
Category	Probability	Impact
Technical and operational	Medium	Medium

Description		
252. The key goal of increasing tree cover in arid and semiarid regions has shown very varied results in some projects. In regions with medium to low desertification risk, local governments have reported larger areas as having been afforested than were actually confirmed. This triggered an intensive debate that questioned the sustainability of overambitious tree-planting projects in arid and semiarid regions where the survival rates of tree seedlings is often as low as 15%. This is especially true in areas receiving less than 500-mm annual rainfall.[145]		
Mitigation Measure(s)		
253. FAO's experience with land rehabilitation has shown that through a consultative process that engages local communities, technical specialists and local and regional research institutions, a strategy of species selection can be identified that is appropriate and effective with high rates of survival. FAO's experience in the AAD demonstrated that vegetation cover was successfully established, land was rehabilitated, and the density of trees and shrubs increased dramatically at rehabilitation sites. However, projects did not implement measurement mechanisms to establish how much of the increase of vegetation can be attributed to donor programs versus other important variables including changing rainfall. A strong monitoring and tracking mechanism will be established under SURAGGWA and local capacity built in its use and sharing the findings.		
Selected Risk Factor 7: Limited Access of Women		
Category	Probability	Impact
Prohibited practices	Medium	Medium
Description		
254. Typically, women's access to the common lands and farm lands is restricted due to social and patriarchal norms. There is a danger that the rights of women can be further compromised when degraded lands are rehabilitated due to the intense competition for programme benefits. Because land restoration mainly benefits those individuals who have access to land, women can be especially disadvantaged in the Sahel and are forced to fend for themselves and their families because either their husbands and sons have migrated to other West African countries to look for work or they belong to households in which they are the head of the household.		
Mitigation Measure(s)		
255. SURAGGWA has been designed to ensure that the rights of women are not compromised and specific targets are identified for them in the selection of farm lands belonging to women or women headed households and ensuring their participation in the access to and use of communal lands that are rehabilitated. Women will be included in the seed and seedling production groups as a source of income and will be provided access to forage in the common property resources. In addition, women are expected to be the main beneficiaries in the development of the incomes and livelihoods from the non-forest timber forest products.		
Selected Risk Factor 8: Limited Participation of Financial Institutions		
Category	Probability	Impact
Technical and Operational	Medium	Medium
Description		
256. The main risks for the access to finance investments and its outcomes stem from two factors. First, the limited participation of financial institutions could hinder programme implementation given their role as key stakeholders. Second, insufficient liquidity in countries could hinder the financial institution's abilities to lend, particularly if IGREENFIN does not start its implementation.		
Mitigation		
257. SURAGGWA will mitigate these risks through several channels. First, the programme's access to finance activities are designed in consultation with key financial institutions in the countries, in response to their existing situation and needs, and securing their participation. Attached letters of support corroborate the strong support of financial institutions. Second, the programme has consulted with the IGREENFIN executing agency, IFAD, on		

[145] https://www.annualreviews.org/doi/10.1146/annurev-environ-033110-092827

harnessing the complementarities between the two programme's activities and takes into account its planned implementation dates.

Selected Risk Factor 9: Anti-Money Laundering and Countering Financial Terrorism Risk		
Category	Probability	Impact
Governance	Low	Low
Description		
Risks of money laundering and countering the financing of terrorism. The inherent risk is assessed as high, but the following mitigation measures would lower the risk to low.		
Mitigation		

FAO includes in the project agreement signed between FAO and the eight Governments clauses related to AML/CFT, as follows:

a) The Government shall comply, and shall require all persons and entities engaged in its activities under the Programme to comply, with all internal anti-money laundering, counter-terrorism financing laws, rules, and regulations;

b) The Government confirms it has obtained sufficient undertakings from all persons and entities involved in its activities under the Project that they shall not engage in any prohibited practices; the Government undertakes and confirm that it shall comply with the substantive objectives of the GCF's Policy on Prohibited Practices;

c) Consistent with numerous United Nations Security Council resolutions adopted under Chapter VII of the UN Charter, the Government and FAO are firmly committed to the international fight against terrorism and, in particular, against the financing of terrorism. It is the policy of the Government and FAO to seek to ensure that none of their funds are used, directly or indirectly, to provide support to individuals or entities: i) associated with terrorism, as included in the list maintained by the Security Council Committee established pursuant to its Resolutions 1267 (1999) and 1989 (2011); or ii) that are the subject of sanctions or other enforcement measures promulgated by the United Nations Security Council. This provision must be included in all agreements that may be concluded with third parties for the implementation of activities under the Programme.

During programme implementation FAO, as AE, will ensure close monitoring and supervision through its offices in the regional office and HQ in order to ensure that the activities are implemented in full compliance with the signed project agreement.

FAO will establish appropriate fiduciary management and control measures to ensure that materials or technology procured under this programme are used only for the purposes intended and are not diverted or misused for unauthorized, improper or illicit purposes based on its institutional and programme-level grievance redress mechanisms, corporate policies on fraud and other corrupt practices, FAO Vendor Sanctions Policy (Admin circular 2014/27), FAO Whistleblower Protection Policy (Admin Circular 2011/05) and others listed here: http://intranet.fao.org/departments/oig/investigations/ including those of the Office of the Inspector General (OIG) will be in place to address this risk

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

Assignment of the Environmental & Social Risk Category

258. The proposed environmental and social (ES) risk assignment for the programme is “Category B” (Moderate Risk) based on the screening of ES risks and impacts associated with the: (i) type, nature, and scale of proposed programme activities; (ii) affected stakeholders and results from stakeholder consultations; and (iii) socioeconomic and environmental settings (covering inherent risks) of the potential intervention sites. The risk assignment accounts for the likelihood of each risk materializing, cumulative environmental and social impacts, and the severity of a given risk’s impact on achievement of the intended programme results. It considers the effect(s) of ES risk mitigation measures already utilized in the programme areas, but does not assume any future additional mitigation measures outside of those proposed in SURAGGWA’s ES instruments.

Summary of the main outcomes of ES instruments

259. Based on the screening, the design team prepared a(n): (i) Environmental and Social Management Framework (ESMF); (ii) Gender Assessment (including the accompanying Gender Action Plan); and (iii) Stakeholder Engagement Plan (SEP) (including a Grievance Redress Mechanism). The instruments describe and respond to the assessed environmental and social impacts/risks, as well as related feedback obtained during field visits, key informant interviews, and focus group discussions with programme stakeholders. Quantitative and qualitative data collected through primary and secondary sources were used to assist the analysis.

Social risks, impacts, and mitigation measures.

260. Social impacts of the programme are largely positive. The programme activities aim to improve the livelihoods and resilience of communities who rely on common and private lands in the Sahel region, with a focus on increasing inter-community collaboration (e.g. transhumant groups with agro-pastoralists). Participatory planning and community engagement under Component 1 is expected to increase collaboration and potentially improve conflict resolution around land use/management. Component 2 activities are expected to increase community resilience, incomes, and access to financial credit. Institutional capacities built under Component 3 will support improved coordination, collaboration, and management.

261. Key social risks pertain to (i) potential conflict as the value of restored lands increase; (ii) management and/or conflict concerns relating to the land tenure arrangements; and (iii) engaging with vulnerable populations, including traditionally underserved Sub-Saharan African communities. To mitigate these concerns, the programme has built principles and best practices of the *Technical Guide on the Integration of the Voluntary Guidelines on the Responsible Governance of Tenure of Land, Fisheries and Forests in the Context of National Food Security into the Implementation of the United Nations Convention to Combat Desertification and Land Degradation Neutrality* into the engagement process and programme activities, and utilizes a participatory approach (also detailed in the SEP) with feedback loops through the programme’s Grievance Redress Mechanism to avoid, and mitigate (when impossible to avoid) the risk of potential conflict. The ESMF also details the programme’s approach to engaging with vulnerable communities and use of Free Prior Informed Consent in relevant programme areas. The Gender Analysis and Action Plan provide guidance, activities, and targets for the programme (particularly under Component 1 and Component 2) so that women are equitably accounted for in the restoration efforts and NTFP benefits (see section G2 for more information).

Environmental risks, impacts, and mitigation measures.

262. The programme is expected to have largely positive environmental impacts, including improved climate resilience, soil fertility/soil health, increased soil water retention and groundwater recharge, and natural resources management. Potential negative environmental impacts are expected to be minor, limited in time/scale, and reversible, as they relate to (i) provision of seeds/seedlings/outputs to farmers to support the landscape restoration activities; (ii) potential indirect increase of pesticide use due to increased production; (iii) increased water consumption due to increased production; and (iv) overlap of programme restoration activities with national parks. These risks are managed through overall programme design (e.g. training under Components 1 and 2 which will ensure understanding of restoration principles and agroecology/integrated pest management/etc.), limitations to

the types of activities held in existing parks (e.g. restoration activities only); and (iii) Environmental and Social Management Plans (ESMP) for programme sites. The ESMP will also take into account the territorial and environmental knowledge of the affected ethnic minorities/vulnerable populations. Inherent environmental risks include the presence of unexploded ordinances (UXOs) and/or Improvised Explosive Devices (IEDs) in some areas across the Sahel. The ESMF provides the screening tools to be used at programme sites to ensure that these risks are sufficiently addressed. For UXOs, in particular, sites with UXOs/IEDs will either be avoided completely (as part of the negative list), or, if deemed a critical location for restoration activities, will involve identification and removal (if any) by experts as part of site clearance prior to commencing any activities on the programme sites. Cumulative environmental impacts are assessed to be largely positive.

Engagement and Information Disclosure during Implementation:

263. Stakeholder engagement will depend on the target communities and is designed, as per the SEP, to be sensitive to vulnerable minority community needs and preferences (as well as differentiated to the needs of vulnerable persons in the programme area. It will build on principles of the VGGT and disclosure of programme-related information will occur on a regular, rolling basis. Gender and youth targets are also mainstreamed into the Monitoring & Evaluation framework and programme indicators to ensure targets are achieved over the programme's life cycle and monitored and evaluated at the programme's annual reports, midterm, and conclusion. All ES instruments, once approved, will be disclosed at least 30 days prior to the GCF Board meeting date. The documents will be posted on the websites of the FAO, GGW agencies, and national government websites. Summaries of the documents will be disclosed in English, French and/or local language, depending on the country. The documents will also be disclosed on the GCF website.

Programme-level Grievance Redress Mechanism (GRM)

264. The programme has designed a GRM to provide complainants with redress procedures that are accessible, easily used, and free of charge to enable affected peoples to raise programme related concerns and grievances. The GRMs provide information on how to lodge complaints, including the forms, channels, steps, and processing/turn-around time associated with each step (e.g. time-limit for acknowledging receipt of the complaint, notification of the resolution decision, etc.). The GRM accounts for local practices of ethnic minorities in relation to grievance redress/conflict resolution, including the use of local ethnic minority language, and builds on existing GRMs already established and successfully functioning. Details on the GRM can be found in the SEP and ESMF.

Capacity of the Executing Entities to implement the ESMP and ESMF and arrangements for compliance monitoring, supervision and reporting.

265. The Executing Entity in each country would either be: (i) FAO; or (ii) a national entity which has passed through the assessment and evaluation of FAO's Operational Partners Implementation Modality (OPIM). As such, and with the support of the programme's safeguards personnel in each country, the EEs have capacity to implement the safeguards. In addition, implementation teams will be trained during the first year of programme implementation on the FAO and GCF's environmental and social safeguard standards (ESS), particularly on how to effectively implement the relevant ESS standards during the project cycle. Moreover, refresher trainings will be held as needed throughout the project cycle.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

266. As part of the formulation process, the Gender Assessment & Action Plan (GAP) were prepared for SURAGGWA (see Annex 8) based on: (i) desk review of relevant literature on gender, forestry, and climate change in GGW countries; (ii) reports by different national and local institutions as well as by international organizations and NGOs; and (iii) primary data collected through field-level consultations. Field level consultations supporting the GAP were held with a variety of stakeholders, including women in local communities; women's associations; civil society organizations; representatives from relevant technical departments of counterpart ministries; the GGW Agency; the National Designated Authority of the GGW; and other project teams operating in the area. These consultations held at the country level (see the SEP – Annex 7 for a full overview of consultations), coupled with the desk review, deepened understanding of the role of women in the GGW area and the constraints women face. Based on these findings, the GAP provides specific actions to assist women in building their resilience to climate change and how to enhance their livelihoods opportunities in the face of climate risks. It integrates women into the SURAGGWA programme to ensure equitable participation and benefits from the programme investment accrue for women. Key highlights are explained below.

267. Within the Sahel, women make up anywhere between 40 to 80 percent of the agricultural labour and play a critical role in enhancing food security and nutrition. They are less likely to own land throughout Africa, but this gender discrepancy is particularly pronounced in Sahelian countries, where only 8% of women own agricultural land.[1] Gender indicators for the region remain very low, revealing the presence of deep inequalities between men and women. The African Development Bank's Gender Equality in Africa Index indicates that the gender gap in the Sahel region remains the widest – with an average of 31.9%, which is below the continental average of 48.4%. This gives an overall gender gap of 68.1% across the three dimensions of economic, social and representation. Social and patriarchal norms often restrict women's decision-making roles. High prevalence of early marriage coupled with low access to maternal and reproductive health care disadvantages women from an early age. Gender indicators for education are among the worst in the world. This inequality, combined with rapid population growth, will accentuate gender gaps in access to opportunities for human capital development for them. In a security context marked by violent extremism and armed conflict, women and girls experience the double impact of an unstable security context and social discrimination.

268. Climate change increasingly and disproportionately affects women, as they are primarily responsible for collection of water and fuel, preparation of food, and feeding and caring for small ruminants, etc. Climate variability makes reliance upon natural resources for sustenance a challenge, particularly for women who rely more heavily on natural resources when there are few alternative livelihood options in rural areas. In most places with high prevalence of undernourishment, women farmers have significantly less access to land, information, finance and agricultural inputs. This makes them more vulnerable to climate shocks, and affects their health and the food security and nutrition of the entire household. Moreover, even though women contribute to household incomes, they do not necessarily retain ownership of that income or control over how that income is used, depending on the country/community. Without the resources and networks to make informed decisions about growing crops or adequate decision-making space in the local value chains, women are less equipped than their male counterparts to adequately plan for a climate resilient future. As long as women's roles in agricultural production remain undervalued it can place them at a disadvantageous position in terms of market negotiation and their ability to wield power within the value chain of the economy they are engaged with vis-à-vis men.

269. Women's participation can be a transformative and driving force, particularly in the Sahel countries where they are at the heart of food security and community resilience. Economic, security, climate and structural crises have often pushed women into the informal economy (e.g. agro-pastoral production, processing of agricultural and livestock products, petty trade, etc.). Whether in conflict zones, refugee camps or during resettlement, women are at the heart of family resilience and help reduce their vulnerability to shocks. In terms of resilience to climate change, women can play a critical role in local adaptation strategies because of their traditional role within households and the community, but also because they possess environmentally friendly local expertise. These practices are the focus of support from the SURAGGWA programme, as they are entry points to increase sustainability and ownership within the communities. Targets for women (and youth) engagement under SURAGGWA are based around: (i) inclusion and active participation in Component 1 restoration activities; and (ii) provision of support to women involved with NTFP value chains as a source of resilience and income. The targets indicated in the GAP, and the ways in which women's engagement is written directly into the programme components, have been developed to consider, also, the time poverty experienced by women in the region. In short, SURAGGWA aims to engage and support women in a way that does not worsen their time poverty. Successful implementation of the GAP relies on (i) support of the social safeguards and gender specialist in each country; and (ii) careful selection of implementing partners, such that women's inclusion and equity is upheld.

270. SURAGGWA is aligned with the GCF Gender Policy, which recognizes that gender relations, roles, and responsibilities exercise important influences on women's and men's access to and control over decisions, assets and resources, information, and knowledge. An analysis of the gender roles and responsibilities in the Sahel confirms that the impacts of climate change can exacerbate existing gender inequalities. SURAGGWA integrates gender equality and women's empowerment considerations into the design and implementation of its country level investments. SURAGGWA's GAP outlines the clear strategy for the participation of women and vulnerable communities in discussions and decisions that affect them. The GAP outlines SURAGGWA's specific targets for women, which will be closely monitored and reported upon regularly with support of the national: (i) social safeguards and gender specialist; and (ii) M&E specialist. These technical experts, in collaboration with local programme implementation teams, will guide implementation of the GAP and track performance on the participation of women/GAP implementation.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

271. FAO will be the Accredited Entity and the FAO country offices in the Sahel will be the Executing Entities together with other partners and will execute the programme in accordance with FAO rules, regulations, policies and procedures. FAO will take the direct responsibility for all funds flow under the SURAGWWA programme. All funds from GCF will come to the FAO-Head Office in Rome and from there be directed to the country offices. FAO-Country Offices will be responsible for the payment of the staff salaries, local procurement, contracting and paying service providers, recording and managing operating costs. In case where LOA, OPIM or other modalities are used, the specific funds flow arrangements will be outlined based on FAO procedures.

272. Financial management and procurement under this programme will be guided by relevant FAO rules and regulations based on the Accreditation Master Agreement (AMA) signed between FAO and GCF. Procurement by FAO will be undertaken in accordance with the FAO manual of procedures for procurement of Goods, Works and Services (502) and for sub-contracting the delivery of specific activities to service providers and ensuring value for money (507). The programme will be subject to FAO's audit regime including the external audit and internal audit function. A 24-month procurement plan has been provided based on the programme budget (Annex 4) and the expected timelines (Annex 5) using the FAO procurement procedures and thresholds and is given in Annex 10.

273. FAO has deployed an Oracle based resource planning system termed the Global Resources Management System (GRMS) which provides FAO personnel with travel, human resources, procurement and finance functionalities. The system has improved the flow of financial information and supports financial monitoring and reporting, increases transparency and visibility and strengthens internal control. FAO maintains a Chart of Accounts that allows for the separation of income and expenditure by donor and programme. The system is based on a standardized coding structure that enables data to be recorded, classified and summarized to facilitate internal management and meet external reporting and audit requirements.

G.4. Disclosure of funding proposal

☒ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.
☐ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

☒	Annex 1	NDA no-objection letter(s) **(template provided)**
☒	Annex 2	Feasibility study - and a market study, if applicable
☒	Annex 3	Economic and/or financial analyses in spreadsheet format
☒	Annex 4	Detailed budget plan **(template provided)**
☒	Annex 5	Implementation timetable including key project/programme milestones **(template provided)**
☒	Annex 6	E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3): **(ESS disclosure form provided)** ☐ Environmental and Social Impact Assessment (ESIA) or ☐ Environmental and Social Management Plan (ESMP) or ☐ Environmental and Social Management System (ESMS) ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
☒	Annex 7	Summary of consultations and stakeholder engagement plan
☒	Annex 8	Gender assessment and project/programme-level action plan **(template provided)**
☒	Annex 9	Legal due diligence (regulation, taxation and insurance)
☒	Annex 10	Procurement plan **(template provided)**
☒	Annex 11	Monitoring and evaluation plan **(template provided)**
☒	Annex 12	AE fee request **(template provided)**
☒	Annex 13	Co-financing commitment letter, if applicable **(template provided)**
☒	Annex 14	Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

☒	Annex 15	Evidence of internal approval **(template provided)**
☒	Annex 16	Map(s) indicating the location of proposed interventions
☒	Annex 17	Multi-country project/programme information **(template provided)**
☐	Annex 18	Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
☐	Annex 19	Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
☐	Annex 20	First level AML/CFT (KYC) assessment
☐	Annex 21	Operations manual (Operations and maintenance)
☒	Annex 22	Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)
☒	Annex 23	Estimates of Direct and Indirect Beneficiaries

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

No-objection letter issued by the national designated authority(ies) or focal point(s)

PRIMATURE

SECRETARIAT GENERAL

SECRETARIAT EXECUTIF DU FONDS VERT POUR LE CLIMAT AU BURKINA FASO

BURKINA FASO

Unité-Progrès-Justice

Ouagadougou, 20 June 2023

N° 2023-036 */PM/SG/SE-FVC/BF*

National Designated Authority

To

The Green Climate Fund

("GCF")

Republic of Korea

Re: Funding proposal for the GCF by Food and Agriculture Organization regarding **Scaling Up Resilience in Africa's Great Green Wall (SURAGGWA)**

Dear Madam, Sir,

We refer to **the project titled *Scaling up Resilience in Africa's Great Green Wall (SURAGGWA)*** in Burkina Faso as included in the funding proposal submitted by Food and Agriculture Organization to us on 1 June 2023.

The undersigned is the duly authorized representative of the National Designated Authority of Burkina Faso.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Burkina Faso has no-objection to the project as included in the funding proposal;

(b) The project as included in the funding proposal is in conformity with the national priorities, strategies and plans of Burkina Faso;

(c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards

Secrétariat Exécutif du Fonds Vert Pour le Climat
L'Autorité Nationale Désignée

Nebyida Lamech KABORE

Secrétariat Exécutif du Fonds Vert pour le Climat au Burkina Faso (SE-FVC/BF)
03 BP 7027 OUAGADOUGOU 03
Tel.: + 226 32 49 48 89/91
Site web : www.fondsvertclimat.bf

Republic of Chad
Presidency of transition
Ministry of Environment, Fisheries and Sustainable Development;
General Secretariat;
Designated National Authority

Unit-Work-Progress

N'Djaména, le

Designated National Authority

To: The Executive Secretary of the Green Climate Fund (GCF)
175, Art Center-Daero,Yeonsou-gu, Incheon 405-840, Korea Republic

Re: Funding proposal for the GCF by Food and Agriculture Organization regarding Scaling Up Resilience in Africa's Great Green Wall (SURAGGWA)

Dear Madam, Sir,

We refer to the project titled *Scaling up Resilience in Africa's Great Green Wall (SURAGGWA)* in Chad as included in the funding proposal submitted by Food and Agriculture Organization to us on 25 April 2023.

The undersigned is the duly authorized representative of **M. ABAKAR MOURNO ABDOULAYE**, the National Designated Authority of Chad.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Chad has no-objection to the project as included in the funding proposal;
(b) The project as included in the funding proposal is in conformity with the national priorities, strategies and plans of Chad;
(c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

ABAKAR MOURNO ABDOULAYE,
Focal Point the Green Climate Fund of Chad
abmournofils12@gmail.com

Copy: FAO

REPUBLIQUE DE DJIBOUTI
UNITE – ÉGALITE – PAIX

MINISTERE DE L'ENVIRONNEMENT ET DU DEVELOPPEMENT DURABLE

LE SECRÉTAIRE GÉNÉRAL

MEDD

جـمـهـورية جيبوتي
الوحدة - المساواة - السلام

وزارة البـيـئـة
والتنمية المستدامة

الأمين العام

N° 185
Djibouti, le 27 APR 2023

رقم
جيبوتي في

Re: Funding proposal for the GCFby Food and Agriculture Organization regarding Scaling Up Resilience in Africa's Great Green Wall (SURAGGWA)

Dear Madam, Sir,

We refer to the project titled *Scaling up Resilience in Africa's Great Green Wall (SURAGGWA)* in Djibouti as included in the funding proposal submitted by Food and Agriculture Organization to us on25 April 2023.

The undersigned is the duly authorized representative of M. Dini Abdallah Omar, the National Designated Authority of Djibouti.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Djibouti has no-objection to the project as included in the funding proposal;
(b) The project as included in the funding proposal is in conformity with the national priorities, strategies and plans of Djibouti;
(c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

MINISTERE DE L'ENVIRONNEMENT ET DU DEVELOPPEMENT DURABLE
SECRETAIRE GENERAL
★ MEDD ★

M. DINI ABDALLAH OMAR]
Secrétaire général du Ministère de l'Environnement et du Développement Durable (MEDD)
Et l'Autorité Nationale Désignée du GCF
République de Djibouti

MINISTERE DE L'ENVIRONNEMENT,
DE L'ASSAINISSEMENT ET DU
DEVELOPPEMENT DURABLE

Agence de l'Environnement et du Développement Durable (AEDD)

REPUBLIQUE DU MALI
Un Peuple - Un But - Une Foi

Bamako, le 19 DEC 2023

N° 100773 /MEADD/AEDD.

The Director General of the Environment and Sustainable Development Agency

To

The Green Climate Fund (GCF).

Re: Funding proposal for the GCF by Food and Agriculture Organization regarding Scaling Up Resilience in Africa's Great Green Wall (SURAGGWA).

Dear Madam, Sir,

We refer to the project titled Scaling up Resilience in Africa's Great Green Wall (SURAGGWA) in Mali as included in the funding proposal submitted by Food and Agriculture Organization to us on 7 June 2023.

The undersigned is the duly authorized representative of Environment and Sustainable Development Agency, the National Designated Authority (NDA) of Mali.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.
By communicating our no-objection, it is implied that :

(a) The government of Mali has no-objection to the project as included in the funding proposal ;
(b) The project as included in the funding proposal is in conformity with the national priorities, strategies and plans of Mali ;
(c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards.

DATE: 18/12/20
FAO Rép.
ACTION: Progra
Ad

The Director General,

Le Directeur Général

Zantigui Boua KONE
Chevalier de l'Ordre national

REPRESENTATION
FAO au MALI

République Islamique de Mauritanie
Honneur - Fraternité – Justice

الجمهورية الإسلامية الموريتانية
شرف – إخاء – عدل

Ministère de l'Environnement et du Développement Durable

وزارة البيئة والتنمية المستدامة

Direction Climat et Economie Verte

مديرية المناخ والإقتصاد الأخضر

N° : 001/23

Nouakchott, le .. 22/06/2023 أنواكشو

Le Directeur ***المدير***

To: The Green Climate Fund ("GCF")

Re: Funding proposal for the GCF by Food and Agriculture Organization regarding Scaling Up Resilience in Africa's Great Green Wall (SURAGGWA)

Dear Madam, Sir,

We refer to the project titled *Scaling up Resilience in Africa's Great Green Wall (SURAGGWA)* in Mauritania as included in the funding proposal submitted by Food and Agriculture Organization to us on 22 June 2023.

The undersigned is the duly authorized representative of the National Designated Authority of Mauritania.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Mauritania has no-objection to the project as included in the funding proposal;
(b) The project as included in the funding proposal is in conformity with the national priorities, strategies and plans of Mauritania;
(c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Sidi Mohamed EL WAVI

Copy : MEDD

REPUBLIQUE DU NIGER

Fraternité – Travail – Progrès

CABINET DU PREMIER MINISTRE

CONSEIL NATIONAL DE L'ENVIRONNEMENT POUR UN DEVELOPPEMENT DURABLE

SECRETARIAT EXECUTIF

To The Green Climate Fund (« GCF »)

N° 0 0 0 8 9

Niamey, May 4, 2023

Re: Funding proposal for the GCF by Food and Agriculture Organization regarding Scaling Up Resilience in Africa's Great Green Wall (SURAGGWA)

Dear Madam, Sir,

We refer to the project titled *Scaling up Resilience in Africa's Great Green Wall (SURAGGWA)* in Niger as included in the funding proposal submitted by Food and Agriculture Organization to us on 04/14/2023

The undersigned is the duly authorized representative of Dr KAMAYE MAAZOU the National Designated Authority of Niger.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Niger has no-objection to the project as included in the funding proposal;
(b) The project as included in the funding proposal is in conformity with the national priorities, strategies and plans of Niger;
(c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

CNEDD SECRETAIRE EXECUTIF

Dr KAMAYE MAAZOU
Secrétaire Exécutif du CNEDD
Point Focal, Autorité Nationale Désignée (AND)
Niger

FEDERAL REPUBLIC OF NIGERIA
NATIONAL COUNCIL ON CLIMATE CHANGE
OFFICE OF THE DIRECTOR GENERAL
Address: No 56, Mamman Nasir Street, Asokoro, Nigeria
Email: info@natccc.gov.ng

NCCC/ADM/1/VOL.I/32 Date: 16th May, 2023

The Green Climate Fund ("GCF")
Songdo International Business District,
175, Art Centre-daero,
Yeonsu-gu Incheon 406-840
Republic of Korea.

Re: Funding proposal for the GCF by Food and Agriculture Organization regarding Scaling Up Resilience in Africa's Great Green Wall (SURAGGWA)

Dear Madam/Sir,

Executive Director of the Green Climate Fund

We refer to the project titled Scaling up Resilience in Africa's Great Green Wall (SURAGGWA) in Nigeria as included in the funding proposal submitted by Food and Agriculture Organization to us on 29th March, 2023.

The undersigned is the duly authorized representative of the National Designated Authority/Focal Point of the Federal Republic of Nigeria.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Nigeria has no-objection to the project as included in the funding proposal;
(b) The project as included in the funding proposal is in conformity with the national priorities, strategies and plans of Nigeria;
(c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Dr. Salisu Dahiru
Director General

République du Sénégal
Un Peuple - Un But - Une Foi

MINISTERE DE L'ENVIRONNEMENT
ET DU DEVELOPPEMENT DURABLE

DIRECTION DE L'ENVIRONNEMENT
ET DES ETABLISSEMENTS CLASSES

DEEC

N° 35 MEDDTE/DEEC/DCC

Dakar, le 12/10/ 2023

To: The Green Climate Fund ("GCF")

Mr. Henry Gonzalez
Deputy Executive Director
G-Tower, 24-4 Songdo-dong, Yeonsu-gu
Incheon City, Republic of Korea

Re: Funding proposal for the GCF by Food and Agriculture Organization (FAO(regarding Scaling-Up Resilience in Africa's Great Green Wall (SURAGGWA)

Dear Sir,

We refer to the project titled *Scaling-Up Resilience in Africa's Great Green Wall (SURAGGWA)"* in Senegal as included in the funding proposal submitted by FAO to us on 10 October 2022.

The undersigned is the duly authorized representative of Madeleine Diouf Sarr, the National Designated Authority of Senegal.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Senegal has no-objection to the project as included in the funding proposal;
(b) The project as included in the funding proposal is in conformity with the national priorities, strategies and plans of Senegal;
(c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,
Madeleine Diouf Sarr
NDA
Senegal

Ministère de l'Environnement et du Développement
Direction de l'Environnement et des Etablissements Classés
Autorité Nationale
Chef de Division des Changements Climatiques

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Scaling-Up Resilience in Africa's Great Green Wall (SURAGGWA)
Existence of subproject(s) to be identified after GCF Board approval	No
Sector (public or private)	Public
Accredited entity	Food and Agriculture Organisation of the United Nations (FAO)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Burkina Faso, Chad, Djibouti, Mali, Mauritania, Niger, Nigeria, and Senegal
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	An ESMP consistent with the requirements for a Category B project is contained in Annex 6 "Environmental and Social Management Framework (ESMF)" of the Funding Proposal. Risk screening of funded activities rated from low to moderate necessitated the development of an ESMP and not an ESIA.
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Friday, May 16, 2025
Language(s) of disclosure	English, Arabic, and French
Explanation on language	These are official languages spoken by the beneficiaries of the programme.
Link to disclosure	English: https://openknowledge.fao.org/handle/20.500.14283/CD5411EN Arabic: https://openknowledge.fao.org/items/41fbafc8-3f80-4966-9c77-ce8dc49eb1f3 French: https://openknowledge.fao.org/handle/20.500.14283/CD5411FR
Other link(s)	FAO Disclosure Portal: https://www.fao.org/environmental-social-safeguards/disclosure/en

Remarks	An ESMP consistent with the requirements for a Category B project is contained in the Environmental and Social Management Framework (ESMF) of the SURAGGWA programme – Annex 6.
Environmental and Social Management System (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)	
Description of report	Environmental and Social Management Framework - ESMF
Date of disclosure on accredited entity's website	Friday, May 16, 2025
Language(s) of disclosure	English, Arabic, and French
Explanation on language	These are official languages spoken by the beneficiaries of the programme.
Link to disclosure	English: https://openknowledge.fao.org/handle/20.500.14283/CD5411EN Arabic: https://openknowledge.fao.org/items/41fbafc8-3f80-4966-9c77-ce8dc49eb1f3 French: https://openknowledge.fao.org/handle/20.500.14283/CD5411FR
Other link(s)	FAO Disclosure Portal: https://www.fao.org/environmental-social-safeguards/disclosure/en
Remarks	An IPPF consistent with the requirements for a Category B project is contained in the Environmental and Social Management Framework (ESMF) of the SURAGGWA programme – Annex 6 titled as Ethnic Minority Planning Framework.
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Friday, May 16, 2025
Place	Below is a summary, by country, of the websites and physical locations where SURAGGWA's ESMF will be disclosed. The details of the physical locations and radio outreach are provided as an annex. **Burkina Faso** **Online Disclosure:** Ministry of Environment website: http://www.environnement.gov.bf

	National Coordination of the Great Green Wall website: http://www.igmvss-bf.net **FAO Burkina Faso:** www.fao.org/burkinafaso/fr **Hard Copy Distribution Locations** Governorates of Dori, Kaya, Ouahigouya and Fada **Radio Outreach:** Broadcasts in local languages on 4 regional stations. **Chad** **Online Disclosure:** Chad FAO: https://www.fao.org/tchad/fr Chadian Institute for Agricultural Research for Development: www.itrad.td **Hard Copy Disclosure Locations:** ANGMV offices in N'Djamena, and provincial office in Batha Hamatié **Radio outreach:** Broadcasts in seven local radio stations across eight towns. **Djibouti** **Online Disclosure:** Ministry of Environment and Sustainable Development: https://www.facebook.com/profile.php?id=61553005181813&locale=fr_FR Djibouti's Information Agency: https://adi.dj/ FAO Djibouti: www.fao.org/djibouti/fr Newspaper La Nation : www.lanation.dj RTD - Djibouti's Radio and Television diffusionwww.rtd.dj Djibouti's UN website: https://djibouti.un.org/fr **Hard Copy Distribution** Ministry Environment and Sustainable Development **Mali** **Online Disclosure:** Ministry of Agriculture: https://ma.gouv.ml/ National Directorate of Development and Planning: https://dnpd.gouv.ml/ General Directorate of Territorial Collectivities: https://dgct.gouv.ml/ National Agency of the Great Green Wall: http://www.grandemurailleverte.org/index.php Non-Governmental Organization Coordination Secretariat: https://www.secoong.org/ FAO Mali: www.fao.org/mali/fr **Hard Copy Distribution Locations:** National and regional ministries and agencies at MEADD, DNACPN, and AEDD.

	FAO Sub-Offices in Kayes, Ségou, and San. Regional and municipal governments in Kayes, Nioro, Koulikoro, Ségou, San, and Dioïla. Municipal Associations **Radio outreach:** Broadcast in 5 local languages via 7 local radio stations. **Mauritania** **Online Disclosure:** Ministry of Environment and Sustainable Development: https://www.environnement.gov.mr Pan-African Great Green Wall Agency: www.grandemurailleverte.org FAO Mauritania: www.fao.org/mauritania/fr **Hard Copy Disclosure distribution in** MEDD regional delegations (Wilayas), regional offices of Niger's ANGMV regional offices in ASSABA (Kiffa), BRAKNA (Aleg), TRARZA (Rosso). **Niger** **Online Disclosure:** The Ministry of Environment, Water and Sanitation: http://www.hydraulique.gouv.ne The Executive Secretariat of the National Council for the Environment and Sustainable Development: www.cnedd.ne X page of the National Agency for the Great Green Wall (ANGMV): https://x.com/angmvniger The Network of Pastoralist and Livestock Breeders' Organizations in Niger (ROPEN): www.maroobe.org FAO Niger: www.fao.org/niger National Network of Agricultural Chambers - Reca: https://reca-niger.org/ **Hard Copy Disclosure Locations:** Ministry of Environment, SE/CNEDD, FAO, ANGMV, ROPEN, RDFN, RECA Regional Environment Directorates: Dosso, Diffa, Maradi, Niamey, Tahoua, Tillabéry, Zinder General Directorates: DGEF, DGPIA, DGA **Nigeria** **Online Disclosure:** Federal Ministry of Environment: https://environment.gov.ng National Agency for the Great Green Wall: https://ggwnigeria.gov.ng Agro Climatic Resilience in Semi-Arid Landscapes: https://acresal.gov.ng National Council on Climate Change: https://natccc.gov.ng FAO Nigeria: www.fao.org/nigeria/en

	Hard Copy Disclosure Locations: Federal Ministry of Environment, Abuja National Agency for the Great Green Wall, Abuja ACRESAL, Abuja National Council on Climate Change, Abuja **Radio outreach:** Radio Nigeria and other state broadcasting stations **Senegal** **Online Disclosure:** Senegal's Ministry of the Environment and Sustainable Development: https://www.senegel.org/en/administration/executive-power/ministers/orgdetails/177 Senegalese Agency for Reforestation and the Great Green Wall (ASERGMV): https://asergmv.org Department of Water and Forestry, Hunting and Soil Conservation (DEFCCS): https://eaux-forets.sn Senegalese Institute for Agricultural Research (ISRA): https://isra.sn The Ecological Monitoring Center: https://www.cse.sn Agence Nationale de Conseil Agricole et Rural (ANCAR) (National Agency for Agricultural and Rural Consultancy): https://ancar.gouv.sn National framework for rural cooperation and consultation (CNCR): https://cncr.org FAO Senegal: https://www.fao.org/senegal/fr **Hard Copy Disclosure Locations:** Ministries and agencies mentioned above and FAO Office
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, June 30, 2025

* *An annex to the Environmental and Social Safeguards (ESS) Disclosure Report Form, titled "Annex to GCF's ESS disclosure report form regarding the SURAGGWA programme," provides supplementary details on the physical locations where SURAGGWA's ESMF will be disclosed, as well as information on radio outreach.*

Note: This form was prepared by the accredited entity stated above.

Annex to GCF's ESS disclosure report form regarding the SURAGGWA programme

Burkina Faso

Governorates of Covered Regions for Paper Copy Distribution

Cities	Longitude	Latitude
Dori	-0.023341°	14.024570°
Kaya	-1.087399°	13.090800°
Ouahigouya	-2.411059°	13.566729°
Fada	0.365420°	12.059999°

Broadcast on Local Radio Stations

Region	**Radio Name**	**Broadcast Time (GMT)**	**Days**
East / Fada	Radio Tintua	9:00 AM – 7:00 PM (Languages: Gourmanché, French)	Every day, from May 25 to June 30
Sahel / Dori	Radio Kawral	8:30 AM (Languages: Fulfulde, Gourmanché, French)	Every day, from May 25 to June 30
North / Ouahigouya	Radio Voix du Paysan	9:30 AM – 6:30 PM (Languages: Mooré, French)	Every day, from May 25 to June 30
Centre-North / Kaya	Radio OR FM	9:30 AM – 6:30 PM (Languages: Mooré, French)	Every day, from May 25 to June 30

Djibouti

GPS Coordinates in the 5 Interior Regions

Location	Latitude	Longitude
Tadjourah Prefecture	11.786775905781033	42.88462452646462
Tadjourah Regional Council	11.785468376911973	42.88530509120329
Obock Prefecture	11.96156222214518	43.29166799066241
Obock Regional Council	11.960434141186697	43.28935991825944
Arta Prefecture	11.521278475500583	42.844729841215916
Arta Regional Council	11.521743206361988	42.847363965495475

Mali

Local Radio Stations

Rural Radio in the Kayes Region

Radio Jamana in the Nioro Region

Radio Dionakan in the Koulikoro Region

Radio Sanya in the Ségou Region

Radio Lafia FM in the Dioila Region

Radio Bendougou in the Ségou Region

Radio Parana in the San Region

Broadcast Dates and Times

- One broadcast per day from 25 to 06 June 2025, between 8:00 and 10:00.
- One broadcast per day from 25 to 13 June 2025, between 18:00 and 20:00.

Nigeria

Locations of Government Offices for ESMP Dissemination

Location	Coordinates
Federal Ministry of Environment, Mabushi, Abuja, Federal Capital Territory	9.0452° N, 7.4811° E
National Agency for the Great Green Wall, 102 Ebitu Ukiwe Street, Jabi, Abuja	9.07154° N, 7.4328° E
Agro Climatic Resilience in Semi-Arid Landscapes (ACReSAL), 16, Biodun Olorunfemi Street, Wuye, Abuja	9.0578° N, 7.4511° E
National Council on Climate Change, 14 Vistula Close, Maitama, Abuja	9.0833° N, 7.4833° E

Radio Broadcast

Broadcast Schedule

Radio Station	Time (GMT +1)	Date
Radio Nigeria (FRCN)	7:00 AM; 12:00 PM; 4:00 PM	Daily, starting on May 25, 2025
State Broadcasting Stations	8:00 AM; 10:00 AM; 6:00 PM	Daily, starting on May 25, 2025

Chad

National and Community Radio Stations

City	Radio Station	Broadcast hours from 25 May onwards
Bol	Radio KADAYE	6:00–12:00 and 15:00–18:00
Bol	ONAMA/Radio	6:00–10:00 and 14:00–22:00
Mao	Radio FM Djimi	6:00–12:00 and 15:00–18:00
Mao	ONAMA/Radio	6:00–10:00 and 14:00–22:00
Ati	Radio FM Al-NADJA	7:00–10:00 and 15:00–18:00
Ati	ONAMA/Radio	6:00–10:00 and 14:00–22:00
Oum-Hadjer	Radio FM ALMOURHAL	6:00–10:00 and 15:00–18:00
Mongo	Radio Communautaire de Mongo (RCM)	7:00–10:00 and 15:00–18:00
Mongo	ONAMA/Radio	6:00–10:00 and 14:00–20:00
Abéché	Radio la Voix du Ouaddaï	6:00–10:00 and 15:00–20:00

Abéché	Voix du Développement	6:00–10:00 and 15:00–20:00
Abéché	ONAMA/Radio	6:00–10:00 and 14:00–22:00
Biltine	ONAMA/Radio	6:00–10:00 and 14:00–22:00
Massakory	ONAMA/Radio	6:00–10:00 and 14:00–22:00
Sila	Radio Sila	6:00–10:00 and 14:00–20:00

Secretariat's assessment of FP268

Proposal name:	Scaling-Up Resilience in Africa's Great Green Wall (SURAGGWA)
Accredited entity:	Food and Agriculture Organization of the United Nations (FAO)
Country/(ies):	Burkina Faso, Chad, Djibouti, Mali, Mauritania, Niger, Nigeria, Senegal
Project/programme size:	Medium

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The programme directly supports the African Union-led Great Green Wall (GGW) initiative by enhancing ecological and climate resilience. Specifically, it contributes to the GGW goals of restoring 100 million hectares of degraded land, creating 10 million new jobs and sequestering 250 million tonnes of carbon dioxide equivalent by 2030 through large-scale land restoration efforts.	The programme is to be implemented over a decade in eight countries, including several that are fragile or affected by conflict, which might impact the timely delivery of the investments. To address these concerns, the design of the programme includes several adaptive management measures, including a contingency budget.
The programme aims to foster a paradigm shift by capitalizing on existing initiatives, accelerating land restoration and climate change mitigation and developing value chains for non-timber forest products (NTFPs). By improving market access and promoting equitable partnerships with private sector buyers and financial institutions, it supports diversified livelihoods, strengthens food security and boosts local economic development.	

2. The Board may wish to consider approving this funding proposal in accordance with the term sheet agreed between the Secretariat and the accredited entity (AE) and, if considered appropriate, subject to the conditions set out in annex II to document GCF/B.42/02.

II. Summary of the Secretariat's assessment

2.1 Programme background

3. The Sahelian nations within the GGW region encounter significant climate-related vulnerabilities and developmental obstacles. A substantial segment of their populations remains deprived of essential healthcare services and educational opportunities. Their economies are predominantly reliant on agriculture, livestock and forestry, with most rural communities dependent on rainfed agriculture. The region has been particularly affected by climate change, which has caused prolonged droughts, torrential rainfall and ecosystem degradation that has severely impacted food security and livelihoods.

4. In response, the Scaling-Up Resilience in Africa's Great Green Wall (SURAGGWA) programme is a multi-country initiative designed to build ecological and climate resilience across eight Sahelian countries, Burkina Faso, Chad, Djibouti, Mali, Mauritania, Niger, Nigeria and Senegal, while significantly contributing to carbon sequestration in restored landscapes.

5. The programme's interventions focus on scaling up successful land restoration practices, including restoring native species, promoting climate-resilient NTFP and fodder value chains and strengthening institutional capacities to coordinate and monitor restoration efforts. Through community engagement, particularly the empowerment of women, SURAGGWA will enhance financial linkages and address existing bottlenecks to support innovative value chain financing models. This approach aims to stimulate sustainable livelihoods, expand job opportunities and fortify resilience in areas where environmental degradation and competition over scarce resources have historically fuelled conflict.

6. The proposed funding for the programme is USD 221,990,212, utilizing grant instruments. The GCF contribution to the programme is a USD 150 million grant. The programme will also receive co-financing from the Food and Agriculture Organization of the United Nations (FAO) and the Governments of Burkina Faso, Chad, Djibouti, Mali, Mauritania, Niger, Nigeria and Senegal.

7. The programme targets approximately 1.88 million direct beneficiaries and 3.86 million indirect beneficiaries. Together, these beneficiaries comprise 1.8 per cent of the total population of the target eight countries. The mitigation impact over the programme's lifetime of 20 years is 65 million tonnes of carbon dioxide equivalent.

8. GCF categorizes the programme as category B (medium) for its environmental and social safeguards (ESS).

2.2 Component-by-component analysis

9. The programme is structured into three components:

Component 1: Large-scale landscape restoration to increase climate change resilience and mitigation, for and by local communities (total cost: USD 142,843,448; GCF cost: USD 82,098,671 in grants)

10. This component will initiate the process of land rehabilitation by engaging with rural communities and working closely with them to identify specific sites, prepare and carry out interventions, maintain and manage the restored sites and assess the impacts, starting with initial baselines and periodic field assessments.

11. Outputs under this component include that community groups are organized and trained for land restoration activities and monitoring; native seed supply systems have been strengthened to ensure the availability of genetically appropriate seeds that provide increased climate resilience; targeted highly degraded lands are restored through arrangements for rainwater harvesting and enhanced soil permeability and through local communities engaging in planting seedlings and direct seeding; targeted moderately degraded lands are planted with a range of species to restore and enrich the landscapes of agroforestry, agroecology and agro-

silvo-pastoral systems, biomes and protected forest ecosystems; and rural community and local authority capacity is strengthened to verify the results of the restoration and inform the management of restored areas.

Component 2: Supporting the development of climate-resilient, low-emission NTFP value chains benefiting vulnerable communities' livelihoods (total cost: USD 36,490,107; GCF cost: USD 32,327,307 in grants)

12. This component aims to improve the resilience and livelihoods of the local agro-silvo-pastoral communities and smallholder NTFP collectors, processors and sellers in the GGW zones in the selected Sahel countries.

13. Outputs under this component include that the technical and organizational capacities of producer organizations and micro, small and medium-sized enterprises are strengthened in priority NTFP value chains; there is increased access to markets for smallholder NTFP actors (including micro and small enterprises); and suitable credit and insurance products are designed for smallholder NTFP value chain actors (producer organizations, cooperatives and micro, small and medium-sized enterprises) and there is increased stakeholder capacity for using these financial products in their NTFP value chain activities.

Component 3: Strengthening the GGW institutions at the country and regional level (total cost: USD 18,615,500; GCF cost: USD 17,517,500 in grants)

14. This component aims to strengthen the institutional capacities of the institutions officially created and mandated to spearhead the GGW initiative at the national and regional level, namely the national GGW agencies in the eight participating countries and the Pan-African Agency for the Great Green Wall.

15. Outputs under this component include that the GGW land restoration monitoring system at the national and regional level is upgraded and functional; national and regional GGW institution planning and coordination capacities are strengthened; national and regional climate change institutional capacities/frameworks are strengthened to integrate land restoration investments in climate change adaptation and mitigation programmes; and GGW knowledge management and communication capacities are leverage for increasing support to climate-resilient land restoration.

Monitoring and evaluation (total cost: USD 8,760,750; GCF cost: USD 6,304,250)

Project management (total cost: USD 11,028,635; GCF cost: USD 7,500,000)

16. The project management budget will finance project management expenses in various fields, such as finance and procurement.

17. A contingency budget is provided to address unforeseen challenges, such as unexpected cost increases owing to higher inflation or security costs.

III. Assessment of performance against investment criteria

18. The programme consistently scores medium to high against the GCF investment criteria. A detailed assessment of the programme's fit with the GCF investment criteria is provided below.

3.1 Impact potential *Scale: High*

19. The programme aims to reach approximately 1.88 million direct beneficiaries and 3.86 million indirect beneficiaries, essentially communities in an agro-silvo-pastoral system, through large-scale land restoration (133,238 hectares of highly degraded land and 1.14 million hectares of moderately degraded land). Additionally, it also aims to mitigate 65 million tonnes of carbon dioxide equivalent through increased tree canopy cover due to the restoration activities. The programme will also introduce improved climate-resilient livelihood options, for example through increased participation in NTFPs and increased employment, which will especially benefit women.

3.2 Paradigm shift potential *Scale: Medium to high*

20. The programme represents a paradigm shift in the agriculture sector by developing sustainable value chains for NTFPs and creating innovative financing models to attract private sector investment. The programme has the potential to transform the livelihoods of smallholder farmers and pastoralist households by investing in climate-resilient land management, building smallholder organizational capacity, improving market access for NTFPs and fostering more equitable partnerships with private sector buyers and financial institutions.

3.3 Sustainable development potential *Scale: High*

21. The programme will contribute to several Sustainable Development Goals (SDGs): SDG 1 (no poverty), SDG 2 (zero hunger), SDG 8 (decent work and economic growth), SDG 13 (climate action), SDG 15 (life on land) and SDG 17 (partnerships for the goals). It is expected that the programme will realize environmental, social, economic and gender co-benefits.

3.4 Needs of the recipient *Scale: High*

22. The Notre Dame Global Adaptation Index rankings for the SURAGGWA countries show a high degree of vulnerability and lack of economic, governance, and social readiness for climate investment. Seven out of the eight target countries in the programme are least developed countries with a low self-financing capacity.

3.5 Country ownership *Scale: High*

23. No-objection letters have been issued by the Governments of all target countries.

24. In response to severe droughts, the African Union launched the GGW initiative in 2007 with support from FAO. This prompted Sahelian countries to establish GGW institutions, nominate national designated authorities and develop tailored mitigation and adaptation strategies, demonstrating strong country ownership. Building on this, SURAGGWA was developed through extensive regional and national consultations, during which countries identified target areas, financing requirements and potential local contributions. Countries also established partnerships, institutional arrangements and focal points for implementation, and are actively seeking additional climate finance.

3.6 Efficiency and effectiveness *Scale: Medium to high*

25. The programme will provide grants to build the technical, organizational and commercial capacities of farmers and pastoralists. This will enable them to adopt more resilient farming practices and invest in restoring additional land, thereby expanding the programme's

impact after the implementation period. The programme will also facilitate the development of a robust rental market for deep ploughing equipment. In addition, complementary measures, such as value chain activities for NTFPs, will establish effective market linkages and encourage greater private sector participation.

26. The adaptation GCF cost of USD 47.9 per direct beneficiary (and USD 15.7 per total beneficiary) and the mitigation cost of USD 0.92 per tonne of carbon dioxide equivalent is competitive within the GCF portfolio for mitigation and adaptation projects. The co-financing ratio is 1:0.5, which is moderate considering the financial constraints of sub-Saharan countries.

27. An analysis of the programme shows that it is economically viable as a whole, resulting in an economic net present value over a 20-year period of USD 154.2 million and an economic internal rate of return of 19.2 per cent without valuing the economic value of greenhouse gas (GHG) emissions mitigated by the programme. The full economic potential of the programme is significantly higher when accounting for GHG mitigation benefits. Using more conservative prices, at current carbon market rates the estimated NPV rises to USD 754.5 million, with an economic internal rate of return of 57.5 per cent over a 20-year period, reflecting robust economic viability and societal value.

28. The results of the financial analysis also show high profitability for beneficiaries and sustainability, indicating that various activities, including land restoration and the development of the NTFP value chain, yield a financial internal rate of return of 29.6 per cent over the period of 20 years.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

29. The Environmental and Social Management Framework (ESMF) shared by the AE including a Stakeholder Engagement Plan (SEP) and Grievance Redress Mechanism (GRM) aligns with the revised Environmental and Social Policy (ESP). The ESMF was developed in line with the FAO Framework for Environmental and Social Management and GCF's revised ESP. The programme has been classified as category B consistent with environmental and social risk screening. The AE has further developed and shared a conflict assessment that will be guided by FAO tools for conflict analysis and resolutions.

30. The risk screening triggered the following standards relevant to the programme namely;

31. **ESS4 – Animal, Livestock, and Aquatic Genetic Resources for Food and Agriculture:** The programme is envisaged to help rehabilitate and regenerate the land and soil in highly degraded and moderately degraded lands of the Sahel region (i.e. increasing trees and agroforestry in the area). While it will not be implemented inside protected areas, it will be in proximity in some cases. Interaction with protected areas will be exclusively positive; with examples being the restoration of sand dunes to prevent siltation of protected Ramsar wetland sites.

32. **ESS5 – Pest and Pesticide Management:** During programme implementation, no procurement or supply of chemical pesticides is envisaged; however, there is risk that increased production in a given area may result in increased (indirect) use of pesticides. To mitigate this, the principles of agro-ecology and Integrated Pest Management (IPM) are included in the ESMF shared and practices taught under the programme may include some sensitization on ecological pest management. In some communities, there may be use of the bio-pesticide Neem.

33. Additionally, the programme may provide some direct inputs in the form of seeds to farmers, in cases where local community seed production is insufficient. The provision will

come with training on the restoration activities and management of the plant species involved. The programme will not provide any pesticides.

34. **ESS7 – Decent Work:** It is anticipated that many of the beneficiaries will be low-income pastoralists and/or agro-pastoralists. The programme will help build their resilience and increase access to finance. This considers that some youth work is unpaid or family work, leading to youths leaving agriculture in rural areas. The programme targets areas with labour market gender inequality and may involve subcontracting or direct employment. It will focus on men, women, and youth, ensuring only those above the minimum employment age participate in age-appropriate activities to prevent child labour.

35. Review iterations further focused on understanding the scale of the area envisaged for restoration. The AE provided information on how the 1.273 million hectares earmarked for restoration will be distributed across the eight beneficiary countries. The programme will further strengthen the seed supply system by adopting native species to avoid the introduction of invasive species. In areas with highly degraded land, local beneficiary communities will adopt rainwater harvesting and enhance soil permeability. Despite the Sahel region being an arid area, ground-based fallow/trench water harvesting approach will be adopted.

36. **Sexual exploitation, abuse and harassment safeguarding:** The revised GCF Environmental and Social Policy adopted by decision B.BM-2021/18 requires safeguarding from sexual exploitation, abuse and harassment (SEAH) in GCF-financed activities. The programme, which will have a zero tolerance for SEAH, includes activities that involve programme workers engaging with communities in subprojects. Risk factors include a high prevalence of early marriages and forced labour, potentially remote and isolated programme locations with limited access to mitigation services, conflict situations, lack of enforcement of legislation that promotes gender equality, gender-based violence as well as gender and social norms in the Sahel region. SEAH risks will be identified through screening and assessed in additional due diligence at the subproject level. Mitigation measures for SEAH issues are described in the ESMF that include ensuring that codes of conduct on workers' behaviour and prevention of SEAH are incorporated into bidding and contract documents, awareness raising for stakeholders to understand differentiated SEAH-related risks and mitigation measures, and clauses within contracts requiring compliance with requirements. Subproject level measures will be detailed in the site-specific ESMPs. The AE's institutional-level grievance mechanism will be available for receiving and resolving SEAH-related complaints. Its contact details are included in the programme's stakeholder engagement plan. The mechanism will handle SEAH grievances in a prompt and strictly confidential manner in addition to referring survivors to relevant services such as for medical, psychological, social and legal to support survivors. A programme-level mechanism will also be established, building on mechanisms at community level and the country-level implementation units. Processing procedures are described in the stakeholder engagement plan. In addition, the GCF independent redress mechanism will also be available to stakeholders. Detailed monitoring of SEAH mitigation measures will be specified in further due diligence documentation.

37. **Indigenous Peoples Policy and ESS 7: Indigenous Peoples.** The ethnic minorities planning framework aligns with the Indigenous Peoples Policy. The AE must ensure the engagement of all peoples who would fall under the scope of the Indigenous Peoples Policy (i.e. not limited to the groups enumerated in the ethnic minorities planning framework) during implementation. While the SEP does not explicitly state that representatives of Indigenous Peoples and relevant ethnic minorities were consulted during the programme's design, the ESMF notes that "field consultations in February 2023 included the nomadic Peul pastoralists within Senegal". The SEP commits to paying special attention to Indigenous Peoples when disseminating and disclosing project information and when resolving complaints and grievances. In line with its roles and functions, the Indigenous Peoples Advisory Group is available to provide advice to the AE and executing entities. In line with the Indigenous Peoples

Policy, the GCF Indigenous Peoples focal point will be available for assistance at any stage, including before a claim has been made.

4.2 Gender policy

38. The AE has provided a gender assessment and action plan and therefore complies with the requirements with the GCF updated gender policy. The SURAGGWA Gender Assessment highlights the challenges women and marginalized groups face in the Sahel, especially in agriculture, land restoration, and climate adaptation. Women are burdened by socio-cultural norms and legal frameworks that restrict access to land, financial services, education, extension services, and decision-making spaces. They make up 60 to 80% of the agricultural workforce yet remain disproportionately disadvantaged, owning less land, receiving fewer services, and being underrepresented in leadership. Early marriage, time poverty and limited literacy exacerbate these challenges, particularly in rural areas. Climate change further magnifies women's vulnerability due to their reliance on rain-fed agriculture and increased exposure to risk from resource scarcity and gender-based violence. Customary systems and gender norms also limit women's participation in adaptation efforts and land restoration initiatives, where their roles and perspectives are often overlooked or undervalued.

39. The programme level gender action plan (GAP) outlines a suite of gendered activities and the AE has indicated that country-level GAPs will be prepared by a social safeguards specialist within the first six months of their onboarding. The GAP aims to reduce structural inequalities and empower women to lead and benefit from climate resilience and land restoration efforts. Key actions include promoting women's participation in restoration activities and decision-making, enhancing access to land and finance, and supporting gender-sensitive agricultural value chains. The GAP prioritises capacity-building for women's organisations, incorporation of gender criteria in land governance, and deployment of gender-sensitive monitoring frameworks. It also focuses on increasing women's access to training in sustainable practices and climate adaptation technologies. The GAP includes the collection of sex-disaggregated data and performance indicators.

40. The AE is encouraged to strengthen the programme by integrating gender mainstreaming into the strengthening of legal and institutional frameworks. The AE is also encouraged to further strengthen the programme by ensuring gender expertise is represented in all decision-making bodies of the programme.

4.3 Risks

4.3.1. Accredited entity/executing entity capability to execute the current programme (low risk)

41. FAO is a key technical partner of the African Union and its Member States, which have devised a clear and detailed approach and methodology for solutions in the Sahel. FAO has designed a comprehensive approach for large-scale dryland restoration for Great Green Wall backed by the Global guidelines for the restoration of degraded forests and landscapes.

4.3.2. Programme-specific execution risks (medium risk)

42. Co-financing: considering the sizeable commitment from the local governments and FAO (32 per cent of the total programme), the Secretariat is reassured that total government co-financing is spread among the target countries and that the largest government co-financiers (in terms of the size of the grants), based on their credit rating or International Monetary Fund reports (e.g. Mauritania, Nigeria (Caa1/B-/B), Senegal (B3/B)), have some capacity to absorb

cost increases; the Secretariat is also reassured that the standard provisions on co-financing are included in the term sheet, which mitigates the co-financiers' funding risk.

43. Execution risk: Since fragile countries are participating in the programme, FAO will apply multiple risk mitigation strategies. The programme will be the first multi-country programme with FAO and the largest in the GCF portfolio to date.

44. Survival rates of vegetation in restoration activities: in the funding proposal FAO identified the risk of low survival rates of tree seedlings and identified mitigation measures as agreed with the Secretariat.

45. Operational: there is a risk that regional activities are not delivered according to the funding proposal i.e. not all eight countries are starting implementation at the same time, resulting in unequal participation of the countries and delays in implementation.

4.3.3. Compliance risk (medium risk)

46. The programme's proposed activities, with their focus on land restoration, developing low emission NTFP and fodder value chains, and Great Green Wall regional and national institutional capacity-building, generally present a relatively lower inherent risk of money laundering/terrorist financing (ML/TF). The AE will also act as EE, alongside the eight target country governments, acting through their respective ministries of environment.

47. The AE has assessed inherent ML/TF as high due to jurisdictional and programme-related risks but residual risk as low due to mitigation measures including anti-money laundering/countering the financing of terrorism (AML/CFT) clauses in the project agreements with the eight implementation country governments, as well as close monitoring and supervision during programme implementation through its various offices, including its Africa regional office, sub-regional offices, as well as country offices in the eight target countries. Additional mitigants to decrease the inherent ML/TF risk include the AE's existing fiduciary management and control measures to prevent ML/TF or other prohibited practices. These include annual review and monitoring of a fraud prevention plan by the AE's regional office, as well as a risk log that is reviewed and updated every 6 months. The AE conducted capacity assessments of the relevant ministries of environment of the eight target countries which will serve as EEs and found that they presented low to moderate risks with no evidence of elevated ML/TF risks.

48. Considering the size and scope of this multi-jurisdictional programme, including a large number of counterparties/beneficiaries and relatively high levels of procurement, the compliance risk is determined to be medium.

4.3.4. GCF portfolio concentration risk

49. In the event of approval, the impact of this proposal on the GCF concentration risk remains within the monitoring thresholds of the Risk Appetite Statement in terms of results areas, single proposal or AE concentration.

Summary risk assessment	
Overall programme	Medium to high
Accredited entity/executing entity capability	Low
Programme-specific execution	Medium to high
Compliance	Medium
GCF portfolio concentration	Within the monitoring thresholds

4.4 Fiduciary

50. The fiduciary framework of the SURAGGWA programme is structured to ensure strong financial oversight, effective coordination and accountability across all levels of implementation. FAO will serve as the AE, with functions clearly segregated from its role as an executing entity, in order to avoid any conflict of interest. FAO will assume full fiduciary responsibility for GCF-funded activities, including financial management, disbursement, procurement oversight, programme supervision, monitoring, reporting and evaluation, in accordance with its internal control systems and the Accreditation Master Agreement with the GCF. These functions will be operationalized through a programme task force, drawing from headquarters, regional and country offices to ensure compliance and consistency.

51. The programme will be implemented through a multi-tiered governance and delivery structure. A regional programme management unit, based in the FAO Subregional Office for Africa, will oversee regional coordination, technical backstopping, progress monitoring and harmonization across countries. Country-level implementation will be carried out by country implementation units, hosted in FAO country offices or national GGW agencies, with staff qualified in finance, procurement, safeguards and monitoring. Implementation will be supported by national governments, FAO and selected partners through agreed modalities that reflect national systems and the operational capacity of the FAO.

52. Strong governance and oversight will be provided by a regional programme steering committee, chaired by the African Union Commission, and national steering committees in each of the eight participating countries. These committees will ensure alignment with GGW priorities, facilitate multi-stakeholder engagement and support resource mobilization. The Pan-African Agency for the Great Green Wall will serve as the programme steering committee secretariat and will contribute to the coordination of co-financing and reporting.

53. Fund flows will be governed by a Funded Activity Agreement between FAO and GCF. GCF proceeds will be received by FAO and disbursed to co-executing entities (the Governments of Chad, Djibouti, Mauritania and Senegal) under the terms of subsidiary project agreements, which also stipulate co-financing contributions and responsibilities. This fiduciary structure, combined with the tested financial systems of the FAO, layered oversight and country presence, provides a solid basis for ensuring accountable and transparent use of GCF resources, while enabling delivery in complex multi-country environments.

4.5 Results monitoring and reporting

54. The funding proposal articulates a well-defined causal pathway grounded in a theory of change that emphasizes the role of large-scale land restoration, strengthening of NTFP value chains and enhanced institutional coordination and monitoring systems in building climate resilience across eight Sahelian countries. This pathway is clearly mapped through the logical framework in the funding proposal and aligns with the GCF Integrated Results Management Framework, linking activities to long-term adaptation and mitigation outcomes over the 10-year implementation period.

55. The programme contributes to three GCF results areas, namely forestry and land use for mitigation, and the most vulnerable people and communities and ecosystems and ecosystem services for adaptation. Accordingly, it commits to reporting against key GCF core and supplementary indicators:

(a) Core indicator 1: GHG emissions reduced, avoided or removed/sequestered (target: 65 million tonnes of carbon dioxide equivalent);

(b) Core indicator 2: the number of direct (1.88 million) and indirect (3.86 million) beneficiaries reached;

(c) Core indicator 4: the area of natural resources brought under improved low-emission and/or climate-resilient management (1.27 million hectares); and

(d) Supplementary indicators: the number of beneficiaries adopting climate-resilient innovations (e.g. Supplementary indicator 2.5), among others.

56. These indicators are well-integrated into the logical framework with appropriate baselines, targets, assumptions and means of verification.

57. The monitoring and evaluation plan is comprehensive, as outlined in Annex 11 to the funding proposal. It includes provisions for coordinated monitoring and evaluation plan implementation at both the regional and country level. A dedicated monitoring, evaluation and learning (MEL) specialist will oversee quality assurance, the consolidation of reporting and methodological guidance. At the national level, MEL specialists in the country implementation unit are responsible for regular data collection, use of standardized monitoring tools and field verification. The monitoring and evaluation plan leverages the use of digital platforms to allow systematic tracking of results across outputs and outcomes. In addition, the proposal includes plans for biennial outcome surveys, the integration of remote sensing, normalized difference vegetation index assessments and participatory feedback tools, which will further enhance the robustness of the system.

58. As this is a multi-country programme, the funding proposal recognizes the importance of disaggregating results at the national level to enable clear tracking of impact. The proposal includes country-specific targets for key indicators and outlines mechanisms for collecting and managing these data through national MEL focal points. The consolidated dashboard system is designed to maintain distinct data segments for each participating country, enabling performance tracking and decision-making. This approach aligns with GCF expectations for regional programmes and will be critical in ensuring transparency, accountability and effectiveness in tracking impacts across different country contexts.

59. The MEL system is adequately resourced, with a total allocation of USD 11.6 million for MEL. The plan also incorporates independent mid-term and end-term evaluations, in line with the Evaluation Policy for the GCF. Overall, the Secretariat considers the proposed MEL system and budget sufficient and appropriate to enable effective tracking of results, learning and adaptive management throughout the implementation of the programme.

4.6 Legal assessment

60. The Accreditation Master Agreement was signed with the Accredited Entity on 8 June 2018 (the “AMA”), and it became effective on 4 October 2018. The Accredited Entity’s five-year accreditation term lapsed on 3 October 2023 but was extended by decision B.37/18, paragraph (q), until the earlier of three years from the date of lapse of the five-year accreditation term or the date on which a revised accreditation framework is adopted by the Board. The Board approved the re-accreditation of the Accredited Entity pursuant to decision B.37/18, and negotiation of the Amended and Restated Accreditation Master Agreement is ongoing.

61. The Accredited Entity has provided a certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the programme.

62. The proposed programme will be implemented in eight host countries. The GCF has signed a bilateral agreement on privileges and immunities with one of those countries, namely Burkina Faso. In the other seven countries, namely Chad, Djibouti, Mali, Mauritania, Niger, Nigeria and Senegal, GCF is not provided with privileges and immunities. This means that,

among other things, GCF is not protected against litigation or expropriation in these countries, which needs to be further assessed. Moreover, the ability of the GCF to undertake redress activities and/or investigations in such countries may be hindered owing to the absence of privileges and immunities for relevant GCF personnel.

63. Therefore, it is recommended that the Board considers whether disbursements of GCF proceeds should only be made after GCF has obtained satisfactory protection against litigation and expropriation in the seven countries, or has been provided with appropriate privileges and immunities for the GCF and its personnel.

64. To address the matters raised in this section, it is recommended that any approval by the Board is made subject to the following conditions:

(a) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval; and

(b) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP268

Proposal name:	Scaling-Up Resilience in Africa's Great Green Wall (SURAGGWA)
Accredited entity:	Food and Agriculture Organization of the United Nations (FAO)
Country/(ies):	Burkina Faso, Chad, Djibouti, Mali, Mauritania, Niger, Nigeria, Senegal
Project/programme size:	Medium

I. Assessment of the independent Technical Advisory Panel[1]

1. The Great Green Wall for the Sahara and the Sahel project was adopted by the African Union in 2007. Initially, it aimed at combating desertification and holding back desert expansion by planting a wall of trees stretching across the entire Sahel. With the adoption of the view that desert boundaries change based on rainfall, the aim evolved into restoring vegetation, harvesting water and improving land-use techniques to promote green and productive landscapes across the Sahel, in a mosaic of forested land, interspersed with farmland and grassland.

2. The Sahelian countries are all highly vulnerable to climate change, and their adaptive capacity is limited by low incomes. The proposed programme for "Scaling-Up Resilience in Africa's Great Green Wall" (SURAGGWA) supports the African Union-led initiative to transform Sahelian landscapes by restoring 100 million hectares of degraded land and creating 10 million jobs. SURAGGWA will work with smallholder farmers and pastoralist communities to build resilience and contribute to climate change mitigation through carbon sequestration in restored lands across the eight participating countries (Burkina Faso, Chad, Djibouti, Mali, Mauritania, Niger, Nigeria and Senegal).

3. The cross-cutting SURAGGWA programme seeks to: (i) scale up restoration practices with well-adapted biodiversity as a way to restore degraded land and sequester and store carbon; (ii) develop value chains for climate-resilient and low-emission non-timber forest products (including forage) supporting the livelihoods and food security of vulnerable communities; and (iii) strengthen national and regional Great Green Wall institutions to ensure the sustainability of interventions.

4. With a total budget of USD 222 million, the funding proposal seeks a USD 150 million grant from GCF. The co-funding is composed of cash and in-kind contributions from the participating countries and the accredited entity (AE), the Food and Agriculture Organization (FAO). The project is to be implemented in a 10-year period and has a total lifespan of 20 years.

5. The programme will be executed by FAO and the governments of eight countries as executing entities (EEs). Activities funded by GCF proceeds will be executed by FAO and the Governments of Chad, Djibouti, Mauritania and Senegal while activities co-funded by the governments and FAO will be directly executed by each co-financier.

[1] In its assessment, the independent Technical Advisory Panel (iTAP) considered the document package as submitted on 24 March 2025, along with written responses to questions posed by iTAP discussed in a zoom call between the iTAP, the accredited entity (AE) and the GCF Secretariat on 11 March, complemented with follow-up correspondence and necessary corrections.

1.1 Impact potential *Scale: Medium to high*

6. This cross-cutting project contributes to both mitigation and adaptation results.

7. Greenhouse gas (GHG) sequestration and reduced emissions are anticipated through land-use management and large-scale restoration of degraded drylands and agro-silvo-pastoral systems. This involves a small reduction in emissions through a shift from fuelwood to solar for agroprocessing, and a reduction in wood use by introducing efficient cookstoves. More significant is the highly ambitious target for carbon removal through sequestration in restored lands. The programme aims to bring about: i) a shift over 1,140,513 hectares of moderately degraded farmlands from annual cultivation of cereal crops under full tillage to mixed agroforestry with perennial shrubs and trees, increasing both the level of carbon storage in the soil and the sequestration potential in the living biomass; and ii) a shift over 133,238 hectares of highly degraded savanna land to a productive state with local grass, shrub and tree species (such as Balanites and acacia) and improved soil organic carbon. Land restoration efforts taken together aim to sequester over 65 million tonnes of carbon dioxide equivalent (t CO_2 eq) over the programme's 20-year lifespan. Restoring a total land area of nearly 1.3 million hectares corresponds to an anticipated sequestration of 2.56 t CO_2 eq per hectare and per year.

8. Beyond mitigation of global climate change, the programme is expected to benefit over 5.7 million people, of which nearly 1.9 million directly and close to 3.9 million indirectly. Direct beneficiaries are those persons who actively participate in land restoration activities (Component 1) or non-timber forest products (NTFP) value chain development (Component 2), while indirect beneficiaries reflect the spillover beneficial effects. The assumptions and calculations are not entirely clear to the iTAP but the multipliers and adoption rates are taken to be entirely in line with AE experience.

9. The targeted beneficiaries are agropastoral communities, namely communities engaged in both agriculture and livestock. The programme seeks to stabilize crop yields on moderately degraded farmlands, despite increasing temperatures and changing rainfall patterns. This will be achieved with the application of agroecological approaches that reduce soil erosion and soil fertility loss, enhance soil moisture retention and maintain soil temperatures suitable for crop production, as well as through better adapted crop varieties, and improved livestock productivity through access to increased forage crops. Climate resilience will also be enhanced through the selection of more resilient plant species, creating buffers to wind and soil erosion.

10. SURAGGWA will also introduce new livelihood options like NTFP and related increased employment in the NTFP value chain. These aspects are expected to be particularly important for women who are the majority participants in the collection and sale of these agroforestry products. Enhanced food security is expected as a result of increased incomes as well as increased access to food for consumption.

11. The iTAP has expressed concern regarding the potentially unequal opportunities in benefiting from the programme between semi-nomadic pastoralist and settled agropastoralist communities. The AE assures that semi-nomadic pastoralist communities can also be beneficiaries of the SURAGGWA programme, even though most beneficiary communities are mixed herders and farmers. According to the written question and answer exchange, semi-nomadic pastoralist communities will have access to seed supply systems and be technically supported for propagation of fodder species as feed to benefit their livestock in sylvo-pastoral systems. They will be included in Community Restoration Committees, where relevant, and will benefit from restoration of livestock migration corridors.

12. The iTAP remains concerned about the long-term sociocultural effects of restoration and land use and recommends that the AE, together with regional and academic organizations, follows up on this issue. It is suggested the AE should facilitate and contribute to research exercises in the region, related to land control and potential land-use conflicts between

pastoralists and farmers and their associated lifestyles and livelihoods (see Section II: Overall remarks).

13. The project will work with smallholder rural communities and their organizations, as well as the institutions involved in the monitoring and support of the broader Great Green Wall initiative. Whereas the people of the Sahel are low emitters of GHG, at present and in history, the endeavour nonetheless makes a substantive contribution towards carbon sequestration and climate change mitigation. The impact potential is assessed to be medium to high.

1.2 Paradigm shift potential *Scale: High*

14. With the shift in focus of the Great Green Wall initiative from tree planting to more comprehensive rural development, including improved production systems of smallholder farmers and pastoralist communities to increase food security, create jobs and boost the local economy through sustainable land management, countries are now tackling systemic challenges in an integrated and holistic manner to ensure long-term sustainability in the entire Sahel region. The SURAGGWA programme subscribes to the more recently established and holistic developmental view.

15. The funding proposal highlights capacity-building among local communities, to gain improved understanding of how to implement land rehabilitation investments in highly degraded common lands, and moderately degraded farmlands. Broadly, SURAGGWA promotes management practices that support and assist natural recovery processes, using a range of native species, and the capacities of communities and local authorities to counter the drivers of land degradation.

16. For large-scale land restoration, especially on communal lands, SURAGGWA seeks to use modern equipment for mechanized ploughing: "Barren lands are ploughed using tractors and Delfino ploughs, which work by creating a pattern system of micro-basins and underground storage compartments, mimicking the traditional half-moons applied in the Sahel" (funding proposal, paragraph 51).[2]

17. In relation to equipment/machinery, SURAGGWA seeks to facilitate the development of private sector rental markets in an innovative way. Rather than purchasing the equipment for earth excavation and similar works, the programme seeks to support private sector and local business development by renting the equipment needed. This avoids the issue of 'handover' with related potential problems of maintenance neglect. By hiring others to provide equipment and workforce for the project, SURAGGWA improves the chances for long-term livelihood creation and capacity development.[3] Boosting such producer services in the agricultural/land management sector could qualify as a market development or transformation.

18. In this context, however, it can be noted that the achievements in many parts of the Sahel of farmer-managed natural regeneration are mentioned only once in the funding proposal (in paragraph 214). These farmer-managed approaches include, for example, innovative ways to regenerate and multiply valuable trees whose roots already lay under the land surface. It is important that SURAGGWA remains consistent with its participatory approach and allows farmer-led approaches to flourish and benefit from the programme.

19. As FAO is one of the leading agencies producing research on land restoration methodologies, the iTAP suggests that SURAGGWA could contribute to improving knowledge

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[2] See also: FAO (2017) 'Forest and Landscape Mechanism. Restoring Landscapes for Enhanced Livelihoods': Good Practice 1. Mechanized Microcatchments for Harvesting Water in Degraded Drylands – the Vallerani System.

[3] This novel practice has been tested in the Action Against Desertification project, where FAO has been acquiring and/or renting equipment, such as Delfino units in Burkina Faso, Niger, Nigeria and Senegal. FAO has also hired service providers for mechanical digging and equipment services in Burkina Faso, Niger and Senegal.

management. Given the many different, sometimes competing and contested approaches to, and rationales for, land restoration, locally grounded research needs to be promoted. To some extent this is addressed in Component 3, but the iTAP recommends that the AE further develops this aspect and seeks to stimulate more local academic and action research around the social, cultural and economic effects of different land management techniques in the Sahel (see Section II: Overall remarks). This relates to the importance of maintaining intelligence and deepening the understanding of how land use and land control play out in relation to conflicts in the region.

20. The programme includes work to strengthen engagement with carbon markets, and also to establish a carbon accounting framework in each participating country. SURAGGWA will seek to strengthen countries' understanding of and capacity to pursue carbon finance for landscape restoration through regional training, knowledge-sharing and peer to peer exchanges, in collaboration with regional carbon market initiatives such as the West Africa Alliance on Carbon Markets, the Africa Carbon Markets Initiative and the GCF-funded iGREENFIN initiative. SURAGGWA will train policymakers on carbon finance opportunities for restoration based upon assessment of sequestration potential and existing and potential methodologies in the Voluntary Carbon Market and article 6 of the Paris Agreement of the United Nations Framework Convention on Climate Change (UNFCCC).

21. Linking communities with the growing carbon markets can incentivize the protection and growth of biomass, for example through direct payments for carbon off-sets. It is of key importance, and has paradigm-shift potential, to ensure that those who traditionally and historically have contributed the least to the growing global carbon crisis, can gain economically from contributing to mitigation of the crisis. This includes establishing clear mechanisms for community benefit-sharing from carbon payments, a matter which is not directly addressed in activity 3.3.1 of the funding proposal, but will be critical, especially in Nigeria where the establishment of a carbon accounting framework is being piloted. It is recommended that the AE pay additional attention to the issue of how any benefits from carbon market participation be distributed (see Section II: Overall remarks).

22. Altogether, with a set of novel practices and activities, the potential for paradigm shift of the SURAGGWA programme is assessed to be high.

1.3 Sustainable development *Scale: High*

23. This broad programme contributes to a range of Sustainable Development Goals (SDGs), most notably SDG 1 aimed at eradicating poverty, SDG 2 on reduced hunger, SDG 5 on gender equality, SDG 8 on decent work and economic growth, as well as SDG 13 on climate action, SDG 15 on land-based ecosystem protection, and SDG 17 encouraging partnerships.

24. Environmental co-benefits: SURAGGWA invests in participatory land restoration and subsequent sustainable management of natural resources through its protection of drylands against erosion and desertification. The programme will rehabilitate selected sites through deep ploughs to create micro-catchments through the Vallerani system. This is to improve water infiltration and water retention, while loosening the soil and improving nutrient access. Tree and grass seedlings are trapped in micro-catchments which help build the natural vegetation cover with native trees and grass species, providing habitat for pollinators and other animals in the area, increasing biodiversity.

25. The funding proposal indicates that 25 million of the 100 million people who live in the Sahel rely on pastoralist livelihoods. Poverty is a great concern, and social and political tensions stand to be exacerbated by climate change as well as agricultural intensification and expansion if and when there is competition for and alternative uses of the land. The competition between herders and farmers for resources and land intensifies with drought and soil erosion. In times of

feed scarcity, open conflict may be triggered, for example if livestock stray into fields with standing crops.

26. Social co-benefits are critical for poverty alleviation and stabilization in the area. Water conservation investments are expected to improve food security and crop/pasture yields while conserving the natural vegetation and increasing the resilience of both pastoralists' and farmers' livelihoods. Rehabilitation of degraded pastures will include working with communities to identify and demarcate community pasture areas and the direct seeding of naturalized tree, shrub and herbaceous species. Clarifying use rights and allowing people to grow and use the common lands for alternative feed sources should help reduce conflicts. It is expected that that sedentary mixed herder and farming communities will benefit from SURAGGWA's comprehensive and participatory strategy of community involvement. The transhumant pastoralist communities may be more difficult to reach, which is why the iTAP suggests that the programme invest even further in knowledge management, including independent research collaboration (see Section II: Overall remarks).

27. In order to generate economic co-benefits, income diversification will be critical. It is expected that SURAGGWA will enhance the incomes of smallholder farmers, increase productivity of livestock along with the production of forage and fodder crops, and raise production of crops on degraded farmlands and increase the production of NTFPs.

28. NTFPs such as Balanites, gum arabic, baobab leaves, neem, honey or the sale of seeds and seedlings, stand to provide important income diversification opportunities for local communities, including the local private sector in the SURAGGWA intervention areas. The market linkages will help identify the quality and volume of each of the NTFPs and assist with better collection, storage, transport and processing techniques to enhance the value of the production.

29. According to the gender action plan (annex 8), building on gender assessments in the eight countries, women in the Sahel are disproportionally affected by increasing food insecurity, protracted crises, dire socioeconomic status and deteriorating climate changes and frequent shocks. Women provide 80 per cent of agricultural labour but undertake most of the responsibilities at home. Women in the Sahel contribute from 60 per cent to 80 per cent of the food production through informal jobs and subsistence farming. Often unpaid, or poorly compensated, for much of the work they perform, women in West Africa also remain underrepresented in the political sphere: "Women occupy only 421 seats in West African parliaments, representing 16.1% of all lawmakers" (annex 8, page 15).

30. To produce gender-sensitive development impacts, SURAGGWA focuses on enhancing women's income and employment opportunities through NTFP where mostly women are engaged. Specific actions will be developed to strengthen the technical and managerial capacities of women engaged with livestock, crops and NTFPs. The programme will also challenge some of the deep-seated patriarchal norms and perceptions held about girls and women in the region. Women's knowledge systems and preferences for communication should be built upon when constructing a system of information-sharing in the programme.

31. As outlined in the gender action plan (annex 8), including women in land restoration processes implies taking into consideration limitations in terms of access to extension services and institutional credit, as well as barriers related to land tenure. For example, in some contexts women are denied the right to plant trees or take measures to improve soil fertility. On average, the land areas women own or control tend to be smaller and of lower quality. Understanding the roles and responsibilities of men and women, along with power relations in land management, is a primary requirement for achieving effective outcomes in addressing land degradation.

32. The inclusive and participatory identification of specific intervention sites and a diverse range of locally appropriate native species are very important for the sustainable development

potential. There is also a regionally consolidated approach to knowledge management, ensuring learning across participating countries and regional organizations.

33. Altogether, the sustainable development potential is assessed to be high.

1.4 Needs of the recipient *Scale: High*

34. The participating Sahel countries all include semi-arid areas in the Great Green Wall belt that are highly vulnerable to the effects of climate change. They also have small economies and revenue bases, so have little leeway for managing the effects of climate change, including higher temperatures, prolonged droughts and torrential rainfalls. This is echoed in the Notre Dame-Global Adaptation Index showing high vulnerability to climate disruption and simultaneously a lack of readiness.

35. The debt sustainability profile of the Sahel countries is low and the ratios of debt to gross domestic product (GDP) are very high, according to the funding proposal (paragraph 221). Figure 12 (funding proposal page 77) shows debt-to-GDP ratios of above 50 per cent in all programme countries except Nigeria (37 per cent) and Djibouti (43 per cent). The highest ratio is presented by Senegal, with a debt-to-GDP ratio as high as 73 per cent.

36. The livelihoods of the intended programme beneficiaries, agropastoral and pastoral communities living in semi-arid regions, are extremely vulnerable to climate change and climate variability due to their high exposure to the countries' limited adaptive capacity. Climate change has increased the frequency, intensity and duration of droughts and floods, rises in temperatures, desertification, water stress and soil erosion, resulting in the loss of assets, crops and livestock, disruptions in value chains and soaring food prices. Combined with the impact of the coronavirus disease 2019 in the last few years, these events have pushed millions of smallholder farmers and pastoralists further into poverty.

37. Altogether, the needs of the recipient are assessed to be high. The proposed project is also undertaking many important steps towards addressing those needs, including strengthening of institutional and implementation capacity to restore and effectively manage dryland ecosystems to maximize their role in effective adaptation to climate change.

1.5 Country ownership *Scale: Medium to high*

38. Country ownership and countries' capacity to implement development programmes, may be challenging in the Sahel region. Many countries face, or risk facing, significant political instability. Notwithstanding, the proposed programme presents no-objection letters from all participating countries and confirms to be fully in line with climate change policies and commitments.

39. All the countries in the Sahel have ratified the United Nations Convention on Biological Diversity, the Convention to Combat Desertification, the Framework Convention on Climate Change and the Kyoto Agreement. The countries have also signed and ratified the Paris Agreement on climate change. By ratifying the UNFCCC, these countries have committed to implementing measures to adapt to climate change. In addition to the reports and commitments submitted to the UNFCCC, the Sahel countries have prepared national climate change-related strategies, policies and actions (funding proposal, paragraph 224).

40. While the Sahel countries are all low-emission countries, they have still committed to reduce emissions. At the end of 2015, all the Sahel countries submitted their intended nationally determined contributions, and subsequently their nationally determined contributions, where however they stressed their *vulnerability* to climate change, and especially in relation to food security. Proposed adaptation actions reflect each country's priorities, but there are common

needs for the restoration of degraded lands; afforestation/reforestation/agroforestry and improved pastures; climate-smart conservation agriculture (including improved seeds) and crop diversification; rotational grazing for livestock, enhancing productivity through improved husbandry, feed and animal management practices and reducing methane emissions from livestock; water harvesting and water efficiency enhancement; and early warning systems (funding proposal, paragraph 225).

41. The AE and major executing agency, FAO, sees participation as the precursor to ownership. Hence, the main strategy for building ownership at various levels relies on participation. Paragraph 226 of the funding proposal confirms that "At the project inception and formulation stage, there was close participation through regional and country level workshops, individual interviews, and extensive stakeholder consultations (annex 7)." Annex 7 (Summary of consultations and stakeholder engagement plan) further outlines the process for stakeholder identification during concept development in 2018 and 2019, and subsequent consultations at global, regional, national and local levels. Most local-level consultations, however, will take place during implementation in order to identify specific intervention areas, technologies and communication tools.

42. The consultation strategy, while brief, seems to be well thought out. Risks of conflict, most pertinently around land use, control and ownership, can be mitigated and managed with the aid of tools like the guidance produced by FAO and the United Nations Convention to Combat Desertification on the responsible governance of tenure of land, fisheries and forests.[4]

43. At the national levels, there is also a relatively large involvement of the Governments of Chad, Djibouti, Mauritania and Senegal, whose relevant ministries will act as co-executing agencies, in the management of GCF proceeds. All eight governments will however co-execute the programme since all entities manage the activities financed by their own co-funding.

44. FAO has a solid presence in the region and has devised a clear and detailed approach and methodology for solutions in the Sahel. The country ownership is assessed as medium to high.

1.6 Efficiency and effectiveness *Scale: Medium to high*

45. The AE is requesting 100 per cent concessionality for the SURAGGWA programme, with the full GCF investment in the form of grants. According to the AE (funding proposal, paragraph 188): "The level of concessionality requested for the countries under SURAGGWA is justified based on their economic and environmental vulnerability, the high level of country indebtedness and the low level of readiness to adapt to climate impacts." Arguably, the impact of this programme could have been enhanced by the addition of non-grant instruments, in particular insurance, though the iTAP notes this could not have been done through FAO, which is not accredited with GCF for on-lending and blending.

46. Notwithstanding, while not engaging in the extension of credits and insurance, the programme (under output 2.3) will develop suitable credit and insurance products designed for smallholder NTFP value chain actors, such as micro, small, and medium-sized enterprises and cooperatives, In this context, the iTAP would suggest that the programme be much more systemic in its effort to bring in insurance for the purpose of protecting economic activity and access to finance for these communities, also beyond the programme's lifespan. Insurance products can significantly enhance resilience by protecting communities against climate-related

[4] Technical Guide on the Integration of the Voluntary Guidelines on the Responsible Governance of Tenure of Land, Fisheries and Forests in the Context of National Food Security into the Implementation of the United Nations Convention to Combat Desertification and Land Degradation Neutrality (https://openknowledge.fao.org/handle/20.500.14283/cb9656en)

shocks, thereby safeguarding restoration investments and newly developed value chains such as NTFP value chains.

47. In this context, access to insurance may also improve communities' creditworthiness, thus unlocking additional financial resources to expand sustainable land-use practices and market participation. This can be linked to broader insurance-related initiatives, such as the African Risk Capacity (ARC) which provides sovereign climate risk insurance designed specifically to support African governments in managing the impacts of drought and extreme weather. By rapidly disbursing funds when predefined climate thresholds are crossed, ARC enables swift responses, protecting vulnerable livelihoods and preventing reversals in economic and ecological gains. It could be strategic for SURAGGWA to help leverage established mechanisms like ARC. This is inferred in the recommendation to invest further in knowledge management to find ways to reduce climate risks for farmers and/or pastoralist communities (see Section II: Overall remarks).

48. The coherent structure of the funding proposal and the logical way that activities feed into outputs, which in turn produce outcomes and co-benefits, suggest there is a high likelihood of the project achieving its objectives. This contributes to project efficiency and effectiveness.

49. In respect to long-term sustainability, however, the overwhelming presence of FAO itself as the AE, main EE and implementing agent of a great many of the activities, may be a drawback. The iTAP trusts that the implementation, as designed, will be smooth and effective. However, SURRAGWA, as designed, might not fully exhaust possibilities for strengthening capacities and building ownership among national and regional institutions. However, the lack of long-term local/national institution-building through 'learning by doing' (in programme implementation), would seem to be a problem for sustainable development globally, rather than of the present funding proposal. Overall control by FAO of most parts may in fact enhance the efficiency and effectiveness of the whole endeavour.

50. The cost of per t CO_2 eq is USD 3.42 when calculated on the whole programme cost. The cost is even lower when considered in relation to the 40 per cent of the programme costs nominally associated with the mitigation benefits (see section A.4 of the funding proposal). The expected volume of carbon sequestration through landscape restoration and agroforestry appears ambitious, and it will be interesting to see actual results monitored and reported to the GCF.

51. In spite of operating in a politically risky geographic area, the SURAGGWA programme is well thought through and set up for efficient and effective implementation. The efficiency and effectiveness criterion is assessed as medium to high.

II. Overall remarks from the independent Technical Advisory Panel

52. The programme stands to become an important contribution to the Great Green Wall of Africa initiative, and to climate change adaptation and mitigation in the region. The well-structured programme includes stakeholder participation and safeguards to operate in a region with risks of, or ongoing, conflicts.

53. The iTAP recommends that the Board approve this funding proposal.

54. The iTAP also recommends to the AE that it invest more in knowledge development in two areas:

(a) Dynamics of restoration and land-use changes, to be studied preferably in cooperation with researchers in the Sahel, specifically to document and better understand (i) how different nature-based, labour-intensive and capital-intensive land restoration techniques fare in the long run on farms and communal lands; and (ii) how land restoration initiatives more broadly connect with sociopolitical developments and security threats over time.

(b) Ways to reduce risks and uncertainty for communities, and to increase their potential benefits from new financial instruments like carbon credits. More specifically, it is suggested to work with a wider range of relevant institutions to (i) give much greater emphasis to the development of insurance products among the financial products to be developed under output 2.3; and (ii) establish mechanisms for assuring community benefit-sharing from carbon payments under output 3.3.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP268)

Proposal name:	Scaling-Up Resilience in Africa's Great Green Wall (SURAGGWA)
Accredited entity:	Food and Agriculture Organization of the United Nations (FAO)
Country/(ies):	Burkina Faso, Chad, Djibouti, Mali, Mauritania, Niger, Nigeria, Senegal
Project/programme size:	Medium

Impact potential
The AE acknowledges that the overall impact potential is assessed as **medium to high** by ITAP.
Paradigm shift potential
The AE acknowledges that the overall paradigm shift potential is assessed as **high** by ITAP.
Sustainable development potential
The AE acknowledges that the overall sustainable development potential is assessed as **high** by ITAP.
Needs of the recipient
The AE acknowledges that the overall needs of the recipient(s) is assessed as **high** by ITAP.
Country ownership
The AE acknowledges that the overall country ownership is assessed as **medium to high** by ITAP.
Efficiency and effectiveness
The AE acknowledges that the overall efficiency and effectiveness is assessed as **medium to high** by ITAP.
Overall remarks from the independent Technical Advisory Panel: The AE acknowledges and welcomes the iTAP recommendations regarding increased knowledge development.

The AE confirms that collaboration with well-established research institutes and observatories in the Sahelian region will be prioritised to document experiences regarding the different land restoration models. The AE will continue, during implementation to leverage on the existing partnerships with the West Africa Carbon Market Alliance and the Sahel/GGW Carbon Working Group.

GENDER ASSESSMENT AND ACTION PLAN

Supporting a Funding Proposal

Programme Title:

Scaling-Up Resilience in Africa’s Great Green Wall (SURAGGWA)

Table of Contents

List of Figures

List of Tables

Part One

I. Executive Summary

Women in the Sahel are disproportionally affected by increasing food insecurity, protracted crises, dire socio-economic status in the region and deteriorating climate changes and shocks. Women provide 80% of agricultural labour force in the Sahel and, at the same time, they continue to have most of the responsibilities at home and perform most of the unpaid work. African women in the Sahel contribute from 60% to 80% of the food production, mostly working in informal jobs and subsistence farming in comparison to men.

Discrimination and Inequality: The key gender indicators in the Sahel region show that the discrimination against women within the family ranged from 45% to 88%, based on factors such as legal frameworks for violence against women, the sex ratio at birth and reproductive rights. Socio-economic factors show several gender inequalities in the region, such as low mean age for first marriage, female genital mutilation has a high prevalence, education gap and low representation in decision making and leadership. This suggests a lack of law enforcement, and how customary frameworks can dictate people's behaviour. There is also a significant gender gap in terms of education, indicating the disadvantage women face relative to men for learning opportunities. These inequalities have repercussions on health, nutrition, economic opportunities and independence, decisions making and development among other issues.

Representation and Decision-Making Power: Women continue to have a minor representation in of decision-making and leadership positions in the West Africa region. The lack of women's representation in government has negative impacts not only on policies and programs in place at the national level, but also on changes in societal perceptions. When women are underrepresented in political spheres and decision-making levels, the root causes of gender inequality, which are the social norms and rules that govern the country, cannot be challenged.

Access to Resources: Women face the greatest challenges in accessing productive resources although they play an important role in agriculture in the region. Women and specifically rural women face tremendous inequalities in terms of access to and ownership of land; access to inputs, extension services; financial services; technologies, trainings, and information. Also, women dominate employment in off-farm segments. Yet, they rarely have access to the resources needed to develop their activities. These challenges are some of the main drivers that exacerbate gender inequality and disempowering women to improving their livelihoods. Furthermore, the economies of the region and the region's agriculture production is harmed, and women's fair and profitable involvement in intra-African agrarian commerce, regional, and international agricultural value chains is hampered.

Legal Frameworks and Enabling Environment: There has been an improvement in conditions over time regarding the legal and policy frameworks conditions for women, some countries made more progress than other in the region. Despite these legal improvements, women in the Sahel region continue to face significant challenges, particularly in rural areas where traditional practices and beliefs often limit their freedom and opportunities. In many cases, customary law is recognized by national legal systems and can be used to resolve disputes where there is no formal law in place. However, customary laws can also perpetuate discrimination against women.

Vulnerability to Climate and Gender-Specific Climate Impacts: The vulnerable climatic nature of Sahel region has gender implications that exacerbate women's vulnerability and have consequences not only in women's livelihoods but on food security, malnutrition, weakened adaptive tools, work burden and poverty in rural communities. The extreme weather events can cause extensive crop damage and affect more severely female farmers who are usually deprived of several adaptation mechanisms. An essential factor for adaptability to climate change shocks is the time available. Women tend to balance reproductive roles such as cooking, cleaning, and childrearing in addition to agricultural responsibilities. Rural households may shift labor patterns in response to heat stress in ways that increase women's work burden, i.e. women farmers suffer disproportionally from weather-related shocks by maintaining working levels despite heat events. The time constraints women have can limit their ability to engage and participate in decision-making processes and other activities related to climate change adaptation. Also, it may be possible that women's lack of knowledge on climate change is influenced by gender norms that prevent them from engaging in training where men are the main trainers. Climate-induced migration in rural areas often affects women disproportionately and their ability to cope with climate change shocks. Women farmers are more likely to be left behind and struggle to manage agricultural activities independently.

All of the above circumstances hinder women's ability to invest in sustainable agricultural practices, diversify their income, and build resilience to climate shocks. The limited access to agricultural services also makes women more likely to rely on rain-fed agriculture, particularly vulnerable to droughts and other climate-related hazards.

Women and girls bear a disproportionate brunt of the impacts when land is depleted and limited in availability. This is because of their significant involvement in agriculture and food production, reliance on forests, susceptibility to poverty, lack of education, and lower legal protections and social status. Despite of the recognition and acknowledgment of the interconnectedness between agricultural production, poverty reduction, land preservation, and gender integration, the gender aspect in frequently disregarded in the execution of land-restoration initiatives. As a result, gender aspects are either overlooked or neglected in restoration practice.

Including women in land restoration processes implies taking into consideration their limitations in terms of extension services and institutional credit, and barriers related to land tenure. As in much of low-income countries, women’s ability to own, control, benefit from and access land in the Sahel region is significantly more limited than their male counterparts. Although evidence is still limited, available data from the World Bank show that percentage of land owned by women is always smaller than men in the same countries. On average, the land women own, or control tends to be smaller and lower quality. In many countries, women's access to, use and control of land depends on their relations to male relatives as wives, mothers, or daughters. Understanding the roles and responsibilities of men and women, along with power relations in land management, is a primary requirement to achieving effective outcomes when combating land degradation and implementing gender-responsive and sustainable land-restoration initiatives.

II. SURAGGWA Overview

The eight countries of the Great Green Wall (GGW) are among the world's poorest and most vulnerable to climate change, with women and girls often disproportionately affected. These countries ranked at the bottom of the Human Development Index in 2020, and a large majority of their population lacks access to employment, basic health care, education, or access to natural resources. Agriculture, livestock, and forestry activities form the basis of their economies, with over 70% of rural communities, including many women-led households, depending on rainfed agriculture. The Sahel has experienced some of the most extreme climate events on earth in the 20th century. Long-term climate change trends show higher temperatures, uncertain precipitation, prolonged drought, increased winds, and more frequent extreme events. The degradation of ecosystems and depletion of vegetation and biodiversity has undermined livelihoods, disproportionately affecting women, and increased vulnerability. This has severely impacted the Sahelian agro-silvo-pastoral landscapes, exacerbating food and nutrition insecurity and compromising the sustainability of livelihoods.

The Scaling-Up Resilience in Africa's Great Green Wall (SURAGGWA) programme aims to advance the objectives of the GGW, an African-led initiative to protect the continent from desertification by restoring natural vegetation across an 8000 km area and a 15 km belt. The programme has evolved to focus not only on tree planting but also on holistic development in the area, addressing the unique needs and opportunities of women and girls. SURAGGWA will facilitate a major paradigm shift by building ecological and climate resilience in eight Sahel countries recognized as the most vulnerable to climate change in Africa, including Burkina Faso, Chad, Djibouti, Mali, Mauritania, Niger, Nigeria, and Senegal. Additionally, the programme will build capacities for coordination, planning, monitoring, mobilization of financial resources, and knowledge management. SURAGGWA builds on FAO's previous support to Sahelian countries through the Action Against Desertification (AAD) initiative, the GEF/World Bank, African Development Bank, and FAO's Operation Acacia.

SURAGGWA proposes three key interventions designed to address the technical, organizational, and financial barriers to alleviate climate change impacts and heighten resilience of local communities, including women and girls:

- **Develop a programme to scale-up successful restoration practices**, promote biodiversity, regenerate native species, and sequester carbon, with inclusion of gender-responsive activities.

- **Support the development of climate-resilient, low emission non-timber forest product value chains** benefiting vulnerable communities' livelihoods, with inclusion of women's economic empowerment, and food and nutrition security.
- **Strengthening the Great Green Wall's regional and national institutions** to secure the sustainability of interventions and scale up successful practices, with inclusion of gender-responsive policies and programs.

**For a full programme description, please refer to the Funding Proposal document.*

The programme will work with community organizations to build ownership and sustainability, while strengthening community awareness and capacity in understanding ecological elements of their environment. It will emphasize women's participation and leadership, including training and building awareness on addressing climate change and land degradation issues. By connecting with existing financial institutions and addressing the specific bottlenecks faced by the target groups, including women, in the GGW area, the program will establish linkages with existing providers, create innovative models of value chain financing, and de-risk lending to smallholders in the region.

The transformational activities of SURAGGWA will support the implementation of the GGW initiative led by the African Union, aiming to transform landscapes by restoring 100 million hectares, creating 10 million jobs, and sequestering 250 million tons CO_2 by 2030. The SURAGGWA multi-country programme will make a direct contribution to the GGW targets through the Green Climate Fund's pledge at the One Planet Summit (January 2021) and catalyze changes needed to achieve these ambitious targets. The proposed program will restore approximately 1.334 million ha of degraded drylands, benefiting 7.6 million vulnerable people, including women and girls, in the GGW area, while sequestering 22.4 million tons of CO_2 over 10 years and 65 million tons of CO_2 over the programme lifespan (20 years). The program will make a significant contribution to: (i) the countries' Nationally Determined Contributions (NDC) to climate change adaptation and mitigation; (ii) the Bonn Challenge and associated African Forest Landscape Restoration Initiative (AFR100) in committed countries; (iii) the concerted UN system Action plan in support of the GGW; (iv) the UN Decade on Ecosystem Restoration (2021-2030) with GGW as one of its flagship programmes/projects, as well as associated Sustainable Development Goals, including No Poverty (SDG 1), Zero Hunger (SDG 2), Gender Equality (SDG 5), Decent Work and Economic Growth (SDG 8), Climate Action (SDG 13), Life on Land (SDG 15), and Partnerships (SDG 17). This document and the accompanying Gender Action Plan ensure that women and girls are empowered and benefit equitably from the climate change mitigation and adaptation efforts through the programme.

III. Objectives of the gender assessment

Gender equality is a fundamental requirement for achieving sustainable development and climate resilience. Climate change affects women and men differently due to differences in gender roles, responsibilities, and access to resources. Women often face higher risks and greater burdens, particularly in areas like agriculture and water. In addition, for conflict areas like the Sahel, women-led households – which are common – are particularly vulnerable because of this lack of customary rights to land, wells, and livestock.[1,2] Understanding existing gender disparities is vital for effective climate policies.[3,4] The SURAGGWA programme will address these disparities and provide empowerment opportunities for women to enable communities in the Sahel to sustainably adapt to the challenges posed by climate change.

The GCF recognizes the importance of gender equality and women's empowerment in addressing climate change. As a result, GCF has integrated gender perspectives into its operations, funding, and decision-making processes. The FAO also shares the same view and commitments, as indicated in FAO's Policy on Gender Equality 2020-2030, which states that, *"all men and women, in particular the poor and the vulnerable, have equal rights to economic resources, as well as access to basic services"*.[5] This assessment is designed to align with FAO's Policy on Gender Equality 2020-2030 and the Green Climate Fund (GCF) Gender Policy[6], which mandates the integration of gender considerations into all GCF-funded programmes and projects to promote equal access to resources, benefits, and decision-making opportunities. The GCF Gender Policy emphasizes the need for programmes and projects to be gender-responsive, acknowledging that climate change disproportionately affects women and marginalized groups due to existing social and economic inequalities.[7]

This Gender Analysis/Assessment for the GCF programme, "*Scaling Up Resilience in Africa's Great Green Wall (SURAGGWA)*" initiative aims to identify and address gender-specific barriers and opportunities in the agriculture and rural development sectors within the target countries: **Burkina Faso, Chad, Djibouti, Mali, Mauritania, Niger, Nigeria, and Senegal**. This document, alongside the country-level assessments, provide the foundation for effective gender mainstreaming, providing an overview of the status of gender equality across the eight Sahelian countries covered by the SURAGGWA programme. The following chapters examine gender roles and responsibilities in land restoration and climate change adaptation (Component 1) and in agricultural value chains, especially related to Non-Timber Forest Products (NTFP), where data are available (Component 2). The assessment concludes with recommendations for mainstreaming gender within the programme design and during implementation, via the Gender Action

[1] McOmber, C. (2020), "Women and Climate Change in the Sahel", West African Papers, No. 27, OECD Publishing, Paris. https://doi.org/10.1787/e31c77ad-en
[2] Omolo, N.A. (2010), "Gender and climate change-induced conflict in pastoral communities: Case study of Turkana in northwestern Kenya", African Journals Online, Vol. 10 No. 2, https://doi.org/10.4314/ajcr.v10i2.63312
[3] https://read.oecd-ilibrary.org/development/women-and-climate-change-in-the-sahel_e31c77ad-en
[4] https://www.ajol.info/index.php/ajcr/article/view/63312
[5] FAO (2021). FAO Policy on Gender Equality 2020-2030. Rome.
[6] GCF (2019). Green Climate Fund (GCF) Gender Policy.
[7] GCF (2019). Green Climate Fund (GCF) Gender Policy.

Plan (GAP). The GAP details specific activities to address gender inequalities in the region and empower women and ensures that all activities across the three components reach specific gender targets.

The primary objectives of this Gender Assessment are to:

1. **Identify gender disparities** in climate resilience, agriculture, and rural livelihoods, highlighting the unique challenges faced by women, men, and vulnerable groups in the Great Green Wall region.
2. **Assess institutional and policy frameworks** that support or hinder gender equality in climate adaptation and mitigation efforts.
3. **Provide actionable recommendations** to ensure the SURAGGWA programme integrates gender-responsive strategies that enhance women's empowerment and equitable participation in climate resilience initiatives.
4. **Align with international commitments** such as the United Nations Framework Convention on Climate Change (**UNFCCC**) Gender Action Plan and the Sustainable Development Goals (**SDGs 5 & 13**), reinforcing the role of gender equality in climate adaptation efforts.[8]

Methodology

This Gender Assessment is based on a multi-dimensional approach, drawing from:

(i) **Desk review of:**

a. **Relevant literature on gender, forestry, and climate change** in GGW countries (e.g. existing research, reports, and data sources from the United Nations, World Bank, NGOs, and other development agencies addressing gender equality and climate resilience in the GGW region), particularly eight country-specific gender profiles prepared collaboratively by the Food and Agriculture Organization (FAO) and the Economic Community of West African States (ECOWAS)[9];

b. **National policies and legal frameworks** on gender, agriculture, and climate adaptation in the target countries;

(ii) **Primary data collected through field-level consultations.** Field-level consultations supporting this analysis and the related GAP were held with a variety of stakeholders, including women in local communities; women's associations; civil society organizations; representatives from relevant technical departments of counterpart ministries; the GGW Agency; the National Designated Authority of the GGW; ethnic minority groups (including women from those groups); and other programme teams operating in the area. These consultations (detailed in Annex 7 – Stakeholder Engagement Plan), coupled with the desk review, deepened understanding of the role of women in the GGW area and the constraints women face. The consultations and review helped to identify key gender issues, vulnerabilities, and capacities within the context of land restoration and climate adaptation, as they relate to programme components and activities.

[8] UNFCCC (2020). UNFCCC Gender Action Plan.

[9] Please refer to the eight supplementary country-level gender profiles submitted, prepared by FAO between 2018-2022.

Based on the findings from this gender assessment, a set of recommendations were developed to mainstream gender into the activities in the SURAGGWA programme in a way that responds to gender-specific needs, priorities, and capacities. By conducting this Gender Assessment and developing the related GAP, the SURAGGWA programme will ensure the implementation of gender-transformative strategies, helping bridge gender gaps in agriculture, enhance women's participation in decision-making, and build climate resilience across communities in the Great Green Wall region.

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IV. Regional overview of gender inequalities and the status of women in the Sahel

The Sahel region is a vast semi-arid strip of land that stretched across several countries in West and Central Africa. According to recent estimates the region has a total population of approximately 300 million people, with a slightly higher proportion of men than women. Despite high mortality rates of children under five, the Sahel have some the fastest growth rates (between 2.6% and 3.8%) and it predicted to doubled its population by 2050[10]. The Sahel region has a relatively young population, with a 64.5% of individuals under the age of 25. However, there are some variations between countries and regions within the Sahel with implications on gender. For instance, in areas where there are more girls than boys in the youngest age groups, the maternal mortality rate are higher due to a high prevalence of early marriages coupled with low access to maternal and reproductive health care. In Niger, about 76% of girls are married before the age of 18 and female genital mutilation remains recurrent with prevalence rates of up to 9 out of 10 excised women[11].

Regarding the **socio-economic situation**, countries in the Sahel region share common economic characteristics, yet their national economies vary in size. According to the World Bank classification, the programme countries range from low-income economies to lower-middle income economies, with of the former being among the least developed countries in the world[12]. Agriculture represents the main sector of the regional economy, in terms of employment and value addition, contributing to an estimated 35% of the regional gross domestic product (GDP).

Women in the Sahel are more affected by this socio-economic status due to their engagements in informal and subsistence farming in comparison to men who are more likely to be engaged in wage labor and other formal sector activities. According to reports, women provide 80% of agricultural labour force in the Sahel[13], and at the same time, they continue to have most of the responsibilities at home and perform most of the unpaid work, creating the well-known double burden of work of women. Women are estimated to spend at least three time as many hours as men on household-related activities. In addition, African women in the Sahel contribute from 60% to 80% of the food production. Nonetheless, while being a well-known issue, women face the greatest challenges in accessing productive resources (Njobe & Kaaria). Low production and yields are two of the region's most pressing agricultural challenges. It is recorded by the FAO that women's restricted access to production means and inability to leverage prospects in agriculture is to blame for the sector's poor performance in low and middle-income economies (Njobe & Kaaria). Restricted access in this respect, is faced more exceedingly by women than the men, indicating a strong gender gap. As a result, the region's production is harmed, and women's fair

[10] The World Bank, "Population, total – Burkina Faso, Cameroon, Chad, Gambia, The, Guinea, Mali, Mauritania, Niger, Nigeria, Senegal | Data," 2021. https://data.worldbank.org/ indicator/SP.POP.TOTL?locations=BF-CM-TD-GM-GN-ML-MRNE-NG-SN&most_recent_value_desc=false (accessed Sep. 20, 2021).

[11] Alliance Sahel. (2021). "Gender equality and women's empowerment: a unique opportunity for the Sahel". Accessed March 2023: https://www.alliance-sahel.org/en/news/gender-equality-sahel/

[12] Large parts of the Sahelian populations live below 1,90 US dollar a day

[13] Alliance Sahel (2021). "Gender equality and women's empowerment: a unique opportunity for the Sahel". Accessed March 2023: https://www.alliance-sahel.org/en/news/gender-equality-sahel/

and profitable involvement in intra-African agrarian commerce, regional, and international agricultural value chains is hampered (Njobe & Kaaria).

Despite their distinctive role in productive, generative, and communal management actions, women continue to have a minor representation in of decision-making and leadership positions. Laws and policies that regulate formal institutions, as well as gender and cultural norms continue to prevent women to actively participate in the economic and political scene. This is often related to women's lack of education and literacy but also to gender-based violence at home including forced marriages and sexual exploitation (UNDP, 2022)

Women's participation in climate effort both at the regional and national levels is also very limited (Dankelman, 2010). Climate change can thwart progress towards gender equality by exacerbating poverty, reinforcing traditional patterns of discrimination, and directly disturbing gender-defined livelihoods. This is particularly acute in the Sahel due to its high dependence on rain-fed agriculture and therefore vulnerable to climate change. Farmers face growing challenges in maintaining subsistence agriculture due to recurring periods of droughts, desertification and floods. These extreme weather events can cause extensive crop damage and affect more severely female farmers who are usually deprived of several adaptation mechanisms (UNHCR14).

While local **food production** available for consumption has increased, it remains insufficient in a number of countries and rates of undernourishment and malnutrition remain very high. At the end of 2022, provisional Cadre Harmonisé analyses indicate that 28.9 million people are in need of immediate food assistance in the Sahel and in West Africa. Acute malnutrition among children exceeds the emergency threshold (15%) in Senegal, Mauritania, Niger and northeastern Nigeria. Moreover, in this subregion of West Africa, the obesity prevalence rates in 2014 were 5 percent among men and 15 percent among women (adults) (IFPRI, 2015 cited in FAO, 2017). In nearly all countries in the subregion, the prevalence of obesity increased between 2014 and 2017, leading to elevated risks of diabetes, heart disease and hypertension among men and women alike. Security tensions and violence, inflation, extreme poverty, the widespread surge in cereal and fertilizer prices are exacerbating the food and nutrition crisis. Particularly in Central Sahel countries, the food security and nutrition status of women and children is severely compromised and worsening amid colliding crisis of conflict, climate change, and COVID-19.

Despite all the difficult conditions described above, women continue to play a decisive role in household food security, food diversification and the nutritional status and health of children. In nearly all West African countries, women prepare most of the meals consumed in the household, hence their central role in ensuring a healthy and balanced diet for family members, especially in families living in coastal communities.

14https://www.unhcr.org/61a49df44.pdf

Key gender inequalities in the Sahel region

Like many other African countries, gender equality remains a significant issue in the countries of the Sahel region. To gain a better understanding of the situation faced by women in the region, various gender indicators are used. One such indicator is the Social Institutions Gender Index (SIGI) from OECD Stats portal, which assesses the level of discrimination against women in each country. According to the SIGI 2019 report, all Sahel countries had high or medium SIGI values. Mauritania and Niger have a "very high" level of gender-based discrimination across all four areas measured by the SIGI, with particularly high scores for discrimination in the family and restricted access to productive and financial resources. Senegal has a relatively low overall SIGI score of 37 percent, with higher scores for discrimination in the family, which is also high for all the programme countries. Nigeria and Mali have highest overall scores. And Burkina Faso has the lowest overall SIGI score of 32 percent, indicating lower gender-based discrimination compared to other countries. Djibouti has no data available, but its score for discrimination in the family is high. The discrimination against women within the family ranged from 45% to 88%, based on factors such as legal frameworks for violence against women, the sex ratio at birth, and reproductive rights. For instance, Nigeria, with a SIGI value of 46 percent (high), exhibits a 1.5 female-to-male ratio for time spent on unpaid care work, while 90% of landholders and 66% of bank account holders are male.

Another significant gender indicator is the Global Gender Gap Index, which benchmarks gender parity across four dimensions: economic participation, educational attainment, health, and political empowerment. Table 1 provides rankings for seven out of the eight countries targeted. Chad and Mali have the largest gender gaps in the Sahel region ranking 141st and 142nd, out of 146 countries, globally (World Economic Forum (2022).[15]

Table 1: SIGI Index and Global Gender Gap for Sahel region

Sehal region	SIGI 2019*			Global Gender Gap**	
	SIGI Value	Discrimination in the family	Restricted civil liberties	Global Gender Gap Index 2022	Position out of 146 countries
Burkina Faso	32% Medium	45%	14%	0.659	115
Chad	45% High	56%	27%	0.579	142
Djibouti	na	73%	na	na	na
Mali	46% High	64%	29%	0.601	141
Mauritania	*Very high**	88%	52%	0.606	146
Niger	*Very high**	84%	26%	0.635	128
Nigeria	46% High	55%	54%	0.639	123
Senegal	37% Medium	65%	4%	0.668	112

Note: *OECD (2019). Social Institutions & Gender Index. Available at : https://www.genderindex.org/.

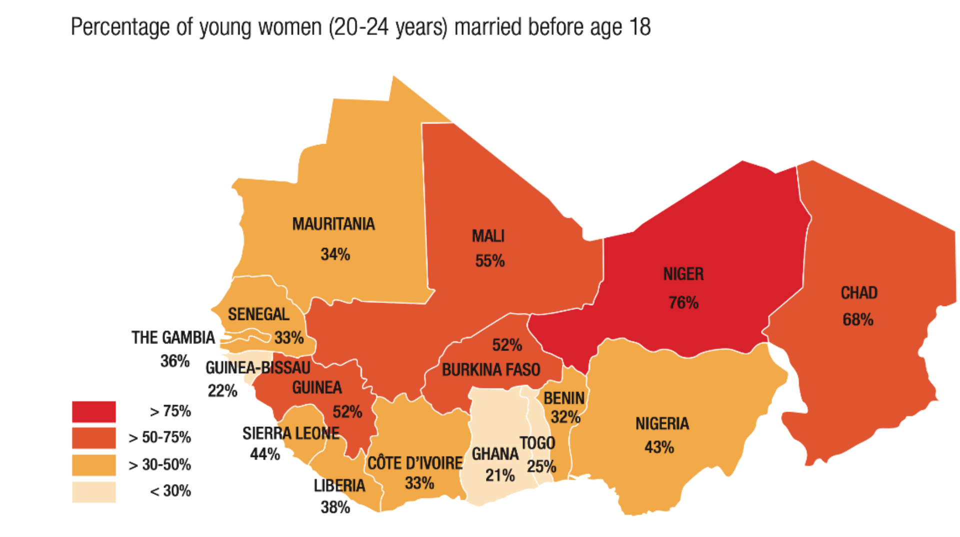
15

**World Economic Forum (2022). Global Gender Gap Report. Available at: https://www3.weforum.org/docs/WEF_GGGR_2022.pdf

Socio-economic indicators

In terms of the mean age for first marriage in the target countries, the percentage of young women (20-24 years) married before age 18 years old varies considerably, ranging from a very high 76% in Niger to a lower 33% in Senegal. For instance in Niger the mean age for first marriage 17 years old, despite of the legal framework which specifies that the minimum age for marriage is 21 years old (Civil Code, Art. 144, 148 and 158). This is because in many cases the people are unaware of the regulations in force, and the customary and religious laws for marriage prevail.

Figure 1: Percentage of young women (20-24 years) married before age 18 years old



Source: OECD

Furthermore, female genital mutilation has a high prevalence among countries in Sahel region. In Djibouti it is a common practice in despite of government's efforts to combat the practice. In fact, in the eight countries female genital mutilation is considered a criminal offense but only in Niger is the percentage of prevalence below five percent. This suggests a lack of law enforcement, and how customary frameworks can dictate people's behaviour. In addition, the adolescent fertility rate is another important concern of the situation of women in the region. In countries such as Mali, there are 162 births per 1,000 adolescent women, having important repercussions on women's health and further on poverty and food security.

There is also a significant gender gap in terms of education. The literacy gender parity index is less than one in all cases, which shows the disadvantage women face relative to men for learning opportunities. Gender gaps in education have repercussions on health, nutrition and the education of other household members. It is also one of the main barriers rural women face in accessing extension services and financial services that are crucial to improving their livelihoods.

Table 2: Status of women in Sahel region: socio-economic indicators

Sahel region	Social Protection	Health/physical integrity		Education	Leadership
	Mean age first marriage	Female genital mutilation prevalence (%	Adolescent fertility rate (Women ages 15-19) is this births per 1 000 women in that age bracket? Pls explain	Literacy rate youth, gender parity index (15-24)	Proportion of seats held by women in national parliaments (%)
Burkina Faso	20	76%	97	0.96	11%
Chad	18	34%	151	0.71	17%
Djibouti	30	95%	17	na	18%
Mali	18	89%	162	0.7	13%
Mauritania	21	64%	67	0.96	23%
Niger	17	2%	177	0.72	16%
Nigeria	21	19%	102	0.84	6%
Senegal	22	25%	67	0.91	43%

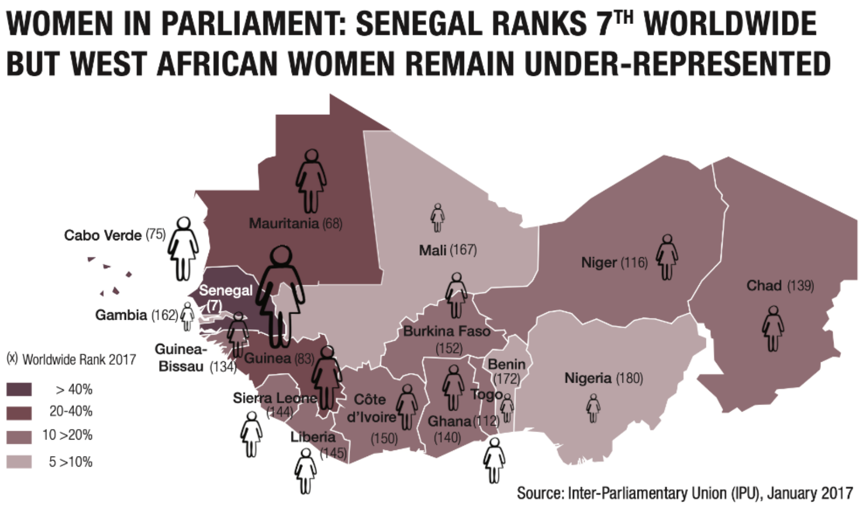
Note: Gender Data Portal of World Bank. Data corresponds to most recent year available. The mean age at marriage, female shows the average length of marriage for women between 15 to 54 years old. Literacy rate is the ratio of female to male ages who can both read and write.

Women's participation in politics and leadership

Women in West Africa remain underrepresented in the political sphere (Figure 2). Women occupy only 421 seats in West African parliaments, representing 16.1% of all lawmakers. In West Africa, 12 out of 17 countries have averages that are blew the world average of 23.3%. Senegal is a notable exception. With women making up 42.7%of its parliament, it ranks as number seven, just behind Sweden. Women occupy on average less than 20% of ministerial posts, and, of those, they are mostly clustered in the ministries covering women's affairs and social issues. Men also outnumber women within ministerial administrations, while women generally occupy secretarial, accounting, human resources and other administrative positions, men hold most technical and managerial positions. To encourage more women to participate in politics, the region adopted the ECOWAS Gender and Election Strategic Framework (GESF) and Action Plan in December 2016. This was followed in February 2017, by a series of recommendations to update the "Supplementary Act relating to Equality of Rights between Men and Women for Sustainable Development in the ECOWAS Region". In terms of top leadership position, the Commission of the African

Union is setting a good example. Its newly elected commission maintains the proper gender balance-four out of eight elected commissioners are women. The previous commission was chaired by Nkosazana Dlamini-Zuma, the first woman to lead the continent-wide organization. Also, the election of the first female head of State in Africa in 2006, the Liberian President Ellen Johnson Sirleaf, inspired many African nations to sought to achieve gender equality and increase women's participation in decision-making and expand women's economic opportunities.[16]

Figure 2: Women in Parliament in West Africa



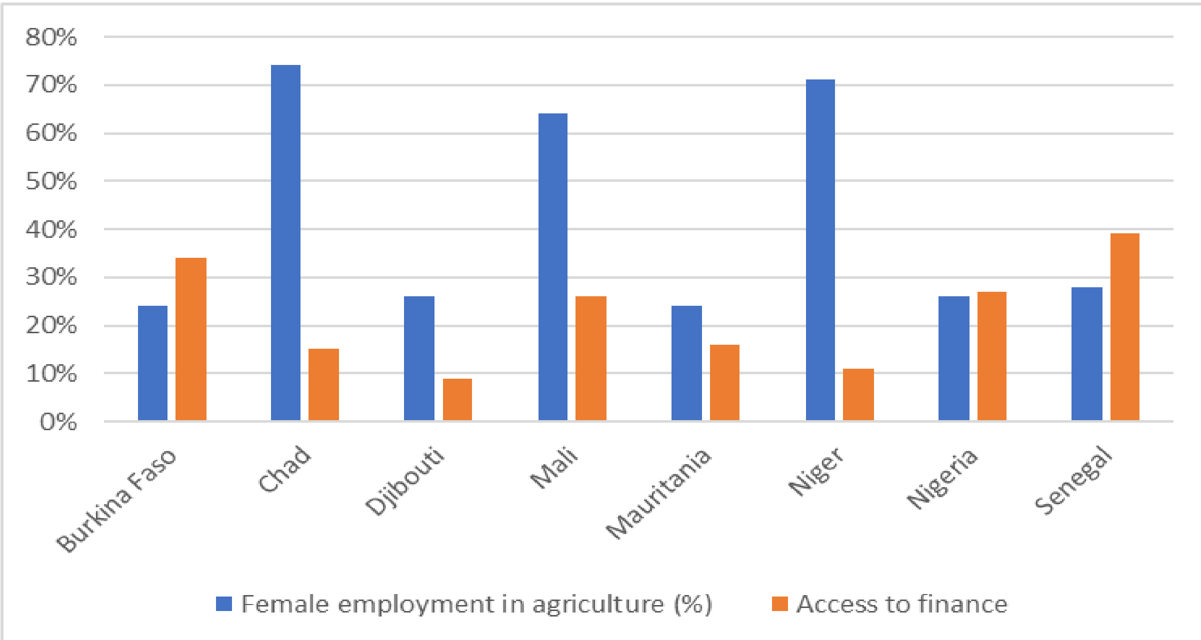
Source: OECD

Figure 2 shows the national seats held by women in each country of this programme, and the differences between the countries, for example in Nigeria only 6% of seats in the parliaments are held by women in comparison with 43% in Senegal. The lack of women's representation in government has negative impacts not only on policies and programs in place at the national level, but also on changes in societal perceptions. When women are underrepresented, the root causes of gender inequality, which are the social norms and rules that govern the country, cannot be challenged.

Female employment and financial inclusion

Figure 3: Female employment in agriculture and access to finance resources

[16] https://www.oecd-ilibrary.org/development/west-african-papers_24142026



Note: Gender Data Portal of World Bank. Data corresponds to most recent year available. Access to finance is measure as percentage of women with account ownership at financial institutions.

As mentioned above, women play an important role in agriculture in the region. However, there are significant disparities from one country to another. For example, in Chad, women account for 74% of employment in agriculture, while in Burkina Faso and Mauritania they account for 24%. And despite women's role in agriculture, very few can access resources such as a bank account in their own name. Interestingly, figure 3 shows that precisely in countries where women's participation in agriculture is higher, the percentage of women who can access a bank account is low. In Mali, for example, women account for 64% of agricultural labor, but only 26% have access to a bank account. This situation jeopardizes women's ability to receive and manage their income.

Women dominate employment in off-farm segments, including food-away-from-home, food processing and food marketing. Yet, they rarely have access to the resources needed to develop their activities. An analysis of the rice sector in Benin, Niger and Nigeria highlights existing gender disparities in trade networks (SWAC/OECD, 2019). Women face a series of obstacles limiting their participation in trade: poorer access to information and markets, male-dominated distribution networks, time and mobility constraints, lower education levels, greater difficulties in complying with regulatory and procedural requirements, etc. Women are thus less likely to hold strategic positions and are less well-connected to central actors within value chains. Supporting employment in food value chains can offer immense opportunities to women and young people. However, more evidence, strategic thinking and integrated approaches are needed for more effective policies.[17]

[17] https://www.oecd.org/swac/maps/Food-systems-Sahel-West-Africa-2021_EN.pdf

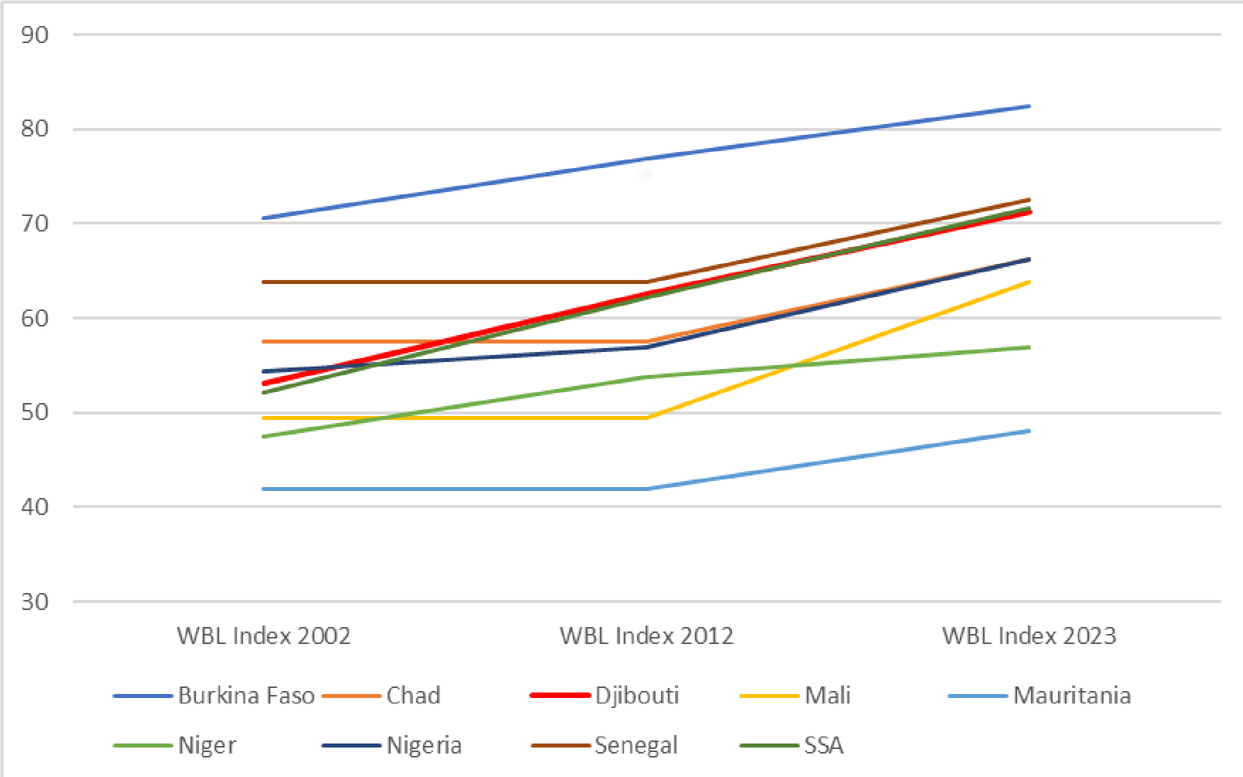
Legal status of women

Policy and legal frameworks that incorporate gender considerations in the design and implementation are fundamental for closing the gender gap in low- and middle-income countries. Evidence suggests that there have been improvements in gender responsiveness in national policies. The Women, Business, and the Law database of the World Bank (WBL) is an indicator for global progress in terms of policy and legal frameworks towards gender equality. This indicator analyzes laws regarding mobility, workplace, pay, marriage, parenthood, entrepreneurship, assets and pension.

In all the countries of the region, there has been an improvement in conditions over time (**Figure 4)**, with the sole exception of Niger. In Niger, there are serious areas for improvement in terms of the legal status of women. For example, to improve on the property indicator, this country could consider granting spouses equal rights to real property, equal inheritance rights for heirs, rights to real property, equalized inheritance rights of sons and daughters, equalized in heritance rights of surviving spouses of male and female surviving spouses, grant spouses equal administrative authority over the property during marriage, and recognize the value of non-monetary contributions.

It is important to highlight the score of Burkina Faso and Senegal, who are the two economies of Sahel with better policy framework conditions for women than the mean of SSA. In Burkina Faso, when it comes to laws affecting women's decisions to work, gender differences in property and inheritance and women's pensions, the improvements have been notable. However, some aspects such as freedom to movements are still affecting gender equality.

Figure 4:Women, Business and Law Indicator for Sahel countries across time



Note: Women, Business and the Law, World Bank Database. Available at: https://wbl.worldbank.org/en/all-topics

Despite some positive development, women in the Sahel region continue to face significant challenges, particularly in rural areas where traditional practices and beliefs often limit their freedom and opportunities. In many cases, customary law is recognized by national legal systems and can be used to resolve disputes where there is no formal law in place. However, customary laws can also perpetuate discrimination against women. As mentioned before, many customary laws contribute to continue practices such as child marriage and female genital mutilation, even though those are considered criminal offenses by the legal statutory systems in the country.

Gender and Climate Adaptation

Climate change is exacerbating gender inequalities.

Gender roles and responsibilities influence the extent to which men and women are affected by climate shocks and stressors. Gender inequality is a critical factor in understanding vulnerability and resilience efforts concerning climate change. Gender differences depend on several factors, such as responsibilities and cultural norms. The Sahel region, characterized by arid and semi-arid conditions, is prone to climate shocks such as droughts, desertification and land degradation. There are important gender implications of these shocks that exacerbate women's vulnerability and have consequences not only in women's livelihoods but in food security, malnutrition and poverty in rural communities. Moreover, social and political factors such as migration, and political instability, present foundational stressors on the region that may intensify due to climate change.

First, since women are often responsible for collecting water and firewood, climate-related shocks resulting in reduced water sources and increased desertification may force women to travel further to collect water. This situation of needing to travel longer distances for water and firewood increases the exposure of women and girls to risks of violence and exploitation, in addition to reducing their time for education, income-generating activities, and caring for their families.

Second, climate shocks such as drought and floods are unfavorable more for women farmers (women in the Sahel region are responsible for 60-80% of food production), who are more likely to grow crops highly vulnerable to climate change. Women's crop failure is expected to result in food insecurity, malnutrition, and poverty for rural households. In addition, recent evidence for Africa suggests that rural households may shift labor patterns in response to heat stress in ways that increase women's work burden, i.e. women farmers suffer disproportionally from weather-related shocks by maintaining working levels despite heat events[18].

Third, as in several African countries, rural women have limited access to agricultural resources and services such as inputs, financial resources, extension services, markets, and land. Likewise, women often hold different roles and responsibilities within the household. Women tend to balance reproductive roles such as cooking, cleaning, and childrearing in addition to agricultural responsibilities. These limitations hinder women's ability to invest in sustainable agricultural practices, diversify their income and build resilience to climate shocks. The limited access to agricultural services also makes women more likely to rely on rain-fed agriculture, particularly vulnerable to droughts and other climate-related hazards.

Moreover, data on mortality and violence following extreme weather events and natural disasters also pointed out that women are more likely to suffer from domestic violence in contexts affected by climate shocks. As climate change exacerbates poverty and food insecurity, it can increase competition over

[18] Azzarri, C., & Nico, G. (2022). "Sex-disaggregated agricultural extension and weather variability in Africa south of the Sahara". World Development, 155, 105897

scarce resources, fueling conflicts and violence, including gender-based violence. This can include cases where women are subjected to violence when they attempt to access natural resources, such as water, or when they engage in agricultural activities that are traditionally seen as male domains.

Women's adaptive capacity

Adapting to climate change depends on people's resilience or adaptive capacities, which are driven by factors such as access to and control over economic and natural resources as well as access to information, education, skills and technologies. As mentioned before, women farmers are more likely to have limited access to these resources than their male counterparts. For instance, women's lack of knowledge about climate change may be due to gender norms that prevent them to engage in training where men are the main trainers.

Another essential factor for adaptability to climate change shocks is the time available. Traditional gender roles and norms often assign women the responsibility for household chores and childcare, limiting their ability to engage and participate in decision-making processes and other activities related to climate change adaptation.

In addition, climate-induced migration in rural areas often affects women disproportionately and their ability to cope with climate change shocks. Women farmers are more likely to be left behind and struggle to manage agricultural activities independently, given the limitations they face due to the factors mentioned above. This has consequences on agricultural productivity and food security. Finally, women are less represented in decision-making within the household and at the community levels. This limits women's ability to advocate for their needs and influence policies affecting their livelihoods.

Several stakeholders have turned their focus to improving gender equality as a central intervention to build climate resilience in the Sahel. **A gendered approach to resilience building focuses on the underlying structural inequalities that make women more vulnerable to climate crises.** Because deeply embedded structural and normative codes of exclusion shape gender inequalities, development institutions are beginning to adopt an approach called gender transformation[19]. To overcome these challenges, it is crucial to promote gender-sensitive policies and programs that address the underlying causes of gender inequality in the agricultural sector. This includes **gender transformative approaches that increase women's access to resources and services, promote gender-responsive agricultural practices, and enhance women's participation in decision-making processes related to climate change adaptation.**

Women and land restoration in Sahel region

[19] McOmber, C. (2020), "Women and Climate Change in the Sahel", West African Papers, No. 27, OECD Publishing, Paris. https://doi.org/10.1787/e31c77ad-en

As mentioned in the previous section, the Sahel is characterized by several vulnerabilities to climate shocks. In response to these challenges, women can play a central role in programmes that aim to restore degraded land. While in the long run the most cost-effective approach to reducing land degradation is to prevent it from occurring, recent findings show that the world is falling short of its targets to combat land degradation and biodiversity loss. Land restoration has been proposed as an important measure since it also contributes to food, and water security. Since women are often responsible for the collection of firewood, water and other natural resources, thus they have a deep understanding of the local environment and the challenges facing it.[20]

Land-restoration programmes or initiatives should not be assumed to be gender-neutral. They should, whether new or old be screened for possible gender-differentiated impacts using sex-disaggregated data. Given the complexity of restoration programmes and projects, their successful implementation requires careful planning and implementation involving all relevant stakeholders[21].

Women and girls bear a disproportionate impact when land is depleted ad limited in availability. This is because of their significant involvement in agriculture and food production, reliance on forests, susceptibility to poverty, lack of education, and lower legal protections and social status. Despite of the recognition and acknowledgment of the interconnectedness between agricultural production, poverty reduction, land preservation, and gender integration, the gender aspect in frequently disregarded in the execution of land-restoration initiatives. As a result, gender aspects are either overlooked or neglected in restoration practice.

Including women in land restoration processes implies taking into consideration their limitations in terms of extension services and institutional credit, and barriers related to land tenure. As in much of low-income countries, women's ability to own, control, benefit from and access land in the Sahel region is significantly more limited than their male counterparts. Although evidence is still limited, available data from the World Bank show that percentage of land owned by women is always smaller than men in the same countries (Figure 5). On average, the land women own, or control tends to be smaller and lower quality. In many countries, women's access to, use and control of land depends on their relations to male relatives as wives, mothers, or daughters[22]. Understanding the roles and responsibilities of men and women, along with power relations in land management, is a primary requirement to achieving effective outcomes when combating land degradation and implementing gender-responsive and sustainable land-restoration initiatives.

In addition, customary systems in many societies may prevent women's participation. For example, in some contexts women are denied the right to plant trees, control soil degradation and take measures to improve soil fertility.

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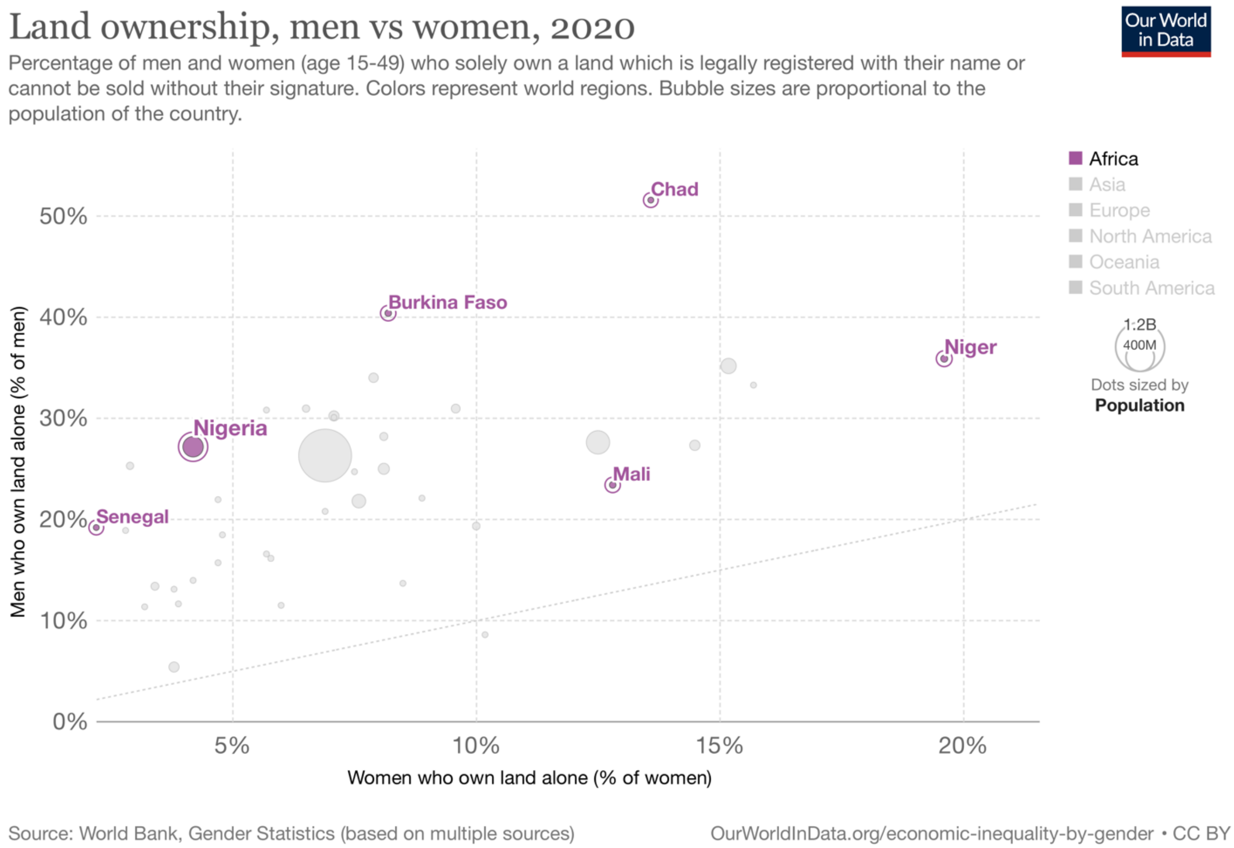
[20] Namubiru-Mwaura, E. (2021). "Gender and Land Restoration". UNCCD Global Land Outlook Working Paper. Bonn.
[21] UNEP (2019). "Women in the Sahel lead on the frontlines of climate change"
[22] SIDA (2015). "Women and Land Rights Brief". Accessed March 2023: https://cdn.sida.se/publications/files/-women-and-land-rights.pdf

Figure 5: Land Ownership, men vs women



Note: World Bank Database. Available at: https://ourworldindata.org/grapher/land-ownership-men-vs-women?zoomToSelection=true&country=BFA~TCD~MLI~NER~NGA~SEN

V. Country Profiles

a) Burkina Faso

Burkina Faso is a low-income, agricultural-based country with a high illiteracy rate and poor housing conditions, ranking low on the Human Development Index, the Institutions Index and Gender, and the Global Innovation Index. There are deep inequalities between men and women, and social institutions discriminate against women. However, the National Economic and Social Development Plan (PNDES) adopted in 2016 aims to reduce social inequalities and increase the number of women entrepreneurs. The Presidential Program also emphasizes the promotion and involvement of women in economic, social, and political life.

Burkina Faso has ratified various international conventions, including the Convention on the Elimination of All Forms of Discrimination against Women and the Sustainable Development Goals. The national legal framework prohibits discrimination and establishes equal rights for women, but customary law often prevails, which is less favorable to women. The country has a policy framework for gender mainstreaming in agricultural policies, including the National Gender Strategy for 2020-2024, and various gender and agriculture action plans. However, gender issues are not adequately integrated into agricultural policies and programmes, budgeting, and indicators.

The Rural Sector Development Strategy (2016-2025) is an exception, which includes a guiding principle on gender and several priority actions for the agriculture sector. However, other sectoral policies, such as the National Rural Land Security Policy incorporates gender and women's rights, including quota measures to promote women's access to irrigated land, and mentions the participation of women's organizations in local processes. The water policy also includes principles of equality and gender but does not sufficiently analyze the relationships between men, women, and water.

Women play a vital role in agriculture, food security, and employment, but they face significant inequalities and disparities. They have less access to natural resources, means of production, and new technologies. They also have a higher illiteracy rate, and female-headed households have limited access to water and energy resources. Women primarily engage in cash crops, livestock breeding, and non-timber forest product development, and they are responsible for the majority of household work.

Livestock is another important sector of the economy in Burkina Faso and is a significant source of income for households, especially for women. Women are mainly engaged in raising pigs, cattle, and goats, and they hold the bulk of the pig herd at the national level. Women also contribute significantly to feeding animals, but their contribution is less on surveillance and veterinary care activities. Women's involvement in livestock activities varies across regions, with some regions having a higher proportion of women in the labor force than others. Livestock breeding is predominantly male, with men owning a larger proportion of livestock than women.

Women also play a significant role in the exploitation of non-timber forest products from communal forests, which are crucial for their income and family needs. However, they are underrepresented in productive aspects of arboriculture due to difficult access to land ownership. Most female-headed households still use solid fuels for cooking, indicating poverty, while the preservation of the environment involves the use of non-solid fuels that are more expensive. The National Action Program to Combat Desertification pays particular attention to gender issues, but land restoration programs assume access to lands is secured, which is a crucial issue for women. The Forest Code recognizes different regimes for forest domains, and the development of local land charters is supposed to contribute to the empowerment and participation of local populations in the management of land and natural resources, but there is no specificity about the inclusion of women.

Household survey from 2014 suggests that access to formal financial services such as bank accounts and credit remains low for households in the lower quintiles (poor households), with only 2% of adults having a bank account. Women tend to save more in small structures or rely on other people, while men borrow more frequently. Lack of supply and credit structures, as well as the complexity of procedures for obtaining credit, are the main constraints to accessing agricultural credit, particularly for women. Credit organizations are often inaccessible to women due to the requirements for guarantees or goods, low literacy, and their sector of activity being considered at risk. While there are credit institutions specific to women and NGOs that can help, they often provide loans that are too small to meet their agriculture investment needs. Such credit institutions are also located far from the poor rural areas where the supply of credit remains insufficient.

Despite the challenges, several good practices and initiatives promote women empowerment, such as school canteens, shea sector development, and credit warrantee programs. Efforts in land tenure have also recognized women's property and use rights, advocating the participation of women's associations. Promising sectors, such as the agrifood sector, are getting organized, with the consideration of gender within the “*Confédération paysanne*” and the “*laboratoires de savoirs locaux*”.

Burkina Faso has taken several measures to address these issues and advance gender equality and empower women, including the creation of a Ministry in charge of the Advancement of Women and Gender and a National Council for the Promotion of Gender. Gender units have also been established in all ministries and public institutions. However, financial and technical resources remain insufficient, and women are underrepresented at higher levels of government and within the agricultural sector.

Overall, while there have been efforts to advance gender equality and empower women in Burkina Faso, there is still much work to be done to address the inequalities and disparities they face, particularly in the agricultural sector. Additional financial and technical resources, as well as greater representation of women in decision-making positions, are crucial to achieving gender equality and promoting sustainable development in the country.

b) Chad

Chad's political, legal, and socio-economic environment is evolving in line with international, regional, and national human rights and gender commitments. The country is committed to promoting gender and women's empowerment through the Millennium Development Goals (MDGs) and the Sustainable Development Goals (SDGs), with mixed results in reducing social inequalities. In fact, the National Gender Policy (PNG) envisions Chad to be free from all forms of inequality and gender inequities by 2063. Sectoral programs and strategies also provide opportunities to address social and gender inequality in rural development through various interventions. Legal frameworks prohibiting child marriage and protecting women's rights further support gender equality.

However, institutional challenges remain, as the Ministry of Women struggles to systematize gender integration across agricultural sub-sectors. Performance indicators in social sectors reveal significant challenges in promoting gender equality in agriculture and rural development. Gender analysis of agricultural policy reveals weaknesses in promoting gender equality and empowering women. Issues include a lack of gender-sensitive planning, inadequate accountability mechanisms, and insufficient collaboration between the Ministry of Women and rural development stakeholders.

Overall, Chad still faces significant gender inequality, ranking 165th out of 191 countries in the 2021 GII. Socio-economic disparities affect various aspects of life, including health, education, and employment.

The country has one of the highest adolescent fertility rates globally and limited access to health services, leading to high infant mortality rates. Educational disparities persist, with lower literacy and schooling rates for women, particularly in rural areas. In employment, women earn less than men and are underrepresented in non-agricultural sectors and political representation.

In addition, Chad faces food and nutrition security challenges due to population growth, conflicts, and climate change. Roughly 47% of the population living below the poverty line, with poverty being more prevalent in rural areas. Food availability and consumption are also affected by climatic shocks, resulting in persistent food insecurity and malnutrition. The education level of mothers greatly impacts child nutrition, with children of uneducated mothers experiencing higher rates of chronic malnutrition. The country also struggles to support displaced populations and refugees, putting pressure on resources such as water and land.

The agriculture sector employs 65% of the working population in Chad, with cereals being the primary food source. Oilseed production, dominated by groundnuts and sesame, serves as cash crops, while legumes, root crops, and tubers contribute to household food security. Women play a crucial role in agricultural production and livestock farming, but their contributions are often undervalued. Women's associations and groups demonstrate their efforts in the field, but women do not fully benefit from the outcomes of their work.

Livestock also contributes significantly to the country’s economy, representing 53% of rural sector GDP and supporting 40% of the rural population. The sector generates substantial revenue and is a major component of Chad's exports. However, there is an imbalance in the distribution of resources and responsibilities between men and women in livestock farming. Men primarily tend to large livestock, while women manage small ruminants and poultry and perform tasks like milking and selling milk. Despite their involvement, women often don't control the income generated from these activities.

In fisheries and aquaculture sector, many fishermen are involved, contributing to the country's food supply. However, the sector is also marked by gender inequality, with men dominating fishing activities and controlling the resources generated from sales. Women are mainly involved in the processing, conservation, and marketing of fish products. Access to these resources can be a source of conflict, and women may face obstacles in the post-harvest phase due to poor application of regulations and conservation measures.

The forestry sector is also important in Chad, with forests covering around 12 million hectares and providing essential resources for rural communities. Access to these resources, such as wood products and non-timber species, varies depending on the agro-ecological zone and socio-cultural context. While both men and women have traditional rights of use, women are primarily responsible for supplying households with wood for cooking. However, government bans on logging - and abuses by enforcement agents - can limit women's access to these resources, affecting their livelihoods. Non-timber resources, like shea, can serve as a source of income, with women being particularly involved in harvesting, processing, and marketing.

In agricultural value chains overall, women's participation is characterized by traditional and artisanal methods, informal channels, and low added value. They are mostly involved in sectors such as peanuts, sesame, fishing, vegetables, milk and dairy products, shea, and gum arabic. In the Sahelian zone, women are primarily responsible for collecting gum arabic, but they have limited representation in marketing. In the Sudanian zone, women collectors of shea nuts have formed associations, strengthening their position in the value chain.

In conclusion, while agriculture and rural development are priority sectors in Chad, social inequality remains a challenge. Interventions targeting women exist and should be sustained, with a focus on equal access to resources and reducing social inequalities in agro-sylvo-pastoral sub-sectors.

c) Djibouti

With a population consisting of 46.21% women, Djibouti experiences gender disparities in urban and rural settings, with more women (51.2%) constituting rural populations. Unemployment affects more women than men, with 70.1% of urban and 64.6% of rural unemployed being women. Differences in access to resources, services, and activities contribute to these disparities, leading to a commitment to equity by public authorities.

Djibouti has established a regulatory and legislative system to promote women's empowerment, including the Poverty Reduction Strategy Paper (2004-2006), the National Initiative for Social Development (2008-2012), the Government Action Plan (2009-2011), and the “Vision Djibouti 2035” strategy. There is also the National Gender Policy (2011-2021) which aims to eliminate gender gaps and establish sustainable development by eradicating gender inequalities. In addition, the country is a party to several international treaties and conventions on gender.

However, women's access to resources and assets remains limited, despite their significant role in creating national wealth. Improved access to production factors would increase women's productivity and contribute to national growth. But the lack of reliable statistical data could hinder the monitoring of gender indicators in the development and implementation of gender-promotion strategies, policies, and programs.

These inequalities are affecting food security in the country, with the country ranking 90th out of 121 on the Global Hunger Index. In Djibouti, 14.5% of households are food insecure, with significant regional differences. Rural areas have higher rates of food insecurity compared to urban areas. While households headed by men in rural areas are more vulnerable to food insecurity, severe food insecurity affects more women-led households. Malnutrition is another significant issue among women of reproductive age, with 14.3% suffering from wasting. Anemia also remains prevalent among women, while overweight and obesity rates are increasing. The main causes of food insecurity in Djibouti include poverty at the household level, weak redistribution, and difficulty accessing basic social services. Women play a crucial role in household food and nutrition security, because they are responsible for domestic tasks related to food preparation and care of children.

Agricultural production in Djibouti is limited due to geo-climatic reasons, and it is a male-dominated sector. The last agricultural census in 1995 revealed only 5% of farms managed by women. Women mostly engage in support tasks, market gardening, and poultry and small ruminant farming, but face challenges in yield, transport, marketing, and access to resources, with a recent trend towards responsibility in market gardening and agricultural by-product marketing.

The country’s agricultural potential relies on oasis agriculture, with only 10% of the 10,000 hectares of cultivable land can be cultivated and easily irrigated. Oasis agriculture is practiced by 1,700 smallholders in Djibouti, using crop association and traditional wells. Agricultural land management varies between pastoral, peri-urban, and pilot community perimeters, with the latter set up by the state for former pastoralists. Rural areas lack industrial equipment and transportation means but have more agricultural land and livestock compared to urban areas.

On livestock, Djibouti has a pastoral tradition with 90% of farms being extensive and itinerant. Sedentary livestock farming, practiced in oases or near cities, is more profitable but faces fodder shortages. A recent survey identified 244 dairy farmers, with 22-48% being women. Among the agricultural households surveyed, 42% practice livestock farming (predominantly goats), with women comprising 40% of these farmers. Among these livestock farmers, 52% say women are responsible for the animals' care, while 37% say it's the husband's duty, 8% attribute it to children, and 4% to a third person.

In fisheries, Djibouti's underused fishing potential is expected to contribute to food security and job creation for youth and women. The government, with FAO's assistance, is working to improve fisheries governance and develop marine aquaculture. Men mostly handle production, while women are responsible for marketing fish. The government aims to settle coastal families to encourage women's income-generating activities and strengthen their commercial capacities.

On the other hand, Djibouti has limited and ecologically fragile forest areas, with 70,000 ha of wooded land, including 22,000 ha of forest formations and 48,000 ha of steppe, tree, and shrub formations. Degradation of these areas is due to natural and anthropogenic factors. The forestry sector contributes to livelihoods and economic development, providing resources such as pasture, fodder, food, medicines, and building materials. Women collect firewood, while charcoal production is a male activity. Data on women's participation in the forestry sector is currently limited, making it difficult to assess their needs, concerns, advantages, and constraints.

Addressing gender inequalities in agricultural value chains is important for three reasons: (i) business opportunities are missed when women's roles are not recognized, (ii) social justice requires equal benefits for men and women from development activities, and (iii) women are crucial for achieving poverty reduction objectives. In Djibouti, the agricultural sector is inefficient, weakly integrated into markets, and offers little local processing, with women producers economically disadvantaged. Agricultural value chains can improve this situation, and efforts have been made to develop promising subsectors, including meat and derivatives, fishery product exports, and date palm cultivation.

d) Mali

Most of Mali's population works in agriculture, mainly family and rural farming, involving socio-economic entities with family relationships. Family members pool their knowledge, skills, and resources to prioritize household needs and produce wealth through surplus marketing. This differs from private agricultural companies that prioritize wealth generation for promoters.

The Malian government recognizes the importance of this agriculture sector and aims to increase production, productivity, and diversification. To support the sector, the government encourages private sector involvement and secure land tenure. Various policies and strategies have been implemented, such as the Development Policy (PDA), National Investment Plan for Agriculture (PNISA), and sub-sectoral policies.

Mali has initiated actions to support women's rights, including legislative, regulatory, and administrative measures that promote gender equality. The country is a stakeholder in ECOWAS instruments on gender equality and women's empowerment. Existing reforms include public finance, gender-responsive planning and budgeting, and sector-specific gender policies. A support Fund for the Empowerment of Women and the Development of Children (FAFE) was also created to finance female entrepreneurship in Mali. Furthermore, the country has been party to the Convention on the Elimination of Discrimination against Women (CEDAW) since 1985 and ratified its additional Protocol in 2000.

However, Malian women still face discrimination due to poverty and cultural and religious traditions. Mali ranks low in human development, and Islam dominates its religious landscape, with around 90% of the population being Muslim. Authorities often use religion and customs to justify the lack of reforms to end gender discrimination, particularly in agriculture.

Both men and women contribute significantly to agriculture and rural enterprises. Women also play a crucial role in fighting rural food insecurities as they are primarily responsible for food selection, preparation, and childcare. However, women still face discrimination in land management, often due to socio-cultural factors, as they cannot inherit land in many communities. This hinders their fight against food insecurity. Despite being present throughout the agricultural chain, women lack the necessary resources to increase production and productivity. Their involvement in food production and access to land is essential for tackling food security challenges.

Mali's land policies have continued to evolve, with the adoption of the new agricultural land policy (PFA) in 2014. However, despite a National Water Policy, access to drinking water remains a challenge, particularly for women and girls, who bear the brunt of water-related chores. Efforts to improve women's access to and control of water points are essential for gender equality in the country.

Malian women encounter challenges in accessing land, leading to their involvement mainly in groundnut and rice production. Factors contributing to this discrimination include traditional customs, household duties, and low-income levels. Although some initiatives have aimed to improve women's access to resources and land, their access to motorized equipment, credit, and agricultural extension services remains limited. The policy developments witnessed in the country since 2015, such as the Agricultural Land Law (“Loi fonciere agricole”), aim to promote women and youth access to land and resources.

Furthermore, access to agricultural equipment has a significant impact on productivity and production in Mali, but fewer women own equipment due to their lower income and limited access to agricultural credit. Women's farms are often under-equipped compared to those of men, which can cause delays in their cropping schedule and lower their productivity. The use of inputs like fertilizers is also essential for increasing productivity, but women often use less input than men. In cotton-growing areas, women also benefit less from equipment as they are less represented in this activity. However, collective activities or joint ventures can improve their access to inputs and other materials.

For livestock, the involvement of different social groups is often influenced by ethnicity, with the possession of animals seen as a source of prestige for some ethnic groups, such as the Fulani. Animal care and veterinary follow-up are typically considered male tasks, while women are more involved in the breeding of small ruminants such as sheep and goats. Women also play a role in the sale of livestock products such as milk, and the breeding of pigs is increasingly popular among women due to the low input required and high reproduction rate.

In terms of inland fisheries, it is important to note that Mali is the third largest producer of freshwater fish in Africa and fishing plays an important role in the country's economy, with 3.6% of the rural population engaged in fishing activities. Men are typically responsible for catching fish while women handle processing and marketing. However, aquaculture remains relatively inexperienced in Mali, with the first aquaculture development plan being funded by USAID in the early 1980s to introduce farmers to fish farming due to deficits in fish production caused by persistent drought.

Finally, forest resources, such as fruits, leaves, and firewood, are another important natural resource, particularly for women and young people who supply families with these products and carry out income-generating activities using forest by-products. The management and conservation of natural resources and the environment are enshrined in the Malian Constitution, but the participation of women in reforestation programs remains low. Factors contributing to this include socio-cultural factors, lack of knowledge of land use rights, limitations on female mobility and communication with male technicians, illiteracy, and low levels of schooling for girls, and a lack of interest or inability of technical supervision staff to address women directly and understand their practices. Efforts have been made to involve women in forestry management, such as through support for women's groups and collective tree plantations, but further action is needed to achieve gender equality in the sector.

e) Mauritania

The latest census of Mauritania in 2013 revealed that the country's population is characterized by a significant proportion of women, with a percentage of 50.8, but they remain marginalized in the rural sector, particularly in terms of access to land. This situation is due to the patriarchal culture that exists in the country and the residues of its matrilineal/matriarchal culture, which causes confusion about gender-specific relations in society.

Women in rural areas struggle to reconstitute themselves in labor in the various sectors of agriculture, trade, and fishing, which leads to their limited participation in decision-making in family and community. Moreover, women in Mauritania remain marginalized in several socio-economic areas, particularly in rural areas. For instance, in Brakna (southern Mauritania), women represent at most 30 percent of landholders, and only 2.6 percent in Trarza, while men represent on average more than 95 percent of title holders in the valley area. Additionally, women landowners usually hold provisional concessions, as land titles as legal acts are rare in the country.

However, agriculture is the main economic activity for rural women in Mauritania, employing more than 90 percent of working women in rural areas. Women participate in all phases of the agricultural cycle, but their contribution to agricultural production, processing, and preservation does not guarantee their food and nutrition security, with women and children being more likely to face malnutrition. Again this is all attributed to the patriarchal nature of rural societies, land insecurity, and limited involvement of women in decision-making. All of this limits women's participation in the fight against food insecurity in the poorest households and regions.

Women also play a significant role in all activities related to livestock, including traditional transhumance activities, dairy product processing, and small stockbreeding. In fact, women make up over 60% of members in local pastoral cooperative associations and lead some associations. However, men mostly hold leadership positions in these organizations.

Another important economic activity for rural women in Mauritania is the marine and inland fisheries. They play a significant role in the value chain that starts from fishing and ends with the sale of fish in the market. While women do not participate in “actual” fishing activities, they are involved in fish landing ports, processing, and sale of fish. However, women face various challenges, including limited access to resources, training, and markets, which could limit the growth and potential of the sector.

In the forestry and environmental sector, women are involved in activities such as picking fruits and leaves but are often excluded from reforestation programmes and projects. Rural women are also vulnerable to environmental degradation and natural disasters but have been historically sidelined in decision-making processes related to climate change.

Finally, the lack of access to credit, training, basic social services, and financial resources further exacerbates the situation for rural women. The limited access to credit, in particular, leads to their inability to use and conserve plots of land. Despite their efforts to derive income from the land, rural women in

Mauritania have limited participation in decision-making, and they face constraints to full involvement in agro-sylvo-pastoral activities.

To address these issues, Mauritania has developed various policies and plans broadly aimed at improving agriculture, rural conditions, and food security, including the National Agricultural Development Plan (2015-2025), Rural Sector Development Strategy for 2025, National Livestock Development Program (2018-2025), and Agropastoral Orientation Law. These policies attempt to integrate the gender aspect, but they are generally not based on prior gender analyses.

f) Niger

According to the gender assessment of the agriculture and rural development sectors in Niger – a 2022 study conducted by Food and Agriculture Organization (FAO) and the ECOWAS Commission – the country has made significant efforts to promote gender equality and combat discrimination through the adoption of various national and international policies and conventions. However, empirical observations and evaluation studies reveal persistent gender inequalities, with the country's Gender Inequality Index (GII) higher than the sub-Saharan Africa average.

Despite the stated commitment to gender equality in public policies, integrating a gender perspective remains inadequate in agricultural and rural development programs, such as the "*Initiative les Nigériens Nourrissent les Nigériens*" (I3N) and "*Programme National d'Investissements Agricole"* (PNIA). Gender-disaggregated data remains scarcely considered in the design of agricultural policies. In addition, social, customary, and religious norms often establish and perpetuate gender differences, with men typically considered the head of the household and the family breadwinner. These norms can hinder women's access to land and other resources. While the Ministry of Agriculture has established a Gender Unit, its activities have been largely stagnant due to a lack of training and sensitization on gender issues among ministry officials.

A deep dive in the agriculture sector showed that gender plays a significant role in food and nutrition security, with women often being marginalized in agricultural production due to limited access to resources like land. Despite this, women still contribute significantly to food security by providing for their families when common resources are depleted. Female-headed households, however, face higher rates of food insecurity due to factors like lower income, limited access to resources, and family responsibilities. Cultural practices can also disadvantage women in terms of food consumption and health.

Gender disparity in agriculture is evident in Niger, where women's involvement has dropped from 40% to 11%, largely due to unequal land distribution. The primary crops grown are identical regardless of the household head's gender, with millet being the most important for both. In general, men own more individual agricultural plots than women – an important factor in the choice of cultivated crops. And they are also more involved horticultural and tree production. Post-harvest management practices vary, with women typically managing their production stock. However, social norms limit women's ability to sell their agricultural stock outside their village, which can impact profitability and reinforce male control over resources. Women's groups and structured agricultural value chains at the local level can help support female producers in selling their harvest.

On livestock, Niger has the largest number of livestock among Sahelian countries, with over 40 million animals. Men own 72% of the livestock, while women own 28%, reflecting inequality similar to that in agriculture. However, women dominate goat farming, owning 54.5% of goats, as it requires less maintenance than cattle farming. Development partners are emphasizing livestock to reduce women's poverty, and programs have positively impacted women's ownership of animals. Inclusion of livestock in agricultural programmes and projects would help fight poverty and support rural household economies in Niger. On the other hand, the fishing and aquaculture sector has not been developed as a specific policy until 2007. This sector, mostly artisanal, directly affects 70,000 people and has an economic potential, despite limited support from PNIA. The sector is however dominated by men in terms of fish capture, while women are more involved in processing and selling, especially in large cities.

In forestry sector, due to poverty and limited access to agricultural land, both men and women increasingly rely on forest resources. Over 90% of Nigerien households rely on wood for energy, leading to a deficit in forest regeneration and fuelwood consumption. Men typically cut tree trunks and branches while women gather dead wood and smaller branches for household use. Nigeriens also utilize gum arabic, doum products, and rônier trees for additional income and subsistence, with men dominating the harvesting and wholesale trade and women focusing on processing and retail marketing.

Overall, women are often less integrated into agricultural value chains than men due to social norms, limited education, lack of working capital, and land issues. They typically participate in lower-value activities, such as collection and primary processing, while men dominate high-value tasks. Efforts are being made to organize women's groups and integrate them into each stage of the value chain, such as the milk chain in Doutchi. The Regional Agricultural Investment Program (PRIA) also emphasizes a gender approach in its interventions. But there remains a need for stronger commitment to integrating gender perspectives in agricultural and rural development programs.

g) Nigeria

In low and middle-income countries, including Nigeria, the gender gap in agriculture remains a persistent issue Women and men exhibit significant differences in vital development indicators such as education, employment, and health. Despite constituting 60 to 70 percent of the rural workforce, women encounter substantial obstacles in formalizing and expanding their businesses. For example, 80-90% of street food vendors in Nigeria are estimated to be women (Nordhagen, 2020). In addition, Nigerian women face numerous challenges, such as low school attendance rates in rural areas, particularly among vulnerable groups, which leads to higher dropout rates for girls nearing the end of primary school. Additionally, Nigeria has the second-highest infant and maternal mortality rates globally, with less than 20 percent of health facilities providing emergency obstetric care and 58 percent of HIV-positive individuals being women (FAO, 2018).

In Nigeria, male farmers still dominate land ownership and control regarding average land size for farming activities, with significant disparities across communities and states. Women's access to land is particularly limited in the Southeast and South regions, where cultural norms and traditions forbid women from owning land. Although the Nigerian Land Use Act of 1978 specified a collectivist framework for property rights systems, women's land tenure concerns persist. Although women inherit land through lease or purchase and sometimes in regions of the country where religion apportions 50 percent of everything men inherit to women, their land rights are still fragile and highly dependent on age and marital status.

Regarding agriculture markets and value chains, FAO' gender country assessment found that women increasingly supply national and international markets with traditional and high-value produce in Nigeria, however men typically control the trade of agricultural products and receive the revenue. This imbalance can be attributed to women's lack of access to and control over critical resources such as land, training, inputs and technologies, equipment, water, and health facilities (FAO Gender Land Database, 2023).Complementary policies must address these challenges and provide women access to essential agricultural resources such as financial markets and information.

Moreover, NTFPs in the country represents an important alternative for seasonal agriculture. However, the collection, processing and marketing of these products is realized informally by member of the family (usually women) and their work is unrecognized.

h) Senegal

In its gender-based analysis of the agriculture and rural development sectors in Senegal, FAO and the ECOWAS Commission (2018) highlighted that Senegal has implemented several strategies and policies to reduce gender inequalities in accessing resources, knowledge, opportunities, and markets.

Senegal has signed and ratified international, continental, and regional conventions promoting women's rights, such as CEDAW, the protocol to the African Charter on Human and Peoples' Rights, and the Supplementary Act on the Rights of Women and Men for Sustainable Development in the ECOWAS region. It has also adopted the Strategy for Equity and Equality of Gender (SNEEG) to eliminate gender inequalities and guarantee women's rights and participation in development.

Gender mainstreaming has been initiated in rural development sectoral policy letters and monitoring instruments. Consequently, Senegal has seen improvements in institutional, macro, and sectoral indicators and gender consideration in policies. Poverty incidence has decreased, and the Human Development Index (HDI) has increased between 2011 and 2014, although the country's ranking has dropped. The Multidimensional Poverty Index and gender inequality index have also decreased.

However, there are still room for improvements and gender disparities persist in terms of resource access and economic participation. In agriculture, which employs roughly 30% of the work force in Senegal, women mostly hold smaller areas than men, and they engage mostly in rainfed rice farming, whereas men dominate in more profitable crops. Also, men primarily care for cattle, sheep, and goats which are working animals, while women focus on traditional poultry and small ruminant breeding. Furthermore, women face challenges in accessing inputs (such as seeds and organic and mineral manure) and using modern production technologies and working animals. With regards to income-generating activities, between 2011 and 2013, women mainly derive their income from millet milling, market gardening, agricultural intensification and arboriculture, while men are more involved in beekeeping. Women are also involved in the traditional processing and conservation of fishery products and environmentally friendly income-generating activities in the forestry sector. However, women face supply difficulties in processing due to unfair competition and national export development policy.

These gender roles in agriculture and forestry contribute to food security and nutrition differences in rural Senegal, with female-headed households (roughly 20% of rural smallholder farms) facing higher food insecurity rates. Women's limited access to factors of production and financing results in less integration and less qualified roles in agricultural value chains. Men, on the other hand, dominate in activities generating higher added value, such as processing and marketing, because of their high purchasing power.

VI. Sexual Exploitation, Abuse, and Harassment (SEAH)

Sexual Exploitation, Abuse, and Harassment (SEAH) are critical issues affecting women and girls globally, including in the SURAGGWA programme countries of Burkina Faso, Chad, Djibouti, Mali, Mauritania, Niger, Nigeria, and Senegal. SEAH encompasses various forms of gender-based violence (GBV), including sexual violence, exploitation, and harassment, perpetrated in both private and public spheres.

Global and Regional Context

Globally, approximately 35% of women have experienced either physical and/or sexual intimate partner violence or non-partner sexual violence in their lifetime. Additionally, 7% of women worldwide have been sexually assaulted by someone other than a partner. In some countries, violence against women is estimated to cost up to 3.7% of their GDP, highlighting the significant economic impact of GBV.[23] The economic impact of GBV is profound, and this includes costs related to healthcare, legal services, and lost productivity.[24,25]

In sub-Saharan Africa, where the GGW countries are located, the prevalence of GBV is notably high. For instance, UNICEF reports that in certain regions, up to 50% of women have experienced physical or sexual violence. Moreover, harmful practices such as female genital mutilation/cutting (FGM/C) remain prevalent, with significant variations across countries[26].

Country-Specific Insights

The following table (Table 3) is meant to provide a snapshot of SEAH-related concerns within the SURAGGWA programme countries as they pertain to agriculture/land restoration/NTFP value chains, albeit not exhaustive. It is a starting point, with the understanding that the National Safeguards Specialist will conduct SEAH screening for country-specific (and site-specific information) during implementation.

Table 3: Snapshot of SEAH Across SURAGGWA Countries

Country	SEAH snapshot relating to agriculture/land restoration/NTFP value chains
Burkina Faso	In Burkina Faso, human trafficking remains a significant issue, with children and women subjected to forced labor and prostitution. Victims often work in agriculture, mining, and domestic services. The government has identified and assisted numerous child-trafficking victims, highlighting the need for continued vigilance in sectors like agriculture and land restoration.
Chad	Specific reports on SEAH within Chad's agricultural and land restoration sectors are scarce. However, the prevalence of human trafficking and forced labor in the region

[23] World Bank (2019). Brief: Gender-Based Violence (Violence Against Women and Girls). URL: https://www.worldbank.org/en/topic/socialsustainability/brief/violence-against-women-and-girls
[24] Idem.
[25] World Bank (2024). Addressing Gender-Based Violence: 16 Days of Activism. URL: https://www.worldbank.org/en/topic/gender/brief/addressing-gender-based-violence
[26] UNICEF (2014). A Statistical Snapshot of Violence Against Adolescent Girls. UNICEF, New York.

	suggests potential risks in these industries, necessitating proactive measures to protect vulnerable populations.
Djibouti	While direct data on SEAH in Djibouti's land restoration and agricultural sectors is limited, the country's strategic location as a transit point for migrants increases the risk of exploitation. Implementing robust safeguarding policies in these sectors is essential to mitigate potential SEAH risks.
Mali	Mali faces challenges related to human trafficking, with children and women subjected to forced labor and sexual exploitation. The agricultural sector, being a significant part of the economy, is susceptible to such abuses, underscoring the importance of integrating SEAH prevention strategies in land restoration and NTFP initiatives.
Mauritania	Mauritania has a history of hereditary slavery practices, affecting marginalized communities. These practices can intersect with land-based industries, highlighting the need for targeted interventions to prevent SEAH in agricultural and land restoration programmes and projects.
Niger	In Niger, human trafficking persists, with children and women exploited in various sectors, including agriculture. The existence of caste-based slavery practices further exacerbates vulnerabilities, making it crucial to address SEAH risks in land restoration and NTFP value chains.
Nigeria	Nigeria's recent World-Bank supported Agro-Climatic Resilience in Semi-Arid Landscapes (ACReSAL) Project acknowledges the risks of sexual exploitation and abuse (SEA) associated with project activities, and highlights the following: *"While gender inequitable norms prevail throughout the country, these vary by region and interact with other structural, community, and individual factors exposing women, girls, and boys to some forms of gender-based violence (GBV) more than others."*[27]
Senegal	Specific data on SEAH within Senegal's land restoration and agricultural sectors is limited. However, the country's commitment to sustainable development necessitates the integration of SEAH prevention measures in these industries to protect vulnerable populations.

Challenges in Data Collection and Reporting

Accurate data on SEAH is often limited due to underreporting, cultural stigmas, and inadequate reporting mechanisms. Victims may fear retaliation or social ostracism, leading to a lack of comprehensive data. For instance, in displacement settings, women may be reluctant to report exploitation due to fear of retribution or lack of trust in reporting systems[28].

[27] World Bank (2021). Nigeria Agro-Climatic Resilience in Semi-Arid Landscapes (ACReSAL) Project.

[28] Associated Press (2024). They fled war in Sudan. Now, women in refugee camps say they're being forced to have sex to survive. URL: https://apnews.com/article/chad-sudan-war-refugees-sexual-exploitation-49b3d344da3573d4abe06bb7c3be965e

SEAH Considerations in SURAGGWA: SEAH remains a pervasive issue in the GGW countries, influenced by cultural, economic, and social factors. Addressing these challenges requires comprehensive strategies, including strengthening legal frameworks, enhancing support services for survivors, and promoting societal change to eliminate harmful practices and norms. For SURAGGWA's approach to addressing SEAH, please refer to Part 2 of this assessment, as well as the Stakeholder Engagement Plan (GRM Chapter) and Chapter 4 of the ESMF.

A note on SEAH-related grievance management and GBV referral pathways: FAO ensures that the programme personnel and the EEs will be trained on prevention of sexual exploitation, abuse, and harassment to achieve maximum prevention of SEAH and GBV. Sensitization campaigns will be carried out to support and catalyze community-driven support measures against SEAH. The programme's Grievance Redress Mechanism will be reinforced to deal effectively with SEAH and GBV incidents (including the development of a procedure to accompany the GRM on SEAH to ensure survivor-centered mechanisms that are gender-responsive and ensure confidentiality, and sensitive and ethical complaint and grievance handling). Referral pathways for GBV will be established and professionals trained for their operationalization, while the FAO E&S and Gender specialist will be engaged in monitoring the process. All SEAH and GBV activities will be inclusive, survivor-centered, and gender-responsive. In case of SEAH/ GBV incidents, the services for survivors will be carefully considered during the implementation.

VII. Recommendations

Focusing on Gender-Transformative Approaches: Many stakeholders are now recognizing that improving gender equality is a central intervention to build climate resilience in the Sahel. A gendered approach to resilience-building requires focusing on the underlying structural inequalities that make women more vulnerable to climate crises. Overcoming these challenges requires the promotion of gender-sensitive policies and programmes that address those underlying causes of gender inequality within the agricultural sector. This includes gender transformative approaches that increase women's access to resources and services, promote gender-responsive agricultural practices, and enhance women's participation in decision-making processes related to climate change adaptation. It is because of how deeply embedded structural and normative codes of exclusion shape gender inequalities that development institutions are beginning to adopt the approach called gender transformation[29].

For SURAGGWA, this means:

- ➔ Making clear that any policies, reports, or activities under Component 3 (focused on institutional strengthening) are gender-sensitive, and that women are engaged within management committees/steering committees/leadership groups/etc.

Time-Savings and Land Use Benefits: Land restoration has been proposed as an important measure since it also contributes to food and water security. Women are often responsible for the collection of firewood,

[29] McOmber, C (2020), "Women and Climate Change in the Sahel", West African Papers, No. 27, OECD Publishing, Paris. https://doi.org/10.1787/e31c77ad-en

water, and other natural resources, and they have a deep understanding of the local environment and the challenges facing it. Land-restoration programmes or initiatives should not be assumed to be gender-neutral. They should, whether new or old, be screened for possible gender-differentiated impacts using sex-disaggregated data. Women and girls bear a disproportionate impact when land is depleted and limited in availability. This is because of their significant involvement in agriculture and food production, reliance on forests, susceptibility to poverty, limited access to/lack of education, and lower legal protections and social status. Despite the recognition and acknowledgment of the interconnectedness between agricultural production, poverty reduction, land preservation, and gender integration, gender aspects are frequently disregarded and overlooked in the execution of land-restoration initiatives. Given the complexity of restoration programmes and projects, their successful implementation requires careful planning and implementation involving all relevant stakeholders[30].

Including women in land restoration processes implies taking into consideration their limitations in terms of extension services and institutional credit, barriers related to land tenure, and constraints in terms of time poverty. As in much of low-income countries, women's ability to own, control, benefit from and access land in the Sahel region is significantly more limited than their male counterparts. Although evidence is still limited, available data from the World Bank show that percentage of land owned by women is always smaller than men in the same countries (see *Figure 5: Land Ownership* in this report). On average, the land women own, or control tends to be smaller and lower quality. In many countries, women's access to, use and control of land depends on their relations to male relatives as wives, mothers, or daughters[31]. Understanding the roles and responsibilities of men and women, along with power relations in land management, is a primary requirement to achieving effective outcomes when combating land degradation and implementing gender-responsive and sustainable land-restoration initiatives.

<u>For SURAGGWA, this means</u>:

- ➔ Extensive consultation during design (refer to SEP for further details) to better align programme activities;
- ➔ Iterative consultation and engagement via the national-level safeguards specialists (and delegated implementation teams, for more day-to-day feedback) to ensure the localized gender context is accounted for in all programme activities;
- ➔ Prioritizing land restoration activities and approaches under Components 1 and 2 which increase women's time savings (ability to spend time on activities *other* than land preparation/etc.) and improve their ability to productively use land in a sustainable way;
- ➔ Engaging and providing training to women-heavy/women-led community groups (particularly under Component 2), and supporting their involvement and voice within decision-making entities (be those community-level groups or higher-level institutions);
- ➔ Focusing on NTFP value chains and areas in which women are highly engaged/reap most benefits from in terms of their livelihoods and wellbeing (Component 2).

[30] UN Environment Programme (2019). "Women in the Sahel lead on the frontlines of climate change".
[31] Sida (2015). "Women and Land Rights Brief". Accessed March 2023: https://cdn.sida.se/publications/files/-women-and-land-rights.pdf

Part Two

Approach to Gender Based Violence and Sexual Exploitation and Abuse

Whilst Gender Based Violence and Sexual Exploitation and/or Abuse are not expected to increase because of the programme's activities, they are inherent risks within the SURAGGWA countries. Given this, the programme will address issues through (i) sensitization and consultation; (ii) clear codes of conduct for all involved with the programme; (iii) SEAH screening, based on the GCF SEAH risk screening guidance, for each country (this will be done at a broader country level within the country-specific Gender Action Plans); and (iv) a specialized procedure for handling sensitive SEAH cases in instances of a grievance (for full details on SEAH grievance redress, please refer to the GRM chapter of the Stakeholder Engagement Plan). A short overview of how SEA grievances will be managed is as follows:

- All grievances of misconduct (such as allegations of fraud or other corrupt practices, harassment or sexual exploitation and abuse) by FAO programme or country office employees, and/or by others as a result of the programme's activities, are to be submitted directly to the Office of the Inspector General (OIG) by: (i) the complainant via the OIG hotline; or (ii) if someone from the office happens to find out about a case (including the PSEA focal point), that person will contact OIG immediately. OIG is responsible for investigating allegations of misconduct.[32] In case of a sexual exploitation and abuse grievance, OIG is responsible for referring the survivor to relevant services (medical, psychological, social, legal, etc.). SEA grievances are handled in a prompt and strictly confidential manner.

Gender Action Plan (GAP)

Programme-level GAP: Based on the findings and recommendations of this Gender Assessment, the SURAGGWA Gender Action Plan was prepared. The GAP provides specific actions/targets under each component to assist women in building their resilience to climate change and how to enhance their livelihoods opportunities in the face of climate risks. Gender considerations have been mainstreamed into the SURAGGWA programme, particularly with the focus on women-dominated NTFP value chains and land-restoration techniques intended to reduce women's time poverty, and these are highlighted throughout the GAP via targets around (i) women's participation; (ii) women's time savings; (iii) gender-sensitivity of reports/policies/governance actions, and more. The SURAGGWA programme and related GAP have been designed to ensure that equitable participation and benefits from the programme investment accrue for women. Please refer to the Excel file (Gender Action Plan) in the Annex 8 Folder for full details.

Country-level GAP: To account for different country contexts, a country-level GAP will be prepared by the National Social Safeguards Specialist within the first six months of their onboarding. The document will be aligned with the programme-level GAP and further refine targets at country-level based on the local

[32] Ensure to inform programme stakeholders that to report possible fraud and unacceptable behaviour including sexual exploitation and abuse, they can contact OIG by confidential hotline (online form & by phone): fao.ethicspoint.com, or by e-mail: Investigations-hotline@fao.org

situation. The country-level GAP will include (but not be limited to): (i) details on SEAH and related grievance redress within the country (utilizing the SEAH screening form guidance of GCF, and in alignment with FAO policies on SEAH and grievance redress); (ii) intersectionality/overlap of ethnic minority groups and women within the given country, highlighting any differentiated needs; (iii) country-specific targets (based on the programme-level GAP and local country context); and (iv) monitoring and implementation timeline for the country-level GAP (including more specific annual/biannual associated milestones to facilitate the tracking of progress towards the gender related targets and goals). Please note that country level GAPs will include targets that are set for female headed households, IP women, youth, and other traditionally marginalized groups.

GAP Budget: For GAP-specific budget (and budget related to gender activities mainstreamed into programme components), please see the following table:

	Burkina Faso	Chad	Djibouti	Mali	Mauritania	Niger	Nigeria	Senegal	Regional	Total
Component/output	Annual (USD)	Annual (USD)	Annual (USD)	Annual (USD)	Annual (USD)	Annual (USD)	Annual (USD)	Annual (USD)	Annual (USD)	Annual (USD)
Gender and safeguards related documents and implementation	295,000	295,000	275,000	370.000	295,000	410,000	410,000	295,000	160.000	2,650,000
Social Safeguards Specialist (shared between comp 1 and comp 2)	270,000	324.000	234.000	198.000	270.000	198.000	270.000	243.000	570.000	2.577.000
	565.000	619.000	509.000	453.000	565.000	568.000	680.000	538.000	730.000	**5,227,000**

In total, the programme has allocated approximately 5.23 million for gender/SEAH related safeguards activities (covering development of the country-specific plans and their implementation), 3.5% of which comes from GCF financing. Budget for development and implementation of country-level GAPs can be found in the overall programme budget under Components 1 and 2 (particularly under activities 1.1.2.5 and 2.1.1.1, for ease of reference), but the issue of gender has also been mainstreamed throughout).

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In addition to the above, references include the eight country-specific gender profiles (submitted to complement this summary assessment).

GCF Expected Outcomes	Indicators and Targets ANNEX 8	Timeline	Responsibilities	Costs
***For activities, responsible people will include: Regional E&S Specialist; National Social Safeguards & Gender Specialist; National M&E Specialist; and/or the Land Tenure Team (supporting during the first 2 years of implementation for each country). All work will require close liaising with Value Chain specialists and site-level implementation teams.*				
ARA1 Most vulnerable people and communities	Core 2: Direct and indirect beneficiaries reached Direct: 1,881,002 (total); of which 937,040 women. Baseline: 0 Target: 937,040 (50% women) Indirect: 3,861,925; of which 1,923,273 women. Baseline: 0 Midterm Target: 35% of final amount Target: 1,923,273 (50% women)	By end of programme year 10	FAO Office of Evaluation and ESA/ESP TA with support from PCU/CIU, Gender specialist, and/or others.	Reaching women beneficiaries requires 50% of total programme budget, i.e. USD 99,561,470
	Supplementary 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options Baseline: 0 Midterm Target: 35% of final amount Target: 480,330 (239,616 women (50%))	By end of programme year 10	FAO Office of Evaluation and ESA/ESP TA with support from PCU/CIU, Gender specialist, and/or others	Reaching women beneficiaries requires 50% of total programme budget related , i.e. USD 89,666,777
	Supplementary indicator 2.5: Beneficiaries (female/male) adopting innovations that strengthen climate change resilience Baseline: 0 Midterm Target: 35% of final amount Target: 1,544,770 million (769,309 women (50%))	By end of programme year 10	FAO Office of Evaluation and ESA/ESP TA with support from PCU/CIU, Gender specialist, and/or others	Reaching women beneficiaries requires 50% of total programme budget, i.e. USD 99,561,470

Activities	Indicators and Targets*	Timeline	Responsibilities	Costs (USD)
Component 1: Large scale landscape restoration to increase climate change resilience and mitigation, for and by local communities.				
Output 1.1: Community groups are organized and trained for land restoration activities (aligned with their NTFP value chain priorities – see component 2) and monitoring	**Share of women participating in community groups/organizations. Baseline: 0% Target: 40%**	By end of programme year 6	programme Implementation Team, Social Safeguards Specialist	1,240,000
Activity 1.1.1: Identify potential sites for restoration and to inform resource base for value chain priorities based on analysis of climate data, GIS maps and potential for impact with local authorities and community.	Site identification is gender sensitive and prioritizes areas that have higher chance for positive gender impacts Baseline: 0% Target: 100%	By end of programme year 2 (for each country)	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget
Activity 1.1.2: Identify and mobilize communities for participatory selection of the specific sites, ensuring inclusion of women.	Share of women participating and/or consulted in identification of sites Baseline: 0% Target: 50%	By end of programme year 2 (for each country)	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget
Activity 1.1.3: Train programme's restoration teams and community management organizations.	Share of women trained as part of the restoration teams and/or community management committees Baseline: 0% Target: 50%	By programme year 6	programme Implementation Team, Social Safeguards Specialist	USD 1,080,000
Output 1.2. Native seed supply systems have been strengthened to ensure the availability of genetically appropriate seeds that provide increased climate resilience.	**Share of women's participation in seed supply systems/networks Baseline: 0% Target: 50%**	**Timeline**	**Responsibilities**	**Costs (USD)**
Activity 1.2.1: Identify and train community technicians on good quality and quantities of restoration seeds and native seed supply.	Share of women in participating seedling production Baseline: 0% Target: 20% for technicians; 50% for organization of community members	By the end of programme year 4.	programme Implementation Team, Social Safeguards Specialist	USD 948,800
Activity 1.2.2: Identify, organize, and train community members involved in seed and seedling production for land restoration activities.				Included in the programme regular budget estimated
Activity 1.2.3: Establish, equip, and operate community nurseries for seedling production and dissemination.	Share of women participating in seed networks (supply and exchange) Baseline: 0% Midterm target: 17.5% Target: 50%	By the end of programme year 10.	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget estimated
Activity 1.2.4: Local communities' seed supply of native seeds is integrated into the national seed system.	N/A	N/A	N/A	N/A
Output 1.3: Targeted highly degraded lands restored through arrangements for rainwater harvesting and enhanced soil permeability and through local communities engaging in planting of seedlings and direct seeding.	**Share of women that have noticed time savings as a result of these land preparation/restoration practices. Baseline: 0% Target: 50%**	**Timeline**	**Responsibilities**	**Costs (USD)**
Activity 1.3.1: Highly degraded land restoration plans prepared and implemented through mechanized, animal, and manual traction techniques.	Share of women that have noticed time savings as a result of these land preparation/restoration practices Baseline: 0% Midterm target: 17.5% Target: 50%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget
Activity 1.3.2: Sowing and planting in prepared sites with communities/villages.	Share of women participating in agricultural activities in prepared sites Baseline: 0% Midterm target: 14% Target: 40%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget

Activity 1.3.3: Training of community members in monitoring & maintenance.	Share of women participating in training for monitoring and maintenance (M+M) of sites Baseline: 0% Midterm target: 14% Target: 40%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget
Output 1.4: Targeted moderately degraded lands planted with a range of species to restore and enrich the landscapes of agro-forestry, agro-ecology and sylvo-pastoral systems, biomes and protected forest ecosystems.	**Share of women that have noticed time savings as a result of these land preparation/restoration practices. Baseline: 0% Midterm target: 17.5% Target: 50%**	**Timeline**	**Responsibilities**	**Costs (USD)**
Activity 1.4.1: Sub-national moderately degraded land restoration plans prepared and implemented	Share of women that have noticed time savings as a result of these land preparation/restoration practices. Baseline: 0% Midterm target: 17.5% Target: 50%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget
Output 1.5: Rural community capacities strengthened for sustainable management and restoration, and verification of restoration results	**Female share of local restoration technicians whose technical capacities have been strengthened for land restoration monitoring. Baseline: 0% Midterm target: 7% Target: 20%**	**Timeline**	**Responsibilities**	**Costs (USD)**
Activity 1.5.1: Village technicians trained and equipped in managing restoration areas and verifying restoration results, in a participatory manner.	Female share of technicians Baseline: 0% Midterm target: 7% Target: 20%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	USD 300,000

Component 2: Supporting the development of climate-resilient, low emission non-timber forest product (NTFP) value chains benefiting vulnerable communities' livelihoods				
Output 2.1. Climate resilient and low carbon production and processing practices in selected NTFP value chains adopted by Producer Organizations and MSMEs.	**Share of women receiving training/capacity development for NTFPs Baseline: 0% Midterm target: 21% Target: 60%**	**Timeline**	**Responsibilities**	**Costs (USD)**
Activity 2.1.1: Train and provide TA to POs and MSMEs in selected NTFP value chains for enhanced organizational and managerial capacities (registration, structuring according to OHADA, training on management, administration etc.)	Percentage of all female actors in the NTFP producer organizations (POs) and MSMEs receiving related training/capacity development Baseline: 0% Midterm target: 35% Target: 100%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget, GAP activity not require significant specific budget
Activity 2.1.2: Train and coach local POs and MSMEs in sustainable production and collection practices to enhance NTFP quality and availability	Percentage of all female actors in the NTFP producer organizations (POs) and MSMEs receiving related training/capacity development Baseline: 0% Midterm target: 35% Target: 100%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget, GAP activity not require significant specific budget
Activity 2.1.3: Train and coach POs and MSMEs to improve the processing practices (including packaging and labelling) of the selected products	Female share of trainees for NFTP processing. Baseline: 0% Midterm target: 21% Target: 60%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget, GAP activity not require significant specific budget
Output 2.2: Increased access to markets for smallholder NTFP actors (including micro- and small enterprises)	**Increase in real gross sales generated by business plans in execution for women-led businesses Baseline: 0% Target: 20%**	**Timeline**	**Responsibilities**	**Costs (USD)**
Activity 2.2.1: Identify NTFPs with market and sustainable production potential that can be integrated in land restoration efforts funded under Component 1	n/a	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget, GAP activity not require significant specific budget
Activity 2.2.2 : Support to marketing and branding of the NTFP (promotional activities, nutritional value pamphlets of NTFP, organization of trade fairs, TV show, etc.).	Share of women/women-led entities in NTFPs benefiting from the marketing/branding. Baseline: 0% Midterm target: 26% Target: 75%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget, GAP activity not require significant specific budget
Activity 2.2.3: Support POs and MSMEs in business plan development and management.	Female share of trainee beneficiaries in business plan development and management. Baseline: 0% Midterm target: 21% Target: 60%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget, GAP activity not require significant specific budget
Activity 2.2.4: Enhance access to national, regional and international markets for NTFP, including through the setting up of norms and standards to improve the marketing of the NTFP.	Female share of individuals involved with setting up of norms and standards to improve the marketing of NTFP. Baseline: 0% Midterm target: 17.5% Target: 50%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget, GAP activity not require significant specific budget
Activity 2.2.5: Identify, train and equip POs and MSMEs with emission-lowering technologies and equipment to enhance NTFP processing.	Female share of trainee beneficiaries. Baseline: 0% Midterm target: 21% Target: 60%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget, GAP activity not require significant specific budget
Output 2.3 Suitable credit and insurance products designed for smallholder NTFP value chain actors (PO, cooperatives, MSMEs) and increased stakeholder capacity for using these financial products in their NTFP value chain activities	**n/a**	**Timeline**	**Responsibilities**	**Costs (USD)**
Activity 2.3.1: Train and sensitize value chain actors on financial institutions' offerings, financial literacy and financial record keeping.	Share of women as trainees in financial trainings. Baseline: 0% Midterm target: 17.5% Target: 50%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget

Activity 2.3.2: Technical assistance provided to develop five new financial products (of which at least one insurance product) tailored to agriculture and NTFP value chains with financial institutions, including those collaborating with iGREENFIN.	Share of women consulted in the development of new products by FIs. Baseline: 0% Midterm target: 17.5% Target: 50%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget
Activity 2.3.3: Train staff (loan officers and others) from financial institutions, including those collaborating with iGREENFIN, on how to assess agricultural and NTFP value chain risks and climate-related risks (national and regional level)	Share of female FI personnel as trainees in risk assessments. Baseline: 0% Midterm target: 7% Target: 20%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget
Activity 2.3.4 : Facilitate linkages between last-mile providers of finance (producer organizations, village savings groups, microfinance networks) and financial institutions, including those participating in iGREENFIN.	Share of female-led last mile providers involved. Baseline: 0% Midterm target: 10.5% Target: 30%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget
Activity 2.3.5 : Pilot access to credit through digital financial solutions in at least one country and facilitate scale up, if successful.	Share of women with bank accounts who are accessing the pilot digital finance solutions. Baseline: 0% Midterm target: 17.5% Target: 50%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget
Activity 2.3.6 : Improve knowledge management and exchanges to increase adoption of best practices across local financial institutions.	Share of women involved in knowledge management and exchange activities of FI solutions. Baseline: 0% Midterm target: 10.5% Target: 30%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget

Component 3: Strengthening the Great Green Wall institutions at country and regional level				
Output 3.1: GGW Land restoration monitoring system at national and regional level upgraded and functional	**100% include special considerations for gender equality and women's empowerment.**	**Timeline**	**Responsibilities**	**Costs (USD)**
Activity 3.1.1: Development, testing and deployment of National GGW land restoration monitoring tools and system	100% include special considerations for gender equality and women's empowerment.	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget
Activity 3.1.2 : Transfer knowledge and skills for the use of the GGW Land restoration monitoring system by national and regional authorities.	100% include special considerations for gender equality and women's empowerment.	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget
Activity 3.1.3: Development and deployment of regional multi-stakeholder Monitoring platform	100% include special considerations for gender equality and women's empowerment.	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget
Activity 3.1.4: Building the capacity of NAGGW and PAGGW to establish databases of ongoing restoration programme s and programmes contributing to national and regional GGW results	100% include special considerations for gender equality and women's empowerment.	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget
Activity 3.1.5 : Prepare regulatory national frameworks for the surveillance of all GGW labelled interventions in all GGW countries	100% include special considerations for gender equality and women's empowerment.	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget
Activity 3.1.6 : Develop operational partnerships between the GGWi and scientific and technical institutions in the region(such as universities, research institutes, CSEs, Aghrymet, ACMAD) on issues of ecological monitoring and adaptation to climate change.	100% include special considerations for gender equality and women's empowerment.	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget
Activity 3.1.7: Strengthen the capacities of national stakeholders in the use of the relevant tools developed by FAO (for example, Collect Earth and Africa OpenDeal database), and other partners such as the Ecological Monitoring Centre (EMC), the Sahara and Sahel Observatory (OSS) and CILSS.	100% include special considerations for gender equality and women's empowerment.	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget
Output 3.2: National and Regional GGW institutions planning and coordination capacities strengthened	**Strengthening of planning and coordination capacity is gender-sensitive. 100% gender sensitive**	**Timeline**	**Responsibilities**	**Costs (USD)**
Activity 3.2.1: Establish National GGW Coalitions to promote coherent coordination and planning at country-level	Percentage of women involved with the National GGW Coalitions. Baseline: 0% Target: 30%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget
Activity 3.2.2: Prepare and issue planning and regulatory frameworks for the coordination of all GGW aligned interventions (programme s, programmes, activities etc…)	Percentage of regulatory frameworks for coordinating GGW interventions which are gender sensitive. Baseline: 0 Target: 100%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget
Output 3.3: National and regional climate change institutional capacities/frameworks are strengthened to integrate land restoration investments in climate change adaptation and mitigation programmes	**n/a**	**Timeline**	**Responsibilities**	**Costs (USD)**
Activity 3.3.1: Strengthen public- and private sector understanding of and capacity to engage with carbon markets for climate change adaptation and mitigation through land restoration	n/a	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget
Activity 3.3.2: Pilot the establishment of domestic carbon accounting framework that integrates agriculture and forestry in Nigeria, and identify additional countries for potential replication.	n/a	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget
Output 3.4 Leverage GGW knowledge management and communication capacities for increasing support to climate-resilient land restoration	**n/a**	**Timeline**	**Responsibilities**	**Costs (USD)**
Activity 3.4.1: Develop innovative and robust methods for evaluating/demonstrating resilience benefits of climate change investments in land restoration and NTFP value chains in GGW	Percentage of women participating in development of these methods. Baseline: 0% Target: 50%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget

Activity 3.4.2 : Communication, visibility and dissemination of knowledge.	Percentage of communication, visibility, and dissemination of knowledge that is gender sensitive. Baseline: 0% Target: 100%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget
Activity 3.4.3 Train communication specialists from GGW's national structures to implement their communication plan and develop tools adapted to different target groups, particularly women and youth.	Share of women involved as communication specialists in GGW structures. Baseline: 0% Target: 20%	By end of programme year 10	programme Implementation Team, Social Safeguards Specialist	Included in the programme regular budget
Project overall management for GAP	**n/a**	**Timeline**	**Responsibilities**	**Costs (USD)**
Gender and Social Safeguards Specialists	n/a	By end of programme year 10	FAO	2,577,000
Development & Implementation of Country-Level GAPs and safeguards documents	8 Country-level GAPs (including country specific target/ milestones)	Q2 year 1 for the development of country-level GAPs Throughout the implementation	FAO	1,410,000
National and regional WCGs workshops to discuss gender issues and Training of WCGs	8 workshops (1 per country). Activity should involve 30 participants over 5 days in each country. The result if the country specifc workshops will be used for a larger regional workshop/conference.	During development of country-level programme	FAO	1,240,000